
Graduate Algebra

Fall 2015 and Spring 2016

Drew Armstrong

Contents

Fall 2015: MTH 761	5
1 Galois theory and the origins of abstract algebra (August 25, 2015)	5
2 The Fundamental Theorem of Galois Theory and Galois connections (August 27, 2015)	7
3 Posets, lattices, Galois connections (September 1, 2015)	12
4 Groups, automorphisms, subgroups (September 3, 2015)	17
5 Subgroup lattices, products, cyclic groups (September 8, 2015)	22
6 Categories and homomorphisms (September 10, 2015)	26
7 Galois connections and quotients (September 15, 2015)	31
8 Quotient groups and normal subgroups (September 17, 2015)	37
9 Normal subgroups and quotient homomorphisms (September 22, 2015)	41
10 The First Isomorphism Theorem and products of groups (September 24, 2015)	46
11 Direct, semidirect, Zappa–Szép products (September 29, 2015)	51
12 Isomorphism theorems and subgroup lattices (October 1, 2015)	56
13 Simple groups and composition series (October 6, 2015)	60
14 Jordan–Hölder, Zassenhaus, Schreier refinement (October 13, 2015)	65
15 Exact sequences and group extensions (October 15, 2015)	71
16 Abel–Ruffini and the symmetric groups (October 20, 2015)	74
17 Functors, natural transformations, free objects (October 22, 2015)	78
18 Midterm review (October 27, 2015)	83
19 Equivalence of categories and group representations (November 3, 2015)	87
20 Group actions, orbits, stabilizers (November 5, 2015)	92
21 The Fundamental Theorem of G-sets and double cosets (November 10, 2015)	97
22 Conjugation, centralizers, the class equation (November 12, 2015)	104
23 Cauchy’s theorem and Sylow theory (November 17, 2015)	108
24 Sylow conjugacy and counting (November 19, 2015)	112

25 Finite simple groups and linear groups (December 1, 2015)	118
26 Finite abelian groups and invariant factors (December 3, 2015)	123
27 Final review (December 8, 2015)	127
 Spring 2016: MTH 762	 132
28 Abelian groups, direct sums, rings (January 12, 2016)	132
29 Rings as endomorphism rings (January 14, 2016)	137
30 Ideals, number fields, Riemann surfaces (January 19, 2016)	142
31 Ideals, quotient rings, isomorphism theorems (January 21, 2016)	146
32 From ring isomorphism theorems to modules (January 26, 2016)	151
33 Motivating modules and the category $R\text{-Mod}$ (January 28, 2016)	154
34 The category $R\text{-Mod}$ (February 2, 2016)	158
35 The category $R\text{-Mod}$ (cont.) (February 4, 2016)	162
36 Modules over a PID (February 9, 2016)	168
37 Free modules (February 11, 2016)	172
38 Bases of free modules (February 16, 2016)	175
39 Independent and spanning sets; fields of fractions (February 18, 2016)	181
40 Localization, torsion, free modules (February 23, 2016)	183
41 Projective modules and the key lemma for PIDs (February 25, 2016)	189
42 Finitely generated modules over a PID, Part I (March 1, 2016)	193
43 Midterm review (March 3, 2016)	197
44 Free resolutions and matrix notation (March 17, 2016)	205
45 Matrix multiplication and change of basis (March 22, 2016)	210
46 R-algebras and elementary matrices (March 24, 2016)	214
47 Row reduction and normal forms (March 29, 2016)	218
48 Smith normal form (March 31, 2016)	222
49 Smith normal form and the structure theorem (April 5, 2016)	229

50 Applications of the structure theorem (April 7, 2016)	233
51 Canonical forms for matrices (April 12, 2016)	237
52 Minimal and characteristic polynomials (April 14, 2016)	243
53 Morita equivalence and base change (April 19, 2016)	250
54 Extension of scalars and uniqueness over a PID (April 21, 2016)	254
55 Final review (April 29, 2016)	258

Fall 2015: MTH 761

1 Galois theory and the origins of abstract algebra

August 25, 2015

BEGIN.

Prior to 1830, “algebra” was dedicated to solving polynomial equations. For example, if

$$ax^2 + bx + c = 0,$$

then

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Over time this subject became too hard and progress stopped. (This often happens in mathematics.) In order to move forward, Évariste Galois (1830) found a completely new way to look at the subject. It took a while for everyone else to catch up. By 1930, Galois’ ideas had completely transformed the subject of “algebra” (solving polynomial equations) into “abstract algebra” (the study of groups, rings, fields, modules, etc.).

I will describe Galois’ ideas in modern language.

Let $f(x) \in K[x]$ be a polynomial in one variable with coefficients in a field K . We will see later that there exists a field extension $K \subseteq E$ such that

- $f(x)$ “splits” over E , i.e., there exist $a_1, a_2, \dots, a_n \in E$ such that

$$f(x) = (x - a_1)(x - a_2) \cdots (x - a_n).$$

- E is “generated over K ” by the roots $a_1, \dots, a_n$, i.e., the smallest subfield of E containing

$$K \cup \{a_1, a_2, \dots, a_n\}$$

is E itself.

This E is unique up to isomorphism and we call it the *splitting field* of the polynomial $f(x)$.

Now the problem of algebra, pre-1830, can be stated as follows:

To “solve” the equation $f(x) = 0$ we should find a chain of field extensions

$$K = F_0 \subseteq F_1 \subseteq F_2 \subseteq \cdots \subseteq F_d = E$$

such that each F_i is obtained from F_{i-1} by adjoining an element $\alpha \in F_i$ such that $\alpha \notin F_{i-1}$ but $\alpha^r \in F_{i-1}$ for some power r . We say

$$F_i = F_{i-1}(\alpha).$$

Basically, to get from F_{i-1} to F_i we allow ourselves to take “the r th root” of α . Since this doesn’t exist in F_{i-1} , we end up with a bigger field F_i .

Now by induction each element of E , including the roots $a_1, \dots, a_n$, can be expressed via elements of the base field K using only field operations $+, -, \times, \div$ and pure radicals $\sqrt{}, \sqrt[3]{}, \sqrt[4]{}, \dots$.

In other words, we have “solved the equation $f(x) = 0$ by radicals.”

OK, but this is just language. The real problem is to find the field extensions

$$K = F_0 \subseteq F_1 \subseteq \dots \subseteq F_d = E,$$

or to prove that no such fields exist.

Usually they won’t exist. Abel and Ruffini had recently proved that a general polynomial equation of degree ≥ 5 is *not solvable* by radicals.

But Abel’s proof was incredibly complicated and hard to read. Galois’ contribution was to significantly clean up the language so we can see why the theorem is true.

To do this, Galois came up with a revolutionary idea: He considered the collection of invertible functions

$$\varphi : E \longrightarrow E$$

satisfying the properties

- $\varphi(a + b) = \varphi(a) + \varphi(b)$ for all $a, b \in E$,
- $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in E$,
- $\varphi(a) = a$ for all $a \in K$.

We will call these “automorphisms” of the field extension E/K and denote the collection by

$$\text{Aut}(E/K).$$

The important thing about automorphisms is that they can be *composed*: given $\varphi, \mu \in \text{Aut}(E/K)$ we have $\varphi \circ \mu \in \text{Aut}(E/K)$.

Galois used the word “group” for this kind of structure and he showed that the group operation on $\text{Aut}(E/K)$ encodes everything we want to know about the structure of the field extension $K \subseteq E$.

To state Galois’ theorem we need a bit of notation:

- For any subgroup $H \leq \text{Aut}(E/K)$, define the set

$$E^H := \{a \in E : \varphi(a) = a \text{ for all } \varphi \in H\}.$$

Note that this is a subfield of E containing K .

- For any intermediate field $K \subseteq F \subseteq E$, define the set

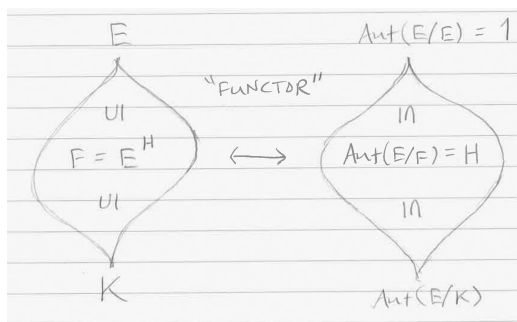
$$\text{Aut}(E/F) := \{\varphi \in \text{Aut}(E/K) : \varphi(a) = a \text{ for all } a \in F\}.$$

Note that this is a subgroup of $\text{Aut}(E/K)$.

Theorem (Fundamental Theorem of Galois Theory). Let $f(x) \in K[x]$ have splitting field E . Under mild conditions on $f(x)$ [its irreducible factors must be “separable”; never mind what that means], the maps

$$H \mapsto E^H \quad \text{and} \quad F \mapsto \text{Aut}(E/F)$$

set up an order-reversing bijection between subgroups of $\text{Aut}(E/K)$ and intermediate fields $K \subseteq F \subseteq E$.



In particular, a chain of subfields

$$K = F_0 \subseteq F_1 \subseteq F_2 \subseteq \cdots \subseteq F_d = E,$$

in which $F_i = F_{i-1}(\alpha)$ for some α such that $\alpha \notin F_{i-1}$ and $\alpha^r \in F_{i-1}$, corresponds to a chain of subgroups

$$\text{Aut}(E/K) = G_0 \supset G_1 \supset G_2 \supset \cdots \supset G_d = 1$$

in which each quotient G_{i-1}/G_i is abelian.

[By analogy, we will say that any such group is “solvable.”]

Finally, if $f(x) \in K[x]$ is a generic polynomial of degree n with splitting field E , then its “Galois group”

$$\text{Aut}(E/K)$$

is isomorphic to the symmetric group S_n of all permutations of n things.

We will prove later that the group S_n is not solvable when $n \geq 5$, from which the Abel–Ruffini theorem follows. In particular, we will show that the alternating subgroup $A_n \leq S_n$ is simple for all $n \geq 5$.

Special case: The group A_5 is isomorphic to the group of rotational symmetries of the regular icosahedron.

That’s quite a revolution.



2 The Fundamental Theorem of Galois Theory and Galois connections

August 27, 2015

Last time we discussed the first theorem of abstract algebra.

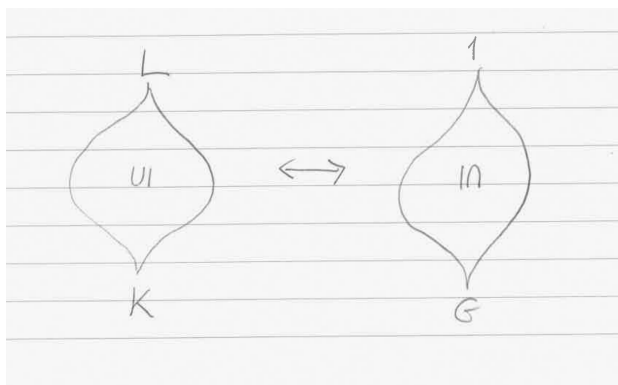
Theorem (F.T.G.T. (~ 1830)). Let $f(x) \in K[x]$ be a polynomial whose irreducible factors are “separable” (never mind what that means; it’s automatically true when $\text{char } K = 0$). Let L be the splitting field of $f(x)$ and consider the group

$$G = \text{Aut}(L/K).$$

Then the maps

$$H \mapsto L^H, \quad F \mapsto \text{Aut}(L/F)$$

set up an order-reversing bijection between subgroups of G and intermediate fields $K \subseteq F \subseteq L$.



The full proof of this theorem is quite involved, but it’s worth noting that certain aspects of the proof are “trivial,” i.e., have nothing whatsoever to do with groups and fields.

The concept of an “abstract Galois connection” will recur throughout the course, so let’s discuss it now.

Here I will follow George Bergman’s book *Invitation to General Algebra and Universal Constructions*, Chap. 6.5.

Let S and T be sets and let $R \subseteq S \times T$ be an arbitrary relation. We will write

$$“aRb” \iff (a, b) \in R.$$

Given any subset $A \subseteq S$ we will define

$$A^* := \{t \in T : aRt \ \forall a \in A\} \subseteq T,$$

and similarly for all $B \subseteq T$ we define

$$B^* := \{s \in S : sRb \ \forall b \in B\} \subseteq S.$$

This gives us a pair of maps between the power sets

$$* : 2^S \longrightarrow 2^T, \quad * : 2^T \longrightarrow 2^S.$$

This is called an *abstract Galois connection*. Let’s explore the basic properties.

(1) $*$ is order-reversing. That is, for all $A_1, A_2 \subseteq S$ we have

$$A_1 \subseteq A_2 \implies A_2^* \subseteq A_1^*,$$

and for all $B_1, B_2 \subseteq T$ we have

$$B_1 \subseteq B_2 \implies B_2^* \subseteq B_1^*.$$

[Since the statements are dual, we only need to prove the first.]

Proof. Assume that $A_1 \subseteq A_2 \subseteq S$ and consider any $t \in A_2^* \subseteq T$. By definition this means that

$$aRt \quad \forall a \in A_2.$$

But since $A_1 \subseteq A_2$ this implies that

$$aRt \quad \forall a \in A_1,$$

and hence $t \in A_1^*$. □

Now we consider the compositions

$$** : 2^S \longrightarrow 2^S, \quad ** : 2^T \longrightarrow 2^T.$$

[Sorry for the abuse of notation.]

(2) $**$ is monotone increasing. That is, for all $A \subseteq S$ and $B \subseteq T$ we have

$$A \subseteq A^{**} \quad \text{and} \quad B \subseteq B^{**}.$$

Proof. Given $a \in A$ we want to show that $a \in A^{**}$. Recall that

$$A^{**} = (A^*)^* = \{s \in S : sRt \quad \forall t \in A^*\}.$$

Hence we want to show that

$$aRt \quad \forall t \in A^*.$$

So consider any $t \in A^*$. By definition this means that sRt for all $s \in A$, and in particular we have aRt . Since $t \in A^*$ was arbitrary we obtain

$$aRt \quad \forall t \in A^*$$

as desired. □

(3) For all $A \subseteq S$ and $B \subseteq T$ we have

$$A^{***} = A^* \quad \text{and} \quad B^{***} = B^*.$$

Proof. Consider any $A \subseteq S$. By (2) we have $A \subseteq A^{**}$. Then by (1) we have

$$(A^{**})^* \subseteq A^*, \quad A^{***} \subseteq A^*.$$

Conversely, applying (2) to the set $A^* \subseteq T$ gives

$$A^* \subseteq (A^*)^{**}, \quad A^* \subseteq A^{***}.$$

□

The function $**$ is often called a “closure operator.” Let’s define this.

Definition. Given a set U we say that

$$\text{cl} : 2^U \longrightarrow 2^U$$

is a closure operator if for all subsets $X, Y \subseteq U$ we have

- i) $X \subseteq \text{cl}(X)$,
- ii) $X \subseteq Y \implies \text{cl}(X) \subseteq \text{cl}(Y)$,
- iii) $\text{cl}(\text{cl}(X)) = \text{cl}(X)$.

We say that the set $X \subseteq U$ is *closed* if $\text{cl}(X) = X$.

(4) $** : 2^S \rightarrow 2^S$ and $** : 2^T \rightarrow 2^T$ are closure operators.

Proof. Part (i) follows from (2). Given $A_1, A_2 \subseteq S$, we apply part (1) twice to get

$$A_1 \subseteq A_2 \implies A_2^* \subseteq A_1^* \implies A_1^{**} \subseteq A_2^{**}.$$

Given $A \subseteq S$ we apply (3) to get

$$(A^{**})^{**} = (A^{***})^* = (A^*)^* = A^{**}.$$

□

Great, now what?

Note that the sets

$$S, \quad S^*, \quad T, \quad T^*$$

are all $**$ -closed. Indeed since $S^{**} \subseteq S$ (by definition) and $S \subseteq S^{**}$ (by part (2)) we must have

$$S^{**} = S,$$

so S is $**$ -closed. And by part (3) we have

$$(S^*)^{**} = S^{***} = S^*,$$

so S^* is $**$ -closed.

Furthermore, note that S^* and T^* are minimal among $**$ -closed sets. Indeed, if $B \subseteq T$ is $**$ -closed, i.e.,

$$B^{**} = B,$$

then since $B^* \subseteq S$ (by definition), part (1) implies that

$$S^* \subseteq (B^*)^* = B^{**} = B,$$

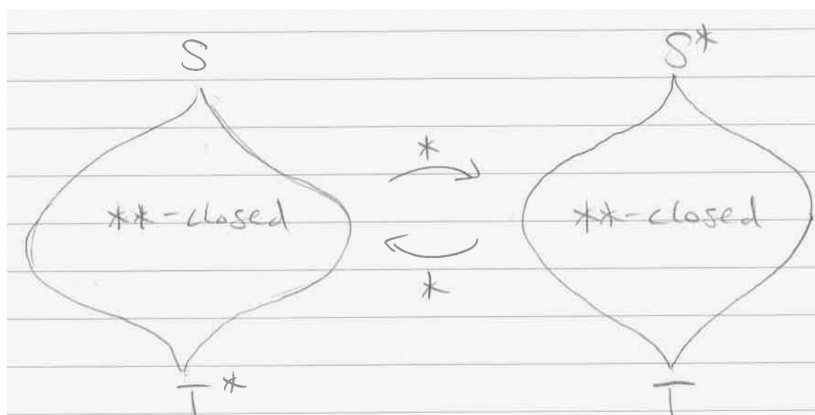
as desired.

Finally, putting everything together gives the following.

Theorem (Abstract Galois Connection). *The maps*

$$* : 2^S \longrightarrow 2^T \quad \text{and} \quad * : 2^T \longrightarrow 2^S$$

*set up an order-reversing bijection between $**$ -closed subsets of S and $**$ -closed subsets of T .*



I apologize for putting you through that extreme abstraction, but I assure you that it is worth doing once in your life (we do it once so that we never have to do it again).

Now let's apply abstract Galois theory to concrete Galois theory.

With the notation as before, let

$$S = L \quad \& \quad T = \text{Aut}(L/K).$$

Let $R \subseteq L \times \text{Aut}(L/K)$ be the relation

$$aR\varphi \iff \varphi(a) = a.$$

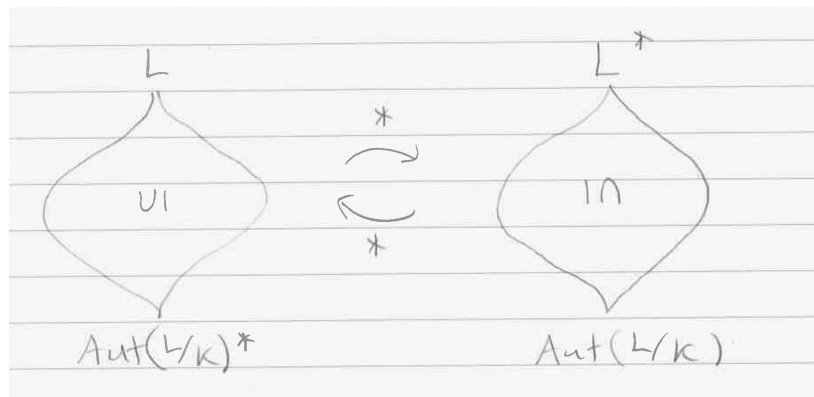
Then for all subsets $H \subseteq \text{Aut}(L/K)$ and $F \subseteq L$ we have

$$H^* = \{a \in L : \varphi(a) = a \forall \varphi \in H\} = L^H$$

and

$$F^* = \{\varphi \in \text{Aut}(L/K) : \varphi(a) = a \forall a \in F\} = \text{Aut}(L/F).$$

We automatically obtain an order-reversing bijection between $**$ -closed sets:



The actual content of the F.T.G.T. is the following pair of statements.

- The $**$ -closed subsets of $\text{Aut}(L/K)$ are precisely the subgroups.
- The $**$ -closed subsets of L are precisely the subfields containing K .

[I hope to prove these next semester as a consequence of something called “Wedderburn Theory.”]

Remark. I assure you that our efforts today will be repaid later. Once you have learned the concept of “Galois connection” you will start to see them everywhere. This will also provide a natural path into the language of categories.

3 Posets, lattices, Galois connections

September 1, 2015

Recall: Last time we discussed the notion of an “abstract Galois connection.”

Given two sets S, T and a relation $R \subseteq S \times T$ we define maps $R : 2^S \rightarrow 2^T$ and $R : 2^T \rightarrow 2^S$ as follows:

- For all $A \subseteq S$ we set

$$A^R := \{t \in T : aRt \forall a \in A\} \subseteq T.$$

- For all $B \subseteq T$ we set

$$B^R := \{s \in S : sRb \forall b \in B\} \subseteq S.$$

The pair of maps

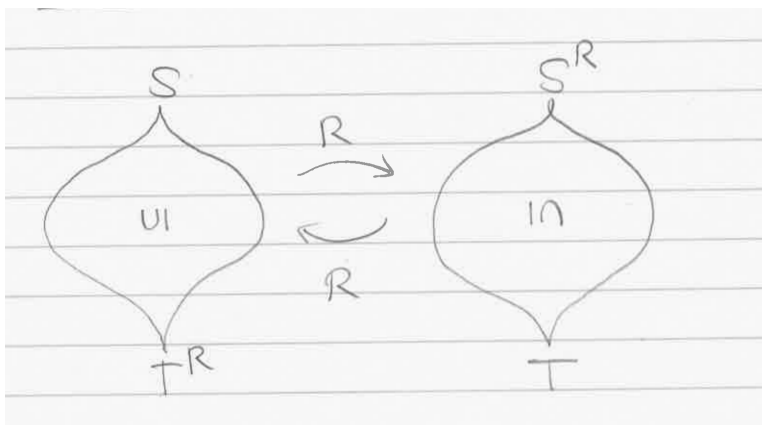
$$R : 2^S \rightrightarrows 2^T : R$$

is called an (abstract) *Galois connection*.

Last time we proved that the maps

$$RR : 2^S \longrightarrow 2^S \quad \text{and} \quad RR : 2^T \longrightarrow 2^T$$

are closure operators. We saw that the Galois connection restricts to an order-reversing bijection between RR -closed sets:



But more is true: You will show on HW1 that this bijection is actually an isomorphism of lattices.

To prepare you for HW1, today I will introduce the concepts of posets and lattices.

Definition. A *poset* (partially-ordered set) is a structure $(P, \leq)$ in which

- P is a set;
- $\leq$ is a relation satisfying
 - $x \leq x$ for all $x \in P$;
 - for all $x, y \in P$, $x \leq y$ and $y \leq x$ implies $x = y$;
 - for all $x, y, z \in P$, $x \leq y$ and $y \leq z$ implies $x \leq z$.

Example. If U is a set then the power set 2^U is partially ordered by inclusion $\subseteq$.

Actually the power set $(2^U, \subseteq)$ has the richer structure of a Boolean algebra. This is encoded by the three functions

$$\begin{aligned} \cup : 2^U \times 2^U &\longrightarrow 2^U \quad (\text{union}), \\ \cap : 2^U \times 2^U &\longrightarrow 2^U \quad (\text{intersection}), \\ c : 2^U &\longrightarrow 2^U \quad (\text{complement}), \end{aligned}$$

and the two special elements

$$\emptyset \in 2^U \quad \text{and} \quad U \in 2^U.$$

More generally, we have the concept of a “lattice.”

Definition. A *lattice* is a structure

$$(\mathcal{L}, \leq, \vee, \wedge, 0, 1)$$

such that

- $(\mathcal{L}, \leq)$ is a poset;
- $\vee : \mathcal{L} \times \mathcal{L} \rightarrow \mathcal{L}$ is a function (called the *join*) satisfying the following “universal property”:
 - $x \leq x \vee y$ and $y \leq x \vee y$;
 - if $x \leq z$ and $y \leq z$ for some z , then we must have $x \vee y \leq z$.

In other words, $x \vee y$ is the “least upper bound” of x and y .

[Note that the join is necessarily unique. Suppose we have another element s such that

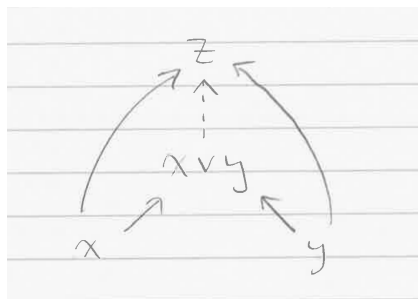
- $x \leq s$ and $y \leq s$;
- if $x \leq z$ and $y \leq z$ then $s \leq z$.

Since $x \leq x \vee y$ and $y \leq x \vee y$, this implies that $s \leq x \vee y$. Conversely, since $x \leq s$ and $y \leq s$, the definition of $x \vee y$ gives $x \vee y \leq s$. Hence $s = x \vee y$. Universal properties like this always lead to uniqueness.]

Let me rephrase the definition of join in a more efficient way. For all $x, y \in \mathcal{L}$ we will write

$$x \longrightarrow y \iff x \leq y.$$

Then we have



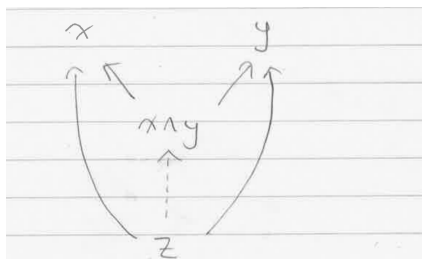
We interpret this to mean: If the solid arrows exist then the dotted arrow necessarily exists. In the language of category theory we say that $x \vee y$ is the “coproduct” of x and y .

- $\wedge : \mathcal{L} \times \mathcal{L} \rightarrow \mathcal{L}$ is a function (called the *meet*) satisfying the universal property:
 - $x \wedge y \leq x$ and $x \wedge y \leq y$;

- if $z \leq x$ and $z \leq y$ then $z \leq x \wedge y$.

In other words, $x \wedge y$ is the “greatest lower bound” of x and y .

The corresponding “commutative diagram” is



This means that $x \wedge y$ is the “product” of x and y in the “category” $(\mathcal{L}, \leq)$.

- $1 \in \mathcal{L}$ is a special element satisfying

$$x \leq 1 \quad \forall x \in \mathcal{L}.$$

It is the “absolute maximum” or the “final object” of $(\mathcal{L}, \leq)$.

[This element is unique: Suppose we have another element $m \in \mathcal{L}$ such that $x \leq m$ for all $x \in \mathcal{L}$. Then $m \leq 1$ and $1 \leq m$ implies $m = 1$.]

- $0 \in \mathcal{L}$ is a special element satisfying

$$0 \leq x \quad \forall x \in \mathcal{L}.$$

It is called the “absolute minimum” or the “initial object” of $(\mathcal{L}, \leq)$.

Example. Let U be a set. Then the poset $(2^U, \subseteq)$ is a lattice with

$$\vee = \cup, \quad \wedge = \cap, \quad 0 = \emptyset, \quad 1 = U.$$

Lattices of the form $(2^U, \subseteq)$ are called Boolean lattices.

Example (Not all lattices are Boolean). Let n be a positive integer and consider its set of positive divisors

$$\text{Div}(n) := \{d \geq 1 : d \mid n\}.$$

This is a lattice under the “divisibility” partial order

$$d_1 \leq d_2 \iff d_1 \mid d_2.$$

The meet and join are given by

$$d_1 \vee d_2 = \text{lcm}(d_1, d_2), \quad d_1 \wedge d_2 = \text{gcd}(d_1, d_2).$$

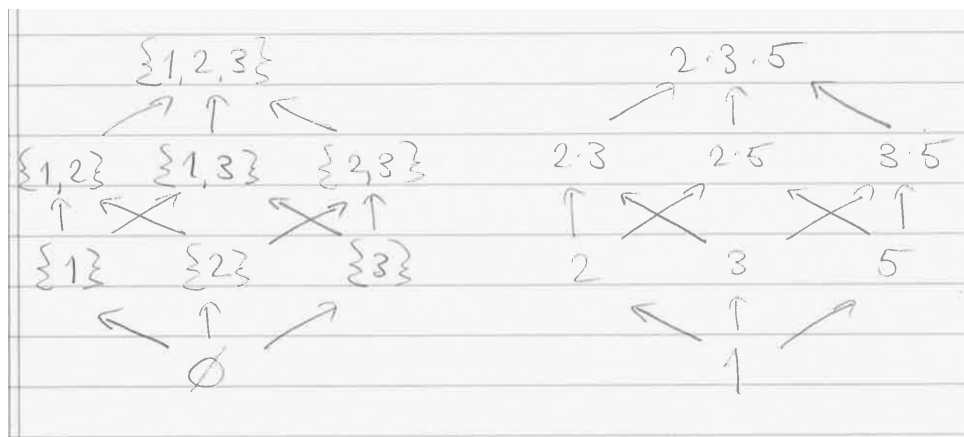
The special elements are

$$0 = 1 \quad \text{and} \quad 1 = n.$$

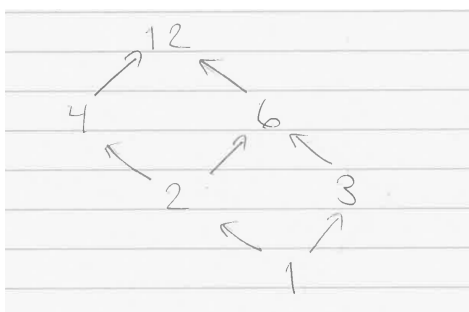
Remark. The lattice $(\text{Div}(n), |)$ is isomorphic to a Boolean lattice if and only if the integer n is “squarefree” (i.e., its prime factors occur with multiplicity 1).

For example, the lattice $\text{Div}(30)$ is isomorphic to the Boolean lattice

$$2^{\{1,2,3\}}.$$



But the lattice $\text{Div}(12)$ is *not* Boolean.



I mentioned the concept of “isomorphism” of lattices. What does this mean?

Definition. Let $(P, \leq)$ and $(Q, \leq)$ be posets.

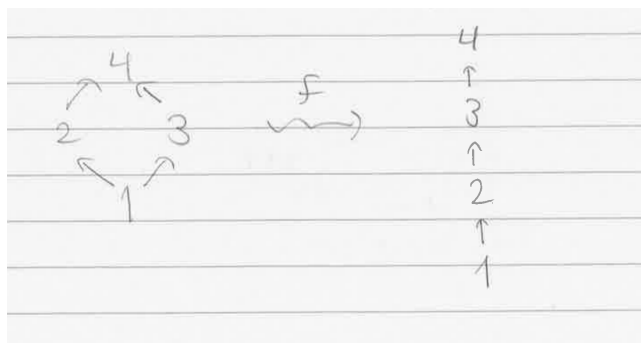
- A function $f : P \rightarrow Q$ is called a *poset homomorphism* if for all $x, y \in P$,

$$x \leq y \implies f(x) \leq f(y).$$

- The function $f : P \rightarrow Q$ is a *poset isomorphism* if

- (1) it is invertible, and
- (2) its inverse is a poset homomorphism.

[Remark: An invertible poset hom is not necessarily an isomorphism. Consider the following.]



- If $(P, \leq)$ and $(Q, \leq)$ are, in addition, lattices we say that $f : P \rightarrow Q$ is a *lattice homomorphism* if
 - f is a poset homomorphism;
 - for all $x, y \in P$ we have

$$f(x \vee y) = f(x) \vee f(y) \quad \text{and} \quad f(x \wedge y) = f(x) \wedge f(y).$$

- We say that a lattice hom $f : P \rightarrow Q$ is a *lattice isomorphism* if
 - (1) it is invertible, and
 - (2) its inverse is also a lattice homomorphism.

[Remark: Actually, any poset isomorphism between lattices will be a lattice isomorphism.
Can you see why?]

4 Groups, automorphisms, subgroups

September 3, 2015

I think we've seen enough abstract nonsense for now. Let's move on to the first real topic of the course:

Group Theory.

What is a group?

There are two different points of view.

(1) A *group* is a structure

$$(G, \circ, 1)$$

such that G is a set, $\circ : G \times G \rightarrow G$ is a function, and $1 \in G$ is a special element satisfying three axioms:

- For all $a, b, c \in G$ we have

$$a \circ (b \circ c) = (a \circ b) \circ c.$$

- For all $a \in G$ we have

$$a \circ 1 = 1 \circ a = a.$$

[Hence the “identity element” is unique. Given any other $1' \in G$ with

$$a \circ 1' = 1' \circ a = a$$

for all $a \in G$, it follows that $1' = 1' \circ 1 = 1$.]

- For all $a \in G$, there exists $b \in G$ such that

$$a \circ b = b \circ a = 1.$$

[This element b is unique. Indeed, given any $b' \in G$ such that $a \circ b' = b' \circ a = 1$, it follows that

$$\begin{aligned} b' &= b' \circ 1 \\ &= b' \circ (a \circ b) \\ &= (b' \circ a) \circ b \\ &= 1 \circ b \\ &= b. \end{aligned}$$

Since the element is unique we can give it a name. We write $b = a^{-1}$ and call it the inverse of a .]

(2) A *group* is the collection of automorphisms (“symmetries”) of a structure X . We say that a function

$$f : X \longrightarrow X$$

is an automorphism if

- f preserves structure;
- f is invertible;
- its inverse preserves structure (in some cases this will be automatic).

Let $\text{Aut}(X)$ be the collection of automorphisms. One can check that this is a group in the sense of (1):

$$1 = \text{the identity function}, \quad \circ = \text{functional composition}.$$

[Note that functional composition is automatically associative.]

Examples:

- $\text{Aut}(\text{set } X) = S(X)$, the permutations of X , called the “symmetric group.”
- $\text{Aut}(\text{vector space } \mathbb{R}^n) = \text{GL}_n(\mathbb{R})$, the invertible $n \times n$ matrices with matrix multiplication, called the “general linear group.”

[Hermann Weyl called it “her all-embracing majesty.”]

- $\text{Aut}(\text{inner product space } \mathbb{R}^n) = O(n)$, the orthogonal $n \times n$ matrices,

$$O(n) = \{A \in \text{GL}_n(\mathbb{R}) : A^T A = I\},$$

called the “orthogonal group.”

- $\text{Aut}(\text{Hermitian space } \mathbb{C}^n) = U(n)$, the unitary matrices,

$$U(n) = \{A \in \text{GL}_n(\mathbb{C}) : A^* A = I\},$$

called the “unitary group.”

$[A^ = \overline{A}^T$ is the complex conjugate transpose matrix.]*

Note that definition (2) motivates the axioms in (1). The interaction between definitions (1) and (2) is called “representation theory.”

Definition. Let $(G, \circ, 1)$ be a group. A subset $H \subseteq G$ is called a *subgroup* if

- $1 \in H$;
- for all $a \in H$ we have $a^{-1} \in H$;
- for all $a, b \in H$ we have $a \circ b \in H$.

In other words, if the triple $(H, \circ, 1)$ is itself a group.

Remark. The three conditions of a subgroup can be rolled into one:

$$\forall a, b \in H \quad a \circ b^{-1} \in H.$$

Definition. Given a group $(G, \circ, 1)$, let

$$\mathcal{L}(G) := \{\text{subgroups of } G\}.$$

I claim that the poset $(\mathcal{L}(G), \subseteq)$ is actually a lattice. What do we need to show?

Clearly the minimum and maximum are

$$0 = \{1\} \quad \text{and} \quad 1 = G.$$

What about $\wedge$ and $\vee$?

Theorem. Given subgroups $H, K \subseteq G$ we have

$$H \wedge K = H \cap K.$$

Proof. It is trivial to check that the intersection $H \cap K$ is indeed a subgroup. We'll show that it satisfies the universal property of the meet. Indeed, we have $H \cap K \subseteq H$ and $H \cap K \subseteq K$. If $L \subseteq G$ is any subgroup satisfying $L \subseteq H$ and $L \subseteq K$, it follows that $L \subseteq H \cap K$ as sets, hence also as groups. $\square$

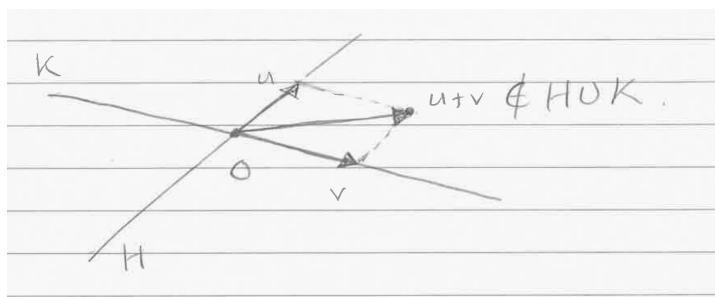
The existence of joins is not so obvious ...

Let $H, K \subseteq G$ be subgroups. If $H \cup K$ is also a subgroup then we will have

$$H \vee K = H \cup K,$$

but this almost never happens!

Example. Let $G = (\mathbb{R}^n, +, 0)$. Let H and K be 1-dimensional subspaces. If $H \neq K$ then $H \cup K$ is not a subgroup.



In linear algebra, we fix this by taking the span of H and K :

$$H \vee K = \text{span}(H, K) = H + K = \{h + k : h \in H, k \in K\}.$$

A similar trick works for any “abelian” group.

Definition. We say that a group $(G, \circ, 1)$ is *abelian* if for all $a, b \in G$ we have

$$a \circ b = b \circ a.$$

When G is abelian we will usually call the operation “+” and the identity “0.”

Given two subgroups H, K of an abelian group $(G, +, 0)$ we define

$$H + K = \{h + k : h \in H, k \in K\}.$$

[Exercise: Check that this is a subgroup.]

Theorem. *In this case we have*

$$H \vee K = H + K.$$

Proof. We will show that $H + K$ satisfies the universal property of join.

Indeed, since $h + 0 = h$ and $0 + k = k$ we have $H \subseteq H + K$ and $K \subseteq H + K$. If L is any subgroup satisfying

$$H \subseteq L \quad \text{and} \quad K \subseteq L,$$

then for all $h \in H$ and $k \in K$ we have $h, k \in L$ and hence $h + k \in L$. It follows that

$$H + K \subseteq L.$$

□

But what if $(G, \circ, 1)$ is nonabelian?

[In a nonabelian group we will write $a \circ b = ab$.]

Given subgroups $H, K \subseteq G$ we can still define the subset

$$HK := \{hk : h \in H, k \in K\} \subseteq G,$$

but this is not always a subgroup.

The best we can do is a sort of formal trick.

Theorem (Complete-lattice trick). *Let $(P, \leq)$ be a poset. If for every subset $S \subseteq P$ (even for infinite subsets) there exists a meet*

$$\bigwedge S \in P,$$

it follows that P also has arbitrary joins. In this case we say that $(P, \leq)$ is a complete lattice.

Proof. Given a subset $S \subseteq P$ we consider its set of upper bounds

$$S^\vee := \{p \in P : s \leq p \ \forall s \in S\}.$$

Now we can define the join of S as the meet of all upper bounds of S ,

$$\bigvee S := \bigwedge (S^\vee).$$

One can check that this satisfies the desired universal property.

□

Finally, we can apply this to define joins in the poset $(\mathcal{L}(G), \subseteq)$.

Definition. Let G be a group and consider any subset $S \subseteq G$. We define

$$\langle S \rangle := \bigcap_{S \subseteq H} H,$$

where the intersection is over all subgroups of G that contain S . One can check that an arbitrary intersection of subgroups is again a subgroup; in particular, $\langle S \rangle$ is a subgroup. We call it the subgroup of G *generated by* S .

In summary, the poset $(\mathcal{L}(G), \subseteq)$ is a complete lattice and the previous lemma tells us that for all subgroups $H, K \subseteq G$ we have

$$H \vee K = \langle H \cup K \rangle.$$

5 Subgroup lattices, products, cyclic groups

September 8, 2015

Recall from last time:

Let $(G, \circ, 1)$ be a group. We say that a subset $H \subseteq G$ is a subgroup if

$$\forall a, b \in H \quad a \circ b^{-1} \in H.$$

It is immediate that an arbitrary intersection of subgroups is again a subgroup. This allows us to make the following definition:

Given any subset $S \subseteq G$, we define

$$\langle S \rangle := \bigcap_{S \subseteq H} H,$$

where the intersection is over all subgroups containing S . This $\langle S \rangle$ is the smallest subgroup of G containing S ; we call it the “subgroup generated by S .”

Now let $(\mathcal{L}(G), \subseteq)$ be the set of subgroups of G , partially ordered by inclusion. The above remarks tell us that $\mathcal{L}(G)$ is a complete lattice.

Indeed, given a family of subgroups $\{H_i\}_{i \in I}$, their meet is just the intersection

$$\bigwedge H_i = \bigcap H_i,$$

and their join is the subgroup generated by their union

$$\bigvee H_i = \langle \bigcup H_i \rangle.$$

[If the index set is empty, $I = \emptyset$, then by convention we have

$$\bigwedge H_i := G \quad \text{and} \quad \bigvee H_i := \{1\}.$$

Compare: “The sum of no numbers is 0 and the product of no numbers is 1.”]

Special case: If $(G, +, 0)$ is an abelian group, we saw that

$$H \vee K = \langle H \cup K \rangle = H + K := \{h + k : h \in H, k \in K\}.$$

If $(G, \circ, 1)$ is nonabelian, we can still define the subset

$$HK := \{hk : h \in H, k \in K\},$$

but in general this is not a subgroup.

Q: When is HK a subgroup?

A: Suppose that HK is a subgroup and consider $h \in H, k \in K$. Since $k \in HK$ and $h \in HK$ this implies that $kh \in HK$. In other words, there exist $h' \in H$ and $k' \in K$ such that

$$kh = h'k'.$$

This implies that $KH \subseteq HK$.

Conversely, given $h \in H$ and $k \in K$ we have $hk \in HK$ and hence $(hk)^{-1} = k^{-1}h^{-1} \in HK$. In other words, there exist $h'' \in H$ and $k'' \in K$ such that

$$k^{-1}h^{-1} = h''k''.$$

Take the inverse of both sides to get

$$hk = (k'')^{-1}(h'')^{-1}.$$

This implies that $HK \subseteq KH$.

Theorem. *The subset HK is a subgroup if and only if*

$$HK = KH.$$

[We will have more to say about this later.]

Let G be a group and consider a subset $S \subseteq G$.

Notation: If $S = \{s_1, s_2, \dots, s_k\}$ is finite we will write

$$\langle S \rangle = \langle s_1, s_2, \dots, s_k \rangle.$$

If $G = \langle s_1, s_2, \dots, s_k \rangle$ we say that G is a finitely generated group.

If $G = \langle a \rangle$ for a single element $a \in G$ we say that G is a cyclic group. In this case we can be quite explicit:

$$G = \{\dots, a^{-2}, a^{-1}, 1, a, a^2, \dots\}.$$

[Here we use the notations $a^{-n} := (a^{-1})^n$ and $a^0 := 1$. This is very convenient because it means that

$$a^m a^n = a^{m+n}$$

for all integers $m, n \in \mathbb{Z}$.]

If G is finite then there must exist integers $m < n$ such that

$$a^n = a^m.$$

Thus $a^n(a^m)^{-1} = 1$, hence $a^{n-m} = 1$.

In other words, there must exist a positive integer $n - m$ such that $a^{n-m} = 1$. If r is the smallest such positive integer then for all $m, n \in \mathbb{Z}$ we have

$$a^n = a^m \iff r \mid n - m,$$

and it follows that

$$G = \{1, a, a^2, \dots, a^{r-1}\}.$$

[If this reminds you of modular arithmetic, good.]

More generally, given any element x of a group G we say that $|\langle x \rangle|$ is the *order* of the element.

The internal structure of cyclic groups is completely described by the following theorem.

Theorem (Fundamental Theorem of Cyclic Groups). *Let G be a cyclic group of order r . Then its subgroup lattice is isomorphic to the lattice of divisors of r ,*

$$(\mathcal{L}(G), \subseteq) \cong (\text{Div}(r), |).$$

We could prove this now, but to give the correct proof we must first discuss the external structure of cyclic groups.

What?

Remarks on Internal vs. External Structure:

Starting in the 1930s, Øystein Ore and Garrett Birkhoff advocated the point of view that the structure of a group G

should be studied through its lattice $\mathcal{L}(G)$ of subgroups. They attempted to say as much as possible about the group without ever referring to its elements, but only to its subgroups and the relationships between them.

It turned out that the subgroup lattice does not contain quite enough information about G . [For example, there are two nonisomorphic groups of order 8 with isomorphic subgroup lattices.] And this point of view has now been absorbed into the theory of “categories.”

We say that $\mathcal{L}(G)$ describes the “internal” structure of G because it encodes the interrelationships between all the subgroups of G . To describe the “external” structure of G we should say how G interacts with all other groups. This information is encoded in the category of groups.

Let Grp denote the category of groups. This consists of

- the collection of all groups;
- the collection of all morphisms between groups.

Definition. Given two groups G and H and a function $\varphi : G \rightarrow H$, we say that φ is a group morphism (or homomorphism) if for all $a, b \in G$ we have

$$\varphi(ab) = \varphi(a)\varphi(b).$$

We say that φ is a group isomorphism if (1) it is a group morphism, and (2) it is invertible.

[In this case it follows automatically that the inverse φ^{-1} is also a morphism. Indeed, let $\varphi(a) = c$ and $\varphi(b) = d$, so that $a = \varphi^{-1}(c)$ and $b = \varphi^{-1}(d)$. Then we have

$$\begin{aligned} \varphi^{-1}(cd) &= \varphi^{-1}(\varphi(a)\varphi(b)) \\ &= \varphi^{-1}(\varphi(ab)) \\ &= ab \\ &= \varphi^{-1}(c)\varphi^{-1}(d). \end{aligned}$$

]

In category theory we draw morphisms as arrows

$$G \xrightarrow{\varphi} H.$$

The “philosophy” of category theory is to argue with arrows and to avoid any mention of the elements of groups.

For example, we can give an arrow-theoretic definition of isomorphism.

Definition. Consider an arrow $G \xrightarrow{\varphi} H$ between groups. We say that this arrow is an isomorphism if there exists another arrow

$$H \xrightarrow{\mu} G$$

such that the following diagrams commute:

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & H \xrightarrow{\mu} G \\ & \searrow \text{id}_G \nearrow & \\ & & \end{array} \qquad \begin{array}{ccc} H & \xrightarrow{\mu} & G \xrightarrow{\varphi} H \\ & \searrow \text{id}_H \nearrow & \\ & & \end{array}$$

In symbols, we have

$$\mu \circ \varphi = \text{id}_G \quad \text{and} \quad \varphi \circ \mu = \text{id}_H.$$

Note that the arrow μ is unique. Indeed, if there exists $H \xrightarrow{\mu'} G$ such that $\mu' \circ \varphi = \text{id}_G$ and $\varphi \circ \mu' = \text{id}_H$, then we have

$$\mu' = \mu' \circ \text{id}_H = \mu' \circ (\varphi \circ \mu) = (\mu' \circ \varphi) \circ \mu = \text{id}_G \circ \mu = \mu.$$

Since μ is unique we will write $\mu = \varphi^{-1}$ and call it the inverse of φ . When there exists an isomorphism between G and H we will write

$$G \cong H.$$

6 Categories and homomorphisms

September 10, 2015

Last time I mentioned the word “category.” Today I’ll define it.

Definition. A category C consists of

- a collection of objects;
- a collection of arrows between objects (sometimes we call the arrows “morphisms” or “homomorphisms”),

satisfying two axioms:

- arrows can be composed and the composition is associative (just like functions);
- every object X has an “identity arrow”

$$X \xrightarrow{\text{id}_X} X$$

such that for all arrows $Z \xrightarrow{f} X$ and $X \xrightarrow{g} Y$ we have

$$\text{id}_X \circ f = f \quad \text{and} \quad g \circ \text{id}_X = g.$$

But we prefer to say this with “commutative diagrams”:

$$\begin{array}{ccc} Z & \xrightarrow{f} & X \xrightarrow{\text{id}_X} X \\ & \searrow f & \nearrow \\ & & \end{array} \qquad \begin{array}{ccc} X & \xrightarrow{\text{id}_X} & X \xrightarrow{g} Y \\ & \searrow g & \nearrow \\ & & \end{array}$$

Last time we saw the definition of isomorphism in a category.

Definition. An arrow $X \xrightarrow{f} Y$ is called an isomorphism if there exists an arrow $Y \xrightarrow{g} X$ such that

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \xrightarrow{g} X \\ & \searrow \text{id}_X & \nearrow \\ & & \end{array} \qquad \begin{array}{ccc} Y & \xrightarrow{g} & X \xrightarrow{f} Y \\ & \searrow \text{id}_Y & \nearrow \\ & & \end{array}$$

The arrow g (if it exists) is unique, so we call it f^{-1} .

Here is some common notation. Let X and Y be objects in a category $\mathcal{C}$.

We define:

- $\text{Hom}_{\mathcal{C}}(X, Y) :=$ the collection of arrows from X to Y ;

[a.k.a. “homomorphisms”]

- $\text{End}_{\mathcal{C}}(X) :=$ the collection of arrows from X to X ;

[a.k.a. “endomorphisms”]

- $\text{Aut}_{\mathcal{C}}(X) :=$ the collection of isomorphisms from X to X .

[a.k.a. “automorphisms”]

[Disclaimer: Without further comment I will always assume that Hom , End , and Aut are sets (i.e., not something bigger than a set). In other words, I will assume that all categories are “locally small.”]

Examples:

- A *monoid* is a category with one object, say, X . In this case we identify the category with the set $\text{End}(X)$.
- A *groupoid* is a category in which every arrow is an isomorphism.
- A *group* is a groupoid with one object, say X . In this case we identify the category with the set $\text{Aut}(X) = \text{End}(X)$.
- A *poset* is a category such that for all objects X, Y we have

$$|\text{Hom}(X, Y)| \in \{0, 1\}.$$

We use the notation

$$“X \leq Y” \iff |\text{Hom}(X, Y)| = 1.$$

[Exercise: Verify the three axioms of partial order.]

- Given a field K , let $K\text{-Vect}$ be the category whose objects are vector spaces over K and whose arrows are K -linear maps.

If V is an n -dimensional K -vector space then we have

$$\text{End}_{K\text{-Vect}}(V) \cong \text{Mat}_n(K) \quad \text{with matrix multiplication}$$

and

$$\text{Aut}_{K\text{-Vect}}(V) \cong \text{GL}_n(K) \quad \text{with matrix multiplication.}$$

Enough general nonsense for now. Let's get back to the category of groups.

Let G, H be groups and consider a group homomorphism

$$\varphi : G \longrightarrow H.$$

This induces two interesting subgroups.

- The *kernel*

$$\ker \varphi := \{g \in G : \varphi(g) = 1_H\}.$$

Note that $\ker \varphi \subseteq G$ is a subgroup. More generally, if $K \subseteq H$ is any subgroup then its *preimage*

$$\varphi^{-1}(K) := \{g \in G : \text{there exists } k \in K, \varphi(g) = k\}$$

is a subgroup of G .

Proof. Let $a, b \in \varphi^{-1}(K)$, so we have $\varphi(a) = c$ and $\varphi(b) = d$ for some $c, d \in K$.

First note that $\varphi(b^{-1}) = \varphi(b)^{-1}$. Indeed, we have $1_G = 1_G 1_G$. Applying φ gives

$$\varphi(1_G) = \varphi(1_G)\varphi(1_G).$$

Then multiply on the left by $\varphi(1_G)^{-1}$ to get

$$\varphi(1_G) = 1_H.$$

Now apply φ to the equation $bb^{-1} = 1_G$ to get

$$\begin{aligned} \varphi(bb^{-1}) &= \varphi(1_G), \\ \varphi(b)\varphi(b^{-1}) &= 1_H. \end{aligned}$$

Multiply on the left by $\varphi(b)^{-1}$ to get

$$\varphi(b^{-1}) = \varphi(b)^{-1}.$$

Finally, since $K \subseteq H$ is a subgroup we have

$$\begin{aligned}\varphi(ab^{-1}) &= \varphi(a)\varphi(b^{-1}) \\ &= \varphi(a)\varphi(b)^{-1} \\ &= cd^{-1} \in K,\end{aligned}$$

and we conclude that $ab^{-1} \in \varphi^{-1}(K)$ as desired. $\square$

[Note that $\ker \varphi = \varphi^{-1}(\{1_H\})$.]

- The *image*

$$\text{im } \varphi := \{h \in H : \text{there exists } g \in G, \varphi(g) = h\}.$$

Note that $\text{im } \varphi \subseteq H$ is a subgroup. More generally, if $K \subseteq G$ is any subgroup then its *image*

$$\varphi(K) := \{h \in H : \text{there exists } k \in K, \varphi(k) = h\}$$

is a subgroup of H .

Proof. Let $c, d \in \varphi(K)$, so we have $\varphi(a) = c$ and $\varphi(b) = d$ for some $a, b \in K$. Then since $K \subseteq G$ is a subgroup we have $ab^{-1} \in K$ and hence

$$\begin{aligned}cd^{-1} &= \varphi(a)\varphi(b)^{-1} \\ &= \varphi(a)\varphi(b^{-1}) \\ &= \varphi(ab^{-1}) \in \varphi(K),\end{aligned}$$

as desired. $\square$

[Note that $\text{im } \varphi = \varphi(G)$.]

Let $\mathcal{L}(G)$ and $\mathcal{L}(H)$ be the subgroup lattices of G and H . We have just defined two maps

$$\varphi : \mathcal{L}(G) \rightleftarrows \mathcal{L}(H) : \varphi^{-1}.$$

Did you notice that this is a Galois connection? Well, it is.

Claim. For all subgroups $A \subseteq G$ and $B \subseteq H$ we have

$$\varphi(A) \subseteq B \iff A \subseteq \varphi^{-1}(B).$$

Proof. Note that

$$\begin{aligned}\varphi(A) \subseteq B &\iff \forall a \in A, \varphi(a) \in B \\ &\iff \forall a \in A, \text{there exists } b \in B, \varphi(a) = b \\ &\iff \forall a \in A, a \in \varphi^{-1}(B) \\ &\iff A \subseteq \varphi^{-1}(B).\end{aligned}$$

$\square$

It follows from HW1 Problem 5 that we get an isomorphism between posets of “closed” subgroups.

Q: Which subgroups are closed?

Let’s examine the “closure” operators.

Claim. For all subgroups $A \subseteq G$, $B \subseteq H$ we have

$$\varphi^{-1}(\varphi(A)) = A \vee \ker \varphi$$

and

$$\varphi(\varphi^{-1}(B)) = B \wedge \operatorname{im} \varphi.$$

[You will prove this on HW2.]

It follows that the “closed” subgroups of G are those containing $\ker \varphi$ and the “closed” subgroups of H are those contained in $\operatorname{im} \varphi$.

Notation: For any subgroup $N \subseteq G$ we will write

$$\mathcal{L}(G, N) := \{A \in \mathcal{L}(G) : N \subseteq A\}.$$

Check that this is a sublattice of $\mathcal{L}(G)$.

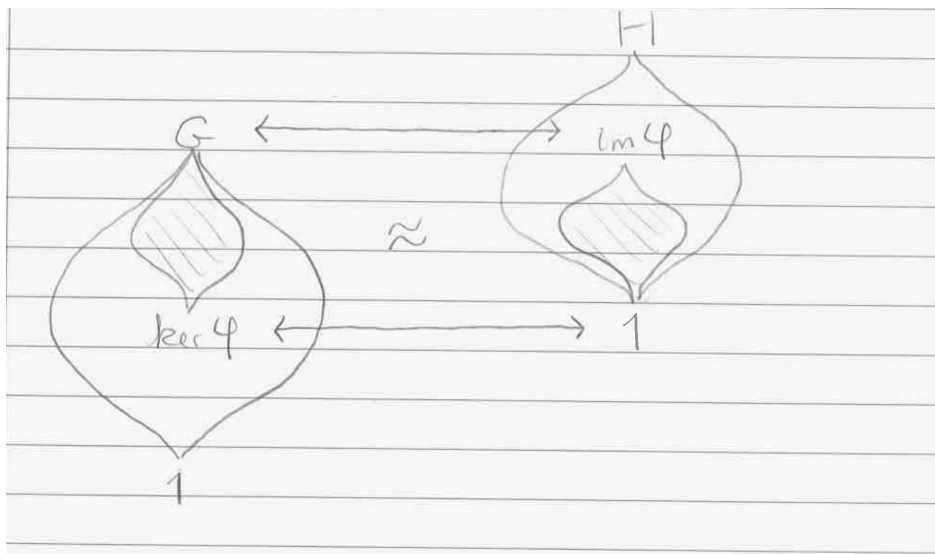
Finally, we obtain the following theorem.

Theorem (Lattice Isomorphism for Groups). *Let $\varphi : G \rightarrow H$ be a group homomorphism. This induces a Galois connection*

$$\varphi : \mathcal{L}(G) \rightleftarrows \mathcal{L}(H) : \varphi^{-1},$$

which restricts to an isomorphism of lattices

$$\varphi : \mathcal{L}(G, \ker \varphi) \rightleftarrows \mathcal{L}(\operatorname{im} \varphi) : \varphi^{-1}.$$



7 Galois connections and quotients

September 15, 2015

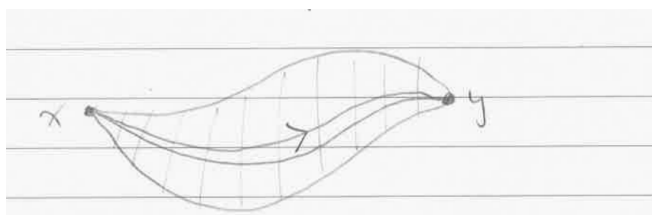
Last time I gave the following definition.

Definition. A *groupoid* is a category in which every arrow is an isomorphism.

But I forgot to mention the most important example.

Example. Let X be a topological space. The *fundamental groupoid* $\Pi_1(X)$ is the category with

- objects = points of X ;
- arrows $x \longrightarrow y$ are homotopy classes of continuous paths.



Arrows are composed by “concatenation.”

If we choose a specific basepoint $x_0 \in X$ then we obtain the *fundamental group*

$$\Pi_1(X, x_0) = \text{Aut}_{\Pi_1(X)}(x_0).$$

[Remark: The fundamental group and groupoid are not “concretizable,” i.e., it is not possible to view their objects as sets and their arrows as functions. Highbrow Translation: There does not exist a “faithful functor” to the category of sets,

$$U : \Pi_1(X) \longrightarrow \text{Set.} \quad]$$

OK, enough of that for now.

Recall from last time:

Let G, H be groups and consider a group homomorphism

$$\varphi : G \longrightarrow H.$$

This induces a pair of maps

$$\varphi : \mathcal{L}(G) \rightleftarrows \mathcal{L}(H) : \varphi^{-1}$$

called image and preimage. We verified that for all subgroups $A \subseteq G$ and $B \subseteq H$ we have

$$\varphi(A) \subseteq B \iff A \subseteq \varphi^{-1}(B),$$

hence this is a Galois connection. You will show on HW2 that the “Galois closures” are

$$\varphi^{-1}(\varphi(A)) = A \vee \ker \varphi$$

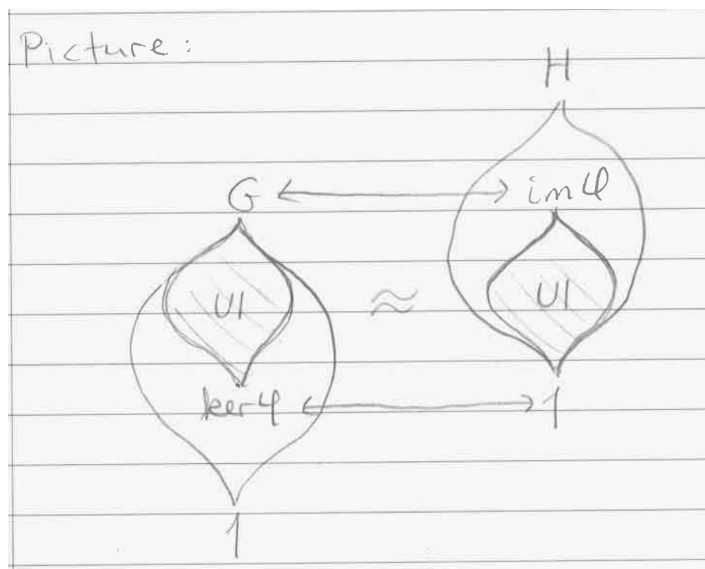
and

$$\varphi(\varphi^{-1}(B)) = B \wedge \text{im } \varphi.$$

Hence we obtain an isomorphism of lattices

$$\varphi : \mathcal{L}(G, \ker \varphi) \rightleftarrows \mathcal{L}(\text{im } \varphi) : \varphi^{-1}.$$

Picture:



More generally, given any subgroup $N \subseteq G$ we can define the “relative” subgroup lattice

$$\mathcal{L}(G, N) := \{A \in \mathcal{L}(G) : N \subseteq A\}.$$

Q: Under what conditions does there exist a group G' such that

$$\mathcal{L}(G, N) \cong \mathcal{L}(G') ?$$

To answer this we must take a detour into the theory of quotients.

Let S be a set. We say that a relation $\sim \subseteq S \times S$ is an equivalence if

- $\forall x \in S, x \sim x$;
- $\forall x, y \in S, x \sim y \implies y \sim x$;
- $\forall x, y, z \in S, x \sim y \text{ and } y \sim z \implies x \sim z$.

For each $x \in S$ we define its *equivalence class*

$$[x]_{\sim} := \{y \in S : x \sim y\}.$$

Given $x, y \in S$, check that the following are equivalent:

- i) $x \sim y$;
- ii) $[x]_{\sim} = [y]_{\sim}$;
- iii) $[x]_{\sim} \cap [y]_{\sim} \neq \emptyset$.

Now it often happens in math that we have a function $\varphi : S \rightarrow T$

that is constant on $\sim$ classes, i.e., for all $x, y \in S$ we have

$$x \sim y \implies \varphi(x) = \varphi(y).$$

We will call any such pair (φ, T) a “ $\sim$ class function.”

Now let’s define a category whose “objects” are class functions (φ, T) and whose “arrows”

$$(\varphi_1, T_1) \xrightarrow{\sigma} (\varphi_2, T_2)$$

are commutative diagrams

$$\begin{array}{ccc} & S & \\ \varphi_1 \swarrow & & \searrow \varphi_2 \\ T_1 & \xrightarrow{\sigma} & T_2 \end{array}$$

It turns out that there is a special object in this category called the *quotient of S by $\sim$* . We define it as follows:

Let

$$S/\sim := \{[x]_\sim : x \in S\}$$

and let $\pi : S \rightarrow S/\sim$ be defined by

$$\pi(x) := [x]_\sim.$$

Theorem. *The object $(\pi, S/\sim)$ is “initial” in the category of class functions. In other words, given any class function (φ, T) , there exists a unique arrow*

$$\begin{array}{ccc} & S & \\ \pi \swarrow & & \searrow \varphi \\ S/\sim & \xrightarrow{\quad \exists! \bar{\varphi} \quad} & T \end{array}$$

Proof. Let $[x]_\sim$ be an arbitrary element of $S/\sim$. By commutativity we must have

$$\bar{\varphi}([x]_\sim) = \varphi(x). \quad (*)$$

Thus if $\bar{\varphi}$ exists at all then it must be unique. To show that $\bar{\varphi}$ exists we just have to show that the prescription $(*)$ is well-defined.

Indeed, for all $[x]_\sim, [y]_\sim \in S/\sim$ we have

$$[x]_\sim = [y]_\sim \implies x \sim y \implies \varphi(x) = \varphi(y).$$

□

Definition. We say that an object X in a category $\mathcal{C}$ is initial if for all objects Y we have

$$|\text{Hom}_{\mathcal{C}}(X, Y)| = 1.$$

Example. The 0 element of a poset (if it exists) is initial.

It follows from general nonsense that initial objects are unique up to isomorphism. Thus the quotient

$$S \xrightarrow{\pi} S/\sim$$

is unique up to isomorphism in the category of class functions.

Application: Let $f : S \rightarrow S'$ be any function of sets. We will define an equivalence relation on S by declaring

$$x \sim_f y \iff f(x) = f(y).$$

Note that the function $S \xrightarrow{f} \text{im } f$ is constant on $\sim_f$ classes (by definition). Note also that for all “ $\sim_f$ class functions” $S \xrightarrow{\varphi} T$ we have

$$\begin{array}{ccc}
 & S & \\
 f \swarrow & & \searrow \varphi \\
 \text{im } f & \xrightarrow{\quad \exists! \bar{\varphi} \quad} & T
 \end{array}$$

where the unique function is (well-)defined by the prescription

$$\bar{\varphi}(f(x)) := \varphi(x).$$

In other words, $S \xrightarrow{f} \text{im } f$ is an initial object in the category of $\sim_f$ class functions.

By uniqueness of initial objects we obtain an isomorphism of $\sim_f$ class functions:

$$\begin{array}{ccc}
 & S & \\
 \pi \swarrow & & \searrow f \\
 S/\sim_f & \xrightarrow[\bar{f}]{\sim} & \text{im } f
 \end{array}$$

In other words we have taken the original set function $f : S \rightarrow S'$ and factored it into a surjection followed by a bijection followed by an injection:

$$\begin{array}{ccccccc}
 & & & f & & & \\
 & & \searrow & \text{arc} & \swarrow & & \\
 S & \xrightarrow{\pi} & S/\sim_f & \xrightarrow[\bar{f}]{\sim} & \text{im } f & \xrightarrow{\iota} & S'
 \end{array}$$

Our goal is to lift this construction from the category of sets to the category of groups.

Let G be a group and consider any equivalence relation $\sim \subseteq G \times G$. Under what conditions is the quotient map

$$G \xrightarrow{\pi} G/\sim$$

a group homomorphism? Note that for all $a, b \in G$ we have

$$\pi(a) = [a]_{\sim}, \quad \pi(b) = [b]_{\sim}, \quad \pi(ab) = [ab]_{\sim}.$$

Thus to turn π into a group hom we must have

$$\pi(a)\pi(b) = \pi(ab), \quad [a]_{\sim} \cdot [b]_{\sim} = [ab]_{\sim}.$$

The problem, of course, is that this “product” operation on $G/\sim$ might not be well-defined. To be well-defined in the first factor we require the following:

$$\text{Given } [a]_{\sim} = [a']_{\sim}, \text{ we have for all } b \in G \text{ that } [ab]_{\sim} = [a'b]_{\sim}.$$

In other words, for all $a, a', b \in G$ we have

$$a \sim a' \implies ab \sim a'b.$$

[We say that the relation is right-invariant under the group operation.]

Similarly, for the operation $[a]_{\sim} \cdot [b]_{\sim} = [ab]_{\sim}$ to be well-defined in the second factor we require that for all $a, b, b' \in G$ we have

$$b \sim b' \implies ab \sim ab'.$$

[We say the relation is left-invariant.]

So let's make a definition.

Definition. An equivalence relation $\sim$ on a group G is called G -invariant if for all $a, b, g \in G$ we have

$$a \sim b \implies ag \sim bg$$

and

$$a \sim b \implies ga \sim gb.$$

And here's the theorem.

Theorem. Let $\sim$ be an equivalence relation on a group G . The prescription

$$[a]_{\sim} \cdot [b]_{\sim} = [ab]_{\sim}$$

defines a group structure on the quotient $G/\sim$ if and only if the relation $\sim$ is G -invariant.

Proof. We already saw that group structure implies G -invariance.

Conversely, assume that $\sim$ is G -invariant. If $[a]_{\sim} = [a']_{\sim}$ and $[b]_{\sim} = [b']_{\sim}$ then by G -invariance we have

$$\begin{aligned} a \sim a' &\implies ab \sim a'b, \\ b \sim b' &\implies a'b \sim a'b', \end{aligned} \implies ab \sim a'b'.$$

Hence $[ab]_{\sim} = [a'b']_{\sim}$. We conclude that the operation is well-defined.

As for group structure, associativity in $G/\sim$ is inherited from G :

For all $a, b, c \in G$ we have

$$\begin{aligned} ([a] \cdot [b]) \cdot [c] &= [ab] \cdot [c] \\ &= [(ab)c] \\ &= [a(bc)] \\ &= [a] \cdot [bc] \\ &= [a] \cdot ([b] \cdot [c]). \end{aligned}$$

The identity is $[1]$. Indeed, $\forall a \in G$ we have

$$[a] \cdot [1] = [a1] = [a], \quad [1] \cdot [a] = [1a] = [a].$$

And for all $a \in G$ we have $[a]^{-1} = [a^{-1}]$. Indeed, note that

$$[a] \cdot [a^{-1}] = [aa^{-1}] = [1], \quad [a^{-1}] \cdot [a] = [a^{-1}a] = [1].$$

□

8 Quotient groups and normal subgroups

September 17, 2015

Let $\sim$ be an equivalence relation on a set S . We say that $\varphi : S \rightarrow T$ is a class function if for all elements $x, y \in S$ we have

$$x \sim y \implies \varphi(x) = \varphi(y).$$

Last time we defined a category whose objects are class functions (φ, T) and whose morphisms

$$(\varphi_1, T_1) \longrightarrow (\varphi_2, T_2)$$

are functions $\sigma : T_1 \rightarrow T_2$ satisfying the commutative diagram

$$\begin{array}{ccc} & S & \\ \varphi_1 \swarrow & & \searrow \varphi_2 \\ T_1 & \xrightarrow{\sigma} & T_2 \end{array}$$

We proved that this category has an initial object $(\pi, S/\sim)$ called the quotient of S by $\sim$.

[An initial object X in a category $\mathcal{C}$ satisfies $|\text{Hom}_{\mathcal{C}}(X, Y)| = 1$ for all objects Y . You will show on HW2 that initial objects are unique up to (a unique) isomorphism.]

In other words, the quotient is defined by the “universal property”

$$\begin{array}{ccc} & S & \\ \pi \swarrow & & \searrow \varphi \\ S/\sim & \xrightarrow{\exists! \bar{\varphi}} & T \end{array}$$

Now if $S = G$ is a group we would like to know whether the “set quotient” $G/\sim$ can be lifted to a “group quotient.”

We can define a category as before where the objects are group homomorphisms $\varphi : G \rightarrow H$ satisfying, for all $a, b \in G$,

$$a \sim b \implies \varphi(a) = \varphi(b).$$

Q: For which equivalence relations $\sim$ does this category have an initial object?

[If an initial object $(\pi, G/\sim)$ exists we call it the quotient group.]

A: We proved last time that the quotient group exists if and only if $\sim$ is “ G -invariant,” i.e., for all $a, b, c \in G$ we have

$$a \sim b \implies ac \sim bc$$

and

$$a \sim b \implies ca \sim cb.$$

For this reason we are very interested in studying G -invariant equivalence relations.

Theorem. *Let $\sim$ be G -invariant. Then*

- $[1]_{\sim} \subseteq G$ is a subgroup.
- For all $a, b \in G$ we have

$$a \sim b \iff a^{-1}b \in [1]_{\sim} \iff ab^{-1} \in [1]_{\sim}.$$

Proof. To show that $[1]_{\sim}$ is a subgroup, consider any $a, b \in [1]_{\sim}$. By definition this means $a \sim 1$ and $b \sim 1$. Applying b^{-1} to $b \sim 1$ gives

$$bb^{-1} \sim b^{-1},$$

hence $b^{-1} \sim 1$. Then applying a to $b^{-1} \sim 1$ gives

$$ab^{-1} \sim a.$$

Since $a \sim 1$, transitivity implies that $ab^{-1} \sim 1$, and hence $ab^{-1} \in [1]_{\sim}$, as desired.

For the second part we will only prove that

$$a \sim b \iff ab^{-1} \in [1]_{\sim}.$$

The proof that $a \sim b \iff a^{-1}b \in [1]_{\sim}$ is similar. Indeed,

$$a \sim b \implies ab^{-1} \sim 1 \implies ab^{-1} \in [1]_{\sim}.$$

Conversely,

$$ab^{-1} \in [1]_{\sim} \implies ab^{-1} \sim 1 \implies a \sim b.$$

□

Notation: Let $\sim$ be G -invariant and define the subgroup $N := [1]_{\sim}$. Then we will write

$$G/N := G/\sim$$

and we will call this the “quotient of G by the subgroup N .” The equivalence relation becomes

$$a \sim_N b \iff a^{-1}b \in N \iff ab^{-1} \in N.$$

The equivalence classes are

$$\begin{aligned}
 [a]_N &= \{b \in G : a \sim_N b\} \\
 &= \{b \in G : a^{-1}b \in N\} \\
 &= \{b \in G : \exists n \in N, a^{-1}b = n\} \\
 &= \{b \in G : \exists n \in N, b = an\} \\
 &= \{an : n \in N\} \\
 &=: aN.
 \end{aligned}$$

We call this the left coset of N generated by $a \in G$. On the other hand, we also have

$$\begin{aligned}
 [a]_N &= \{b \in G : ab^{-1} \in N\} \\
 &= \{na : n \in N\} \\
 &=: Na.
 \end{aligned}$$

We call this the right coset of N generated by $a \in G$.

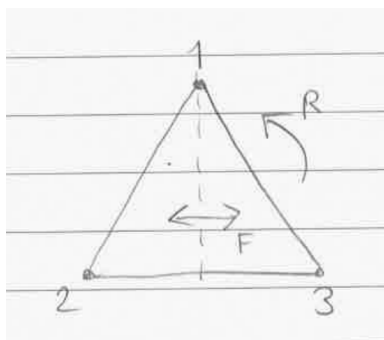
Thus for all $a \in G$ we have

$$aN = Na.$$

Definition. Any subgroup with this property will be called *normal* (following Galois), but a better name might be “ G -invariant subgroup.”

Example. Not all subgroups are normal!

Clearly all subgroups of abelian groups are normal, so we must look to the smallest non-abelian group for an example. This is the dihedral group D_6 of symmetries of an equilateral triangle:



We can identify group elements with permutations of the vertices $\{1, 2, 3\}$. Let F be the “flip” switching $2 \leftrightarrow 3$ and let R be the “rotation”

$$1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 1.$$

The six elements of the group are

$$D_6 = \{1, R, R^2, F, FR, FR^2\}.$$

We observe that the subgroup $\langle F \rangle = \{1, F\}$ is not normal. Indeed, the cosets

$$\langle F \rangle R = \{R, FR\}, \quad R\langle F \rangle = \{R, RF\}$$

are not equal because $RF = FR^2 \neq FR$.

Hence the symbol $D_6/\langle F \rangle$ can mean two different things (the set of left cosets or the set of right cosets), neither of which is a group.

More generally, we can define the dihedral group D_{2n} . It is generated by two elements F and R satisfying

$$F^2 = R^n = 1 \quad \text{and} \quad RF = FR^{n-1}.$$

We can think of D_{2n} as the group of symmetries of a regular n -gon, where F = “flip” and R = “rotate.”

Let's summarize what we've learned.

Theorem (Universal Property of Group Quotient). *Let G be a group with normal subgroup N and consider the quotient homomorphism*

$$G \xrightarrow{\pi} G/N.$$

If $\varphi : G \rightarrow G'$ is any group homomorphism that is constant on cosets of N , then there exists a unique group homomorphism $\bar{\varphi} : G/N \rightarrow G'$ such that the following diagram commutes:

$$\begin{array}{ccc} & G & \\ \pi \swarrow & & \searrow \varphi \\ G/N & \xrightarrow{\quad \bar{\varphi} \quad} & G' \end{array}$$

Recall that in the category of sets we used quotients to factor any set function $f : S \rightarrow S'$ as

$$\begin{array}{ccccccc} & & f & & & & \\ & & \curvearrowright & & & & \\ S & \longrightarrow & S/\sim_f & \xrightarrow{\sim} & \text{im } f & \hookrightarrow & S' \end{array}$$

Next time we will lift this factorization to the category of groups.

9 Normal subgroups and quotient homomorphisms

September 22, 2015

Recall from last time:

Let $H \leq G$ be a subgroup. We can use H to define two equivalence relations on G :

$$a \sim_H b \iff ab^{-1} \in H$$

and

$$a \sim_H b \iff a^{-1}b \in H.$$

The equivalence classes

$$[a]_{\sim_H} = Ha = \{ha : h \in H\}$$

and

$$[a]_{\sim} = aH = \{ah : h \in H\}$$

are called right and left H -cosets, respectively. If the two equivalence relations coincide, that is, if we have $aH = Ha$ for all $a \in G$, then we say H is a normal subgroup and we write

$$H \trianglelefteq G.$$

But to prove that a subgroup is normal, it is more convenient to use the following criterion.

Theorem (Normal Subgroup Criterion). *Let $H \leq G$ be a subgroup. Then we have $H \trianglelefteq G$ if and only if*

$$\boxed{\text{for all } a \in G \text{ and } h \in H, aha^{-1} \in H.}$$

Proof. Suppose that for all $a \in G$ and $h \in H$ we have $aha^{-1} \in H$. Now consider any element $ah \in aH$. Then we have

$$ah = (aha^{-1})a \in Ha.$$

Conversely, given any element $ha \in Ha$ we have

$$ha = a(a^{-1}ha) = a((a^{-1})h(a^{-1})^{-1}) \in aH.$$

Next suppose that $aH = Ha$ for all $a \in G$. Then given $a \in G$ and $h \in H$ we have $ah \in aH = Ha$, hence there exists $h' \in H$ such that $ah = h'a$. We conclude that

$$aha^{-1} = h' \in H.$$

□

But the easiest way to prove that a subgroup is normal is the following.

Theorem (Normal Subgroup Criterion). *Let $H \leq G$ be a subgroup. Then we have $H \trianglelefteq G$ if and only if there exists a group homomorphism*

$$\varphi : G \longrightarrow G'$$

such that $H = \ker \varphi$.

Proof. First suppose that $H = \ker \varphi$ for some homomorphism $\varphi : G \rightarrow G'$.

Then for all $a \in G$ and $h \in \ker \varphi = H$,

$$\begin{aligned}\varphi(aha^{-1}) &= \varphi(a)\varphi(h)\varphi(a^{-1}) \\ &= \varphi(a)\varphi(h)\varphi(a)^{-1} \\ &= \varphi(a)1\varphi(a)^{-1} \\ &= \varphi(a)\varphi(a)^{-1} \\ &= 1,\end{aligned}$$

hence $aha^{-1} \in \ker \varphi = H$.

Conversely, suppose that $H \trianglelefteq G$. We need to construct a group G' and a homomorphism $\varphi : G \rightarrow G'$ such that $H = \ker \varphi$.

We have already done the work for this. Since $H \trianglelefteq G$, the equivalence relation $\sim_H$ is G -invariant and we obtain a canonical quotient map

$$\begin{aligned}\pi : G &\longrightarrow G/H, \\ a &\longmapsto [a]_H.\end{aligned}$$

Note that

$$a \in \ker \pi \iff [a]_H = [1]_H \iff a1^{-1} \in H \iff a \in H.$$

□

We have shown that

$$\text{normal subgroup} \equiv \text{kernel of group hom.}$$

Combining this with HW2 Problem 4 will give a purely arrow-theoretic characterization of normal subgroups.

[Contrast this with the subgroup lattice point of view. If $H \in \mathcal{L}(G)$ is normal then it is a modular element of $\mathcal{L}(G)$. By definition this means that for all $K, L \in \mathcal{L}(G)$ we have

$$\boxed{K \subseteq L \implies (H \vee K) \wedge L = (H \wedge L) \vee K.}$$

However, not all modular elements of $\mathcal{L}(G)$ are normal and in general there is no lattice-theoretic characterization of normal subgroups. This is one good reason why Ore's program was abandoned. However, the concept of "modularity" is still useful to know.]

Now we are ready to prove the category theory analogue of the Lattice Isomorphism Theorem. It is based on the following universal property of group quotients:

Let $N \trianglelefteq G$. Then a group homomorphism $\varphi : G \rightarrow G'$ is constant on N -cosets if and only if $N \subseteq \ker \varphi$.

Indeed, note that

$$[a]_N = [b]_N \iff a^{-1}b \in N.$$

Also, we have

$$\begin{aligned} \varphi(a) = \varphi(b) &\iff \varphi(a)^{-1}\varphi(b) = 1 \\ &\iff \varphi(a^{-1}b) = 1 \\ &\iff a^{-1}b \in \ker \varphi. \end{aligned}$$

Finally, note that the statement

$$\text{for all } a, b \in G, [a]_N = [b]_N \implies \varphi(a) = \varphi(b)$$

is equivalent to the statement

$$\text{for all } a, b \in G, a^{-1}b \in N \implies a^{-1}b \in \ker \varphi,$$

which is equivalent to $N \subseteq \ker \varphi$.

Thus if $\varphi : G \rightarrow G'$ is a group homomorphism such that $N \subseteq \ker \varphi$, we obtain a unique set function $\bar{\varphi} : G/N \rightarrow G'$:

$$\begin{array}{ccc} & G & \\ \pi \swarrow & & \searrow \varphi \\ G/N & \xrightarrow{\exists! \bar{\varphi}} & G' \end{array}$$

defined by

$$\bar{\varphi}([a]_N) = \varphi(a).$$

And this set function is actually a group homomorphism because

$$\begin{aligned} \bar{\varphi}([a]_N[b]_N) &= \bar{\varphi}([ab]_N) \\ &= \varphi(ab) \\ &= \varphi(a)\varphi(b) \\ &= \bar{\varphi}([a]_N)\bar{\varphi}([b]_N). \end{aligned}$$

In summary:

Theorem (Universal Property of Group Quotients). *Let $N \trianglelefteq G$. If $\varphi : G \rightarrow G'$ is a group homomorphism satisfying $N \subseteq \ker \varphi$, then there exists a unique group homomorphism $\bar{\varphi} : G/N \rightarrow G'$ such that*

$$\begin{array}{ccc} & G & \\ \pi \swarrow & & \searrow \varphi \\ G/N & \xrightarrow{\exists! \bar{\varphi}} & G' \end{array}$$

[In other words, the map $\pi : G \rightarrow G/N$ is the initial object in a certain category of “N-homomorphisms” $\varphi : G \rightarrow G'$.]

Finally we have the famous

Theorem (First Isomorphism Theorem (E. Noether)). *Let $\varphi : G \rightarrow G'$ be any group homomorphism. Then this induces a canonical group isomorphism*

$$\bar{\varphi} : G/\ker \varphi \xrightarrow{\sim} \operatorname{im} \varphi.$$

Proof. I went through the universal properties so I could give the “correct” proof of this. We will show that the map $\varphi : G \rightarrow \operatorname{im} \varphi$ is another initial object in the category of $\ker \varphi$ -homomorphisms.

Indeed, $\varphi : G \rightarrow \operatorname{im} \varphi$ is a $\ker \varphi$ -homomorphism because $\ker \varphi \subseteq \ker \varphi$. Now let $\mu : G \rightarrow G'$ be any other group homomorphism satisfying $\ker \varphi \subseteq \ker \mu$. We will show that

$$\begin{array}{ccc} & G & \\ \varphi \swarrow & & \searrow \mu \\ \operatorname{im} \varphi & \overset{\text{-----}}{\longrightarrow} & G' \\ & \exists! \bar{\mu} & \end{array}$$

By commutativity we must have

$$\bar{\mu}(\varphi(a)) = \mu(a),$$

and this is well defined because

$$\begin{aligned} \varphi(a) = \varphi(b) &\implies \varphi(a^{-1}b) = 1 \\ &\implies a^{-1}b \in \ker \varphi \\ &\implies a^{-1}b \in \ker \mu \\ &\implies \mu(a^{-1}b) = 1 \\ &\implies \mu(a) = \mu(b). \end{aligned}$$

By uniqueness of initial objects we obtain a canonical (i.e., unique) isomorphism

$$\begin{array}{ccc} & G & \\ \pi \swarrow & & \searrow \varphi \\ G/\ker \varphi & \xrightarrow{\quad \bar{\varphi} \quad} & \operatorname{im} \varphi \\ & \bar{\varphi} \sim & \end{array}$$

□

In other words, for all morphisms $\varphi : G \rightarrow G'$ in the category of groups there is a canonical factorization

$$\begin{array}{ccccccc}
 & & & \varphi & & & \\
 & & \swarrow & \text{---} & \searrow & & \\
 G & \xrightarrow{\pi} & G/\ker \varphi & \xrightarrow[\overline{\varphi}]{\sim} & \operatorname{im} \varphi & \xrightarrow{\iota} & G' \\
 & & & & & &
 \end{array}$$

I think now maybe I've gone too far in the direction of category theory, so let's take a break and process what we have done.

Let $\varphi : G \rightarrow G'$ be a group homomorphism. Combining the Lattice Isomorphism Theorem

$$\mathcal{L}(G, \ker \varphi) \cong \mathcal{L}(\operatorname{im} \varphi)$$

with the First Isomorphism Theorem

$$\operatorname{im} \varphi \cong G/\ker \varphi$$

gives us an isomorphism of lattices

$$\mathcal{L}(G, \ker \varphi) \cong \mathcal{L}(G/\ker \varphi).$$

“excision”

This is often very useful.

Example. Let $G = \langle a \rangle$ be a cyclic group and consider the group homomorphism

$$\begin{aligned}
 \varphi : (\mathbb{Z}, +, 0) &\longrightarrow \langle a \rangle, \\
 s &\longmapsto a^s.
 \end{aligned}$$

[We defined the exponential notation just for this purpose.]

The kernel is $n\mathbb{Z}$, where

$$n = \begin{cases} |\langle a \rangle|, & \text{if } |\langle a \rangle| < \infty, \\ 0, & \text{if } |\langle a \rangle| = \infty. \end{cases}$$

We obtain an isomorphism of groups

$$\overline{\varphi} : \mathbb{Z}/n\mathbb{Z} \xrightarrow{\sim} \langle a \rangle$$

and an isomorphism of lattices

$$\mathcal{L}(\langle a \rangle) \cong \mathcal{L}(\mathbb{Z}/n\mathbb{Z}) \cong \mathcal{L}(\mathbb{Z}, n\mathbb{Z}).$$

The good thing about this is that we already understand the structure of

$$\mathcal{L}(\mathbb{Z}, n\mathbb{Z}).$$

Recall that the subgroups of $(\mathbb{Z}, +, 0)$ are precisely $d\mathbb{Z}$ for $d \in \mathbb{Z}$ [by the Division Algorithm] and we have

$$n\mathbb{Z} \subseteq d\mathbb{Z} \iff d \mid n.$$

Thus we obtain an isomorphism

$$\mathcal{L}(\mathbb{Z}, n\mathbb{Z}) \cong \text{Div}(n)^{\text{op}}.$$

and hence,

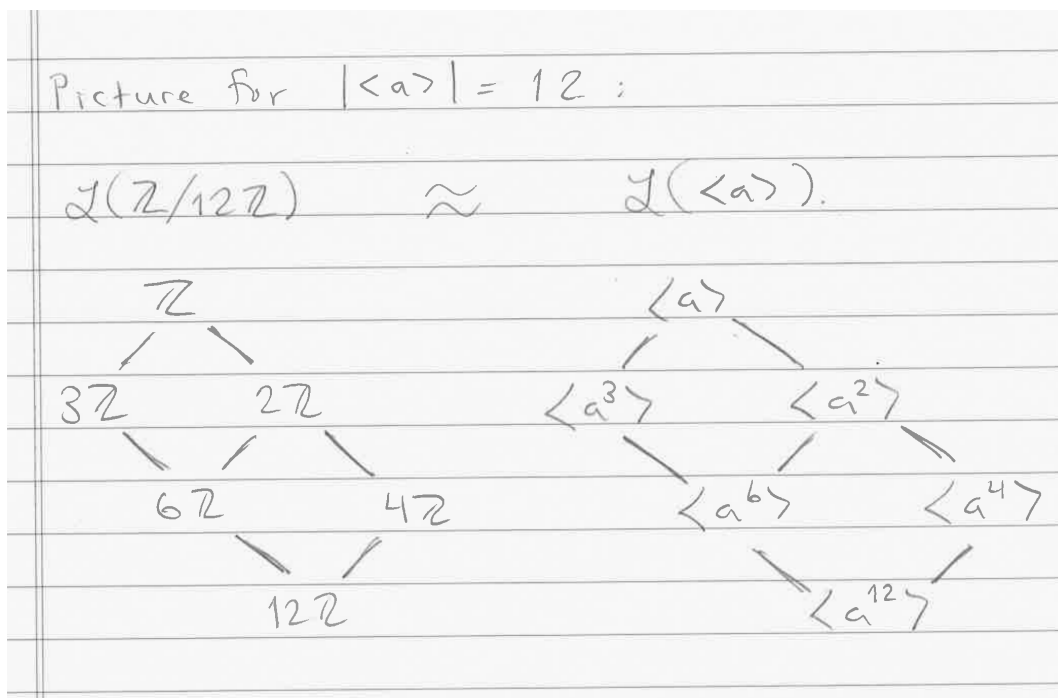
Theorem (Fundamental Theorem of Cyclic Groups). *Every cyclic group is isomorphic to $\mathbb{Z}/n\mathbb{Z}$ for some $n \in \mathbb{Z}$ and we have*

$$\mathcal{L}(\mathbb{Z}/n\mathbb{Z}) \cong \text{Div}(n)^{\text{op}}.$$

For $|\langle a \rangle| = 12$, the isomorphism

$$\mathcal{L}(\mathbb{Z}/12\mathbb{Z}) \cong \mathcal{L}(\langle a \rangle)$$

looks like this:



Remark. Certainly there are quicker ways to prove the Fundamental Theorem of Cyclic Groups, but doing it this way gave us the theorem for free.

I like getting theorems for free.

10 The First Isomorphism Theorem and products of groups

September 24, 2015

Let $\varphi : G \rightarrow G'$ be an arbitrary group homomorphism. It will be an isomorphism in the category of groups if and only if it is bijective as a function.

[Note: this doesn't work in all categories; e.g., it fails in the category of topological spaces.]

We can force φ to be surjective by taking the pullback onto the image:

$$\varphi : G \twoheadrightarrow \text{im } \varphi.$$

Q: How can we force φ to be injective?

A: Note that φ is injective if and only if the kernel is trivial.

Indeed, recall that

$$\varphi(a) = \varphi(b) \iff a^{-1}b \in \ker \varphi.$$

If $\ker \varphi = \{1\}$, then

$$\varphi(a) = \varphi(b) \implies a^{-1}b \in \ker \varphi \implies a^{-1}b = 1 \implies a = b.$$

If φ is injective, then

$$b \in \ker \varphi \implies \varphi(1) = \varphi(b) \implies b = 1.$$

Thus we can make φ injective by “killing the kernel,” i.e., taking the quotient

$$\overline{\varphi} : G/\ker \varphi \xrightarrow{\sim} \text{im } \varphi.$$

Now it's an isomorphism.

Next we will discuss “products” in the category of groups.

Let $H, K \leq G$ be subgroups. Recall that the subset

$$HK = \{hk : h \in H, k \in K\} \subseteq G$$

is a subgroup if and only if

$$HK = KH,$$

and in this case we have

$$H \vee K = HK.$$

Definition. Given a subgroup $H \leq G$, we define its *normalizer* in G to be

$$N_G(H) := \{g \in G : gH = Hg\}.$$

[Exercise: Prove that this is a subgroup of G ; in fact, the largest subgroup of G in which H is normal.]

Claim. Let $H, K \leq G$ be subgroups. If either $H \subseteq N_G(K)$ or $K \subseteq N_G(H)$, then we have $H \vee K = HK$.

Proof. If $H \subseteq N_G(K)$, then we have

$$HK = \bigcup_{h \in H} hK = \bigcup_{h \in H} Kh = KH,$$

and if $K \subseteq N_G(H)$, then we have

$$HK = \bigcup_{k \in K} Hk = \bigcup_{k \in K} kH = KH.$$

In either case we conclude that $H \vee K = HK$. □

Now consider the “multiplication map” from the Cartesian product set $H \times K$ into G ,

$$\begin{aligned} \mu : H \times K &\longrightarrow G, \\ (h, k) &\longmapsto hk. \end{aligned}$$

Note that μ is injective if and only if $H \cap K = 1$.

Indeed, suppose $H \cap K \neq 1$, so there exists $1 \neq g \in H \cap K$. Then we have $g \in H$, $g^{-1} \in K$, and $(g, g^{-1}) \neq (1, 1)$. But

$$\mu(g, g^{-1}) = gg^{-1} = 1 = \mu(1, 1),$$

and hence μ is not injective.

Conversely, suppose that $H \cap K = 1$ and consider $(h_1, k_1), (h_2, k_2) \in H \times K$ such that $h_1k_1 = h_2k_2$. Then we have

$$h_2^{-1}h_1 = k_2k_1^{-1} \in H \cap K,$$

which implies that $h_2^{-1}h_1 = k_2k_1^{-1} = 1$, hence $h_1 = h_2$ and $k_1 = k_2$. Since $(h_1, k_1) = (h_2, k_2)$, we conclude that μ is injective.

Since the image of μ is HK , we conclude that μ is a bijection if and only if

$$\underbrace{H \cap K = 1}_{\text{injective}} \quad \& \quad \underbrace{HK = G}_{\text{surjective}}$$

in other words, if and only if

$$H \wedge K = 1 \quad \& \quad H \vee K = G.$$

In this case we say that H and K are *complementary subgroups* of G .

If H and K are complementary, then there is a unique way to define a group operation on the set $H \times K$ such that the bijection μ becomes a group isomorphism

$$\mu : H \times K \xrightarrow{\sim} G.$$

Let's investigate a few cases.

(1) Suppose that

$$H \wedge K = 1, \quad H \vee K = G, \quad \text{and} \quad H \subseteq N_G(K).$$

[Note that $H \subseteq N_G(K)$ and $H \vee K = G$ together imply $K \trianglelefteq G$.]

Then for all $(h_1, k_1), (h_2, k_2) \in H \times K$ we must define a product

$$(h_1, k_1) \bullet (h_2, k_2) = (h, k)$$

such that

$$\mu(h_1, k_1)\mu(h_2, k_2) = \mu(h, k), \quad h_1 k_1 h_2 k_2 = h k.$$

Since $H \subseteq N_G(K)$, we have $h_2^{-1} k_1 h_2 \in K$. If we define

$$h = h_1 h_2 \in H \quad \text{and} \quad k = (h_2^{-1} k_1 h_2) k_2 \in K,$$

then we have

$$h k = h_1 h_2 (h_2^{-1} k_1 h_2) k_2 = h_1 k_1 h_2 k_2,$$

as desired. By uniqueness, this is the only solution for (h, k) .

The set $H \times K$ together with the operation

$$(h_1, k_1) \bullet (h_2, k_2) := (h_1 h_2, (h_2^{-1} k_1 h_2) k_2)$$

is called the (internal) *semi-direct product* of H and K . We will denote it by

$$H \rtimes K = (H \times K, \bullet, (1, 1)).$$

[Mnemonic: The triangle pointing at K indicates that K is normal.]

(2) Suppose that

$$H \wedge K = 1, \quad H \vee K = G, \quad \text{and} \quad K \subseteq N_G(H).$$

[Hence also $H \trianglelefteq G$.]

Similar reasoning shows that the group operation on $H \times K$ is

$$(h_1, k_1) \bullet (h_2, k_2) = (h_1 (k_1 h_2 k_1^{-1}), k_1 k_2).$$

In this case we write

$$H \rtimes K = (H \times K, \bullet, (1, 1)).$$

This is also called an (internal) semi-direct product.

(3) Suppose that

$$H \wedge K = 1, \quad H \vee K = G, \quad H \subseteq N_G(K), \quad \text{and} \quad K \subseteq N_G(H).$$

[Hence also $H \trianglelefteq G$ and $K \trianglelefteq G$.]

Then for all $h \in H, k \in K$, we have $h k h^{-1} \in K$ (because $H \subseteq N_G(K)$) and $k h^{-1} k^{-1} \in H$ (because $K \subseteq N_G(H)$), hence

$$h(k h^{-1} k^{-1}) = (h k h^{-1}) k^{-1} \in H \cap K.$$

Since $H \wedge K = 1$, we conclude that

$$hkh^{-1}k^{-1} = 1, \quad hk = kh.$$

This implies that the group operation on $H \times K$ is the most obvious one,

$$(h_1, k_1) \bullet (h_2, k_2) = (h_1 h_2, k_1 k_2).$$

In this case we write

$$H \times K = (H \times K, \bullet, (1, 1))$$

and call this the (internal) *direct product* of H and K .

What do I mean by “internal”? This suggests that there is also an “external” perspective on direct and semi-direct products.

Problem. Given two (isomorphism classes of) groups H and K , construct a group G (up to isomorphism) containing (isomorphic copies of) H and K such that

$$G = H \rtimes K, \quad G = H \ltimes K, \quad \text{or} \quad G = H \times K,$$

respectively. This G will be called the “external” direct or semi-direct product of H and K .

Solution: See HW2 Problem 6.

So “internal” means that H and K are given as subgroups of some group G . “External” means that H and K are given abstractly and we have to construct the group G for ourselves out of thin air.

Discussion:

- Let $H, K \leq G$ be subgroups such that

$$H \wedge K = 1 \quad \& \quad H \vee K = G.$$

In this case we say that G is the internal product of H and K . Not all internal products are semi-direct. [I’ll show you an example next time.] After doing HW2 Problem 6 you might wonder if internal products can be characterized “externally.”

Yes they can. This concept is called the “knot product” or the “Zappa–Szép product.” It starts with two group homomorphisms

$$\alpha : K \longrightarrow \text{Aut}(H), \quad \beta : H \longrightarrow \text{Aut}(K),$$

[subject to certain conditions] and then defines

$$(h_1, k_1) \bullet (h_2, k_2) = (h_1 \alpha_{k_1}(h_2), \beta_{h_2}(k_1) k_2).$$

The notation

$$H \bowtie_{(\alpha, \beta)} K = (H \times K, \bullet, (1_H, 1_K))$$

is used, which I think is really terrible since the symbol $\bowtie$ seems to suggest that both subgroups are normal, when in reality neither of them is normal.

If I ran the world I would switch the symbols

$$\times \longleftrightarrow \bowtie.$$

(Sigh...)

- The situation is much easier in the category of abelian groups. Let G be abelian and let $H, K \leq G$ be subgroups such that

$$H \wedge K = 1 \quad \& \quad H \vee K = G.$$

Since every subgroup of an abelian group is normal, we must have $H \trianglelefteq G$ and $K \trianglelefteq G$, hence G is the direct product of H and K . In this case we write

$$G = H \oplus K$$

and we say G is the *direct sum* of H and K .

[Sadly, we can't use the symbol $\otimes$ for nonabelian direct products because this symbol is being used for other purposes...]

11 Direct, semidirect, Zappa–Szép products

September 29, 2015

Recall: Let $H, K \leq G$ be subgroups. The following are equivalent:

- $HK \subseteq G$ is a subgroup.
- $H \vee K = HK$.
- $HK = KH$.

Any of these is implied by

$$H \subseteq N_G(K) \quad \text{or} \quad K \subseteq N_G(H),$$

but the last statement is not equivalent to the others [as we will see].

Consider the multiplication function

$$\begin{aligned} \mu : H \times K &\longrightarrow G, \\ (h, k) &\longmapsto hk. \end{aligned}$$

It is a bijection if and only if

$$\underbrace{H \wedge K = 1}_{\text{injective}} \quad \& \quad \underbrace{H \vee K = G}_{\text{surjective}}.$$

This bijection induces a group structure on the set $H \times K$.

There are four cases. Note that, since $G = H \vee K = HK$, we have

$$H \subseteq N_G(K) \iff K \trianglelefteq G, \quad K \subseteq N_G(H) \iff H \trianglelefteq G.$$

Case 1. If $H \trianglelefteq G$ and $K \trianglelefteq G$, we write

$$G = H \times K$$

and call this an *internal direct product*.

Case 2. If $H \trianglelefteq G$ and $K \not\trianglelefteq G$, we write

$$G = H \rtimes K$$

and call this an *internal semi-direct product*.

Case 3. If $H \not\trianglelefteq G$ and $K \trianglelefteq G$, we write

$$G = H \ltimes K.$$

This is also called a semi-direct product.

Case 4. If $H \not\trianglelefteq G$ and $K \not\trianglelefteq G$, we write

$$G = H \ltimes K$$

and call this an *internal Zappa–Szép product*.

[This notation is bad, but kind of standard. Mnemonic: An open side means normal; a closed side means not normal.]

Finally, if G is abelian, then all four cases collapse into the direct sum

$$G = H \oplus K.$$

Each situation has an “external” characterization: Given abstract groups H, K , construct a group G containing isomorphic copies of H, K such that, respectively,

$$G = H \times K, \quad G = H \rtimes K, \quad G = H \ltimes K, \quad \text{or} \quad G = H \ltimes K.$$

The construction of $H \times K$ is unique up to isomorphism, but there are many ways to construct the others.

Case 1.

Let H, K be groups. Define a group structure on $H \times K$ by

$$(h_1, k_1) \bullet (h_2, k_2) := (h_1 h_2, k_1 k_2).$$

Call this group G . We have natural injective morphisms

$$\begin{aligned} i_1 : H &\longrightarrow G, & h &\longmapsto (h, 1_K), \\ i_2 : K &\longrightarrow G, & k &\longmapsto (1_H, k). \end{aligned}$$

Denote their images by $\tilde{H}$ and $\tilde{K}$ inside G .

Check that

$$G = \tilde{H} \times \tilde{K}.$$

Cases 2 & 3.

Let H, K be groups and consider any group homomorphism

$$\varphi : K \longrightarrow \text{Aut}_{\text{Grp}}(H).$$

For each $k \in K$ we obtain a bijection $\varphi_k : H \rightarrow H$ such that

$$\begin{aligned} \varphi_k(h_1 h_2) &= \varphi_k(h_1) \varphi_k(h_2) && \text{for all } k \in K \text{ and } h_1, h_2 \in H, \\ \varphi_{k_1 k_2}(h) &= \varphi_{k_1}(\varphi_{k_2}(h)) && \text{for all } k_1, k_2 \in K \text{ and } h \in H. \end{aligned}$$

Define a group structure on $H \times K$ by

$$(h_1, k_1) \bullet (h_2, k_2) := (h_1 \varphi_{k_1}(h_2), k_1 k_2).$$

Let G denote this group, and let $\tilde{H}, \tilde{K}$ denote the images of H, K as before.

You will show on HW2 Problem 6 that

$$G = \tilde{H} \rtimes \tilde{K}.$$

Moreover, for all $h \in H$ and $k \in K$, with images $\tilde{h}$ and $\tilde{k}$, we have

$$\tilde{\varphi}_k(\tilde{h}) = \tilde{k} \tilde{h} \tilde{k}^{-1}.$$

[Thus we have turned the abstract homomorphism φ into “conjugation.”]

Case 4.

Let H, K be groups and let

$$\alpha : K \longrightarrow \text{Aut}_{\text{set}}(H), \quad \beta : H \longrightarrow \text{Aut}_{\text{set}}(K)^{\text{op}}$$

be two group homomorphisms.

[“op” means that the order of the group operation is reversed.]

Suppose that the following compatibility conditions hold:

$$\begin{aligned} \alpha_k(h_1 h_2) &= \alpha_k(h_1) \alpha_{\beta_{h_1}(k)}(h_2), \\ \beta_h(k_1 k_2) &= \beta_{\alpha_{k_2}(h)}(k_1) \beta_h(k_2). \end{aligned}$$

Define an operation on $H \times K$ by

$$(h_1, k_1) \bullet (h_2, k_2) := (h_1 \alpha_{k_1}(h_2), \beta_{h_2}(k_1)k_2).$$

Let G denote this group. We use the notation

$$G = H \rtimes_{\alpha \bowtie \beta} K$$

and call this the *external Zappa–Szép product with respect to α, β* .

One can show [we won't!] that

$$G = \tilde{H} \rtimes \tilde{K}$$

and that every internal Zappa–Szép product arises in this way.

Examples:

1. Consider the cyclic groups

$$\mathbb{Z}/m\mathbb{Z} = \langle a \rangle, \quad \mathbb{Z}/n\mathbb{Z} = \langle b \rangle.$$

A group homomorphism

$$\varphi : \langle b \rangle \longrightarrow \langle b \rangle$$

is determined

by its action on the generator b , say

$$\varphi(b) = b^k$$

for some $k \in \mathbb{Z}$. This homomorphism is invertible [and hence an automorphism] if and only if $\gcd(k, n) = 1$. Thus

$$\text{Aut}_{\text{Grp}}(\mathbb{Z}/n\mathbb{Z}) \cong (\mathbb{Z}/n\mathbb{Z})^\times, \quad (b \mapsto b^k) \longleftrightarrow k.$$

A homomorphism

$$\varphi : \langle a \rangle \longrightarrow \text{Aut}_{\text{Grp}}(\langle b \rangle)$$

is determined by how it acts on the generator a . Choose $k \in (\mathbb{Z}/n\mathbb{Z})^\times$ and define

$$\varphi_a(b) = b^k.$$

One can check that the resulting semi-direct product has the presentation

$$\langle a, b : a^m = b^n = 1, aba^{-1} = b^k \rangle.$$

[Think: What happens if $n = 0$?]

2. Dihedral groups are a special case. Let $m = 2$ and $k = -1$. Then

$$D_{2n} = \langle a \rangle \rtimes \langle b \rangle = \langle a, b : a^2 = b^n = 1, aba^{-1} = b^{-1} \rangle.$$

This is isomorphic to the group of symmetries of a regular n -gon in the plane.

3. Let $[4] = \{1, 2, 3, 4\}$ and consider its automorphism group

$$S_4 := \text{Aut}_{\text{set}}([4]),$$

called the *symmetric group*. Its elements are bijections, also called permutations.

For example, in cycle notation,

$$\begin{aligned} (1234) : & \quad 1 \mapsto 2 \mapsto 3 \mapsto 4 \mapsto 1, \\ (12)(34) : & \quad 1 \leftrightarrow 2, \quad 3 \leftrightarrow 4, \\ (123)(4) : & \quad 1 \mapsto 2 \mapsto 3 \mapsto 1, \quad 4 \mapsto 4. \end{aligned}$$

Normally we drop singleton cycles, so

$$(123) = (123)(4).$$

Consider the subgroups

$$H = \langle (1234), (12)(34) \rangle, \quad K = \langle (123) \rangle.$$

One can show that

$$H \cong D_8, \quad K \cong \mathbb{Z}/3\mathbb{Z},$$

and

$$S_4 = H \rtimes K,$$

with neither subgroup normal.

4. Let V be a vector space over a field $\mathbb{K}$; in particular, V is an abelian group. Define the forgetful functor

$$\text{grp} : \mathbb{K}\text{-Vec} \longrightarrow \text{Grp},$$

which sends each $\mathbb{K}$ -vector space to its underlying abelian group.

[It forgets scalar multiplication.]

Recall the general linear group

$$\text{GL}(V) := \text{Aut}_{\mathbb{K}\text{-Vec}}(V).$$

This automatically gives a group homomorphism

$$\varphi : \text{GL}(V) \longrightarrow \text{Aut}_{\text{Grp}}(\text{grp}(V)),$$

since for all $X \in \text{GL}(V)$ and $\alpha, \beta \in V$ we have

$$X(\alpha + \beta) = X\alpha + X\beta.$$

The resulting semi-direct product is called the *general affine group*:

$$\boxed{\text{GA}(V) := \text{GL}(V) \rtimes_{\varphi} \text{grp}(V)}.$$

We will discuss its geometric meaning later.

12 Isomorphism theorems and subgroup lattices

October 1, 2015

Today we are going to upgrade the First Isomorphism Theorem for groups.

Let $N \trianglelefteq G$ be a normal subgroup and consider the canonical quotient map

$$\pi : G \twoheadrightarrow G/N.$$

This induces a Galois connection

$$\mathcal{L}(G) \begin{matrix} \xrightarrow{\pi} \\ \xleftarrow{\pi^{-1}} \end{matrix} \mathcal{L}(G/N),$$

which restricts to an isomorphism of lattices

$$\mathcal{L}(G, N) \begin{matrix} \xrightarrow{\pi} \\ \xleftarrow{\pi^{-1}} \end{matrix} \mathcal{L}(G/N).$$

Let's examine this isomorphism. Suppose we have $H \in \mathcal{L}(G, N)$, i.e., H is a subgroup of G containing N .

Since $N \trianglelefteq G$ we also have $N \trianglelefteq H$, so we can define the quotient group H/N . The quotient map $H \twoheadrightarrow H/N$ is defined by

$$h \mapsto hN.$$

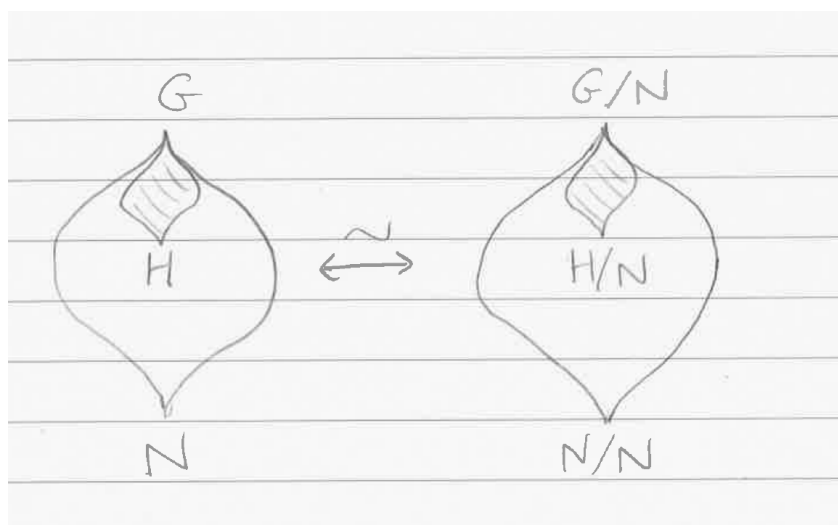
But note that this is the map $\pi : G \twoheadrightarrow G/N$ restricted to H . We conclude that

$$\pi(H) = H/N,$$

and it follows that

$$\pi^{-1}(H/N) = H.$$

Picture:



Now suppose we also have $H \trianglelefteq G$, so we can define the quotient group G/H .

By the previous isomorphism we know that

$$\mathcal{L}(G, H) \cong \mathcal{L}(G/N, H/N).$$

Does this lattice isomorphism come from an isomorphism of groups

$$\frac{G}{H} \cong \frac{G/N}{H/N} ?$$

What do we need to check?

Lemma. Consider a surjective group homomorphism

$$\varphi : G \twoheadrightarrow G'$$

and two subgroups $H \leq G, H' \leq G'$.

- If $H \trianglelefteq G$, then $\varphi(H) \trianglelefteq G'$.
- If $H' \trianglelefteq G'$, then $\varphi^{-1}(H') \trianglelefteq G$.

Proof. For the first part, suppose that $H \trianglelefteq G$ and consider $g' \in G', h' \in \varphi(H)$. Since φ is surjective, there exist $g \in G, h \in H$ such that $\varphi(g) = g'$ and $\varphi(h) = h'$. Then, since $H \trianglelefteq G$, we have $ghg^{-1} \in H$, hence

$$g'h'(g')^{-1} = \varphi(g)\varphi(h)\varphi(g)^{-1} = \varphi(ghg^{-1}) \in \varphi(H).$$

For the second part, suppose that $H' \trianglelefteq G'$ and consider any $h \in \varphi^{-1}(H'), g \in G$. Since $\varphi(h) \in H'$ and $H' \trianglelefteq G'$, we have

$$\varphi(ghg^{-1}) = \varphi(g)\varphi(h)\varphi(g)^{-1} \in H',$$

and hence $ghg^{-1} \in \varphi^{-1}(H')$. □

Returning to the situation

$$N \trianglelefteq G \quad \text{and} \quad H \in \mathcal{L}(G, N),$$

we have

$$H \trianglelefteq G \iff H/N \trianglelefteq G/N.$$

Proof. Consider the surjective homomorphism

$$\pi : G \twoheadrightarrow G/N$$

and note that $\pi(H) = H/N$ and $\pi^{-1}(H/N) = H$. □

Theorem (Isomorphism Theorem (either the Second or Third, depending on who you ask)). *Let $N \trianglelefteq G$ and $H \in \mathcal{L}(G, N)$. Then we have $H \trianglelefteq G$ if and only if $H/N \trianglelefteq G/N$, in which case we get a group isomorphism*

$$\frac{G}{H} \cong \frac{G/N}{H/N}.$$

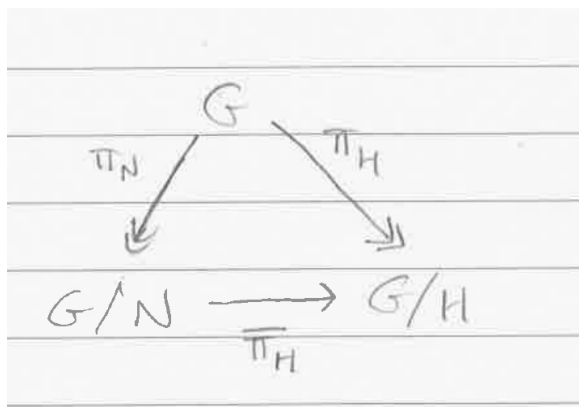
Proof. We will use the universal property of group quotients. Let $H \trianglelefteq G$ and consider the quotient homomorphisms

$$\pi_N : G \twoheadrightarrow G/N, \quad \pi_H : G \twoheadrightarrow G/H.$$

Since

$$\ker \pi_N = N \subseteq H = \ker \pi_H,$$

the universal property of π_N says that π_H factors through π_N as follows:



Since π_H is surjective, we conclude that

$$\bar{\pi}_H : G/N \longrightarrow G/H$$

is surjective.

To finish the proof we will show that $\ker \bar{\pi}_H = H/N$, and hence

$$\frac{G/N}{H/N} = \frac{G/N}{\ker \bar{\pi}_H} \cong \text{im } \bar{\pi}_H = G/H$$

by the First Isomorphism Theorem.

Recall that $\bar{\pi}_H$ is defined (uniquely) by

$$\bar{\pi}_H(gN) = \pi_H(g) = gH.$$

We conclude that

$$\bar{\pi}_H(gN) = 1_{G/H} \iff gH = 1_{G/H} \iff g \in H,$$

and hence

$$\ker \bar{\pi}_H = \{gN : g \in H\} = H/N.$$

□

Remark. This theorem actually goes back to Galois. Let F be a field and consider a finite subgroup

$$G \leq \text{Aut}(F).$$

Recall the classical Galois connection

$$\mathcal{L}(F) \begin{matrix} \xrightarrow{E \mapsto \text{Aut}(F/E)} \\ \xleftrightarrow{\quad} \\ \xleftarrow{H \mapsto F^H} \end{matrix} \mathcal{L}(\text{Aut}(F)).$$

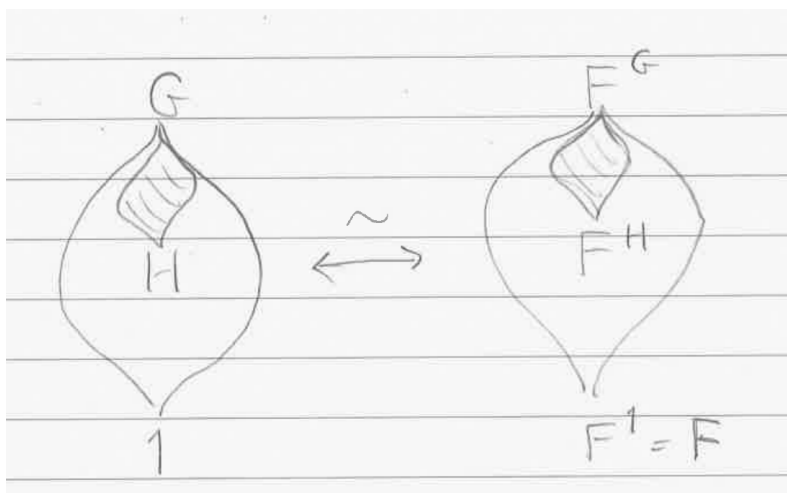
An important property I didn't mention the first time is that, given a subgroup $H \leq G$, we have

$$H \trianglelefteq G \iff \text{the field extension } F^G \subseteq F^H \text{ is "Galois" (whatever that means),}$$

in which case we get an isomorphism of groups

$$\frac{G}{H} = \frac{\text{Aut}(F/F^G)}{\text{Aut}(F/F^H)} \cong \text{Aut}(F^H/F^G).$$

Picture:



In fact, this is the reason that Galois invented the concept of normal subgroups. Recall that the fundamental theorem says the following:

Theorem (F.T.G.T.). *If $f(x) \in F^G[x]$ is a polynomial with splitting field F , then the polynomial is "solvable by radicals" if and only if there exists a chain of subgroups*

$$G = G_0 \triangleright G_1 \triangleright G_2 \triangleright \cdots \triangleright G_m = 1$$

such that each quotient G_i/G_{i+1} is an abelian group. In this case we say by analogy that the group G is solvable.

My next goal is to examine some issues surrounding solvable groups.

Definition. We say that a group G is *simple* if it has no normal subgroups except 1 and G .

Example. Consider $n \in \mathbb{Z}$. Recall the Fundamental Theorem of Cyclic Groups:

$$\mathcal{L}(\mathbb{Z}/n\mathbb{Z}) \cong \text{Div}(n)^{\text{op}}.$$

Since every subgroup of $\mathbb{Z}/n\mathbb{Z}$ is normal, we conclude that $\mathbb{Z}/n\mathbb{Z}$ is a simple group if and only if n is prime.

Thus, in some sense, simple groups are a generalization of prime numbers.

Now suppose that G is not simple. In this case we would like to “factorize” G into simple groups.

Here’s how we will proceed: Let $G_0 := G$. Since G is not simple we can find a nontrivial normal subgroup. Let G_1 be a maximal such group.

[Warning: This G_1 might not exist if G is badly infinite. Let’s just assume that G_1 exists.]

Claim. Then the quotient G_0/G_1 is a simple group.

Proof. We have a lattice isomorphism

$$\mathcal{L}(G_0, G_1) \cong \mathcal{L}(G_0/G_1)$$

that preserves normality.

Since there are no normal subgroups between G_0 and G_1 (by definition), we conclude that there are no normal subgroups between G_0/G_1 and 1. $\square$

Now we continue the process. Let G_2 be a maximal proper normal subgroup of G_1 so that the quotient G_1/G_2 is simple.

If this process is able to continue, and if it terminates, then we obtain a chain of subgroups

$$G = G_0 \supseteq G_1 \supseteq G_2 \supseteq \cdots \supseteq G_m = 1 \tag{*}$$

such that G_i/G_{i+1} is simple for all i . We call such a chain $(*)$ a *composition series* for G , and the simple quotients G_i/G_{i+1} are called “the” *composition factors* of G .

[Why did I say “the”?]

13 Simple groups and composition series

October 6, 2015

Recall: We say that a group G is *simple* if the trivial subgroup $1 \leq G$ is maximal among proper normal subgroups. [In particular, we must have $1 \neq G$.]

Theorem. The simple abelian groups are just $\mathbb{Z}/p\mathbb{Z}$ for p prime.

Proof. Let G be a simple abelian group. Since $G \neq 1$, there exists an element $1 \neq a \in G$. Since $1 \neq \langle a \rangle \leq G$, and since 1 is maximal normal, we have $\langle a \rangle = G$, hence G is cyclic.

Note that $\mathbb{Z}$ is not simple, so G must be finite, say $|G| = |\langle a \rangle| = n$. If there exists $1 < k < n$ with $k \mid n$, then we obtain a proper normal subgroup

$$1 = \langle a^n \rangle \subsetneq \langle a^k \rangle \subsetneq \langle a^1 \rangle = G,$$

which contradicts the fact that G is simple. We conclude that n is prime. $\square$

Thus simple groups are a generalization of prime numbers, and the interesting simple groups are non-abelian.

Examples:

1. The group of rotational symmetries of the regular dodecahedron/icosahedron is a simple group of order

$$60 = 2^2 \cdot 3 \cdot 5.$$

It is the smallest non-abelian simple group.

2. If we view S_n as the group of $n \times n$ permutation matrices, then the determinant is a surjective group homomorphism onto the group of units $\mathbb{Z}^\times := \{\pm 1\}$:

$$\det : S_n \twoheadrightarrow \mathbb{Z}^\times.$$

The kernel is a normal subgroup,

called the *alternating group*:

$$A_n := \ker(\det) \leq S_n.$$

By the First Isomorphism Theorem and Lagrange's Theorem (i.e., the fact that there is a bijection between any two cosets) we have

$$\frac{|S_n|}{|A_n|} = |\mathbb{Z}^\times|.$$

Hence

$$|A_n| = \frac{|S_n|}{|\mathbb{Z}^\times|} = \frac{n!}{2} = 3 \cdot 4 \cdot 5 \cdots n.$$

Theorem. *The group A_n is simple for all $n \geq 5$. Moreover, A_5 is isomorphic to the unique simple group of order 60.*

Corollary. There exist infinitely many non-abelian simple groups.

3. Let K be a field and consider the general linear group $GL_n(K)$. The determinant is a surjective homomorphism onto the group of units $K^\times = K \setminus \{0\}$:

$$\det : GL_n(K) \twoheadrightarrow K^\times.$$

The kernel is a normal subgroup, called the *special linear group*:

$$SL_n(K) := \ker(\det) \leq GL_n(K).$$

The center of $GL_n(K)$ is given by scalar matrices

$$Z(GL_n(K)) = \{\alpha I : \alpha \in K, \alpha \neq 0\},$$

and the center of $SL_n(K)$ is given by n th roots of unity

$$Z(SL_n(K)) = \{\alpha I : \alpha \in K, \alpha^n = 1\}.$$

The center is always normal. We kill it by defining the *projective special linear group*:

$$PSL_n(K) := \frac{SL_n(K)}{Z(SL_n(K))}.$$

Theorem. *The group $PSL_n(K)$ is simple for all $n \geq 2$ and for all fields K , except when $n = 2$ and $K = \mathbb{F}_2$ or $\mathbb{F}_3$.*

Remark. I proved this theorem last time I taught the course and it ate up too much time, so I won't prove it this time. I might (or you might) prove that A_n is simple.

When $K = \mathbb{F}_q$ is a finite field of order q , we usually just write

$$PSL_n(\mathbb{F}_q) = PSL_n(q).$$

This gives us another infinite family of finite simple groups. One can check that

$$|PSL_n(q)| = \frac{q^{\binom{n}{2}}(q^2 - 1)(q^3 - 1) \cdots (q^n - 1)}{\gcd(n, q - 1)}.$$

There is a sense in which

$$PSL_n(q) \longrightarrow A_n \quad \text{as } q \longrightarrow 1,$$

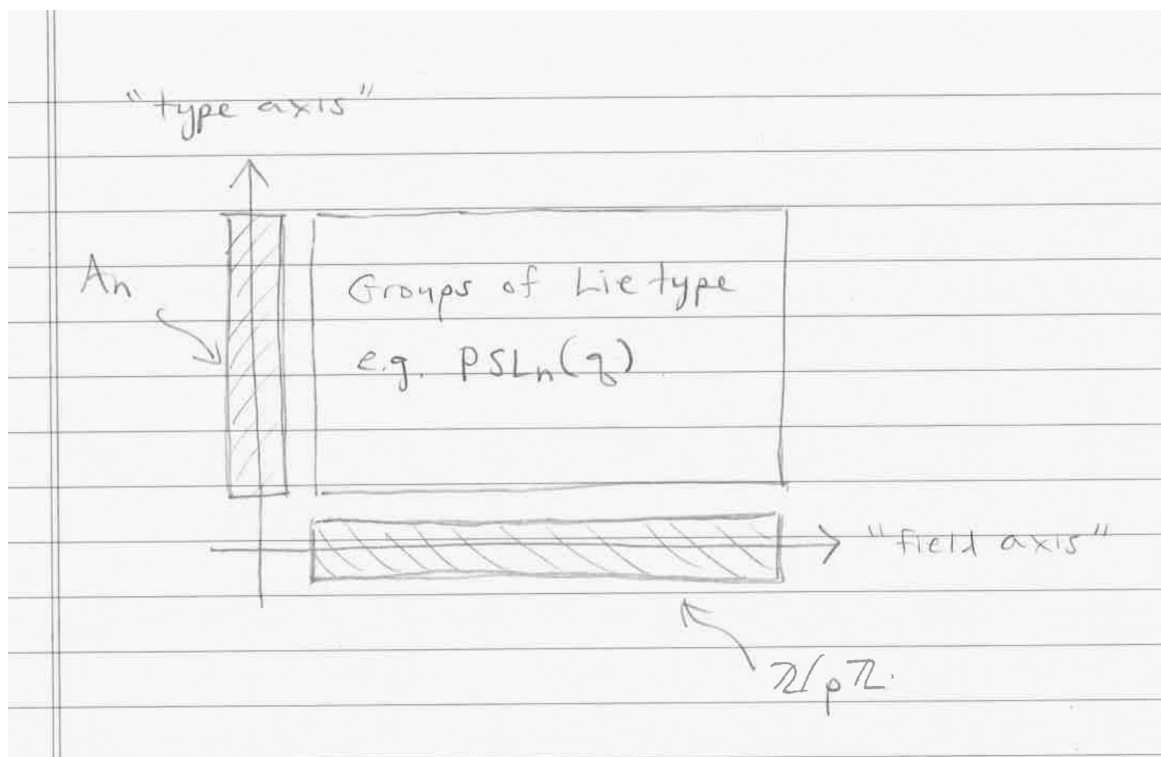
but nobody really knows what this means because there is no such thing as a “field of order 1.”

4. After a century of work on thousands of pages of theorems, the finite simple groups have been classified.

HUGE THEOREM. There are 18 infinite families and 26 “sporadic” finite simple groups. The largest sporadic group is called the Monster M . It has order

$$|M| \approx 8 \cdot 10^{53}.$$

The 18 infinite families can be visualized as follows:



The classification of finite groups of Lie type depends on the classification of semisimple Lie algebras.

Apart from classifying all simple groups, the other major problem is to show how to “put together” simple groups to obtain all other groups.

We began to discuss this last time.

Let G be a group and suppose that we can find a chain of subgroups

$$G = G_0 \supseteq G_1 \supseteq G_2 \supseteq \cdots \supseteq G_m = 1 \quad (*)$$

such that G_{i+1} is a maximal proper normal subgroup in G_i for all i . In this case we say that $(*)$ is a *composition series* for G . Since G_{i+1} is maximal normal in G_i , we know that the quotient G_i/G_{i+1} is a simple group. The groups G_i/G_{i+1} are called “the” *composition factors* of G .

Q: Why “the”?

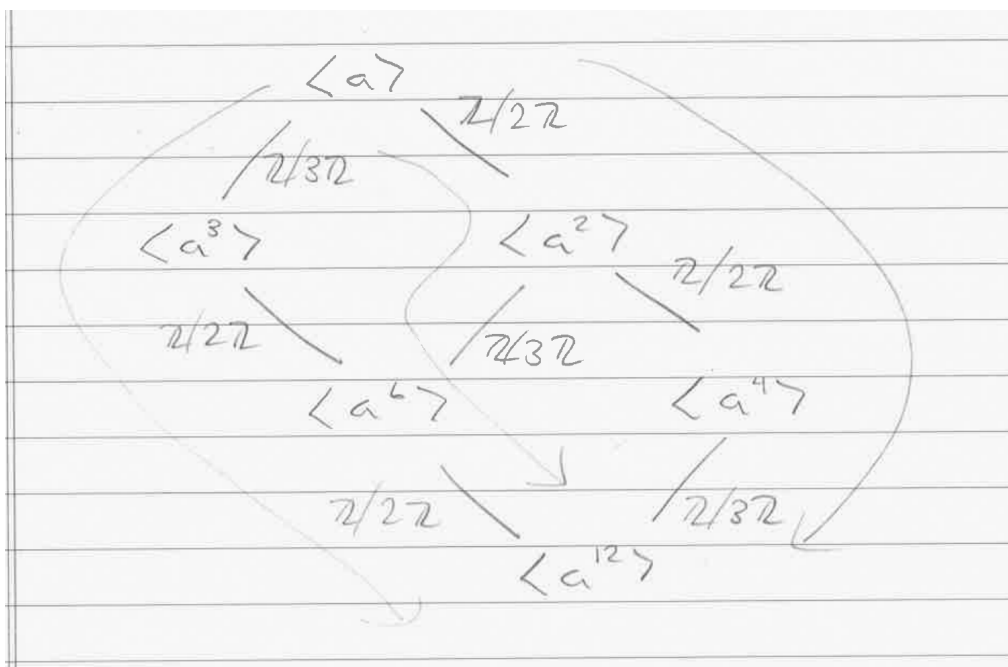
Theorem (Jordan–Hölder). *Let G be a group. If G has a composition series, then any two composition series are “equivalent” in the sense that*

- *they have the same length;*

- the composition factors are isomorphic in pairs (up to a permutation).

Remark. If $G = \mathbb{Z}/n\mathbb{Z}$, then the Jordan–Hölder Theorem just says that n has a unique prime factorization.

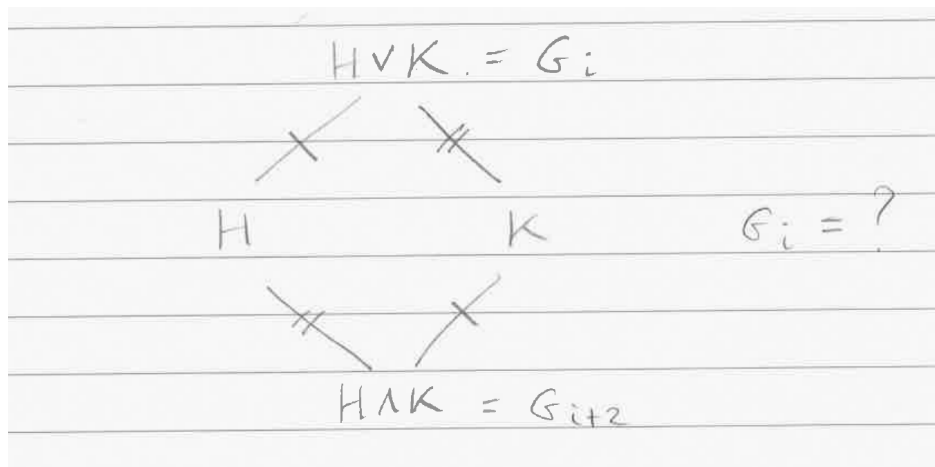
For example, if $n = 12$, then there are three composition series. Let $\mathbb{Z}/12\mathbb{Z} = \langle a \rangle$.



The composition factors are

- $\mathbb{Z}/3\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}$;
- $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}$;
- $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z}$.

Note that the composition series are related by “local swaps.” Suppose we have two maximal normal subgroups H, K in G_i such that $H \vee K = G_i$ and $H \wedge K = G_{i+2}$:



We would like to show that

$$\frac{H \vee K}{H} \cong \frac{K}{H \wedge K} \quad \text{and} \quad \frac{H \vee K}{K} \cong \frac{H}{H \wedge K},$$

so we could move from one composition series to another without changing the composition factors. But is this true?

A: Yes. You will prove it on HW3...

We could give a proof of Jordan–Hölder using just this fact, but to get a very slick proof we will use two lemmas that are perfectly defined for the task:

1. Zassenhaus (“Butterfly”) Lemma.
2. Schreier Refinement Theorem.

Stay tuned...

14 Jordan–Hölder, Zassenhaus, Schreier refinement

October 13, 2015

Recall the statement of the Jordan–Hölder Theorem for groups:

Let G be a group. If G has a composition series, i.e., a chain of subgroups

$$G = G_0 \supseteq G_1 \supseteq G_2 \supseteq \cdots \supseteq G_m = 1$$

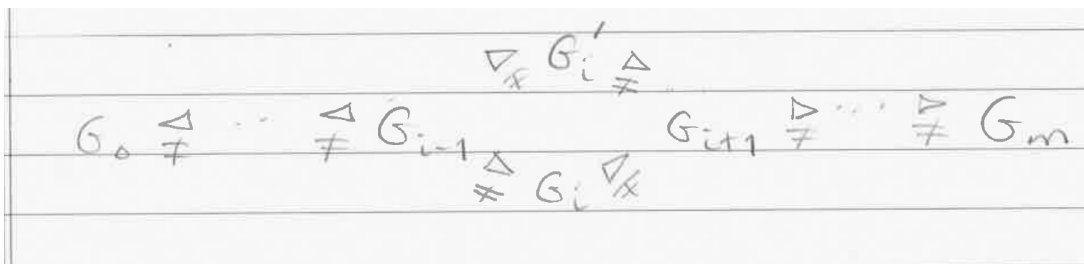
such that G_i/G_{i+1} is simple for all i , then any two composition series are equivalent in the sense that

- they have the same length;

- the composition factors G_i/G_{i+1} are isomorphic up to a permutation.

One can prove the theorem using HW3 Problem 2 by showing that

any two composition series are related by “local swaps” of the form



where

$$G_i \vee G'_i = G_{i-1} \quad \text{and} \quad G_i \wedge G'_i = G_{i+1}.$$

In this case HW3.2 says that

$$\frac{G_{i-1}}{G_i} \cong \frac{G'_i}{G_{i+1}} \quad \text{and} \quad \frac{G_i}{G_{i+1}} \cong \frac{G_{i-1}}{G'_i},$$

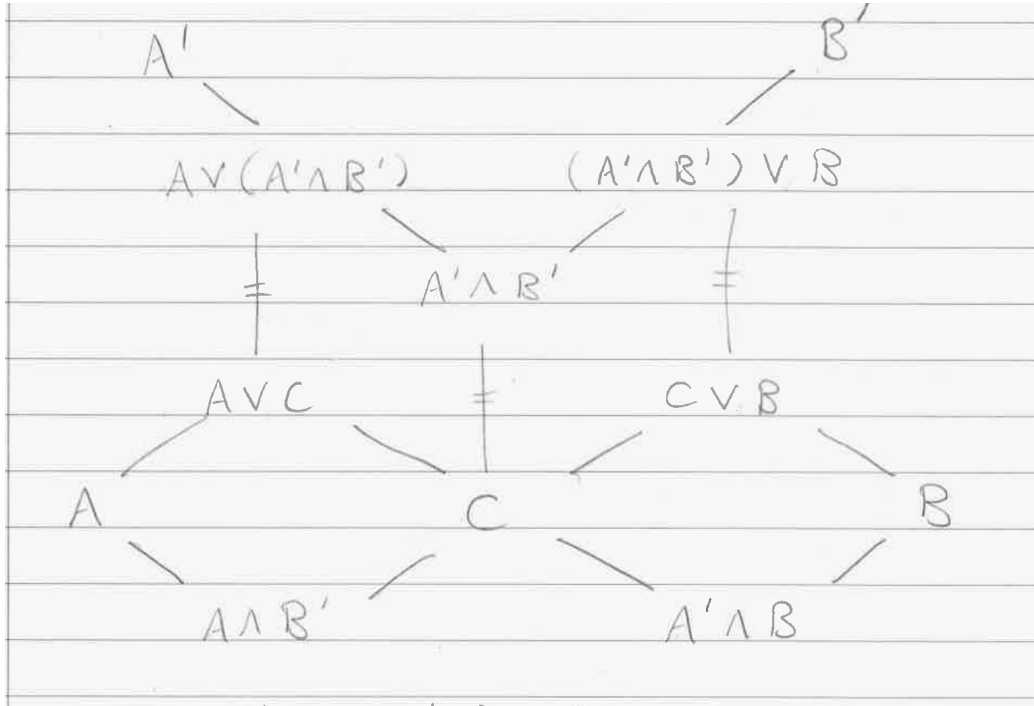
so the composition factors are preserved.

But I prefer to give the “book proof,” using two lemmas that were explicitly designed to give the smoothest proof of Jordan–Hölder.

1. Zassenhaus (“Butterfly”) Lemma.
2. Schreier Refinement Theorem.

1. Zassenhaus Lemma

If $A, A', B, B' \leq G$ are subgroups with $A \trianglelefteq A'$ and $B \trianglelefteq B'$, then we obtain a diagram that looks like an upside-down butterfly:



Here we have defined

$$C := (A \wedge B') \vee (A' \wedge B).$$

The lemma says that if $A \leq A'$ and $B \leq B'$, then we also have

$$A \vee C \leq A \vee (A' \wedge B') \quad \text{and} \quad C \vee B \leq (A' \wedge B') \vee B,$$

and an isomorphism of groups

$$\frac{A \vee (A' \wedge B')}{A \vee C} \cong \frac{(A' \wedge B') \vee B}{C \vee B}.$$

Finally, since

$$\begin{aligned} A \vee C &= A \vee [(A \wedge B') \vee (A' \wedge B)] \\ &= [A \vee (A \wedge B')] \vee (A' \wedge B) \\ &= A \vee (A' \wedge B), \end{aligned}$$

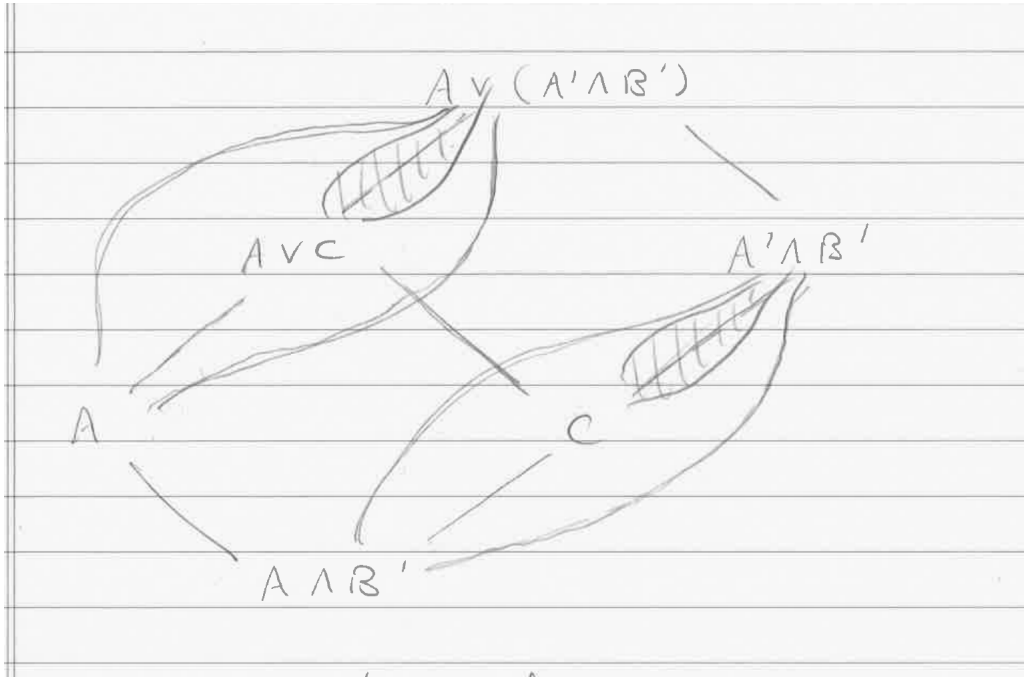
and

$$\begin{aligned} C \vee B &= [(A \wedge B') \vee (A' \wedge B)] \vee B \\ &= (A \wedge B') \vee [(A' \wedge B) \vee B] \\ &= (A \wedge B') \vee B, \end{aligned}$$

we obtain an isomorphism

$$\frac{A \vee (A' \wedge B')}{A \vee (A' \wedge B)} \cong \frac{(A' \wedge B') \vee B}{(A \wedge B') \vee B}.$$

Proof. Let $A \leq A'$ and $B \leq B'$ and consider the diagram.



Since $A \leq A'$, we have

$$A \leq A \vee (A' \wedge B') \quad \text{and} \quad A \wedge B' \leq A' \wedge B'.$$

Then since

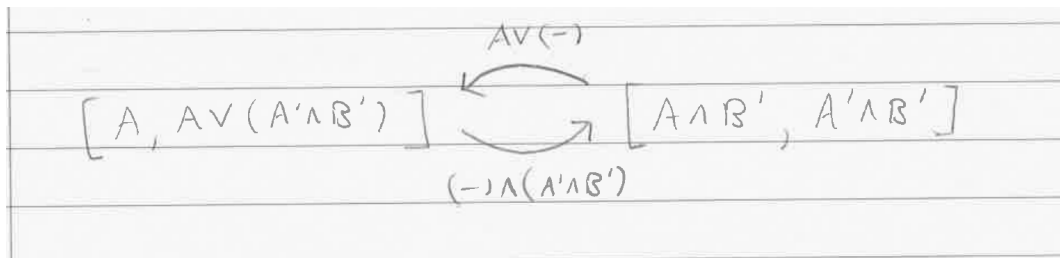
$$A \wedge (A' \wedge B') = (A \wedge A') \wedge B' = A \wedge B',$$

HW3.2 gives an isomorphism

$$\frac{A \vee (A' \wedge B')}{A} \cong \frac{A' \wedge B'}{A \wedge B'}.$$

HW3.1 says that the resulting isomorphism of lattices

is given explicitly by



Note that this isomorphism identifies $A \vee C$ with C . This means that they are both normal or both non-normal. One can check that they are both normal, so we obtain an isomorphism

$$\frac{A \vee (A' \wedge B')}{A \vee C} \cong \frac{A' \wedge B'}{C}.$$

A parallel argument gives

$$\frac{A' \wedge B'}{C} \cong \frac{(A' \wedge B') \vee B}{C \vee B}.$$

□

2. Schreier Refinement Theorem

Theorem. *Let G be a group. Then any two chains of subgroups*

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_m = 1, \quad G = H_0 \supseteq H_1 \supseteq \cdots \supseteq H_n = 1$$

have equivalent refinements.

Proof. Refine the first chain by inserting between each $G_i \supseteq G_{i+1}$ the chain of subgroups

$$G_i := G_{i+1} \vee (G_i \wedge H_0) \supseteq \cdots \supseteq G_{i+1} \vee (G_i \wedge H_n) = G_{i+1}.$$

Similarly, refine the second chain by inserting between $H_j \supseteq H_{j+1}$ the chain

$$H_j := (G_0 \wedge H_j) \vee H_{j+1} \supseteq \cdots \supseteq (G_m \wedge H_j) \vee H_{j+1} = H_{j+1}.$$

Now the subquotients of the first refined chain have the form

$$\frac{G_{i+1} \vee (G_i \wedge H_j)}{G_{i+1} \vee (G_i \wedge H_{j+1})} \tag{*}$$

for all $0 \leq i < m$ and $0 \leq j < n$. And the subquotients of the second refined chain have the form

$$\frac{(G_i \wedge H_j) \vee H_{j+1}}{(G_{i+1} \wedge H_j) \vee H_{j+1}} \tag{**}$$

for all $0 \leq i < m$ and $0 \leq j < n$. Finally, the Zassenhaus Lemma says that the groups (*) and (**) are isomorphic. □

Proof of Jordan–Hölder. Consider any two composition series

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_m = 1, \quad G = H_0 \supseteq H_1 \supseteq \cdots \supseteq H_n = 1.$$

Since the subquotients are simple, the only way to refine either chain is by repeating some of the groups. The Schreier Refinement Theorem says it is possible to do this so that the composition factors are the same.

If we ignore the trivial subquotients, it follows that the original subquotients were the same. □

Remarks:

- As I mentioned last time, Jordan–Hölder applied to cyclic groups is just the Fundamental Theorem of Arithmetic (unique prime factorization of integers).
- The original motivation for this theorem comes from Galois Theory. Recall: If $f(x) \in K[x]$ has irreducible factors that are separable, and splitting field L , then $f(x) = 0$ is solvable by radicals if and only if the group $\text{Aut}(L/K)$ is “solvable.” We can now rephrase the definition of solvable groups.

Definition. A group G is *solvable* if its composition factors are abelian; equivalently, if its composition factors are $\mathbb{Z}/p\mathbb{Z}$ for various primes p .

An interesting special case of this says that $f(x) = 0$ is solvable with ruler and compass constructions if and only if the composition factors of $\text{Aut}(L/K)$ are all $\mathbb{Z}/2\mathbb{Z}$. [This is a generalization of the Gauss–Wantzel Theorem, which says that the regular n -gon is constructible if and only if $\varphi(n)$ is a power of 2.]

- Otto Hölder (1859–1937) was led to the Jordan–Hölder Theorem by his study of Galois Theory. Hölder was also the first person to seriously attempt a classification of finite groups. The two steps of this “Hölder Programme” are
 1. classify finite simple groups;
 2. determine how every finite group can be built up from its simple composition factors.

Part (1) is now completed, as you know. Part (2) is now called the “extension problem” for groups. It is somehow “dual” to the “factorization problem” that we studied before.

Definition. Let N and Q be groups. We say that a group G is an *extension of N by Q* if G has a subgroup $\tilde{N}$ isomorphic to N such that $G/\tilde{N}$ is isomorphic to Q ; equivalently, if there exists a short exact sequence

$$1 \longrightarrow N \longrightarrow G \longrightarrow Q \longrightarrow 1.$$

You will show on HW3 that the easiest kind of group extension is a semi-direct product

$$G = N \rtimes_{\varphi} Q.$$

The general problem of extensions is very difficult and leads to the subject of “group cohomology.”

- The Jordan–Hölder Theorem is essentially a theorem of lattice theory. It holds in any modular or semi-modular lattice. The concept of a modular lattice was invented by Dedekind to model the lattice of submodules of a module over a commutative ring. We’ll return to this later.

15 Exact sequences and group extensions

October 15, 2015

Last time we proved that the (simple) composition factors of a group are unique. This leads to a program of research sometimes referred to as “Hölder’s Program”:

1. Classify finite simple groups.
2. Describe how to reconstruct every finite group from its simple factors.

Part (i) is complete, but part (ii) is far from complete even in the case of solvable groups (with factors of the form $\mathbb{Z}/p\mathbb{Z}$ for prime p).

For example, suppose the finite group G has composition factors H and K . This means that there exists a chain

$$G = G_0 \supseteq G_1 \supseteq G_2 = 1$$

such that

$$\{G/G_1, G_1\} \cong \{H, K\}.$$

Without loss of generality, suppose that $G_1 \cong K$ and $G/G_1 \cong H$. Now consider the maps

The pair of maps

$$K \xrightarrow{\alpha} G \xrightarrow{\beta} H$$

is called exact because $\text{im } \alpha = \ker \beta$ (both are equal to G_1). The notation of “exactness” also gives us a fancy way to say that α is injective and β is surjective. Recall that the trivial group 1 is the zero object in the category of groups (meaning it has a unique map to and from each other group). We will say that a sequence of groups and maps

$$\cdots \xrightarrow{\alpha_{-2}} A_{-1} \xrightarrow{\alpha_{-1}} A_0 \xrightarrow{\alpha_0} A_1 \xrightarrow{\alpha_1} A_2 \xrightarrow{\alpha_2} \cdots$$

is *exact at A_i* if

$$\text{im } \alpha_{i-1} = \ker \alpha_i.$$

We say that the sequence is *exact* if it is exact at every position. Now we can rephrase the relationship between K, G, H by saying that the sequence

$$1 \longrightarrow K \xrightarrow{\alpha} G \xrightarrow{\beta} H \longrightarrow 1$$

is exact. Exactness at K means that $\ker \alpha = 1$ (i.e., α is injective) and exactness at H means that $\operatorname{im} \beta = H$ (i.e., β is surjective). A sequence of this form is called a *short exact sequence*.

Back to the problem at hand: We want to describe G in terms of K and H . There are two “relatively” easy cases:

1. Suppose there exists a group homomorphism $r : G \rightarrow K$ such that $r \circ \alpha = \operatorname{id}_K$ (this is called a *left-inverse* or a *retract* of α). If such an r exists we say that the short exact sequence is “left-split.”

In this case I claim that $G \cong K \times H$.

Proof. Define a function $\gamma : G \rightarrow K \times H$ by

$$\gamma(g) := (r(g), \beta(g)).$$

This is a group homomorphism because r and β are group homomorphisms.

To see that γ is injective, suppose that

$$\gamma(g) = (1_K, 1_H),$$

i.e., $r(g) = 1_K$ and $\beta(g) = 1_H$. Since $\beta(g) = 1_H$, we have $g \in \ker \beta$. The condition $\operatorname{im} \alpha = \ker \beta$ implies that $g = \alpha(k)$ for some $k \in K$. Then since

$$k = r(\alpha(k)) = r(g) = 1_K$$

we conclude that $g = \alpha(k) = \alpha(1_K) = 1_G$, as desired.

To see that γ is surjective, consider any $(k, h) \in K \times H$. Since β is surjective, there exists $g \in G$ such that $\beta(g) = h$. Now the inverse image $\beta^{-1}(h)$ contains g and is a coset of the kernel, hence

$$\beta^{-1}(h) = g \ker \beta = g \operatorname{im} \alpha = \{g\alpha(k') : k' \in K\}.$$

To finish, we want to find $k' \in K$ such that

$$r(g\alpha(k')) = k,$$

so then we will have

$$\begin{aligned} \gamma(g\alpha(k')) &= (r(g\alpha(k')), \beta(g\alpha(k'))) \\ &= (k, h), \end{aligned}$$

as desired. Since r is a group homomorphism, we need

$$k = r(g\alpha(k')) = r(g)r(\alpha(k')) = r(g)k'.$$

So we can take $k' = r(g)^{-1}k$.

We conclude that $\gamma : G \rightarrow K \times H$ is a group isomorphism. $\square$

2. Suppose there exists a group homomorphism $s : H \rightarrow G$ such that $\beta \circ s = \operatorname{id}_H$ (this is called a *right-inverse* or a *section* of β). If such an s exists then we say the short exact sequence is “right-split.”

Note that left-split $\implies$ right-split. Indeed, given a retract $r : G \rightarrow K$ we can define a section $s : H \rightarrow G$ via the isomorphism $\gamma : G \xrightarrow{\sim} K \times H$ by taking

$$s(h) := \gamma^{-1}(1_K, h).$$

Note that by definition of γ the following square commutes:

$$\begin{array}{ccc} \gamma^{-1}(1_K, h) \in G & \xrightarrow{\beta} & H \ni \beta(\gamma^{-1}(1_K, h)) \\ \gamma \downarrow & & \downarrow \text{id}_H \\ (1_K, h) \in K \times H & \xrightarrow{\pi_2} & H \ni h \end{array}$$

It follows that

$$\beta(s(h)) = \beta(\gamma^{-1}(1_K, h)) = h \quad \text{for all } h \in H.$$

In the category of abelian groups, we also have

$$\text{right-split} \implies \text{left-split},$$

but for general groups this is not true. You will prove on HW3 Problem 5 that in general we have

$$\text{right-split} \implies G \cong K \rtimes_{\varphi} H$$

for some $\varphi : H \rightarrow \text{Aut}(K)$.

Believe it or not, split exact sequences are regarded as “trivial” in a certain sense (but I don’t find the proof trivial).

The easiest example of a non-split s.e.s. is the following short exact sequence of abelian groups

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \longrightarrow 0. \quad (*)$$

Note that

$$\ker(\cdot 2) = 0, \quad \text{im } \pi = \mathbb{Z}/2\mathbb{Z}, \quad \text{im}(\cdot 2) = \ker \pi = 2\mathbb{Z}.$$

Now suppose we have a homomorphism $s : \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}$. If s is not the zero map then it must send $1 \in \mathbb{Z}/2\mathbb{Z}$ to some $0 \neq a \in \mathbb{Z}$. But then since s is a homomorphism we have

$$\begin{aligned} 2a &= a + a = s(1) + s(1) \\ &= s(1 + 1) = s(0) = 0. \end{aligned}$$

Contradiction. We conclude that s must be the zero map,

which is certainly not a right-inverse of $\pi : \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$. Therefore the sequence $(*)$ is not right-split. Since the groups are abelian it follows that $(*)$ is also not left-split, but this can also be checked directly.

Remark. The algebraic fact that $(*)$ doesn't split has a topological interpretation: It says that a Möbius band can not be retracted onto its boundary circle.

Proof (for those who have seen some algebraic topology). Consider the Möbius band M and its boundary circle ∂M . Let $i : \partial M \hookrightarrow M$ be the inclusion map. A retraction is a continuous map $r : M \rightarrow \partial M$ such that

$$r \circ i := \text{id}_{\partial M}.$$

Recall that the fundamental groups are

$$\pi_1(M) = \pi_1(\partial M) = \mathbb{Z}.$$

Applying the π_1 functor to the maps r and i gives homomorphisms

$$i_* : \pi_1(\partial M) \longrightarrow \pi_1(M) \quad \text{and} \quad r_* : \pi_1(M) \longrightarrow \pi_1(\partial M)$$

such that

$$r_* \circ i_* = \text{id}_{\pi_1(\partial M)}.$$

Since the boundary wraps around twice, we also know that

$$i_* : \mathbb{Z} \longrightarrow \mathbb{Z}$$

is multiplication by 2. Finally, the fact that no such map r_* exists is equivalent to the fact that the sequence $(*)$ doesn't split. $\square$

The language of exact sequences emerged from topology but it has also become an important language for algebra.

16 Abel–Ruffini and the symmetric groups

October 20, 2015

We have discussed the notions of “simplicity” and “solvability” of groups, which emerged from the study of Galois Theory. Before moving on, I would like to finish the proof of the Abel–Ruffini Theorem (unsolvability of the quintic). For now I will just assume the Fundamental Theorem of Galois Theory [I hope to prove it next semester].

Consider a polynomial $f(x) \in \mathbb{Q}[x]$ of degree n having n distinct roots

$$r_1, r_2, \dots, r_n \in \mathbb{C}.$$

Let

$$K = \mathbb{Q}(r_1, r_2, \dots, r_n)$$

be the smallest subfield of $\mathbb{C}$ containing $\{r_1, r_2, \dots, r_n\}$; i.e., let K be the splitting field of the polynomial $f(x)$.

Then we define the *Galois group* of the polynomial as

$$G = \text{Aut}(K/\mathbb{Q}).$$

Actually, the “/ $\mathbb{Q}$ ” notation here is superfluous because any field automorphism $\sigma : K \rightarrow K$ automatically fixes $\mathbb{Q}$.

I claim that G is finite. Indeed, let $\sigma \in G$ and let r_i be any root of $f(x)$. Then we have

$$f(\sigma(r_i)) = \sigma(f(r_i)) = \sigma(0) = 0,$$

so that $\sigma(r_i) \in \mathbb{C}$ is another root. Thus σ defines a permutation of the set $\{r_1, \dots, r_n\}$. By abuse of notation, we will also write $\sigma \in S_n$ for the permutation of $\{1, \dots, n\}$ defined by

$$\sigma(i) = j \iff \sigma(r_i) = r_j.$$

This defines a group homomorphism

$$G \longrightarrow S_n, \quad \sigma \longmapsto \sigma.$$

Note that the homomorphism is injective since if σ fixes the roots then, since $K = \mathbb{Q}(r_1, \dots, r_n)$ is generated by the roots, we conclude that $\sigma : K \rightarrow K$ is the identity automorphism.

We have now identified G with a subgroup of S_n , hence

$$|G| \leq |S_n| = n!.$$

Remark. In Galois’ original definition he explicitly thought of G as a group of permutations, which he called “substitutions.” Even until Camille Jordan’s *Traité des substitutions* (1870), all groups in mathematics were thought of as subgroups of S_n .

Furthermore, it turns out that for all n there exist (many) polynomials $f(x) \in \mathbb{Q}[x]$ of degree n with

Galois group $G = S_n$.

Now, assuming the FTGT, we can prove the Abel–Ruffini Theorem by proving that for all $n \geq 5$, S_n is not a solvable group.

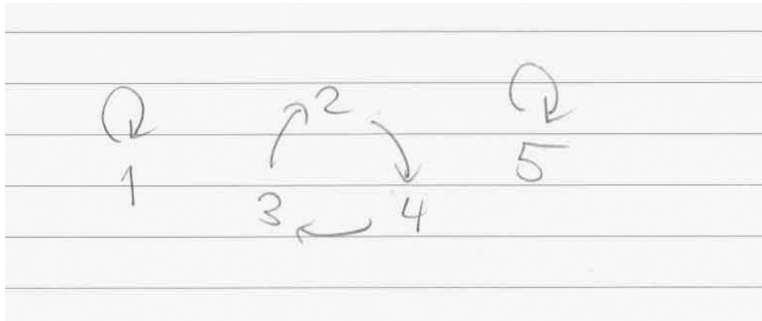
Recall the cycle notation for permutations. Let

$$c = (a_1 \ a_2 \ \cdots \ a_k) \in S_n$$

be the permutation defined by

$$c(a_i) = a_{i+1 \bmod k} \quad \text{and} \quad c(b) = b \quad \text{for } b \notin \{a_1, a_2, \dots, a_k\}.$$

For example, the permutation $(243) \in S_5$ has the following diagram:



Every permutation $\sigma \in S_n$ can be written uniquely as a product of commuting cycles

$$\sigma = c_1 c_2 \cdots c_m.$$

By convention, we will omit 1-cycles from the notation. This allows us to think of S_n as a subgroup of S_{n+1} .

The following Lemma is the foundation for the structure theory of S_n .

Lemma. Let $c = (a_1 a_2 \cdots a_k) \in S_n$. Then for all $\pi \in S_n$ we have

$$\pi c \pi^{-1} = (\pi(a_1) \pi(a_2) \cdots \pi(a_k)).$$

Proof. Since the group operation in S_n is composition of functions, we will read products from right to left. Note that $\pi c \pi^{-1}$ acts on the element $\pi(a_i)$ by

$$\pi(a_i) \xrightarrow{\pi^{-1}} a_i \xrightarrow{c} a_{i+1 \bmod k} \xrightarrow{\pi} \pi(a_{i+1 \bmod k}),$$

and for $\pi(b) \notin \{\pi(a_1), \pi(a_2), \dots, \pi(a_k)\}$ we have

$$\pi c \pi^{-1}(\pi(b)) = \pi(c(b)) = \pi(b).$$

□

More generally, if $\sigma = c_1 c_2 \cdots c_m$ is a product of cycles then we have

$$\begin{aligned} \pi \sigma \pi^{-1} &= \pi c_1 c_2 \cdots c_m \pi^{-1} \\ &= (\pi c_1 \pi^{-1})(\pi c_2 \pi^{-1}) \cdots (\pi c_m \pi^{-1}), \end{aligned}$$

which is a product of cycles of the same lengths.

Notation. Given $\sigma \in S_n$, define

$$\text{sh}(\sigma) = (\text{sh}_1, \text{sh}_2, \text{sh}_3, \dots),$$

where sh_k is the number of cycles of length k in the unique cycle decomposition of σ .

We call $\text{sh}(\sigma)$ the *shape* or the *cycle-type* of σ .

Now the above Lemma has the following corollary: For all $\sigma, \pi \in S_n$ we have

$$\text{sh}(\pi \sigma \pi^{-1}) = \text{sh}(\sigma). \quad (\text{"Jordan canonical form"})$$

In other words, conjugate permutations have the same shape. Conversely, it is not difficult to prove that permutations of the same shape are conjugate. Thus, conjugacy classes in S_n are parametrized by “valid shapes,” i.e., decompositions of n as a sum of unordered positive integers.

Remark. “Valid shapes” are more commonly called *integer partitions*. Let $p(n)$ be the number of partitions of n . Hardy & Ramanujan proved in 1918 that

$$p(n) \sim \frac{1}{4n\sqrt{3}} e^{\pi\sqrt{2n/3}}.$$

Now we can prove the Abel–Ruffini Theorem.

Theorem. For $n \geq 5$, S_n is not solvable.

Proof. Let $n \geq 5$ and assume that S_n is solvable, with composition series

$$S_n = G_0 \geq G_1 \geq \cdots \geq G_m = 1.$$

Let $X \subseteq S_n$ be the set of 3-cycles. We will show that

$$X \subseteq G_i \implies X \subseteq G_{i+1},$$

which implies by induction that $X \subseteq G_m = 1$. Contradiction.

So suppose that $X \subseteq G_i$. If $c_1, c_2 \in X$ then since G_i/G_{i+1} is abelian we have

$$\begin{aligned} & (c_1 c_2 c_1^{-1} c_2^{-1}) G_{i+1} \\ &= (c_1 G_{i+1})(c_2 G_{i+1})(c_1^{-1} G_{i+1})(c_2^{-1} G_{i+1}) \\ &= (c_1 G_{i+1})(c_1 G_{i+1})^{-1}(c_2 G_{i+1})(c_2 G_{i+1})^{-1} \\ &= G_{i+1}, \end{aligned}$$

and hence

$$c_1 c_2 c_1^{-1} c_2^{-1} \in G_{i+1}.$$

Now consider an arbitrary 3-cycle (ijk) . Since $n \geq 5$, we can choose $l, m \notin \{i, j, k\}$, and then we have

$$\begin{aligned} & (jkm)(ilj)(jkm)^{-1}(ilj)^{-1} \\ &= (jkm)(ilj)(jmk)(ijl) \\ &= (ijk)(l)(m) = (ijk). \end{aligned}$$

We conclude that $(ijk) \in G_{i+1}$, and hence $X \subseteq G_{i+1}$, as desired. $\square$

Remark. It's sort of shocking that in the end the proof of the unsolvability of the quintic looks like that. Thank you Galois.

In fact, we can be more specific. It turns out that the alternating subgroup $A_n \subseteq S_n$ is simple, so the composition factors of S_n are

$$A_n \quad \text{and} \quad \mathbb{Z}^\times = \{\pm 1\}.$$

Furthermore, the s.e.s.

$$1 \longrightarrow A_n \longrightarrow S_n \xrightarrow{\text{sgn}} \mathbb{Z}^\times \longrightarrow 1$$

splits. Proof: Define $s : \mathbb{Z}^\times \rightarrow S_n$ by

$$s(-1) = (12) \quad [\text{or any 2-cycle}].$$

Then we have

$$\text{sgn}(s(-1)) = \text{sgn}((12)) = -1.$$

It follows that

$$S_n = A_n \rtimes \mathbb{Z}^\times.$$

[I (or you) will prove the simplicity of A_n after we have discussed group actions.]

17 Functors, natural transformations, free objects

October 22, 2015

Having proved the Abel–Ruffini Theorem, we will now shift gears. Recall that I originally gave two definitions of a group:

1. A structure $(G, \circ, 1)$ where

- G is a set,
- $\circ : G \times G \rightarrow G$ is a function,
- $1 \in G$ is a special element,

satisfying three axioms [omitted].

2. A set G of automorphisms of some object X in a category $\mathcal{C}$ satisfying

$$\forall \alpha, \beta \in G, \quad \alpha \circ \beta^{-1} \in G.$$

All groups that we care about are of Type (2). Definition (1) is just a convenient device that allows us to see patterns and to prove general theorems.

It is clear how to go from Type (2) to Type (1), i.e., just “forget” about the object X and the category $\mathcal{C}$. It is less clear how to go from (1) to (2). Today I’ll show you how to do this.

But first, please let me test your patience with some more abstract nonsense.

Definition. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A (covariant) *functor*

$$F : \mathcal{C} \longrightarrow \mathcal{D}$$

assigns to each object X in $\mathcal{C}$ an object $F(X)$ in $\mathcal{D}$, and assigns to each morphism $\alpha \in \text{Hom}_{\mathcal{C}}(X, Y)$ a morphism

$$F(\alpha) \in \text{Hom}_{\mathcal{D}}(F(X), F(Y))$$

such that

- for all objects X in $\mathcal{C}$, $F(\text{id}_X) = \text{id}_{F(X)}$;
- for all $\alpha \in \text{Hom}_{\mathcal{C}}(X, Y)$ and $\beta \in \text{Hom}_{\mathcal{C}}(Y, Z)$, we have

$$\begin{array}{c}
 X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z \implies F(X) \xrightarrow{F(\alpha)} F(Y) \xrightarrow{F(\beta)} F(Z) \\
 \text{--- } \beta \circ \alpha \text{ ---} \qquad \qquad \qquad \text{--- } F(\beta \circ \alpha) \text{ ---}
 \end{array}$$

It follows that F sends commutative diagrams in $\mathcal{C}$ to commutative diagrams in $\mathcal{D}$.

A contravariant functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is the same thing as a covariant functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$, where $\mathcal{C}^{\text{op}}$ is the

opposite category of $\mathcal{C}$ (with the same objects, and reversed arrows). That is, if F is contravariant then we have

$$\begin{array}{c}
 X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z \implies F(X) \xleftarrow{F(\alpha)} F(Y) \xleftarrow{F(\beta)} F(Z) \\
 \text{--- } \beta \circ \alpha \text{ ---} \qquad \qquad \qquad \text{--- } F(\beta \circ \alpha) \text{ ---}
 \end{array}$$

We have already seen several examples of functors:

- If $\mathcal{C}$ and $\mathcal{D}$ are posets (i.e., with $|\text{Hom}(X, Y)| \in \{0, 1\}$ for all X, Y), then a functor $F : \mathcal{D} \rightarrow \mathcal{C}$ is just a morphism of posets. A contravariant Galois connection [the kind I originally defined] is a pair of contravariant functors

$$F : \mathcal{D} \rightrightarrows \mathcal{C} : G$$

such that for all objects X in $\mathcal{D}$ and Y in $\mathcal{C}$ there exists a bijection of hom sets

$$\text{Hom}_{\mathcal{D}}(X, G(Y)) \longleftrightarrow \text{Hom}_{\mathcal{C}}(Y, F(X)).$$

If we replace C by C^{op} then this becomes a covariant Galois connection

$$F : \mathcal{D} \rightleftarrows C^{\text{op}} : G.$$

[Note that a functor $F : \mathcal{D} \rightarrow C^{\text{op}}$ is the same thing as a functor $F : \mathcal{D}^{\text{op}} \rightarrow C$.]

- We have seen functors that forget structure. The functor

$$\text{set} : \mathbf{Grp} \longrightarrow \mathbf{Set}$$

forgets the identity element and product function of a group. The functor

$$\text{ab} : K\text{-}\mathbf{Vec} \longrightarrow \mathbf{Ab}$$

forgets how to do scalar multiplication. The technical name for these is “forgetful functors.”

Remark. Many forgetful functors $C \rightarrow \mathcal{D}$ have an associated “free functor”

$$\text{free} : \mathcal{D} \rightleftarrows C : \text{forget}$$

which sends an object X of $\mathcal{D}$ to the object $\text{free}(X)$ of C that is “freely generated” by X . In our first example

$$\text{free} : \mathbf{Set} \rightleftarrows \mathbf{Grp} : \text{forget},$$

$\text{free}(X)$ is called the free group generated by the set X . It satisfies the following universal property: For all sets X and groups G there is a “natural” bijection of hom sets

$$\text{Hom}_{\mathbf{Set}}(X, \text{forget}(G)) \cong \text{Hom}_{\mathbf{Grp}}(\text{free}(X), G).$$

This probably seems intuitive: a set function $X \rightarrow \text{forget}(G)$ is the same thing as a group hom $\text{free}(X) \rightarrow G$. But what is the word “natural” supposed to mean?

“I didn’t invent categories to study functors; I invented them to study natural transformations.”

—Saunders Mac Lane

Definition. Let C and $\mathcal{D}$ be categories and let F and G be functors $C \rightarrow \mathcal{D}$. A *natural transformation*

$$\nu : F \longrightarrow G$$

assigns to each object X of C a morphism

$$\nu_X \in \text{Hom}_{\mathcal{D}}(F(X), G(X))$$

such that for all morphisms $\alpha \in \text{Hom}_C(X, Y)$ the following diagram commutes:

$$\begin{array}{ccc}
 F(X) & \xrightarrow{F(\alpha)} & F(Y) \\
 \downarrow v_X & & \downarrow v_Y \\
 G(X) & \xrightarrow{G(\alpha)} & G(Y)
 \end{array}$$

We call v a *natural isomorphism* if each v_X is an isomorphism in $\mathcal{D}$.

Q: Isn't this an unnecessary level of abstraction?

A: Not according to Mac Lane.

More seriously, we will see that the notion of natural transformation will help us better understand even mundane situations.

The mother of all natural transformations is the following construction.

Definition. Let $\mathcal{C}$ and $\mathcal{D}$ be categories and consider a pair of functors

$$F : \mathcal{D} \rightleftarrows \mathcal{C} : G.$$

We can use this to define a pair of functors

$$\begin{aligned}
 \mathrm{Hom}_{\mathcal{C}}(F-, -) : \mathcal{D}^{\mathrm{op}} \times \mathcal{C} &\longrightarrow \mathbf{Set}, \\
 \mathrm{Hom}_{\mathcal{D}}(-, G-) : \mathcal{D}^{\mathrm{op}} \times \mathcal{C} &\longrightarrow \mathbf{Set},
 \end{aligned}$$

where the product category $\mathcal{D}^{\mathrm{op}} \times \mathcal{C}$ is defined in the obvious way.

Now suppose that there exists a natural isomorphism

$$\Phi : \mathrm{Hom}_{\mathcal{C}}(F-, -) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{D}}(-, G-).$$

In this case we will say that

- F is left adjoint to G ;
- G is right adjoint to F ;
- (F, G) is an adjoint pair;
- $F : \mathcal{D} \rightleftarrows \mathcal{C} : G$ is an adjunction.

Examples:

- A Galois connection is nothing but an adjunction of posets.
- If $\mathrm{forget} : \mathcal{D} \rightarrow \mathcal{C}$ is a forgetful functor, then we say $\mathrm{free} : \mathcal{C} \rightarrow \mathcal{D}$ is the associated free functor if $(\mathrm{free}, \mathrm{forget})$ is an adjoint pair.

Okay; I think that's enough abstract nonsense for now. I promised that I would tell you how to go from Definition (1) to Definition (2) of groups.

Let G be a group. Recall that we can think of G as a category with one object (call it $*$) and such that every morphism is an isomorphism. In this case we can identify

$$G = \text{Aut}_G(*) = \text{End}_G(*).$$

But what is this mysterious object $*$? Well, it could be anything. To turn $*$ into something interesting, choose a nice category $\mathcal{C}$ and consider a functor

$$F : G \longrightarrow \mathcal{C}.$$

In other words, choose an object X in $\mathcal{C}$ and for every element $\alpha \in G = \text{End}_G(*)$ consider an element $F(\alpha) \in \text{End}_{\mathcal{C}}(X)$ such that

- $F(\text{id}_*) = \text{id}_{F(*)} = \text{id}_X$;
- for all $\alpha, \beta \in G$ we have

$$F(\alpha \circ \beta) = F(\alpha) \circ F(\beta).$$

Since every $\alpha \in G$ is invertible we have

$$\begin{aligned} F(\alpha) \circ F(\alpha^{-1}) &= F(\alpha \circ \alpha^{-1}) \\ &= F(\text{id}_*) \\ &= \text{id}_X, \end{aligned}$$

and similarly $F(\alpha^{-1}) \circ F(\alpha) = \text{id}_X$. It follows that $F(\alpha) \in \text{End}_{\mathcal{C}}(X)$ is actually an automorphism, so we obtain a homomorphism of groups

$$F : G \longrightarrow \text{Aut}_{\mathcal{C}}(X),$$

which by abuse of notation we will also call F .

[You might also like to force this F to be injective, but we will get a better theory if we don't require this.]

Jargon. Let G be a group and let $\mathcal{C}$ be a category. A functor

$$F : G \longrightarrow \mathcal{C}$$

is called a *representation* of G in the category $\mathcal{C}$. The flavor of representation theory depends on the choice of category.

We will begin next time by choosing $\mathcal{C} = \text{Set}$, in which case the representations are called “ G -sets.”

18 Midterm review

October 27, 2015

I didn't find any really appropriate previous exams to study from, so today I'll give a thorough list of the topics for the midterm.

1. Categories

A category C consists of

- a "collection" $\text{Obj}(C)$ of objects,
- for all $X, Y \in \text{Obj}(C)$ a set of morphisms $\text{Hom}_C(X, Y)$,
- for all $X, Y, Z \in \text{Obj}(C)$ a function

$$\text{Hom}_C(X, Y) \times \text{Hom}_C(Y, Z) \longrightarrow \text{Hom}_C(X, Z)$$

called composition,

satisfying two axioms:

- for all $\alpha : A \rightarrow B, \beta : B \rightarrow C, \gamma : C \rightarrow D$ we have

$$\gamma \circ (\beta \circ \alpha) = (\gamma \circ \beta) \circ \alpha;$$

- for all $X \in \text{Obj}(C)$ there exists

$$\text{id}_X \in \text{Hom}_C(X, X) = \text{End}_C(X)$$

such that for all $\alpha : X \rightarrow Y$ we have

$$\alpha \circ \text{id}_X = \alpha.$$

You should know the definitions of

- isomorphism/automorphism,
- initial/final object,
- zero object/morphism,
- product/coproduct,
- kernel/cokernel,

and some examples in specific categories.

2. Posets & Lattices

A poset is a pair $(P, \leq)$ where P is a set and $\leq$ is a partial order:

- $x \leq x$,
- $x \leq y$ and $y \leq z$ imply $x \leq z$,
- $x \leq y$ and $y \leq x$ imply $x = y$.

Alternatively, a poset P is a (small) category in which

$$|\mathrm{Hom}_P(x, y)| \in \{0, 1\} \quad \forall x, y \in \mathrm{Obj}(P).$$

We will write

$$x \leq y \iff |\mathrm{Hom}_P(x, y)| = 1$$

and $x = y$ if x, y are isomorphic in P .

The product/coproduct if they exist are called meet/join. The initial/final objects if they exist are called 0/1.

3. Galois Connections

Let P, Q be posets and consider a pair of maps

$$L : P \rightleftarrows Q : R.$$

We say (L, R) is a covariant Galois connection if for all $x \in P, y \in Q$ we have a bijection

$$\mathrm{Hom}_P(x, R(y)) \longleftrightarrow \mathrm{Hom}_Q(L(x), y),$$

i.e.,

$$x \leq R(y) \iff L(x) \leq y.$$

[We use the word covariant because it follows that L, R are covariant functors.]

We say $L : P \rightleftarrows Q : R$ is a contravariant Galois connection if and only if

$$L : P \rightleftarrows Q^{\mathrm{op}} : R$$

is a covariant Galois connection. The notation becomes easier if we write

$$* : P \rightleftarrows Q : *$$

for a contravariant connection, i.e.,

$$x \leq y^* \iff y \leq x^*.$$

You should know the results of HW1 and be able to prove the more basic facts.

You should know the following examples of Galois connections:

- fields & automorphisms,
- image/preimage of a group hom,
- join/meet fixed elements of a lattice.

4. Equivalence & Quotient

Let $\sim$ be an equivalence relation on a set S :

- $x \sim x$,
- $x \sim y$ implies $y \sim x$,
- $x \sim y$ and $y \sim z$ imply $x \sim z$.

We say $f : S \rightarrow T$ is a class function if for all $x, y \in S$ we have

$$x \sim y \implies f(x) = f(y).$$

We say $\pi : S \rightarrow Q$ is a quotient if

- $\pi : S \rightarrow Q$ is a class function,
- for all class functions $f : S \rightarrow T$ there exists a unique function $\bar{f} : Q \rightarrow T$ such that

$$\begin{array}{ccc} & S & \\ \pi \swarrow & & \searrow f \\ Q & \xrightarrow{\quad \bar{f} \quad} & T \end{array}$$

$\exists! \bar{f}$

Prove that the quotient exists and say in what sense it is unique. We call “the” quotient

$$\pi : S \longrightarrow S/\sim.$$

Application: If $f : S \rightarrow T$ is any function, then we define an equivalence on S by

$$x \sim y \iff f(x) = f(y),$$

and we call the quotient $\pi : S \rightarrow S/f$.

Prove that this leads to a canonical factorization:

$$\begin{array}{ccccccc} & & f & & & & \\ & \nearrow & & \searrow & & & \\ S & \xrightarrow{\pi} & S/f & \xrightarrow[\tilde{f}]{\sim} & \text{im } f & \xrightarrow{i} & T \\ & & & & & & \end{array}$$

Show how to lift this factorization to the category of groups:

- $\pi : G \rightarrow G/\sim$ is a group hom if and only if $\sim$ is G -invariant, in which case $N := [1]_\sim$ is a normal subgroup.

- For all $g \in G$ we have

$$[g]_\sim = gN = Ng.$$

- If $\varphi : G \rightarrow H$ is a group hom, then the equivalence

$$x \sim y \iff \varphi(x) = \varphi(y)$$

is G -invariant with $[1]_\sim = \ker \varphi$.

- Universal Property of Group Quotient:

Let $N \trianglelefteq G$. If $\varphi : G \rightarrow H$ is a group hom with $N \subseteq \ker \varphi$, then there exists a unique group hom $\bar{\varphi} : G/N \rightarrow H$ such that

$$\begin{array}{ccc} & G & \\ \pi \swarrow & & \searrow \varphi \\ G/N & \xrightarrow{\quad \bar{\varphi} \quad} & H \end{array}$$

$\exists! \bar{\varphi}$

5. Isomorphism Theorems

Apply the above universal property to prove the following:

- Given $\varphi : G \rightarrow G'$ we have an isomorphism of groups

$$\bar{\varphi} : G/\ker \varphi \xrightarrow{\sim} \text{im } \varphi.$$

- Let $N \trianglelefteq G$ and $N \trianglelefteq H \trianglelefteq G$. Then we have $H/N \trianglelefteq G/N$ and an isomorphism

$$\frac{G}{H} \cong \frac{G/N}{H/N}.$$

- Let $H, K \leq G$ with $K \trianglelefteq G$. Then we have $H \cap K \trianglelefteq H$, $K \trianglelefteq HK$, and an isomorphism

$$\frac{H}{H \cap K} \cong \frac{HK}{K}.$$

Be able to state and use (but not prove) the Jordan–Hölder Theorem.

6. Products of Groups

Let $H, K \leq G$ be arbitrary subgroups. If H, K are finite then we have

$$|H||K| = |HK||H \cap K|,$$

even though HK might not be a group.

Prove that HK is a group if and only if $HK = KH$, in which case we have

$$H \vee K = HK.$$

If $H \wedge K = H \cap K = 1$ and $H \vee K = HK = G$, we say that G is a product of H and K . There are three cases:

(i) In general we write

$$G = H \rtimes K$$

and call this a Zappa–Szép product.

(ii) If $H \trianglelefteq G$ or $K \trianglelefteq G$ we write

$$G = H \rtimes K \quad \text{or} \quad G = H \ltimes K,$$

respectively, and call this a semi-direct product. Know the “external” characterization and the theorem on right-split short exact sequences.

(iii) If $H \trianglelefteq G$ and $K \trianglelefteq G$ we write

$$G = H \times K$$

and call this the direct product. In this case we have

$$hk = kh \quad \forall h \in H, k \in K.$$

Know the external characterization as the product object in the category of groups.

7. Specific Examples of Groups

- cyclic,
- dihedral,
- symmetric/alternating,
- general/special linear.

19 Equivalence of categories and group representations

November 3, 2015

Having finished the abstract structure theory of groups, our next topic will be the “representation theory” of groups.

This will require some new categorical ideas, so let me recall the definitions from last time.

Definition. A covariant functor of categories

$$F : \mathcal{C} \longrightarrow \mathcal{D}$$

sends objects to objects and arrows to arrows such that

- $F(\text{id}_X) = \text{id}_{F(X)}$ for all $X \in \mathcal{C}$;
- $F(\alpha \circ \beta) = F(\alpha) \circ F(\beta)$ for all objects $X, Y, Z \in \mathcal{C}$ and morphisms $\beta : X \rightarrow Y, \alpha : Y \rightarrow Z$.

A contravariant functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is the same thing as a covariant functor $F : \mathcal{C} \rightarrow \mathcal{D}^{\text{op}}$ [equivalently, $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$], i.e., for all compositions $\alpha \circ \beta$ we have

$$F(\alpha \circ \beta) = F(\beta) \circ F(\alpha).$$

One can check that categories are the objects of a category (called “Cat”) with functors as arrows. The identity arrows are given by the identity functor

$$\text{id}_{\mathcal{C}} : \mathcal{C} \longrightarrow \mathcal{C}$$

that sends each object and arrow in $\mathcal{C}$ to itself. Then by definition we say that categories $\mathcal{C}$ and $\mathcal{D}$ are isomorphic if there exist functors

$$F : \mathcal{C} \rightleftarrows \mathcal{D} : G$$

such that $G \circ F = \text{id}_{\mathcal{C}}$ and $F \circ G = \text{id}_{\mathcal{D}}$.

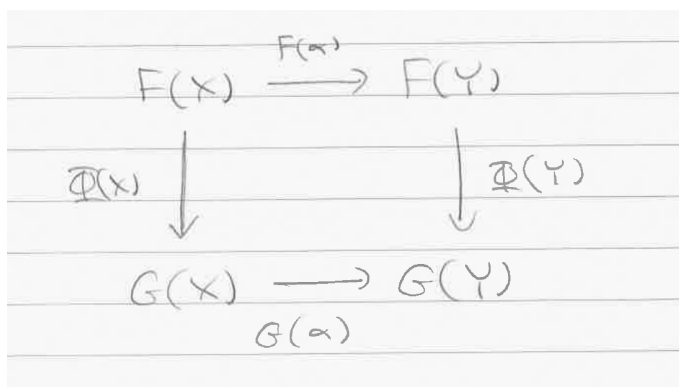
However, isomorphism of categories is usually too strong to be useful. Since we really only care about objects up to isomorphism we should look for a notion of “equivalence” for categories that matches up isomorphism classes of objects.

For this we need another definition.

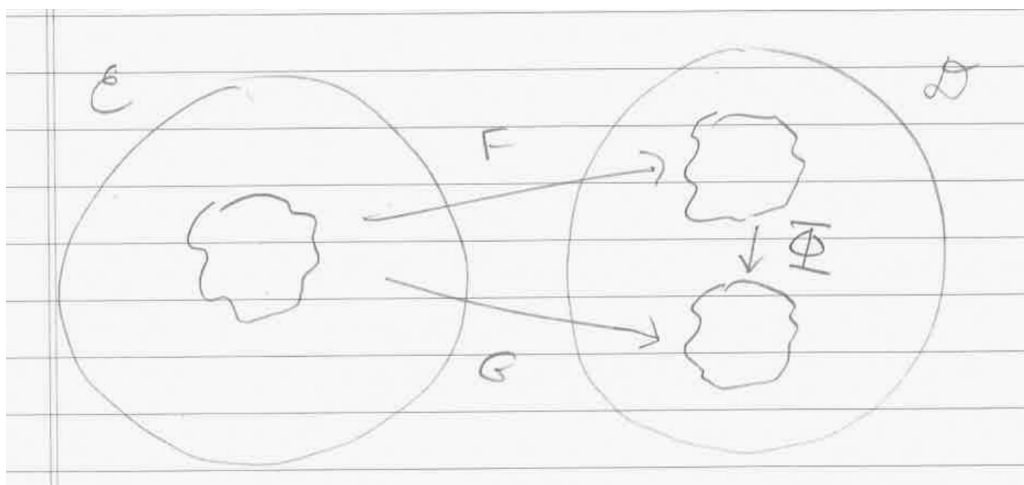
Definition. Given two functors F and G from $\mathcal{C}$ to $\mathcal{D}$, a natural transformation

$$\Phi : F \longrightarrow G$$

assigns to each object $X \in \mathcal{C}$ an arrow $\Phi(X) : F(X) \rightarrow G(X)$ in $\mathcal{D}$ such that for all objects $X, Y \in \mathcal{C}$ and arrows $\alpha \in \text{Hom}_{\mathcal{C}}(X, Y)$ the following diagram commutes:



Intuition. Given a commutative diagram in $\mathcal{C}$, F and G produce two commutative diagrams in $\mathcal{D}$ and Φ produces a “morphism” of commutative diagrams.



We will say that $\Phi : F \rightarrow G$ is a natural isomorphism if the arrow $\Phi(X) : F(X) \rightarrow G(X)$ is an isomorphism in $\mathcal{D}$ for all $X \in \mathcal{C}$. [Thus F and G are “equivalent up to isomorphism.”]

Now we can state the correct definition for equivalence of categories.

Definition. We say that categories $\mathcal{C}$ and $\mathcal{D}$ are equivalent if there exist functors

$$F : \mathcal{C} \rightleftarrows \mathcal{D} : G$$

and natural isomorphisms

$$\varepsilon : F \circ G \xrightarrow{\sim} \text{id}_{\mathcal{D}} \quad \text{and} \quad \eta : \text{id}_{\mathcal{C}} \xrightarrow{\sim} G \circ F.$$

[Motivating Example: Let X be a topological space and consider its “fundamental groupoid” $\pi_1(X)$. This is a category whose objects are the points of X and whose morphisms are homotopy equivalence classes of paths (thus every morphism is an isomorphism; hence “groupoid”).

If X and Y are homotopy equivalent spaces then their groupoids $\pi_1(X)$ and $\pi_1(Y)$ are equivalent as categories, but not necessarily isomorphic.]

I’ll present just one more definition today.

Since functors can be thought of as morphisms in a category of categories, you might wonder if natural transformations can be thought of as morphisms in a category of functors ...

Definition. Let $\mathcal{C}$ and $\mathcal{D}$ be categories and let $\mathcal{C}$ be small (i.e., it has only a set of objects). Then we can define the *functor category*

$$\mathcal{D}^{\mathcal{C}}$$

whose objects are functors $F : \mathcal{C} \rightarrow \mathcal{D}$ and whose morphisms are natural transformations $\Phi : F \rightarrow G$.

[Remark: The notation $\mathcal{D}^{\mathcal{C}}$ agrees with most of the other uses of exponential notation that you know.]

Okay, that’s enough abstract nonsense for next several days. It’s time to begin the subject of “representation theory.”

Let G be a group thought of as a (small) category with one object, and let $\mathcal{C}$ be any category.

A functor $\psi : G \rightarrow \mathcal{C}$ is called a representation of G in $\mathcal{C}$. If $\text{Obj}(G) \equiv \{*\}$ and $X := \psi(*) \in \mathcal{C}$ then ψ induces a homomorphism of groups

$$\psi : G \longrightarrow \text{Aut}_{\mathcal{C}}(X).$$

If the objects of $\mathcal{C}$ are called “things” then we often refer to the pair (X, ψ) as a “ G -thing” and we refer to the functor category

$$\mathcal{C}^G$$

as the category of G -things.

The prototypical example is the category Set^G of G -sets. Suppose that (X, ψ) is a G -set so that

$$\psi : G \longrightarrow \text{Aut}_{\text{Set}}(X)$$

is a homomorphism from G into the group of permutations of the set X .

Let's unpack what this means:

Given $g \in G$ we obtain a function $\psi_g : X \rightarrow X$ satisfying

1. $\psi_1(x) = x$ for all $x \in X$;
2. $\psi_{gh}(x) = \psi_g(\psi_h(x))$ for all $g, h \in G, x \in X$.

Surprisingly, these two properties are sufficient to define a group homomorphism $\psi : G \rightarrow \text{Aut}(X)$. This makes working in the category of G -sets very concrete.

In fact, let's record a definition.

Definition. Consider a group G and a set X . A left action of G on X is a function

$$\varphi : G \times X \longrightarrow X$$

satisfying the two properties

1. $\varphi(1, x) = x$ for all $x \in X$;
2. $\varphi(gh, x) = \varphi(g, \varphi(h, x))$ for all $g, h \in G, x \in X$.

To simplify notation we will often use the metaphor

$$“\varphi(g, x) = g(x)”$$

and simply say that $g : X \rightarrow X$ is a function. Then the properties become

1. $1(x) = x$;
2. $(gh)(x) = g(h(x))$.

This notation makes our lives easier, as long as we're careful.

Okay, so we have a very concrete notation for G -sets. Is there also a concrete notation for morphisms of G -sets?

Let $\alpha : G \rightarrow \mathbf{Set}$ and $\beta : G \rightarrow \mathbf{Set}$ be functors, i.e., choose sets $X, Y \in \mathbf{Set}$ and group homomorphisms

$$\alpha : G \longrightarrow \text{Aut}(X) \quad \text{and} \quad \beta : G \longrightarrow \text{Aut}(Y).$$

Since G has just one object, a natural transformation $f : \alpha \rightarrow \beta$ is just a function $f : X \rightarrow Y$ such that for all “arrows” $g \in G$ the following diagram commutes:

$$\begin{array}{ccc}
 X & \xrightarrow{f} & Y \\
 \alpha_g \downarrow & & \downarrow \beta_g \\
 X & \xrightarrow{f} & Y
 \end{array}$$

If you're willing to use the metaphors

$$“\alpha_g(x) = g(x)” \quad \text{and} \quad “\beta_g(y) = g(y)”$$

then we can express the definition succinctly as follows.

Definition. Let X and Y be G -sets. We say that a function $f : X \rightarrow Y$ is G -equivariant (or just a G -function) if for all $x \in X$, $g \in G$ we have

$$\boxed{f(g(x)) = g(f(x))}.$$

Pretty simple, right? So why did I make all the fuss?

Because sometimes this notation is too simple. It's a great notation as long as you remember the implicit definitions behind it. But if you forget the definitions and start taking the metaphor literally, you'll be lost.

The same remark applies to any kind of representation theory: use the notation responsibly.

20 Group actions, orbits, stabilizers

November 5, 2015

Let G be a group. Last time we defined the category of G -sets as the “functor category”

$$\mathbf{Set}^G.$$

Today we will work with G -sets in a concrete way and we will prove a fundamental decomposition theorem ...

Recall the following concrete interpretation of G -sets:

Definition. Let G be a group and X a set. A left action of G on X is a function

$$\begin{aligned} G \times X &\longrightarrow X, \\ (g, x) &\longmapsto g(x), \end{aligned}$$

satisfying the following two properties

- $1(x) = x$ for all $x \in X$;
- $(gh)(x) = g(h(x))$ for all $x \in X$, $g, h \in G$.

Sometimes we just write “ $G \curvearrowright X$ ” to indicate the existence of a left action.

A right action (written “ $X \curvearrowleft G$ ”) is defined by

- $1(x) = x$ for all $x \in X$;

- $(gh)(x) = h(g(x))$ for all $x \in X, g, h \in G$.

(Or you could say $(x)1 = x$ and $(x)(gh) = ((x)g)h$, but I prefer not to do that, thank you very much.)

[Remark: A left G -action is the same as a covariant functor $G \rightarrow \mathbf{Set}$ and a right action is a contravariant functor $G \rightarrow \mathbf{Set}$. The left/right terminology is arbitrary, but keep in mind that they are distinct concepts.]

Definition. Let $G \curvearrowright X$ be a G -set and consider a subset $Y \subseteq X$. We will say that Y is a G -subset if

$$\text{for all } g \in G, y \in Y, \quad g(y) \in Y.$$

Note that $\emptyset, X \subseteq X$ are trivially G -subsets. We will say that $G \curvearrowright X$ is a *simple* (or *transitive*) G -set if it has no nontrivial G -subsets.

A G -set is analyzed by means of “orbits” and “stabilizers.”

Definition. Let $G \curvearrowright X$ and consider an element $x \in X$. Then we define

the *orbit* of x ,

$$\text{Orb}_G(x) = G(x) = \{g(x) \in X : g \in G\},$$

and the *stabilizer* of x ,

$$\text{Stab}_G(x) = G_x = \{g \in G : g(x) = x\}.$$

Orbits and stabilizers are intimately connected through the following lemma.

Orbit–Stabilizer Lemma. Let $G \curvearrowright X$. Then for all $x \in X$, the stabilizer $\text{Stab}_G(x) \subseteq G$ is a subgroup and we have a bijection of sets

$$\begin{aligned} \text{Orb}_G(x) &\longleftrightarrow G/\text{Stab}_G(x), \\ g(x) &\longleftrightarrow g \text{Stab}_G(x). \end{aligned}$$

Proof. For all $g, h \in G$ we have

$$\begin{aligned} g(x) = h(x) &\iff g^{-1}h(x) = x \\ &\iff g^{-1}h \in \text{Stab}_G(x) \\ &\iff g \text{Stab}_G(x) = h \text{Stab}_G(x), \end{aligned}$$

so the function is well-defined and injective in both directions. □

In fact this is more than a bijection:

Given two G -sets X and Y and a function $f : X \rightarrow Y$, we say that f is a G -equivariant function (or just a G -function) if for all $x \in X, g \in G$, we have

$$f(g(x)) = g(f(x)).$$

[Remark: This is the same as a natural transformation between functors $G \rightarrow \mathbf{Set}$.]

We will say that G -sets X and Y are isomorphic if there exists a G -equivariant bijection $X \xrightarrow{\sim} Y$. [The inverse will automatically be equivariant.]

Orbit–Stabilizer Theorem. Let $G \curvearrowright X$. Then for all $x \in X$ we have an isomorphism of G -sets,

$$\text{Orb}_G(x) \cong G/\text{Stab}_G(x).$$

Proof. We should first say in what sense these are G -sets:

- $\text{Orb}_G(x)$ is a G -subset of X .
- We let $G \curvearrowright G/\text{Stab}_G(x)$ by left multiplication:

$$g(h \text{Stab}_G(x)) := (gh) \text{Stab}_G(x).$$

Then the bijection

$$\Phi(h(x)) = h \text{Stab}_G(x)$$

is equivariant since for all $g \in G$ and $h(x) \in \text{Orb}_G(x)$ we have

$$\begin{aligned} \Phi(g(h(x))) &= \Phi((gh)(x)) \\ &= (gh) \text{Stab}_G(x) \\ &= g(h \text{Stab}_G(x)) \\ &= g(\Phi(h(x))). \end{aligned}$$

□

This theorem is the key to the structure theory of G -sets:

Recall that a G -set X is simple if it has no nontrivial G -subsets. Since every orbit is a G -subset we conclude that

$$G \curvearrowright X \text{ is simple} \iff \text{for all } x \in X, \text{Orb}_G(x) = X.$$

There is a convenient way to rephrase this.

Lemma. $G \curvearrowright X$ is simple if and only if for all $x, y \in X$ there exists $g \in G$ such that

$$g(x) = y.$$

Proof. Let $G \curvearrowright X$ be simple and consider any $x, y \in X$. Since $\text{Orb}_G(x) = X$ we have $y \in \text{Orb}_G(x)$, as desired.

Conversely, suppose $\forall x, y \in X, \exists g \in G, g(x) = y$. Then for all $x \in X$ we have $y \in \text{Orb}_G(x)$ for all $y \in X$ and hence $\text{Orb}_G(x) = X$. We conclude that $G \curvearrowright X$ is simple. □

We can also think of orbits as equivalence classes for the following equivalence relation on X :

$$x \sim_G y \iff \text{there exists } g \in G, g(x) = y.$$

Then I guess we would write

$$\text{Orb}_G(x) = [x]_G.$$

This makes it clear that X is a disjoint union of orbits.

Theorem. Every G -set can be expressed uniquely as a disjoint union of simple G -sets.

Proof. Let $G \curvearrowright X$. Then we have

$$X = \bigsqcup_{i \in I} [x_i]_G, \quad (*)$$

where $\{x_i : i \in I\}$ is a set of representatives for the orbits. To show that this expression is unique, let $Y \subseteq X$ be any simple G -subset.

Now choose any element $y \in Y$, which generates a G -subset $[y]_G \subseteq Y$. Since Y is simple we must have $Y = [y]_G$. Then from the disjoint union (*), there exists i such that $y \in [x_i]_G$ and it follows that

$$Y = [y]_G = [x_i]_G.$$

□

It remains to classify the simple G -sets up to isomorphism.

If $G \curvearrowright X$ is simple, then by the Orbit–Stabilizer Theorem we have an isomorphism of G -sets

$$X \cong G/H,$$

where $H := \text{Stab}_G(x)$ for any $x \in X$. Certainly, every subgroup $H \leq G$ occurs because we could just define $X := G/H$ with left multiplication.

The classification will be completed by the following result.

Theorem. Let $H, K \leq G$ be subgroups. Then we have an isomorphism of G -sets

$$G/H \cong G/K$$

if and only if $H = gKg^{-1}$ for some $g \in G$.

Proof. We'll give the proof in three steps:

(1) Let $G \curvearrowright X$. Then for all $g \in G$ and $x \in X$ we have

$$\text{Stab}_G(g(x)) = g \text{Stab}_G(x) g^{-1}.$$

Indeed, for all $h \in G$ we have

$$\begin{aligned} h \in \text{Stab}_G(g(x)) &\iff h(g(x)) = g(x) \\ &\iff g^{-1}hg(x) = x \\ &\iff g^{-1}hg \in \text{Stab}_G(x) \\ &\iff h \in g \text{Stab}_G(x) g^{-1}. \end{aligned}$$

(2) If $\varphi : X \rightarrow Y$ is an isomorphism of G -sets, then for all $x \in X$ we have

$$\text{Stab}_G(x) = \text{Stab}_G(\varphi(x)).$$

Indeed, let $g \in \text{Stab}_G(x)$. Then

$$g(\varphi(x)) = \varphi(g(x)) = \varphi(x)$$

implies that $g \in \text{Stab}_G(\varphi(x))$. We conclude that

$$\text{Stab}_G(x) \subseteq \text{Stab}_G(\varphi(x)).$$

Now using the fact that φ^{-1} is also a G -function gives

$$\begin{aligned} \text{Stab}_G(\varphi(x)) &\subseteq \text{Stab}_G(\varphi^{-1}(\varphi(x))) \\ &= \text{Stab}_G(x). \end{aligned}$$

(3) Let $H, K \leq G$ be subgroups.

If there exists an isomorphism of G -sets $\varphi : G/H \rightarrow G/K$, then let $\varphi(H) = gK$. It follows from (1) and (2) that

$$\begin{aligned} H &= \{h \in G : hH = H\} \\ &= \text{Stab}_G(H) \\ &= \text{Stab}_G(\varphi(H)) \\ &= \text{Stab}_G(gK) \\ &= g \text{Stab}_G(K)g^{-1} \\ &= g\{h \in G : hK = K\}g^{-1} \\ &= gKg^{-1}. \end{aligned}$$

Conversely, suppose that $H = gKg^{-1}$ for some $g \in G$. Then we have

$$\begin{aligned} H &= gKg^{-1} \\ &= g \text{Stab}_G(K)g^{-1} \\ &= \text{Stab}_G(gK), \end{aligned}$$

and the Orbit–Stabilizer Theorem gives

$$\begin{aligned} G/H &= G/\text{Stab}_G(gK) \\ &\cong \text{Orb}_G(gK) \\ &= \{hgK : h \in G\} \\ &= \{gK : g \in G\} \\ &= G/K. \end{aligned}$$

□

In summary, let me state the full result:

Fundamental Theorem of G -sets.

- Every G -set has a unique expression as a disjoint union of simple G -sets.
- Every simple G -set has the form G/H for some subgroup $H \leq G$ and we have

$$G/H \cong G/K$$

if and only if H and K are conjugate.

This inspires a very deep idea:

There is no such thing as a G -set; there is only G .

21 The Fundamental Theorem of G -sets and double cosets

November 10, 2015

Last time we proved the following result.

Fundamental Theorem of G -sets.

- Every G -set can be expressed uniquely as a disjoint union of simple G -sets.
- The simple G -sets are just G/H , where H is a subgroup. Furthermore, we have

$$G/H \cong G/K$$

if and only if $H = gKg^{-1}$ for some $g \in G$.

This theorem suggests a deep idea:

There is no such thing as a G -set; there is only G acting on itself.

This idea was famously expressed by Felix Klein in his “Erlangen Program.”

So let’s investigate how G acts on itself.

Given a subgroup $H \leq G$, we can define a left action of H on G by right multiplication as follows:

$$\begin{aligned} H \times G &\longrightarrow G, \\ (h, g) &\longmapsto gh^{-1}. \end{aligned}$$

Check the axioms: We have

- $g1^{-1} = g$ for all $g \in G$;
- $g(h_1h_2)^{-1} = (gh_2^{-1})h_1^{-1}$ for all $g \in G, h_1, h_2 \in H$,

as desired.

The orbits of this action are the left cosets of H in G :

$$\begin{aligned}\text{Orb}_H(g) &= \{gh^{-1} : h \in H\} \\ &= \{gh : h \in H\} = gH.\end{aligned}$$

Now we can define a left action of G on the set G/H of H -orbits by left multiplication as follows:

$$\begin{aligned}G \times G/H &\longrightarrow G/H, \\ (g_1, g_2H) &\longmapsto (g_1g_2)H.\end{aligned}$$

Check: For all $gH \in G/H$ and $g_1, g_2 \in G$,

- $(1g)H = gH$;

•

$$\begin{aligned}(g_1g_2)(gH) &= ((g_1g_2)g)H \\ &= (g_1(g_2g))H \\ &= g_1(g_2(gH)).\end{aligned}$$

Note that the G -set G/H is simple because G acts transitively: For all $g_1H, g_2H \in G/H$ there exists $g_2g_1^{-1} \in G$ such that

$$(g_2g_1^{-1})(g_1H) = (g_2g_1^{-1}g_1)H = g_2H.$$

Note that for all $gH \in G/H$ we have

$$\text{Orb}_G(gH) = G/H \quad \text{and} \quad \text{Stab}_G(gH) = gHg^{-1},$$

hence the Orbit–Stabilizer Theorem says

$$G/H = \text{Orb}_G(gH) \cong G/\text{Stab}_G(gH) = G/(gHg^{-1}),$$

which we already knew. So let's consider a more general situation.

Definition. Let $H, K \leq G$ be subgroups. Then we can define a left action of the direct product $H \times K$ on G by

$$\begin{aligned}(H \times K) \times G &\longrightarrow G, \\ ((h, k), g) &\longmapsto h g k^{-1}.\end{aligned}$$

The orbits of this action are called *double cosets*. For all $g \in G$ we have

$$\begin{aligned}\text{Orb}_{H \times K}(g) &= \{h g k^{-1} : h \in H, k \in K\} \\ &= \{h g k : h \in H, k \in K\} \\ &=: HgK.\end{aligned}$$

We will denote the set of double cosets by

$$H \backslash G / K := \{HgK : g \in G\}.$$

What does Orbit-Stabilizer say here?

First consider the action $H \curvearrowright G/K$ defined by

$$h(gK) = (hg)K$$

and note that the double coset HgK is given by

$$HgK = \bigcup_{h \in H} (hg)K = \bigsqcup_{C \in \text{Orb}_H(gK)} C,$$

where the second union is disjoint by definition. Since any two cosets g_1K, g_2K are in bijection, we obtain a bijection

$$HgK \longleftrightarrow \text{Orb}_H(gK) \times K.$$

Then since

$$\begin{aligned} \text{Stab}_H(gK) &= \{h \in H : hgK = gK\} \\ &= \{h \in H : g^{-1}hgK = K\} \\ &= \{h \in H : g^{-1}hg \in K\} \\ &= \{h \in H : h \in gKg^{-1}\} \\ &= H \cap gKg^{-1}, \end{aligned}$$

Orbit-Stabilizer gives a bijection of H -sets:

$$\text{Orb}_H(gK) \longleftrightarrow H/(H \cap gKg^{-1}).$$

Putting the two bijections together gives

$$HgK \longleftrightarrow (H/(H \cap gKg^{-1})) \times K.$$

[Remark: Starting instead with the right action $H \backslash G \curvearrowright K$ gives a bijection

$$HgK \longleftrightarrow H \times (K/(g^{-1}Hg \cap K)).$$

]

Finally, if H and K are both finite, the above bijection implies that

$$\begin{aligned} |HgK| &= \left| \frac{H}{H \cap gKg^{-1}} \right| \cdot |K| \\ &= \frac{|H|}{|H \cap gKg^{-1}|} \cdot |K| \\ &= \frac{|H| \cdot |K|}{|H \cap gKg^{-1}|}. \end{aligned}$$

In summary, we have the following.

Theorem. Let $H, K \leq G$ be finite subgroups. Then for all $g \in G$ we have

$$|HgK| = \frac{|H| \cdot |K|}{|H \cap gKg^{-1}|} = \frac{|H| \cdot |K|}{|g^{-1}Hg \cap K|}.$$

Corollary. Setting $g = 1$ gives

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|},$$

even when HK is not a group.

The most interesting examples of double cosets come from matrix groups.

For example, let's consider the "LU decomposition" of an invertible matrix.

Definition. Given a field K and a nonzero scalar $\alpha \in K^\times$, we define the elementary matrices

$$E_{ij}(\alpha), \quad E_{ii}(\alpha), \quad P_{ij} \in GL_n(K)$$

as follows:

$$E_{ij}(\alpha) = \begin{pmatrix} & & & j & \\ & 1 & & & 0 \\ & & \ddots & & \\ i & \cdots & 1 & \cdots & \alpha \\ & & & \ddots & \\ & 0 & & & 1 \end{pmatrix}$$

$$E_{ii}(\alpha) = \begin{pmatrix} & & & i & \\ & 1 & & & 0 \\ & & \ddots & & \\ i & \cdots & & \alpha & \\ & & & \ddots & \\ & 0 & & & 1 \end{pmatrix}$$

$$P_{ij} = \begin{pmatrix} 1 & & & & & & 0 \\ & \ddots & & & & & \\ & & \ddots & & & & \\ i & & & 0 & & 1 & \\ & & & & \ddots & & \\ j & & & 1 & & 0 & \\ & & & & & & \ddots \\ 0 & & & & & & & 1 \end{pmatrix}$$

Now suppose $A \in GL_n(K)$ has k th row a_k . Then we have

- the k th row of $E_{ij}(\alpha)A$ is

$$\begin{cases} a_k, & k \neq i, \\ a_k + \alpha a_j, & k = i; \end{cases}$$

- the k th row of $E_{ii}(\alpha)A$ is

$$\begin{cases} a_k, & k \neq i, \\ \alpha a_k, & k = i; \end{cases}$$

- the k th row of $P_{ij}A$ is

$$\begin{cases} a_k, & k \notin \{i, j\}, \\ a_j, & k = i, \\ a_i, & k = j. \end{cases}$$

Thus multiplying on the left of A by elementary matrices performs “elementary row operations.” Similarly, multiplying on the right of A performs “elementary column operations.”

Now the familiar Gaussian elimination algorithm can be phrased as follows.

Gaussian Elimination. Let $A \in GL_n(K)$. Then we can reduce A to the identity matrix in three steps:

- (1) Multiply on the left by lower triangular matrices $E_{ij}(\alpha)$ ($j \leq i$).
- (2) Multiply on the left by permutation matrices P_{ij} .
- (3) Multiply on the left by upper triangular matrices $E_{ij}(\alpha)$ ($i \leq j$).

In summary, we obtain

$$UPLA = I, \quad A = L^{-1}P^{-1}U^{-1},$$

where L^{-1} is lower triangular, P^{-1} is a permutation, and U^{-1} is upper triangular.

For example, consider the matrix

$$A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ -1 & 0 & 4 \end{pmatrix}.$$

Then we have

$$\begin{aligned} \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ -1 & 0 & 4 \end{pmatrix} &\xrightarrow{E_{21}(-2)} \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ -1 & 0 & 4 \end{pmatrix} \xrightarrow{E_{32}(1)} \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ 0 & 0 & 3 \end{pmatrix} \\ &\xrightarrow{E_{33}(1/3)} \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{P_{12}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \\ &\xrightarrow{E_{23}(-2)} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{E_{13}(1)} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned}$$

We conclude that

$$E_{13}(1)E_{23}(-2)P_{12}E_{33}(1/3)E_{32}(1)E_{21}(-2)A = I,$$

and hence $A = LPU$, where

$$L = E_{21}(-2)^{-1}E_{32}(1)^{-1}E_{33}(1/3)^{-1}$$

$$\begin{aligned} L &= E_{21}(2)E_{32}(-1)E_{33}(3) \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -1 & 3 \end{pmatrix}, \end{aligned}$$

$$P = P_{12}^{-1} = P_{12} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and

$$\begin{aligned} U &= E_{23}(-2)^{-1}E_{13}(1)^{-1} \\ &= E_{23}(2)E_{13}(-1) \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned}$$

To be fancy, we will let

$$B^+, B^- \leq GL_n(K)$$

be the subgroups of upper and lower triangular matrices, and we will call them positive and negative *Borel subgroups*. We will denote the subgroup of permutation matrices by

$$W \leq GL_n(K)$$

and will call it a *Weyl subgroup*.

Now let the direct product $B^- \times B^+$ act on $GL_n(K)$ by

$$\begin{aligned} (B^- \times B^+) \times GL_n(K) &\longrightarrow GL_n(K), \\ ((\ell, u), g) &\longmapsto \ell g u^{-1}. \end{aligned}$$

Gaussian elimination tells us that every double coset $B^- g B^+$ contains a permutation matrix.

Proof. Consider an element $\ell g u \in B^- g B^+$. Applying Gaussian elimination gives

$$\ell g u = \ell_1 w u_1$$

for some $\ell_1 \in B^-$, $w \in W$, $u_1 \in B^+$, and then we have

$$w = \ell_1^{-1} \ell g u u_1^{-1} \in B^- g B^+.$$

□

In fact it is not too difficult to show that each double coset contains a unique permutation, so we can express $GL_n(K)$ as a disjoint union parametrized by W :

$$GL_n(K) = \bigsqcup_{w \in W} B^- w B^+.$$

For some reason algebraists prefer to use a modified form of Gaussian elimination and to use double cosets of $B^+ \times B^+$ to show that

$$GL_n(K) = \bigsqcup_{w \in W} B^+ w B^+,$$

and they call this a *Bruhat decomposition* of $GL_n(K)$. This decomposition is an important tool in studying $GL_n(K)$ as an algebraic variety.

Remark. The Bruhat decomposition of $GL_n(K)$ also induces a partial order on the set W of permutations.

Given $u, v \in W$, the cosets $B^+ u B^+$ and $B^+ v B^+$ are disjoint, but their closures (in a certain topology) might be related. So we define

$$u \leq_B v \iff \overline{B^+ u B^+} \subseteq \overline{B^+ v B^+}.$$

When $K = \mathbb{C}$ the topology is the obvious one (i.e., we think of $GL_n(\mathbb{C})$ as a subset of $\mathbb{C}^{n \times n}$).

22 Conjugation, centralizers, the class equation

November 12, 2015

In the philosophy of Klein’s “Erlangen Program” there is no such thing as a G -set; there is only G acting on itself.

Last time we discussed the “regular” action of G on itself by left or right multiplication (or both).

Today we’ll discuss a very important special case, called “conjugation.”

Recall that we have a group homomorphism

$$\varphi : G \times G \longrightarrow \text{Sym}(G)$$

defined by

$$\varphi_{g,h}(a) := gah^{-1}.$$

If we compose this with the left or right embedding

$$G \longrightarrow G \times G, \quad g \longmapsto (g, 1), \quad \text{and} \quad G \longrightarrow G \times G, \quad g \longmapsto (1, g),$$

we obtain the left or right regular action. However, if we compose with the diagonal embedding

$$\begin{aligned} \Delta : G &\longrightarrow G \times G, \\ g &\longmapsto (g, g), \end{aligned}$$

then we obtain the action of G on itself by conjugation:

$$\begin{aligned} G &\xrightarrow{\Delta} G \times G \xrightarrow{\varphi} \text{Aut}(G), \\ g &\longmapsto (g, g) \longmapsto \varphi_{g,g}, \end{aligned}$$

defined by

$$\varphi_{g,g}(a) = g a g^{-1}.$$

Let’s investigate what orbit–stabilizer says about conjugation.

Definition. Let $G \curvearrowright G$ by conjugation. For all $a \in G$ we define its *conjugacy class*

$$K_G(a) := \text{Orb}_G(a) = \{g a g^{-1} : g \in G\}$$

and its *centralizer*

$$\begin{aligned} Z_G(a) &:= \text{Stab}_G(a) = \{g \in G : g a g^{-1} = a\} \\ &= \{g \in G : g a = a g\}. \end{aligned}$$

[K is for “Klasse,” Z is for “Zentrum.”]

Then orbit–stabilizer gives a bijection

$$K(a) \longleftrightarrow G/Z(a).$$

We also define the *center* of G

$$Z(G) := \{a \in G : ga = ag \text{ for all } g \in G\}$$

and note that

$$\begin{aligned} |K(a)| = 1 &\iff K(a) = \{a\} \\ &\iff gag^{-1} = a \quad \text{for all } g \in G \\ &\iff ga = ag \quad \text{for all } g \in G \\ &\iff a \in Z(G). \end{aligned}$$

If we let $K_i, i \in I$, be the conjugacy classes of G with corresponding centralizers Z_i (defined up to conjugacy), then we obtain the orbit decomposition

$$\begin{aligned} G &= \bigsqcup_i K_i \\ &= \left(\bigsqcup_{|K_i|=1} K_i \right) \sqcup \left(\bigsqcup_{|K_i| \neq 1} K_i \right) \\ &= \left(\bigsqcup_{a \in Z(G)} \{a\} \right) \sqcup \left(\bigsqcup_{|K_i| \neq 1} K_i \right) \\ &= Z(G) \sqcup \left(\bigsqcup_{|K_i| \neq 1} K_i \right), \end{aligned}$$

and orbit–stabilizer gives a bijection

$$G \longleftrightarrow Z(G) \sqcup \left(\bigsqcup_{Z_i \neq G} G/Z_i \right).$$

If G is finite we obtain

$$\boxed{|G| = |Z(G)| + \sum_{Z_i \neq G} \frac{|G|}{|Z_i|}},$$

which is called the *class equation* of G .

Example: The symmetric group S_n .

Recall that conjugacy classes of S_n are parametrized by “integer partitions” of n , i.e., sequences

$$\lambda = (\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_\ell)$$

such that

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_\ell \geq 1 \quad \text{and} \quad \sum_i \lambda_i = n.$$

Let K_λ be the class of permutations of “cycle type” λ and let Z_λ be the corresponding centralizer. If λ has m_i parts of size i , then one can check that

$$\begin{aligned} |Z_\lambda| &= 1^{m_1} m_1! 2^{m_2} m_2! 3^{m_3} m_3! \dots \\ &= \prod_{i \geq 1} i^{m_i} m_i!, \end{aligned}$$

and hence

$$|K_\lambda| = \frac{n!}{\prod_{i \geq 1} i^{m_i} m_i!}.$$

For example, when $n = 5$ we have

λ	m	$ Z_\lambda $	$ K_\lambda $
5	00001	$5^1 \cdot 1! = 5$	24
41	1001	$1^1 \cdot 1! 4^1 \cdot 1! = 4$	30
32	011	$2^1 \cdot 1! 3^1 \cdot 1! = 6$	20
311	201	$1^2 \cdot 2! 3^1 \cdot 1! = 6$	20
221	12	$1^1 \cdot 1! 2^2 \cdot 2! = 8$	15
2111	31	$1^3 \cdot 3! 2^1 \cdot 1! = 12$	10
11111	5	$1^5 \cdot 5! = 120$	1

Note that

$$|K_\lambda| = 1 \iff \lambda = (1, 1, 1, 1, 1),$$

and hence

$$Z(S_5) = \{1\}.$$

In general we have $Z(S_n) = \{1\}$, which shows that S_n is “very” non-abelian.

Note also that

$$1 + 10 + 15 + 20 + 20 + 30 + 24 = 120,$$

so we probably didn’t make a mistake.

One application of the class equation is that every normal subgroup is a union of conjugacy classes. If we know the sizes of the conjugacy classes, then this greatly restricts the sizes of possible normal subgroups.

Thus, a normal subgroup $N \trianglelefteq S_5$ must have size

$$1 + (\text{some of } 10, 15, 20, 20, 30, 24)$$

and the size must divide $|S_5| = 120$. It turns out there are only three nontrivial possibilities:

$$N_1 = K_{11111} \sqcup K_{221} \sqcup K_5,$$

$$N_2 = K_{11111} \sqcup K_{221} \sqcup K_{311} \sqcup K_5,$$

$$N_3 = K_{11111} \sqcup K_{221} \sqcup K_{32} \sqcup K_5.$$

It turns out that $N_2 = A_5 \trianglelefteq S_5$ and that N_1 and N_3 are not subgroups.

We conclude that A_5 is the only nontrivial normal subgroup of S_5 . However, this does not quite prove that A_5 is simple. [For that, see HW4.]

The other major application of the class equation is to Sylow Theory.

If G is a finite group, recall that “Lagrange’s Theorem” says:

If there exists a subgroup $H \leq G$ of size n , then $n \mid |G|$.

Unfortunately, the converse statement,

If $n \mid |G|$, then there exists a subgroup $H \leq G$ of order n ,

is FALSE.

Proof. We will show that the alternating group of order 12,

$$A_4 = K_{1111} \sqcup K_{22} \sqcup K_3 \trianglelefteq S_4,$$

has no subgroup of order 6.

So suppose $H \leq A_4$ is a subgroup with $|H| = 6$. Then $H \cong \mathbb{Z}/6\mathbb{Z}$ or D_6 . Since A_4 has no element of order 6, we must have

$$H \cong D_6.$$

Since D_6 has three elements of order 2, we must have

$$K_{22} = \{(12)(34), (13)(24), (14)(23)\} \subseteq H.$$

But the elements of order 2 in D_6 (i.e., the reflections) don't commute, and the elements of K_{22} do commute. Contradiction.

Sylow Theory is the attempt to find the strongest possible approximation to the converse that is true, i.e., to determine all properties of G that depend only on its size $|G|$.

Here is the main result.

Sylow's Theorem. Suppose that $|G| = p^\alpha m$ with p prime and $p \nmid m$. We say that $H \leq G$ is a *Sylow p -subgroup* if $|H| = p^\alpha$. Let $\text{Syl}_p(G)$ be the set of Sylow p -subgroups. Then:

1. $\text{Syl}_p(G) \neq \emptyset$.
2. If $P \in \text{Syl}_p(G)$ and $Q \leq G$ is any p -subgroup (i.e., $|Q| = p^\beta$, $\beta \leq \alpha$), then there exists $g \in G$ such that

$$gQg^{-1} \subseteq P.$$

In particular, any two Sylow p -subgroups are conjugate.

3.

$$|\text{Syl}_p(G)| \equiv 1 \pmod{p} \quad \text{and} \quad |\text{Syl}_p(G)| \mid m.$$

We'll prove the theorem next week. [Surprisingly, it turns out that part (1) is the hardest.]

For now, here's a cool analogy:

Let $G = GL_n(K)$. We say $g \in G$ is *unipotent* if all of its eigenvalues are 1. A subgroup $H \leq G$ is called *unipotent* if all of its elements are unipotent.

It turns out that the group of upper unitriangular matrices

$$U = \left\{ \begin{pmatrix} 1 & * & \cdots & * \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & 1 \end{pmatrix} \in G \right\}$$

is a maximal unipotent subgroup and, furthermore, if $H \leq G$ is any unipotent subgroup, then there exists $g \in G$ such that

$$gHg^{-1} \subseteq U.$$

Over a general field K this is called “Kolchin’s Theorem,” and it is fairly tricky to prove.

However, if K is a finite field of characteristic p , it turns out that a subgroup

$$H \leq GL_n(K)$$

is unipotent if and only if it is a p -group, so in this case Kolchin’s Theorem is equivalent to part (2) of Sylow’s Theorem.

Cool, right?

23 Cauchy’s theorem and Sylow theory

November 17, 2015

Last time I stated Sylow’s Theorem. It requires some terminology.

Definition. Let p be prime. A p -group is a finite group of size p^α for some α . We say that a subgroup $H \leq G$ is a p -subgroup if H is a p -group.

Now let $|G| = p^\alpha m$ with $\gcd(p, m) = 1$. A Sylow p -subgroup is a p -subgroup $H \leq G$ with size $|H| = p^\alpha$.

Let $\text{Syl}_p(G)$ be the set of Sylow p -subgroups of G .

Sylow’s Theorem (1872). Let $|G| = p^\alpha m$, with $\gcd(p, m) = 1$. Then we have:

1. $\text{Syl}_p(G) \neq \emptyset$, i.e., a Sylow p -subgroup exists.
2. If $P \in \text{Syl}_p(G)$ and $Q \leq G$ is any p -subgroup, then there exists $g \in G$ such that

$$gQg^{-1} \leq P.$$

This implies:

- Every p -subgroup is contained in a Sylow p -subgroup.
- Any two Sylow p -subgroups are conjugate; hence if there exists a unique Sylow p -subgroup it must be normal.

3.

$$|\mathrm{Syl}_p(G)| \equiv 1 \pmod{p} \quad \text{and} \quad |\mathrm{Syl}_p(G)| \mid m.$$

Before proving the theorem, let me show you an application.

Theorem. Let $|G| = pqr$, with $p < q < r$ prime. Then G is not simple.

Proof. By part (1),

$$n_p := |\mathrm{Syl}_p(G)| \geq 1, \quad n_q := |\mathrm{Syl}_q(G)| \geq 1, \quad n_r := |\mathrm{Syl}_r(G)| \geq 1.$$

If any of n_p, n_q, n_r equals 1, then we obtain a normal Sylow subgroup by (2).

So assume for contradiction that $n_p, n_q, n_r > 1$. By (3) we must have

$$n_r = pq, \quad n_q \in \{r, pr\}, \quad n_p \in \{q, r, qr\}.$$

Note that the Sylow subgroups of G are isomorphic to

$$\mathbb{Z}/p\mathbb{Z}, \quad \mathbb{Z}/q\mathbb{Z}, \quad \text{or} \quad \mathbb{Z}/r\mathbb{Z}.$$

Since p, q, r are prime, any two of these subgroups intersect trivially.

Now let's count the elements of G with orders $p, q, r, 1$, and show that there are too many.

Each of the n_p copies of $\mathbb{Z}/p\mathbb{Z}$ contains $p - 1$ elements of order p , and similarly for q and r . Thus we have

$$\begin{aligned} |G| &\geq n_r(r - 1) + n_q(q - 1) + n_p(p - 1) + 1 \\ &\geq pq(r - 1) + r(q - 1) + q(p - 1) + 1 \\ &= pqr - pq + qr - r + pq - q + 1 \\ &= pqr + (q - 1)(r - 1). \end{aligned}$$

Since $|G| = pqr$, this implies

$$(q - 1)(r - 1) \leq 0,$$

which contradicts the fact that $q - 1 > 0$ and $r - 1 > 0$. □

Pretty slick, right? That's the typical way that Sylow's Theorem gets applied.

Now let's prove it. We'll start small and work our way up:

Theorem. Let A be a finite abelian group and let p be prime. If $p \mid |A|$, then A has an element of order p .

Proof. Proof by induction on $|A|$: If A is cyclic we're done. So consider an element $1 \neq \alpha \in A$ such that

$$H := \langle \alpha \rangle \neq A.$$

If $p \mid |H|$, then by induction H has an element of order p , hence so does A .

So assume that $p \nmid |H|$ and consider the quotient group A/H .

[Note that $H \leq A$ because A is abelian.] Since $p \mid |A|$ and $p \nmid |H|$, we have

$$p \mid \frac{|A|}{|H|} = |A/H|.$$

By induction this implies that A/H has an element of order p , say aH . Then we have

$$aH \neq H \quad \text{and} \quad (aH)^p = a^p H = H,$$

and hence

$$a \notin H \quad \text{and} \quad a^p \in H.$$

Now let $m = |H|$, so by Lagrange we have

$$a^p \in H \implies (a^p)^m = 1.$$

But

$$(a^m)^p = (a^p)^m = 1,$$

so that a^m has order dividing p . If a^m has order p we're done, so assume that a^m has order 1, i.e., $a^m = 1$.

Then we have

$$(aH)^m = a^m H = H.$$

This implies that the order of aH divides m . But the order of aH is p , and we assumed that $p \nmid |H| = m$. Contradiction. $\square$

[Remark: This result could be proved more easily with stronger technology, e.g., the Fundamental Theorem of Finite Abelian Groups, but we don't have that yet. We'll prove it next semester.]

We have shown

$$p \mid |A| \implies \exists \text{ element of order } p$$

when A is an abelian group. Next we'll prove it for non-abelian groups.

Cauchy's Theorem (1875). Consider a finite group G and a prime p . If $p \mid |G|$, then G has an element of order p .

Proof. Proof by induction on $|G|$:

Let $p \mid |G|$ and consider the class equation

$$|G| = |Z(G)| + \sum_{Z_i \neq G} \frac{|G|}{|Z_i|},$$

where the Z_i are the non-trivial stabilizer subgroups.

If $p \mid |Z_i|$ for some $Z_i \neq G$, then by induction Z_i has an element of order p , hence so does G .

So assume that $p \nmid |Z_i|$ for all $Z_i \neq G$. Then we have

$$p \mid \frac{|G|}{|Z_i|},$$

so the class equation implies that $p \mid |Z(G)|$.

If $Z(G) \neq G$, then by induction $Z(G)$ has an element of order p , hence so does G .

So assume that $Z(G) = G$. Then G is abelian, so the previous theorem implies that G has an element of order p . $\square$

In particular, this shows that if $p \mid |G|$, then G has a subgroup of order p (the cyclic group generated by an element of order p).

Our next job is to lift this to higher powers of p :

Theorem (Sylow 1). Let G be finite and let p be prime. If $p^n \mid |G|$, then G has subgroups of order p^m for all $1 \leq m \leq n$.

Proof. Proof by induction on $|G|$:

Let $p^n \mid |G|$ and consider the class equation

$$|G| = |Z(G)| + \sum_{Z_i \neq G} \frac{|G|}{|Z_i|}.$$

If $p^n \mid |Z_i|$ for some $Z_i \neq G$, then we're done by induction. So assume that $p^n \nmid |Z_i|$ for all $Z_i \neq G$. Since $p^n \mid |G|$, this implies that

$$p \mid \frac{|G|}{|Z_i|}$$

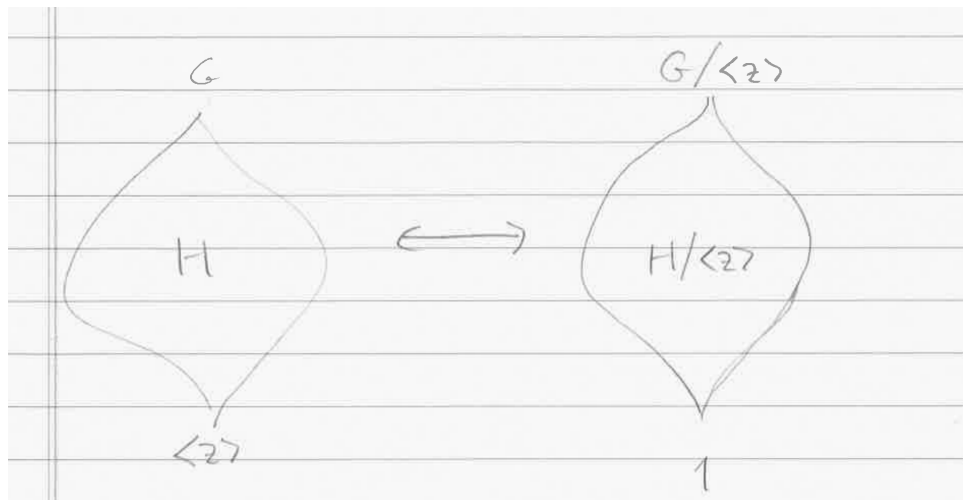
for all $Z_i \neq G$, and the class equation then implies that $p \mid |Z(G)|$. Hence $Z(G)$ has an element of order p , say $z \in Z(G)$.

Now consider the group $\langle z \rangle \leq G$, with $|\langle z \rangle| = p$. Since $z \in Z(G)$ we have $\langle z \rangle \trianglelefteq G$, and then

$$p^{n-1} \mid |G/\langle z \rangle| = \frac{|G|}{p}.$$

By induction, $G/\langle z \rangle$ has subgroups of size p^m for all $1 \leq m \leq n-1$. By the Correspondence Theorem, every subgroup of $G/\langle z \rangle$ has the form $H/\langle z \rangle$ for some subgroup

$$\langle z \rangle \leq H \leq G.$$



Finally, note that

$$p^m = |H/\langle z \rangle| \iff |H| = p^{m+1},$$

so G has subgroups of size p^m for all $1 \leq m \leq n$. $\square$

Remark: We only used the abelian version of Cauchy's Theorem in this proof. So maybe I was just wasting your time when we proved Cauchy's Theorem. Or maybe there was some pedagogical reason to do it.

We have shown that Sylow p -subgroups exist. Our next job is to investigate how G acts on its p -subgroups by conjugation.

The main tool will be double cosets.

(Nothing is wasted.)

24 Sylow conjugacy and counting

November 19, 2015

Last time we proved Sylow (1):

If $|G| = p^\alpha m$ with $\gcd(p, m) = 1$, then G has subgroups of order p^β for all $1 \leq \beta \leq \alpha$.

The subgroups of size p^α are called Sylow p -subgroups. Let $\text{Syl}_p(G)$ be the set of these. Now suppose that

$$|G| = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_n^{\alpha_n},$$

where $p_1, p_2, \dots, p_n$ are distinct primes, and for each p_i choose a Sylow p_i -subgroup

$$P_i \in \text{Syl}_{p_i}(G).$$

Then it is possible to express G as some kind of product

$$G = "P_1 P_2 P_3 \cdots P_n,"$$

but it is not clear what kind of product this is because the partial products $P_i P_j$ might not be subgroups.

However, if $G = A$ is abelian, then every Sylow subgroup is normal and we can express A as a direct product

$$A = P_1 \times P_2 \times P_3 \times \cdots \times P_n.$$

This is called the *primary decomposition* of a finite abelian group.

We will see today that the primary decomposition is unique because there exists a unique Sylow p_i -subgroup for each prime $p_i \mid |A|$.

Our goal is to prove the remaining pieces of Sylow's Theorem:

2. If $P \in \text{Syl}_p(G)$ and if Q is any p -subgroup, then there exists $g \in G$ such that

$$gQg^{-1} \subseteq P.$$

3.

$$|\text{Syl}_p(G)| \equiv 1 \pmod{p} \quad \text{and} \quad |\text{Syl}_p(G)| \mid m,$$

where $|G| = p^\alpha m$.

Our tool will be the action of G on its subgroups by conjugation.

Definition. Let G be a group and define an action $G \curvearrowright \mathcal{L}(G)$ by

$$\begin{aligned} G \times \mathcal{L}(G) &\longrightarrow \mathcal{L}(G), \\ (g, H) &\longmapsto gHg^{-1}. \end{aligned}$$

Observe that $H \trianglelefteq G$ if and only if

$$\text{Stab}_G(H) = \{g \in G : gHg^{-1} = H\} = G.$$

For a general subgroup $H \in \mathcal{L}(G)$ we define its *normalizer*

$$N_G(H) := \text{Stab}_G(H).$$

This is the largest subgroup of G in which H is normal. In other words, if $K \leq G$ is a subgroup such that $H \trianglelefteq K$, then we must have

$$K \subseteq N_G(H).$$

Proof of Sylow (2). Let $|G| = p^\alpha m$ with $\gcd(p, m) = 1$. Let $P, Q \leq G$ be subgroups with $|P| = p^\alpha$ and $|Q| = p^\beta$ for some $1 \leq \beta \leq \alpha$.

Consider the decomposition of G into double cosets

$$G = \bigsqcup_i P g_i Q.$$

Recall from last week that we proved

$$|P g_i Q| = \frac{|P| |Q|}{|P \cap g_i Q g_i^{-1}|},$$

so we have

$$|G| = \sum_i \frac{|P| |Q|}{|P \cap g_i Q g_i^{-1}|}. \quad (*)$$

Assume for contradiction that

$$P \cap g_i Q g_i^{-1} \neq g_i Q g_i^{-1}$$

for all i , so that

$$|P \cap g_i Q g_i^{-1}| < |g_i Q g_i^{-1}| = |Q| = p^\beta.$$

The intersection of p -groups is a p -group, so we have

$$|P \cap g_i Q g_i^{-1}| = p^{\gamma_i} \quad \text{with} \quad \gamma_i < \beta$$

for all i . But then

$$\frac{|P| |Q|}{|P \cap g_i Q g_i^{-1}|} = \frac{p^\alpha p^\beta}{p^{\gamma_i}} = p^{\alpha + (\beta - \gamma_i)}$$

is divisible by $p^{\alpha+1}$ for all i , and from $(*)$ we conclude that

$$p^{\alpha+1} \mid |G|.$$

Contradiction.

We conclude that there exists $g_i \in G$ such that

$$P \cap g_i Q g_i^{-1} = g_i Q g_i^{-1},$$

hence

$$g_i Q g_i^{-1} \subseteq P.$$

□

[I once had a professor who called this a “sleazy” proof. I don’t know what he meant by that.]

Remarks:

- Sylow (2) implies that every p -subgroup is contained in a Sylow p -subgroup.

Proof:

$$g Q g^{-1} \subseteq P \implies Q \subseteq g^{-1} P g.$$

Thus the concept of “Sylow p -subgroup” is the same as “maximal p -subgroup.”

- It also implies that any two Sylow p -subgroups are conjugate.

Proof: $gQg^{-1} \subseteq P$ and $|gQg^{-1}| = |P|$ imply that $gQg^{-1} = P$.

If G is abelian (hence every subgroup is normal), this implies that there is a unique Sylow p -subgroup for each prime $p \mid |G|$. This proves the uniqueness of the primary decomposition.

Proof of Sylow (3). Let $|G| = p^\alpha m$ with $\gcd(p, m) = 1$. Now consider the conjugation action $G \curvearrowright \mathcal{L}(G)$ restricted to the set of Sylow p -subgroups:

$$G \curvearrowright \text{Syl}_p(G).$$

By Sylow (2) we know that this action is transitive, so for all $P \in \text{Syl}_p(G)$ we have a bijection

$$\text{Syl}_p(G) = \text{Orb}_G(P) \longleftrightarrow G/N_G(P),$$

and hence

$$|\text{Syl}_p(G)| = \frac{|G|}{|N_G(P)|}.$$

Since $P \subseteq N_G(P)$ has order p^α , Lagrange tells us that

$$p^\alpha \mid |N_G(P)|, \quad \text{say} \quad |N_G(P)| = p^\alpha n.$$

Then we have

$$|\text{Syl}_p(G)| = \frac{|G|}{|N_G(P)|} = \frac{p^\alpha m}{p^\alpha n} = \frac{m}{n} \mid m.$$

To prove $|\text{Syl}_p(G)| \equiv 1 \pmod{p}$ we need a lemma.

Lemma. Let $P, Q \in \text{Syl}_p(G)$. Then we have

$$P \subseteq N_G(Q) \iff P = Q.$$

Proof. Suppose that $P \subseteq N_G(Q)$. This implies that the product set PQ is a subgroup of G , and hence $|PQ|$ divides $|G|$. But note that

$$|PQ| = \frac{|P||Q|}{|P \cap Q|} = p^{2\alpha - \beta},$$

where $|P \cap Q| = p^\beta$ for some $0 \leq \beta \leq \alpha$. If $\beta < \alpha$, then we conclude that $|PQ|$, and hence $|G|$, is divisible by $p^{\alpha+1}$. Contradiction. This implies that

$$|P \cap Q| = p^\alpha, \quad \text{and hence} \quad P = P \cap Q = Q.$$

□

Finally, let $P \in \text{Syl}_p(G)$ and consider the action of P on $\text{Syl}_p(G)$ by conjugation. We obtain an orbit decomposition

$$\text{Syl}_p(G) = \bigsqcup_i \text{Orb}_P(Q_i).$$

By orbit–stabilizer we have

$$|\text{Orb}_P(Q_i)| = \frac{|P|}{|\text{Stab}_P(Q_i)|} = p^{\gamma_i}$$

for some $0 \leq \gamma_i \leq \alpha$. Furthermore, we have

$$\begin{aligned} \gamma_i = 0 &\iff |\text{Orb}_P(Q_i)| = 1 \\ &\iff gQ_ig^{-1} = Q_i \quad \text{for all } g \in P \\ &\iff P \subseteq N_G(Q_i) \\ &\iff P = Q_i \quad (\text{lemma}). \end{aligned}$$

We conclude that

$$\begin{aligned} |\text{Syl}_p(G)| &= 1 + \sum_{Q_i \neq P} |\text{Orb}_P(Q_i)| \\ &\equiv 1 \pmod{p}. \end{aligned}$$

□

On Tuesday we applied Sylow’s Theorem to show that a group of size pqr is not simple. Now let’s look at some more applications.

Theorem. If $|G| = p^\alpha$, with $\alpha \geq 2$, then G is not simple.

Proof. If G is abelian, then Sylow (1) gives a subgroup of order p , which must be normal. So assume that G is non-abelian, i.e.,

$$1 \leq Z(G) \subsetneq G.$$

Then since

$$p \mid |G| \quad \text{and} \quad p \mid \frac{|G|}{|Z_i|}$$

for all Z_i , the class equation says that $p \mid |Z(G)|$, hence

$$1 \neq Z(G) \subsetneq G.$$

□

Theorem. Let $|G| = p^\alpha m$ with $m > 1$ and $\gcd(p, m) = 1$. If m has no divisor $d \mid m$ such that

- $d \neq 1$,
- $d \equiv 1 \pmod{p}$,

then G is not simple.

Proof. By Sylow (3) we have

$$|\text{Syl}_p(G)| \mid m \quad \text{and} \quad |\text{Syl}_p(G)| \equiv 1 \pmod{p}.$$

By assumption this implies that $|\text{Syl}_p(G)| = 1$, so G has a unique Sylow p -subgroup, which is normal by part (2). Finally, since $m > 1$, this subgroup is proper. □

Theorem. Let $p < q$ be prime.

- If $q \not\equiv 1 \pmod{p}$, then

$$|G| = pq \implies G \text{ is cyclic.}$$

- If $q \equiv 1 \pmod{p}$, then there are exactly two groups of size pq : one cyclic and one non-abelian.

Proof Sketch. Let

$$n_q = |\text{Syl}_q(G)| \quad \text{and} \quad n_p = |\text{Syl}_p(G)|.$$

Sylow (3) says

$$n_q \mid p \quad \text{and} \quad n_q \equiv 1 \pmod{q}.$$

Since $p < q$, this implies $n_q = 1$.

Hence we obtain a normal Sylow q -subgroup

$$Q \cong \mathbb{Z}/q\mathbb{Z}.$$

We also have

$$n_p \mid q \quad \text{and} \quad n_p \equiv 1 \pmod{p}.$$

- If $q \not\equiv 1 \pmod{p}$, this implies that $n_p = 1$ and we obtain a normal Sylow p -subgroup

$$P \cong \mathbb{Z}/p\mathbb{Z}.$$

Since $|G| = |P||Q|$, with P and Q both normal, we have

$$G = P \times Q.$$

Furthermore, if $P = \langle x \rangle$ and $Q = \langle y \rangle$, then we have

$$\begin{aligned} \text{ord}_G(x, y) &= \text{lcm}(\text{ord}_P(x), \text{ord}_Q(y)) \\ &= pq, \end{aligned}$$

hence $G = \langle (x, y) \rangle$ is cyclic.

- If $q \equiv 1 \pmod{p}$, then we may have $n_p = 1$ (in which case G is again cyclic), or we may have $n_p = q$.

In the second case, let $P \cong \mathbb{Z}/p\mathbb{Z}$ be any Sylow p -subgroup. Then we have

$$G = P \rtimes_{\varphi} Q,$$

where the action

$$\varphi : P \longrightarrow \text{Aut}(Q)$$

is by conjugation. We know that φ is not trivial because $n_p = q \neq 1$ implies that P is not normal.

What could φ be? Recall that

$$\text{Aut}(Q) = \text{Aut}(\mathbb{Z}/q\mathbb{Z}) = (\mathbb{Z}/q\mathbb{Z})^{\times},$$

which is a group of size $q - 1$. In fact, the “primitive root theorem” [proof omitted] says that

$$(\mathbb{Z}/q\mathbb{Z})^{\times} \cong \mathbb{Z}/(q-1)\mathbb{Z}.$$

Now let $P = \langle x \rangle$ and consider any two nontrivial automorphisms

$$\varphi_1, \varphi_2 : P \longrightarrow \text{Aut}(Q).$$

By assumption $\ker \varphi_1, \ker \varphi_2 \neq P$. Since $|P|$ is prime, this implies

$$\ker \varphi_1 = \ker \varphi_2 = 1,$$

hence

$$|\varphi_1(P)| = |\varphi_2(P)| = p.$$

Since $\text{Aut}(Q)$ is cyclic, it has a unique subgroup of order p , which is also cyclic, hence

$$\varphi_1(P) = \varphi_2(P) = \langle \varphi_1(x) \rangle = \langle \varphi_2(x) \rangle.$$

Let

$$\varphi_1(x) = \varphi_2(x)^m.$$

Then the map

$$\begin{aligned} \theta : P &\longrightarrow P, \\ x &\longmapsto x^m \end{aligned}$$

is a group automorphism. Finally, one can use the fact $\varphi_1 = \varphi_2 \circ \theta$ to prove that

$$P \rtimes_{\varphi_1} Q \cong P \rtimes_{\varphi_2} Q. \quad \square$$

Remarks:

- Let q be an odd prime. Then the unique non-abelian group of size $2q$ is the dihedral group.
- Is there an explicit description of the non-abelian group for general $p < q$ with $q \equiv 1 \pmod{p}$?

Yes. Think of $\mathbb{Z}/q\mathbb{Z}$ as a one-dimensional vector space and consider the “general affine” group $GA(\mathbb{Z}/q\mathbb{Z})$, as in HW4 Problem 5. We can view this as a matrix group

$$GA(\mathbb{Z}/q\mathbb{Z}) = \left\{ \begin{pmatrix} x & y \\ 0 & 1 \end{pmatrix} : x, y \in \mathbb{Z}/q\mathbb{Z}, x \neq 0 \right\}.$$

The unique non-abelian group of order pq is isomorphic to the subgroup

$$\left\{ \begin{pmatrix} x & y \\ 0 & 1 \end{pmatrix} \in GA(\mathbb{Z}/q\mathbb{Z}) : x^p = 1 \right\}.$$

25 Finite simple groups and linear groups

December 1, 2015

This Week: Epilogue.

Recall the Sylow Theorems: Let G be a finite group and let $p \in \mathbb{Z}$ be prime. Suppose that $|G| = p^\alpha m$ with $p \nmid m$.

1. There exists a subgroup $P \subseteq G$ with size $|P| = p^\alpha$.
2. Given subgroups $P, Q \subseteq G$ with $|P| = p^\alpha$ and $|Q| = p^\beta$ ($\beta \leq \alpha$), there exists a group element $g \in G$ such that

$$gQg^{-1} \subseteq P.$$

3. If

$$n_p = \#\{P \subseteq G : |P| = p^\alpha\},$$

then we have

$$n_p \equiv 1 \pmod{p}, \quad n_p \mid m.$$

We saw last time that these results are a powerful tool for studying the structure of finite groups. Let me summarize some results.

- If $|G| = p^\alpha$ then G is solvable with composition length α . It is simple if and only if $\alpha = 1$.
- Let $|G| = pq$ with $p < q$ prime. If $p \nmid (q-1)$ then G is cyclic. If $p \mid (q-1)$ then G is either cyclic or

$$G \cong \left\{ \begin{pmatrix} x & y \\ 0 & 1 \end{pmatrix} : x, y \in \mathbb{Z}/q\mathbb{Z}, x^p = 1 \right\},$$

which is non-abelian.

- If $|G| = pqr$ with $p < q < r$ prime then G is not simple. More generally, if

$$|G| = p_1 p_2 \cdots p_k \quad \text{with} \quad p_1 < p_2 < \cdots < p_k$$

prime, then G is solvable. [You can prove this by induction.]

- Let $|G| = p^\alpha m$ with $p \nmid m$. If m has no divisor $d \mid m$ with $d \neq 1$ and $d \equiv 1 \pmod{p}$, then G is not simple.

- Burnside's Theorem (1904):

If $|G| = p^\alpha q^\beta$ with $p < q$ prime then G is solvable.

[The proof uses representation theory and is beyond the scope of the course.]

- Feit-Thompson Theorem (1962):

If $|G|$ is odd then G is solvable.

[The proof is ~ 250 pages and is way beyond the scope of the course!]

We thus have a lot of restrictions on the possible sizes of finite simple groups.

Apart from the abelian simple groups (i.e., $\mathbb{Z}/p\mathbb{Z}$ for prime p), the smallest possible size of a simple group is

$$60 = 2^2 \cdot 3 \cdot 5.$$

We know that such a simple group does exist, and its existence is the reason that polynomial equations of degree ≥ 5 are not solvable by radicals.

This suggests that finite simple groups might have important connections to other parts of mathematics, and indeed they do.

Let me list some of the smallest simple groups. Recall that A_n is the alternating subgroup of S_n defined by

$$1 \longrightarrow A_n \longrightarrow S_n \xrightarrow{\det} \mathbb{Z}_2 \longrightarrow 1,$$

and

$$\mathrm{PSL}_n(q) := \mathrm{SL}_n(\mathbb{F}_q)/Z(\mathrm{SL}_n(\mathbb{F}_q)),$$

where $q = p^k$ with p prime, $\mathbb{F}_q$ is the field of size q , and

$\mathrm{SL}_n(\mathbb{F}_q)$ is the special linear group defined by

$$1 \longrightarrow \mathrm{SL}_n(\mathbb{F}_q) \longrightarrow \mathrm{GL}_n(\mathbb{F}_q) \xrightarrow{\det} \mathbb{F}_q^\times \longrightarrow 1.$$

These definitions give us all of the simple groups up to size 6000:

$ G $	simple group G
60	$A_5 = \mathrm{PSL}_2(4) = \mathrm{PSL}_2(5)$
168	$\mathrm{PSL}_2(7) = \mathrm{PSL}_3(2)$
360	$A_6 = \mathrm{PSL}_2(9)$
504	$\mathrm{PSL}_2(8)$
660	$\mathrm{PSL}_2(11)$
1092	$\mathrm{PSL}_2(13)$
2448	$\mathrm{PSL}_2(17)$
2520	A_7
$\vdots$	$\vdots$

There are other families of finite simple groups coming from the simple Lie algebras. For example, if q is a prime power then there exists a family of simple groups $E_8(q)$ with sizes

$$|E_8(q)| = q^{120} \prod_{i \in I} (q^i - 1), \quad I = \{2, 8, 12, 14, 18, 20, 24, 30\}.$$

And then there are 26 sporadic simple groups that are not part of infinite families. The largest of these is called the monster $\mathbb{M}$. It has size

$$|\mathbb{M}| \approx 8 \cdot 10^{53}.$$

It seems that there is a huge amount of mathematics hidden inside these groups. For example, Borcherds won the Fields medal in 1992 for proving the “monstrous moonshine” conjecture relating $\mathbb{M}$ to the mathematics of string theory and modular forms.

We know that

$$|A_n| = |S_n|/2 = n!/2 = 3 \cdot 4 \cdot 5 \cdots (n-1) \cdot n,$$

and I mentioned previously that

$$|\mathrm{PSL}_n(q)| = \frac{q^{\binom{n}{2}}(q^2 - 1)(q^3 - 1) \cdots (q^n - 1)}{\gcd(n, q - 1)},$$

but I never proved this. Let me prove it now, since it is relevant to HW6 Problem 4.

Theorem.

$$|\mathrm{GL}_n(q)| = q^{\binom{n}{2}}(q - 1)(q^2 - 1) \cdots (q^n - 1).$$

Proof. The columns of an invertible $n \times n$ matrix over $\mathbb{F}_q$ form an ordered basis of the vector space $(\mathbb{F}_q)^n$,

and it is not difficult to count these.

To create an ordered basis $b_1, b_2, \dots, b_n \in \mathbb{F}_q^n$:

- first choose $b_1 \in \mathbb{F}_q^n - \{0\}$ in $q^n - 1$ ways,
- then choose $b_2 \in \mathbb{F}_q^n - \langle b_1 \rangle$ in $q^n - q^1$ ways,
- then choose $b_3 \in \mathbb{F}_q^n - \langle b_1, b_2 \rangle$ in $q^n - q^2$ ways.

Continuing in this way gives

$$\begin{aligned} |\mathrm{GL}_n(q)| &= (q^n - 1)(q^n - q^1)(q^n - q^2) \cdots (q^n - q^{n-1}) \\ &= (q^n - 1)q(q^{n-1} - 1)q^2(q^{n-2} - 1) \cdots q^{n-1}(q^1 - 1) \\ &= q^{1+2+\cdots+(n-1)}(q - 1)(q^2 - 1) \cdots (q^n - 1) \\ &= q^{n(n-1)/2}(q - 1)(q^2 - 1) \cdots (q^n - 1) \\ &= q^{\binom{n}{2}}(q - 1)(q^2 - 1) \cdots (q^n - 1), \end{aligned}$$

as desired. □

[Remark: There is alternative notation for this that emphasizes the analogy between S_n and GL_n . For all $m \in \mathbb{N}$ we define

$$[m]_q := 1 + q + q^2 + \cdots + q^{m-1} = \frac{q^m - 1}{q - 1}.$$

Note that $[m]_q \rightarrow m$ as $q \rightarrow 1$. We call this a “ q -analogue” or a “quantum analogue” of the integer m . Then for $n \in \mathbb{N}$ we define the q -factorial

$$\begin{aligned} [n]_q! &:= [1]_q[2]_q \cdots [n-1]_q[n]_q \\ &= \frac{q - 1}{q - 1} \cdot \frac{q^2 - 1}{q - 1} \cdots \frac{q^{n-1} - 1}{q - 1} \cdot \frac{q^n - 1}{q - 1}. \end{aligned}$$

Now we can easily see that

$$|\mathrm{GL}_n(q)| = q^{\binom{n}{2}}(q - 1)^n [n]_q!.$$

]

Corollary.

$$|\mathrm{SL}_n(q)| = q^{\binom{n}{2}}(q^2 - 1)(q^3 - 1) \cdots (q^n - 1).$$

Proof. Since the sequence

$$1 \longrightarrow \mathrm{SL}_n(q) \longrightarrow \mathrm{GL}_n(q) \xrightarrow{\det} \mathbb{F}_q^\times \longrightarrow 1$$

is exact we have

$$\begin{aligned} |\mathrm{SL}_n(q)| &= |\mathrm{GL}_n(q)| / |\mathbb{F}_q^\times| \\ &= |\mathrm{GL}_n(q)| / (q - 1). \end{aligned}$$

□Next, we need to compute the size of the center $Z(\mathrm{SL}_n(q))$. Here is a general fact.

Theorem. Let K be a field. Then we have

$$\begin{aligned} Z(\mathrm{GL}_n(K)) &= \{\alpha I : \alpha \in K^\times\}, \\ Z(\mathrm{SL}_n(K)) &= \{\alpha I : \alpha \in K^\times, \alpha^n = 1\}. \end{aligned}$$

Proof. Suppose that $A = (a_{ij}) \in Z(\mathrm{GL}_n(K))$. Let $e_{ij}(\alpha)$ be the matrix with (i, j) entry α and zeroes elsewhere, and define the elementary matrix

$$E_{ij}(\alpha) := I + e_{ij}(\alpha).$$

For $i \neq j$ note that $E_{ij}(\alpha)$ is invertible with $\det(E_{ij}(\alpha)) = 1$, hence $E_{ij}(\alpha) \in \mathrm{SL}_n(K)$.

Since A commutes with $E_{ij}(1)$ it must also commute with $e_{ij}(1)$. Then the equation

$$Ae_{ij}(1) = e_{ij}(1)A$$

says

$$a_{ri}\delta_{js} = \delta_{ri}a_{js} \quad \text{for all } r, s,$$

which implies that $a_{ij} = 0$ and $a_{ii} = a_{jj}$ for all $i \neq j$. We conclude that

$$A = \alpha I \quad \text{for some } \alpha \in K^\times.$$

If $A \in \mathrm{SL}_n(K)$ we must also have

$$\alpha^n = \det(\alpha I) = 1.$$

□

Remark: This proof works more generally when K is a commutative ring with 1.

Finally, we need to count the n th roots of unity in the field $\mathbb{F}_q$.

Theorem.

$$|Z(\mathrm{SL}_n(q))| = \gcd(n, q - 1).$$

Proof. By the previous theorem we have

$$|Z(\mathrm{SL}_n(q))| = \#\{\alpha \in \mathbb{F}_q^\times : \alpha^n = 1\}.$$

Recall that the primitive root theorem says that $\mathbb{F}_q^\times = \langle g \rangle$ is cyclic of order $q - 1$.

Thus we want to count integers $x \in \mathbb{Z}$ such that $0 \leq x < q - 1$ and $(g^x)^n = 1$. Note that

$$(g^x)^n = 1 \iff g^{xn} = 1 \iff xn \equiv 0 \pmod{q-1}.$$

To find such x we will solve the linear Diophantine equation

$$xn + y(q - 1) = 0 \tag{*}$$

for $x, y \in \mathbb{Z}$. The equation translates to

$$-\frac{x}{y} = \frac{q-1}{n}.$$

If $d = \gcd(n, q - 1)$ then the most general way to write this fraction is

$$-\frac{x}{y} = \frac{k(q-1)/d}{kn/d},$$

and it follows that the general solution of (*) is

$$(x, y) = \left(\frac{k(q-1)}{d}, -\frac{kn}{d} \right), \quad k \in \mathbb{Z}.$$

After reducing mod $q - 1$ we find that the general solution of $xn \equiv 0 \pmod{q-1}$ is given by

$$x = \frac{k(q-1)}{d} \pmod{q-1},$$

and there are d distinct solutions:

$$x = 0, \frac{q-1}{d}, 2\frac{q-1}{d}, \dots, (d-1)\frac{q-1}{d}.$$

□

26 Finite abelian groups and invariant factors

December 3, 2015

Today: Epilogue Part 2.

Last time I discussed the classification of finite simple groups. To classify all finite groups we would need to describe all the ways to put groups together (the “extension problem”), but this turns out to be extremely complicated.

Example: If $|G| = p^k$ where p is prime then we know that the composition factors of G are just

$$\underbrace{\mathbb{Z}/p, \mathbb{Z}/p, \mathbb{Z}/p, \dots, \mathbb{Z}/p}_{k \text{ times}}.$$

However, there are a ridiculous number of ways to put the composition factors together.

Specifically, Higman proved in 1960 that the number of groups of size p^k is

$$\geq p^{\frac{2}{27}k^2(k-6)}.$$

For example, the number of groups of size $1024 = 2^{10}$ is

$$49,487,365,422.$$

[Remark: In fact, over 99% of the groups of size ≤ 2000 have size 1024.]

No one really expects to “solve” the problem of p -groups, and since every finite group is built out of its Sylow p -subgroups, no one really expects to “solve” the problem of finite groups.

However, the problem of finite abelian groups is much easier. For example, there are only 42 abelian groups of size 1024. We’ll prove this next semester; today I’ll just state the theorem and discuss some applications.

Let G be an abelian group of size

$$|G| = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k},$$

where $p_1, p_2, \dots, p_k$ are distinct primes. For each i let $P_i \subseteq G$ be a subgroup of size $|P_i| = p_i^{\alpha_i}$, i.e., a Sylow p_i -subgroup.

Since G is abelian each $P_i \subseteq G$ is normal. Since the sizes of the P_i are coprime we obtain a direct product decomposition

$$G = P_1 \times P_2 \times \cdots \times P_k. \quad (*)$$

Furthermore, this expression is unique since each normal Sylow subgroup is unique. We call $(*)$ the *primary decomposition* of G .

Thus to understand the structure of finite abelian groups we only need to investigate the structure of abelian p -groups. The answer turns out to be remarkably simple:

Fundamental Theorem of Finite Abelian Groups. Every abelian p -group is a direct product of cyclic groups, hence the same is true of any finite abelian group.

However, the F.T.F.A.G. is not easy to prove with our current technology. [I tried to present the proof from Aluffi pages 236–237 but it was too complicated for my taste and it resulted in a bad lecture ☹.] The correct proof is based on a completely new point of view: thinking of a finite abelian group as a “finite-dimensional vector space over $\mathbb{Z}$ ” and then using the machinery of Gaussian elimination. We will develop this point of view in the spring and it will allow us to see that the F.T.F.A.G. is really the “same thing” as the Jordan Canonical Form of a matrix.

For now let’s see some examples and a fun application.

Example: The abelian groups of size $32 = 2^5$ are

- $\mathbb{Z}/(2^5),$
- $\mathbb{Z}/(2^4) \times \mathbb{Z}/(2^1),$
- $\mathbb{Z}/(2^3) \times \mathbb{Z}/(2^2),$

- $\mathbb{Z}/(2^3) \times \mathbb{Z}/(2^1) \times \mathbb{Z}/(2^1)$,
- $\mathbb{Z}/(2^2) \times \mathbb{Z}/(2^2) \times \mathbb{Z}/(2^1)$,
- $\mathbb{Z}/(2^2) \times \mathbb{Z}/(2^1) \times \mathbb{Z}/(2^1) \times \mathbb{Z}/(2^1)$,
- $\mathbb{Z}/(2^1) \times \mathbb{Z}/(2^1) \times \mathbb{Z}/(2^1) \times \mathbb{Z}/(2^1) \times \mathbb{Z}/(2^1)$.

Note that these are in bijection with partitions of the integer 5:

$$5, \quad 4+1, \quad 3+2, \quad 3+1+1, \quad 2+2+1, \quad 2+1+1+1, \quad 1+1+1+1+1.$$

For any prime p , the abelian groups of size p^k are in bijection with integer partitions of k . Recall that the number of these is given by the Hardy–Ramanujan asymptotic formula

$$\frac{1}{4k\sqrt{3}} \exp\left(\pi\sqrt{\frac{2k}{3}}\right),$$

which is big, but still *much* smaller than Higman’s bound.

Example: Consider the following random abelian group of size $360 = 2^3 \cdot 3^2 \cdot 5^1$:

$$G = \mathbb{Z}/(2) \times \mathbb{Z}/(2) \times \mathbb{Z}/(2) \times \mathbb{Z}/(3) \times \mathbb{Z}/(3) \times \mathbb{Z}/(5).$$

Is there a more efficient way to describe this group? Yes. To see this we first recall the following lemma.

Lemma. Given $a, b \in \mathbb{Z}$ with $\gcd(a, b) = 1$ we have

$$\mathbb{Z}/(a) \times \mathbb{Z}/(b) \cong \mathbb{Z}/(ab).$$

Proof. Let $\mathbb{Z}/(a) = \langle x \rangle$ and $\mathbb{Z}/(b) = \langle y \rangle$. Then the element $(x, y) \in \langle x \rangle \times \langle y \rangle$ has order

$$|\langle (x, y) \rangle| = \text{lcm}(|\langle x \rangle|, |\langle y \rangle|) = \text{lcm}(a, b) = \frac{ab}{\gcd(a, b)} = ab,$$

and hence

$$\mathbb{Z}/(a) \times \mathbb{Z}/(b) = \langle (x, y) \rangle$$

is cyclic. □

There are several ways to apply this to our group G but one is more efficient than all the others:

$$\begin{aligned} G &= (\mathbb{Z}/(2)) \times (\mathbb{Z}/(2) \times \mathbb{Z}/(3)) \times (\mathbb{Z}/(2) \times \mathbb{Z}/(3) \times \mathbb{Z}/(5)) \\ &\cong \mathbb{Z}/(2) \times \mathbb{Z}/(6) \times \mathbb{Z}/(30). \end{aligned}$$

This most efficient encoding is characterized by the fact that

$$2 \mid 6 \mid 30.$$

These three numbers are called the “invariant factors” of G .

In general, given a finite abelian group G there exist unique integers $2 \leq d_1, d_2, \dots, d_m$ (called the *invariant factors* of G) such that

- $d_1 \mid d_2 \mid \cdots \mid d_m$,
- $|G| = d_1 d_2 \cdots d_m$,
- $G \cong \mathbb{Z}/(d_1) \times \mathbb{Z}/(d_2) \times \cdots \times \mathbb{Z}/(d_m)$.

The proof is algorithmic, so let's just do an example.

Example: Find the invariant factors of

$$G = \prod_{i,j} \mathbb{Z}/(p_i^{\alpha_{ij}}),$$

when

$$\{p_i^{\alpha_{ij}}\} = \{2^1, 2^3, 3^1, 3^1, 3^4, 5^1, 5^1, 7^2\}.$$

To do this we arrange the numbers $p_i^{\alpha_{ij}}$ (called the *elementary divisors* of G) in a table so that exponents decrease along columns:

$$\begin{array}{cccc|c} 2^3 & 3^4 & 5^1 & 7^2 & 158760 \\ 2^1 & 3^1 & 5^1 & & 30 \\ & 3^1 & & & 3 \end{array}$$

The invariant factors are given by the products of the rows:

$$3 \mid 30 \mid 158760.$$

The F.T.F.A.G. is quite useful because it gives us shortcuts to lots of theorems about abelian groups. Here is one striking example.

Lemma. Let G be a finite abelian group and assume that for all integers $n > 0$ the number of elements $g \in G$ such that $g^n = 1$ is at most n . Then G is cyclic.

Proof. Suppose G has invariant factors

$$d_1 \mid d_2 \mid \cdots \mid d_m$$

and assume for contradiction that $m \geq 2$.

Then since $|G| = d_1 d_2 \cdots d_m$ and $2 \leq d_1, \dots, d_m$ we have $d_m < |G|$. Furthermore, since

$$G \cong \mathbb{Z}/(d_1) \times \mathbb{Z}/(d_2) \times \cdots \times \mathbb{Z}/(d_m)$$

and $d_1 \mid d_2 \mid \cdots \mid d_m$ we have $g^{d_m} = 1$ for all $g \in G$. We have found an integer $n := d_m$ such that there exist more than n elements $g \in G$ with $g^n = 1$. $\square$

Primitive Root Theorem. Let K be a field. Then any finite subgroup of $K^\times$ is cyclic. In particular, if K is finite then $K^\times$ is cyclic.

Proof. Let n be a positive integer and consider the polynomial $x^n - 1 \in K[x]$. Now recall from your previous life that a polynomial of degree n over a field has at most n roots. This completes the proof. $\square$

That was cute right? I like this proof much better than messing around with the Euler totient function and I think it illustrates the power of the F.T.F.A.G.

Remark: The theory of abelian groups is really very different from the theory of general groups. It is more rightly seen as an aspect of the theory of modules, which will be our main topic in MTH 762.

See you then!

27 Final review

December 8, 2015

Here are the topics discussed since the Midterm.

1. Functors & Natural Transformations

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A covariant functor $F : \mathcal{C} \rightarrow \mathcal{D}$ consists of

- a function

$$F : \text{Obj}(\mathcal{C}) \longrightarrow \text{Obj}(\mathcal{D}),$$

- for each pair $X, Y \in \text{Obj}(\mathcal{C})$ a function

$$F : \text{Hom}_{\mathcal{C}}(X, Y) \longrightarrow \text{Hom}_{\mathcal{D}}(F(X), F(Y)),$$

satisfying the two rules

- for all $X \in \mathcal{C}$,

$$F(\text{id}_X) = \text{id}_{F(X)},$$

- for all $\alpha : X \rightarrow Y$ and $\beta : Y \rightarrow Z$,

$$F(\beta \circ \alpha) = F(\beta) \circ F(\alpha).$$

A contravariant functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is the same as a covariant functor $F : \mathcal{C} \rightarrow \mathcal{D}^{\text{op}}$ or $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$. That is, for all $\alpha : X \rightarrow Y$ and $\beta : Y \rightarrow Z$ we have

$$F(\beta \circ \alpha) = F(\alpha) \circ F(\beta).$$

Now let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be covariant functors. A natural transformation

$$\Phi : F \Longrightarrow G$$

assigns to each object $X \in \mathcal{C}$ a morphism $\Phi(X) : F(X) \rightarrow G(X)$ such that for all objects $X, Y \in \mathcal{C}$ and morphisms $\alpha : X \rightarrow Y$, the following square commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{\Phi(X)} & G(X) \\ F(\alpha) \downarrow & & \downarrow G(\alpha) \\ F(Y) & \xrightarrow{\Phi(Y)} & G(Y) \end{array}$$

We say that Φ is a natural isomorphism, written " $F \cong G$ ", if $\Phi(X)$ is an isomorphism for all $X \in \mathcal{C}$.

2. The Category of G -sets

Let $\mathcal{C}$ and $\mathcal{D}$ be categories with $\mathcal{C}$ small. Then we can define the functor category

$$\mathcal{D}^{\mathcal{C}}$$

whose objects are functors $F : \mathcal{C} \rightarrow \mathcal{D}$ and whose morphisms are natural transformations.

For example, let G be a group thought of as a category with one object. Then a functor $F : G \rightarrow \mathbf{Set}$ is called a " G -set".

A G -set $F : G \rightarrow \mathbf{Set}$ consists of a set

$$X := F(*)$$

and a function

$$F : G = \text{Aut}_G(*) \longrightarrow \text{End}_{\mathbf{Set}}(X).$$

The axioms of a functor imply that this is actually a group homomorphism

$$F : G \longrightarrow \text{Aut}_{\mathbf{Set}}(X).$$

To be more concrete, a G -set consists of a set X and a function

$$G \times X \longrightarrow X, \quad (g, x) \longmapsto g(x),$$

satisfying the two axioms

- for all $x \in X$, $1_G(x) = x$,
- for all $x \in X$ and $g, h \in G$, $g(h(x)) = (gh)(x)$.

This is equivalent to the previous definition via the identification

$$g(x) = F(g)(x).$$

Now let X and Y be two G -sets with the actions $G \times X \rightarrow X$ and $G \times Y \rightarrow Y$ written implicitly. Then a morphism of G -sets is just a function $\Phi : X \rightarrow Y$ such that for all $x \in X$ and $g \in G$ we have

$$\Phi(g(x)) = g(\Phi(x)).$$

Check that this is the same as a natural transformation of functors $G \rightarrow \mathbf{Set}$.

3. Fundamental Theorem of G -sets

Given a G -set X and an element $x \in X$ we define

$$\begin{aligned}\text{Orb}_G(x) &:= \{y \in X : \exists g \in G, y = g(x)\}, \\ \text{Stab}_G(x) &:= \{g \in G : g(x) = x\}.\end{aligned}$$

Theorem (FTGS).

(i) For each $x \in X$, $\text{Stab}_G(x) \subseteq G$ is a subgroup and we have an isomorphism of G -sets

$$\text{Orb}_G(x) \cong_G G/\text{Stab}_G(x), \quad g(x) \longleftrightarrow g \text{Stab}_G(x).$$

(ii) Given two subgroups $H, K \subseteq G$ we have

$$G/H \cong_G G/K$$

if and only if there exists $g \in G$ such that $gHg^{-1} = K$.

Know how to prove the FTGS. Part (i) is straightforward. The key to part (ii) is the identity

$$\text{Stab}_G(g(x)) = g \text{Stab}_G(x)g^{-1}.$$

Two Examples from HW4:

- Let $\text{Gr}_1(r, n)$ be the set of r -element subsets of $\{1, 2, \dots, n\}$. Then we have an isomorphism of S_n -sets

$$\text{Gr}_1(r, n) \cong S_n/(S_r \times S_{n-r}).$$

- Let $\text{Gr}_K(r, n)$ be the set of r -dimensional subspaces of K^n . Then we have an isomorphism of $\text{GL}_n(K)$ -sets

$$\text{Gr}_K(r, n) \cong \frac{\text{GL}_n(K)}{\text{Mat}_{r, n-r}(K) \rtimes (\text{GL}_r(K) \times \text{GL}_{n-r}(K))}.$$

General Example:

Let $H, K \subseteq G$ be subgroups. Then we define an action of $H \times K$ on G by

$$(H \times K) \times G \longrightarrow G, \quad ((h, k), g) \longmapsto h g k^{-1}.$$

The orbits are called double cosets

$$\text{Orb}_{H \times K}(g) := H g K,$$

and we denote the set of orbits by

$$H \backslash G / K := \{H g K : g \in G\}.$$

If H and K are finite, be able to prove that

$$|H g K| = \frac{|H| |K|}{|H \cap g K g^{-1}|}.$$

Hint: Show that

$$H g K \longleftrightarrow \text{Orb}_H(gK) \times K$$

and

$$\text{Stab}_H(gK) = H \cap g K g^{-1}.$$

4. The Class Equation

Let G act on itself by $g(h) := ghg^{-1}$. The orbits are called conjugacy classes

$$K_G(a) := \{b \in G : \exists g \in G, b = gag^{-1}\},$$

and the stabilizers are called centralizers

$$Z_G(a) := \{g \in G : gag^{-1} = a\}.$$

The intersection of all centralizers is called the center of G :

$$Z(G) := \bigcap_{a \in G} Z_G(a).$$

If $K_1, K_2, \dots, K_n$ are the classes of G and $Z_1, Z_2, \dots, Z_n$ are the corresponding centralizers (up to isomorphism), use the FTGS to prove that we have an isomorphism of G -sets

$$G \cong Z(G) \sqcup \left(\bigsqcup_{Z_i \neq G} G/Z_i \right).$$

Hint:

$$K_G(a) = \{a\} \iff a \in Z(G).$$

If G is finite, we obtain

$$|G| = |Z(G)| + \sum_{Z_i \neq G} \frac{|G|}{|Z_i|},$$

which is called the class equation.

5. Application: Sylow Theory

Let $|G| = p^\alpha m$ with p prime and $p \nmid m$.

Theorem (Sylow).

- (i) For all $0 \leq \beta \leq \alpha$, there exists a subgroup $H \subseteq G$ with $|H| = p^\beta$.
- (ii) If $H, K \subseteq G$ are subgroups with $|K| = p^\alpha$ and $|H| = p^\beta$ ($\beta \leq \alpha$), then there exists $g \in G$ such that $gHg^{-1} \subseteq K$.
- (iii) If

$$n_p := \#\{K \subseteq G : |K| = p^\alpha\},$$

then

$$n_p \mid m, \quad n_p \equiv 1 \pmod{p}.$$

You do not need to memorize the proof but you should know what goes into it.

The hardest part is the following lemma.

Lemma. Let A be a finite abelian group and let p be prime. If $p \mid |A|$ then A has an element of order p .

[We proved this with a tricky argument. We'll see the correct proof next semester when we prove the Fundamental Theorem of Finitely Generated $\mathbb{Z}$ -modules.]

The rest of the proof is more straightforward.

Proof of (i): If $p \mid |Z_i|$ for some $Z_i \neq G$, then we're done by induction. Otherwise, the class equation says that $p \mid |Z(G)|$, hence $Z(G)$ has an element $z \in Z(G)$ of order p by the lemma. By induction $G/\langle z \rangle$ has p -subgroups of all orders and we can lift these up to G .

Proof of (ii): Let $H, K \subseteq G$ with $|K| = p^\alpha$ and $|H| = p^\beta$ ($\beta \leq \alpha$). Decompose G into double cosets to get

$$|G| = \sum_i \frac{|H||K|}{|K \cap g_i^{-1}Hg_i|}.$$

Show by contradiction that one of the denominators has size p^β , hence

$$g_i^{-1}Hg_i \subseteq K.$$

Proof of (iii): Let G act on $\text{Syl}_p(G)$ by conjugation and fix $K \in \text{Syl}_p(G)$.

- By (ii) and FTGS we know that

$$|\text{Syl}_p(G)| = |\text{Orb}_G(K)| = |G|/|N_G(K)|.$$

Since $K \subseteq N_G(K)$ we have $p^\alpha \mid |N_G(K)|$ and it follows that $|\text{Syl}_p(G)| \mid m$.

- Now let K act on $\text{Syl}_p(G)$ by conjugation, so

$$|\text{Syl}_p(G)| = \sum_i \frac{|K|}{|\text{Stab}_K(H_i)|}$$

for some $H_i \in \text{Syl}_p(G)$. Show that

$$\text{Stab}_K(H_i) = K \iff H_i = K,$$

and hence

$$|\text{Syl}_p(G)| = 1 + \sum_{H_i \neq K} \frac{|K|}{|\text{Stab}_K(H_i)|} \equiv 1 \pmod{p}.$$

You should know how to apply Sylow to study groups of small order.

Example: Prove that there is no simple group of order 12.

Proof. Let $|G| = 12 = 2^2 \cdot 3$.

Let $n_2 = |\text{Syl}_2(G)|$ and $n_3 = |\text{Syl}_3(G)|$. By Sylow (i) we have $n_3 \geq 1$. If $n_3 = 1$ then by Sylow (ii) we obtain a normal subgroup of size 3. So assume that $n_3 \geq 2$. Then by Sylow (iii) we have

$$n_3 \equiv 1 \pmod{3}, \quad \text{hence } n_3 \geq 4.$$

Since these subgroups intersect trivially (they are cyclic), G must contain at least 8 elements of order 3.

But then there is room for only one Sylow 2-subgroup (of size 4), which must therefore be normal. $\square$

6. Finite Matrix Groups

You should also remember that

$$\begin{aligned} |\mathrm{GL}_n(q)| &= q^{\binom{n}{2}}(q-1)^n[n]_q!, \\ |\mathrm{SL}_n(q)| &= q^{\binom{n}{2}}(q-1)^{n-1}[n]_q!, \\ |\mathrm{PSL}_n(q)| &= \frac{q^{\binom{n}{2}}(q-1)^{n-1}[n]_q!}{\gcd(n, q-1)}. \end{aligned}$$

In case I want to use these as an example somewhere.

Spring 2016: MTH 762

28 Abelian groups, direct sums, rings

January 12, 2016

Last semester in MTH 761 (Grad Algebra I) we discussed

- the category **Set**,
- the category **Grp**,
- the category $\mathbf{Set}^G$, for $G \in \mathbf{Grp}$.

This had to do with the algebra, combinatorics and geometry of symmetry (i.e., groups acting on things).

Recall that a group has only one algebraic operation, which we may refer to as $+$, $\times$, or $\circ$, depending on the situation.

This semester in MTH 762 (Grad Algebra II) we will add another layer of complexity by considering structures with at least *two* algebraic operations.

Let me motivate this by considering a category that we neglected last semester: the category of abelian groups.

The category **Ab**

This semester we will always denote the algebraic operation in an abelian group $G \in \mathbf{Ab}$ by “+” and the identity element by “0”.

We have already seen some hints that the theory of abelian groups is very different from the theory of general groups. In particular, finite abelian groups are much less complicated than general finite groups [although we didn’t prove this yet].

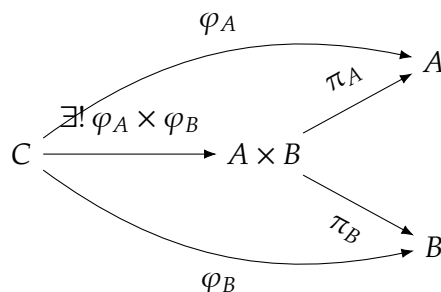
Here are two key differences between the categories **Grp** and **Ab**.

(1) Coproduct

Given $A, B \in \mathbf{Grp}$ recall that the direct product $A \times B$, with operation

$$(a_1, b_1) \cdot (a_2, b_2) = (a_1 a_2, b_1 b_2),$$

satisfies the universal property



That is, given a group $C \in \mathbf{Grp}$ and group homomorphisms $\varphi_A : C \rightarrow A$ and $\varphi_B : C \rightarrow B$, there exists a unique group homomorphism, say

$$\varphi_A \times \varphi_B : C \longrightarrow A \times B,$$

such that the above diagram commutes.

This means that direct product is the categorical product in **Grp**. Now suppose that $A, B \in \mathbf{Ab}$. Since the above property holds for all $(C, \varphi_A, \varphi_B)$ in **Grp**, it certainly holds for all $(C, \varphi_A, \varphi_B)$ in the (full) subcategory $\mathbf{Ab} \subseteq \mathbf{Grp}$.

Remark. “Full” in this case means that

$$\mathrm{Hom}_{\mathbf{Ab}}(A, B) = \mathrm{Hom}_{\mathbf{Grp}}(A, B)$$

for all $A, B \in \mathbf{Ab}$.

So we conclude that direct product is also the categorical product in **Ab**. However, we will use a different notation.

Definition. Given $A, B \in \mathbf{Ab}$ we define the *direct sum* by

$$A \oplus B := \{(a, b) : a \in A, b \in B\},$$

with operation

$$(a_1, b_1) + (a_2, b_2) = (a_1 + a_2, b_1 + b_2).$$

Recall the following characterization of direct sums: Given $A \in \mathbf{Ab}$ and subgroups $B_1, B_2 \subseteq A$ we have

$$A \cong B_1 \oplus B_2$$

if and only if

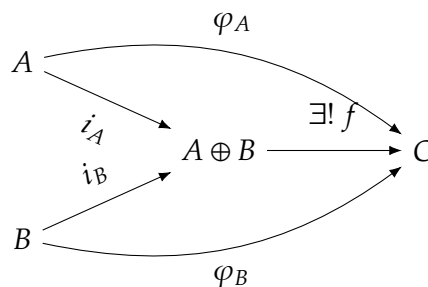
$$B_1 + B_2 = A \quad \text{and} \quad B_1 \cap B_2 = 0.$$

Now recall that the categorical coproduct in $\mathbf{Grp}$ is given by the “free product” $*$, which is big and complicated.

The situation is much nicer in $\mathbf{Ab}$.

Theorem. *The product and coproduct coincide in the category $\mathbf{Ab}$.*

Proof. Given $A, B \in \mathbf{Ab}$ we must show that the direct sum $A \oplus B \in \mathbf{Ab}$ satisfies the universal property of coproducts:



Suppose $f : A \oplus B \rightarrow C$ is some homomorphism making the diagram commute. Then for all $a \in A$ and $b \in B$ it must satisfy

$$\begin{aligned} f(a, b) &= f[(a, 0_B) + (0_A, b)] \\ &= f(a, 0_B) + f(0_A, b) \\ &= f(i_A(a)) + f(i_B(b)) \\ &= \varphi_A(a) + \varphi_B(b). \end{aligned}$$

The only question is whether this function

$$f(a, b) = \varphi_A(a) + \varphi_B(b)$$

is actually a homomorphism. Here is where we need the commutativity of addition: For all (a_1, b_1) and (a_2, b_2) in $A \oplus B$ we have

$$\begin{aligned} f(a_1 + a_2, b_1 + b_2) &= \varphi_A(a_1 + a_2) + \varphi_B(b_1 + b_2) \\ &= \varphi_A(a_1) + \varphi_A(a_2) + \varphi_B(b_1) + \varphi_B(b_2) \\ &= \varphi_A(a_1) + \varphi_B(b_1) + \varphi_A(a_2) + \varphi_B(b_2) \\ &= f(a_1, b_1) + f(a_2, b_2). \end{aligned}$$

This completes the proof. □

We have shown that (finite) products and coproducts coincide in **Ab**. That's nice. 

(2) “Enriched” Structure

Let $A, B \in \mathbf{Ab}$ and consider two homomorphisms $\varphi_1, \varphi_2 : A \rightarrow B$.

We have seen above that commutativity (in B) implies that the function

$$a \mapsto \varphi_1(a) + \varphi_2(a)$$

is actually a group homomorphism $A \rightarrow B$. Let's give it a name.

Definition. Given abelian groups A, B and homomorphisms $\varphi_1, \varphi_2 \in \text{Hom}_{\mathbf{Ab}}(A, B)$ we define the homomorphism

$$\varphi_1 + \varphi_2 \in \text{Hom}_{\mathbf{Ab}}(A, B)$$

by

$$(\varphi_1 + \varphi_2)(a) := \varphi_1(a) + \varphi_2(a).$$

So what? Well, this tells us that the “hom set” $\text{Hom}_{\mathbf{Ab}}(A, B)$ is actually more than a set; it's an abelian group.

*[The identity element is the “zero morphism” $0 : A \rightarrow B$ from **Grp**.]*

Jargon: If the hom sets in a (locally small) category $\mathcal{C}$ are objects in a category $\mathcal{D}$, we say that

“ $\mathcal{C}$ is enriched over $\mathcal{D}$.”

Thus we have shown that

*“**Ab** is enriched over **Ab**.”*

There is a very interesting special case of this. Recall that for any object X in a (locally small) category C , the set of endomorphisms

$$\text{End}_C(X) := \text{Hom}_C(X, X)$$

is a monoid (group without inverses) under composition of arrows. Thus, if A is an abelian group then the set

$$\text{End}_{\text{Ab}}(A)$$

actually has *two* algebraic operations: addition and composition.

You might wonder how these two operations interact.

Theorem. For all $F_1, F_2, F_3 \in \text{End}_{\text{Ab}}(A)$,

$$\begin{aligned} F_1 \circ (F_2 + F_3) &= (F_1 \circ F_2) + (F_1 \circ F_3), \\ (F_1 + F_2) \circ F_3 &= (F_1 \circ F_3) + (F_2 \circ F_3). \end{aligned}$$

Proof. The proofs are similar so we'll just show the first. For all $a \in A$ we have

$$\begin{aligned} [F_1 \circ (F_2 + F_3)](a) &= F_1((F_2 + F_3)(a)) \\ &= F_1(F_2(a) + F_3(a)) \\ &= F_1(F_2(a)) + F_1(F_3(a)) \\ &= (F_1 \circ F_2)(a) + (F_1 \circ F_3)(a) \\ &= [(F_1 \circ F_2) + (F_1 \circ F_3)](a). \end{aligned}$$

□

In other words, we conclude that

$$\text{End}_{\text{Ab}}(A)$$

is a RING.

Just as $\text{Aut}_C(X)$ is the prototypical example of a “group”, $\text{End}_{\text{Ab}}(A)$ is the prototypical example of a “ring”.

We will think of $\text{End}_{\text{Ab}}(A)$ as a “concrete ring” and we will define “abstract rings” by capturing the relevant abstract properties.

Definition. A *ring* is a structure

$$(R, +, \cdot, 0, 1),$$

where R is a set, $+$ and $\cdot$ are functions $R \times R \rightarrow R$, and $0, 1 \in R$ are special elements satisfying three axioms:

- $(R, +, 0)$ is an abelian group,
- $(R, \cdot, 1)$ is a monoid,

- for all $a, b, c \in R$ we have

$$\begin{aligned} a \cdot (b + c) &= (a \cdot b) + (a \cdot c), \\ (a + b) \cdot c &= (a \cdot c) + (b \cdot c). \end{aligned}$$

Stay tuned for why I believe this is the correct way to think about rings.

29 Rings as endomorphism rings

January 14, 2016

Last time I tried to make the case that the “ring” concept emerges naturally from the “abelian group” concept. In particular, let $A \in \mathbf{Ab}$ be an abelian group. Then the set of endomorphisms

$$\text{End}_{\mathbf{Ab}}(A),$$

which is always a monoid under composition, *also* has the structure of an abelian group.

Proof. Given $\varphi_1, \varphi_2 \in \text{End}_{\mathbf{Ab}}(A)$ and $a, b \in A$, the commutativity of A tells us that

$$\begin{aligned} \varphi_1(a + b) + \varphi_2(a + b) &= [\varphi_1(a) + \varphi_1(b)] + [\varphi_2(a) + \varphi_2(b)] \\ &= [\varphi_1(a) + \varphi_2(a)] + [\varphi_1(b) + \varphi_2(b)]. \end{aligned}$$

This says that the function $\varphi_1 + \varphi_2 : A \rightarrow A$ defined by

$$(\varphi_1 + \varphi_2)(a) := \varphi_1(a) + \varphi_2(a)$$

is also an endomorphism. □

Moreover, we showed that the operation of $\circ$ distributes over $+$, so $\text{End}_{\mathbf{Ab}}(A)$ becomes a RING with compositional identity $\text{id}_A : A \rightarrow A$ and additive identity given by the zero morphism $0 : A \rightarrow A$.

Given an abelian group $A \in \mathbf{Ab}$ I will call $\text{End}_{\mathbf{Ab}}(A)$ a “concrete ring”. The definition of “abstract rings” is meant to model this.

Definition. A ring is a structure

$$(R, +, \circ, 0, 1)$$

such that

- $(R, +, 0)$ is an abelian group,
- $(R, \circ, 1)$ is a monoid,
- $\circ$ distributes over $+$ (from the left and the right).

You might wonder how your favorite ring fits into this picture.

Example ($\mathbb{Z}$). First let's think of $(\mathbb{Z}, +, 0)$ as an abelian group. I claim that the “multiplication” operation $\mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ emerges naturally from the abelian group structure. To see this let's consider the ring of endomorphisms

$$R := \text{End}_{\text{Ab}}(\mathbb{Z}).$$

I claim that we have a bijection $R \leftrightarrow \mathbb{Z}$. Indeed, since $\mathbb{Z} = \langle 1 \rangle$ is a cyclic group any group homomorphism $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}$ is determined by the value

$$a := \varphi(1).$$

Then since φ is a homomorphism we have

- $\varphi(0) = 0$;
- for all $n > 0$,

$$\begin{aligned} \varphi(n) &= \varphi(\underbrace{1 + 1 + \cdots + 1}_{n \text{ times}}) \\ &= \underbrace{\varphi(1) + \varphi(1) + \cdots + \varphi(1)}_{n \text{ times}} \\ &= \varphi(1)n = an; \end{aligned}$$

- for all $n < 0$,

$$\varphi(n) = -\varphi(-n) = -a(-n) = an.$$

We conclude that φ is just the function

$$\lambda_a(n) := an.$$

Then the map $\lambda : \mathbb{Z} \rightarrow R$ defined by $a \mapsto \lambda_a$ is a bijection.

So what? Well, it's actually more than a bijection. For all $a, b, n \in \mathbb{Z}$ we have

•

$$\begin{aligned} \lambda_{a+b}(n) &= (a+b)n \\ &= an + bn \\ &= \lambda_a(n) + \lambda_b(n) \\ &= (\lambda_a + \lambda_b)(n); \end{aligned}$$

•

$$\begin{aligned} \lambda_{ab}(n) &= (ab)n \\ &= a(bn) \\ &= \lambda_a(\lambda_b(n)) \\ &= (\lambda_a \circ \lambda_b)(n). \end{aligned}$$

Furthermore we have

$$\lambda_0 = 0_{\mathbb{Z}} : \mathbb{Z} \rightarrow \mathbb{Z}$$

and

$$\lambda_1 = \text{id}_{\mathbb{Z}} : \mathbb{Z} \rightarrow \mathbb{Z}.$$

In summary, we have shown that the function

$$\begin{aligned} \mathbb{Z} &\longrightarrow \text{End}_{\mathbf{Ab}}(\mathbb{Z}), \\ a &\longmapsto (n \longmapsto an) \end{aligned}$$

is an isomorphism of rings,

$$\mathbb{Z} \cong \text{End}_{\mathbf{Ab}}(\mathbb{Z}).$$

This suggests that the additive structure of $\mathbb{Z}$ is primary and that the multiplicative structure emerges as composition of additive homomorphisms.

So there is really no such thing as “multiplication”.

More generally, if R is any abstract ring, one can show that the function

$$\lambda : R \longrightarrow \text{End}_{\mathbf{Ab}}(R)$$

defined by

$$\lambda_a(b) := ab \quad \text{for all } a, b \in R$$

is an injective ring homomorphism.

This establishes the fact that the concept of “abstract ring” is not more general than the concept of “concrete ring”.

The importance of this idea will become clear soon when we discuss “modules”. But first let’s discuss abstract rings for their own sake [just as we discussed abstract groups before we discussed G -sets].

The category $\mathbf{Rng}$

Let R and S be abstract rings.

Definition. We say that $\varphi : R \rightarrow S$ is a *ring homomorphism* if

- φ is a homomorphism of abelian groups, i.e.,

$$\text{for all } a, b \in R, \quad \varphi(a + b) = \varphi(a) + \varphi(b);$$

- φ is a homomorphism of monoids, i.e.,

$$\text{for all } a, b \in R, \quad \varphi(a \circ b) = \varphi(a) \circ \varphi(b),$$

and

$$\varphi(1_R) = 1_S.$$

[Note that the condition $\varphi(a \circ b) = \varphi(a) \circ \varphi(b)$ does not imply $\varphi(1_R) = 1_S$, because of the lack of inverses in a monoid.]

One can check that the collection of rings and homomorphisms form a category, which we call **Rng**.

The basic properties of **Rng** are similar to **Grp**, so we'll go over them fairly quickly. The analogous concept to "normal subgroup" is the concept of "ideal".

Definition. Given a ring homomorphism $\varphi : R \rightarrow S$ we define the *image*

$$\text{im } \varphi := \{s \in S : \text{there exists } r \in R, \varphi(r) = s\}$$

and the *kernel*

$$\ker \varphi := \{r \in R : \varphi(r) = 0_S\}.$$

Note that setwise these are just the image and kernel of φ as a homomorphism of abelian groups. The question is: how do they interact with the monoid structure?

Claim. $\text{im } \varphi \subseteq S$ is a subring.

Indeed, we already know that it's a subgroup of $(S, +, 0_S)$. To see that it's a subring we note that

- $1_S = \varphi(1_R) \in \text{im } \varphi$, by definition;
- for $s_1 = \varphi(r_1)$ and $s_2 = \varphi(r_2)$ in $\text{im } \varphi$ we have

$$s_1 \circ s_2 = \varphi(r_1) \circ \varphi(r_2) = \varphi(r_1 \circ r_2) \in \text{im } \varphi.$$

Furthermore, any subring $S' \subseteq S$ is the image of the natural inclusion homomorphism $i : S' \hookrightarrow S$. We can summarize this with the following slogan:

"image $\iff$ subring".

The kernel is also a subgroup of R , but it is not in general a subring because it probably doesn't contain 1_R .

Instead, the kernel satisfies the following defining(?) property:

$$\text{for all } a, r \in R \text{ with } a \in \ker \varphi \text{ we have} \quad r \circ a \in \ker \varphi \quad \text{and} \quad a \circ r \in \ker \varphi.$$

Proof. If $\varphi(a) = 0_S$ then we have

$$\begin{aligned}\varphi(r \circ a) &= \varphi(r) \circ \varphi(a) = \varphi(r) \circ 0_S = 0_S, \\ \varphi(a \circ r) &= \varphi(a) \circ \varphi(r) = 0_S \circ \varphi(r) = 0_S.\end{aligned}$$

□

The fact that $s \circ 0_S = 0_S \circ s = 0_S$ for all $s \in S$ is an easy consequence of the ring axioms.

Remark. Suppose that $1_R \in \ker \varphi$. Then

- for all $r \in R$ we have

$$r = r \circ 1_R \in \ker \varphi,$$

and hence $\ker \varphi = R$;

- since

$$1_S = \varphi(1_R) = 0_S,$$

we conclude that S is the *zero ring* $S = \{0_S\}$.

In fact, the zero ring $0 \in \mathbf{Rng}$ is the final object in $\mathbf{Rng}$. The unique maps $R \rightarrow 0$ are the only ring maps whose kernels are subrings.

We will try to capture the notion of kernels with the following definition.

Definition. Let R be an abstract ring and let $I \subseteq R$ be a subgroup of $(R, +, 0)$.

- (i) We say I is a *left ideal* if

$$a \in I \text{ and } r \in R \implies r \circ a \in I.$$

- (ii) We say I is a *right ideal* if

$$a \in I \text{ and } r \in R \implies a \circ r \in I.$$

- (iii) We say that I is a *two-sided ideal* (or just an *ideal*) if it is both left and right.

Now we hope that the following slogan holds:

$$\text{"kernel} \iff \text{ideal"}.$$

Does it?

30 Ideals, number fields, Riemann surfaces

January 19, 2016

Pause: I should note that the definition/motivation for rings I'm giving here is far from the original version.

The notion of a ring was introduced by Richard Dedekind in 1871 in order to generalize the properties of $\mathbb{Z}$ to so-called “integral extensions” $\mathbb{Z}[\alpha]$, $\alpha \in \mathbb{C}$. Dedekind used the term “Ordnung”; our term “ring” comes from David Hilbert’s “Zahlring” (1897).

Now let $\omega_n := e^{2\pi i/n}$. In 1847 Gabriel Lamé observed that if the ring

$$\mathbb{Z}[\omega_n] := \{a_0 + a_1\omega_n + \cdots + a_{n-1}\omega_n^{n-1} : a_i \in \mathbb{Z}\}$$

has the property of unique prime factorization (UF) for all $n \in \mathbb{Z}$ then Fermat’s Last Theorem is true, i.e., if $x, y, z, n \in \mathbb{Z}$ with $n \geq 3$ and $xyz \neq 0$ then

$$x^n + y^n \neq z^n.$$

Very soon, though, Ernst Kummer showed that $\mathbb{Z}[\omega_n]$ usually does *not* have UF.

To recover UF, Dedekind and Kummer both developed a theory of “ideal numbers” [Dedekind shortened the term to “ideals”].

To be specific, here is Dedekind’s main result.

Dedekind’s Theorem (1871). Let $\mathbb{Q} \subseteq K$ be a finite degree field extension and define the corresponding “ring of integers” in K ,

$$\mathcal{O}_K := \{\alpha \in K : \alpha \text{ is the root of a monic polynomial in } \mathbb{Z}[x]\}.$$

Picture:

$$\begin{array}{ccc} \mathcal{O}_K & \hookrightarrow & K \\ \uparrow & & \uparrow \\ \mathbb{Z} & \hookrightarrow & \mathbb{Q} \end{array}$$

Then every ideal in $\mathcal{O}_K$ can be expressed uniquely as a “product of prime ideals” (whatever that means).

Example. Consider the field $\mathbb{Q} \subseteq K$, where

$$K = \mathbb{Q}(\sqrt{-5}) := \{\alpha + \beta\sqrt{-5} : \alpha, \beta \in \mathbb{Q}\}.$$

In this case the ring of integers is

$$\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}] = \{\alpha + \beta\sqrt{-5} : \alpha, \beta \in \mathbb{Z}\}.$$

This ring does not have unique factorization of elements. Indeed, we have

$$2 \cdot 3 = 6 = (1 + \sqrt{-5})(1 - \sqrt{-5}),$$

where $2, 3, (1 + \sqrt{-5})$, and $(1 - \sqrt{-5})$ are all prime.

What's the problem? Well, recall that if $a, b \in \mathbb{Z}$ then we have

$$\begin{aligned} a\mathbb{Z} + b\mathbb{Z} &:= \{ax + by : x, y \in \mathbb{Z}\} \\ &= d\mathbb{Z}, \end{aligned}$$

where $d \in \mathbb{Z}$ is the greatest common divisor of a and b .

Unfortunately, there does *not* exist an element $d \in \mathbb{Z}[\sqrt{-5}]$ such that

$$\begin{aligned} &2 \cdot \mathbb{Z}[\sqrt{-5}] + (1 + \sqrt{-5}) \cdot \mathbb{Z}[\sqrt{-5}] \\ &:= \{2x + (1 + \sqrt{-5})y : x, y \in \mathbb{Z}[\sqrt{-5}]\} \\ &= d \cdot \mathbb{Z}[\sqrt{-5}]. \end{aligned}$$

Dedekind's solution is to regard the whole set as an "ideal number". Let's call it

$$A := (2, 1 + \sqrt{-5}) := 2 \cdot \mathbb{Z}[\sqrt{-5}] + (1 + \sqrt{-5}) \cdot \mathbb{Z}[\sqrt{-5}].$$

Similarly we will define the ideal numbers

$$\begin{aligned} A^* &:= (2, 1 - \sqrt{-5}), \\ B &:= (3, 1 + \sqrt{-5}), \\ B^* &:= (3, 1 - \sqrt{-5}). \end{aligned}$$

One can check that as sets we have

$$\begin{aligned} AA^* &= (2) := 2 \cdot \mathbb{Z}[\sqrt{-5}], \\ BB^* &= (3), \\ AB &= (1 + \sqrt{-5}), \\ A^*B^* &= (1 - \sqrt{-5}). \end{aligned}$$

One can also check that each of the ideals A, A^*, B, B^* is prime. Thus we have recovered "unique prime factorization":

$$\begin{aligned} (6) &= (2)(3) = (AA^*)(BB^*) \\ &= (1 + \sqrt{-5})(1 - \sqrt{-5}) = (AB)(A^*B^*). \end{aligned}$$

In general, the ring of integers for $K := \mathbb{Q}(\sqrt{d})$ when $d \in \mathbb{Z}$ is

$$O_K = \begin{cases} \mathbb{Z}[\sqrt{d}], & \text{if } d \equiv 2 \text{ or } 3 \pmod{4}, \\ \mathbb{Z}\left[\frac{1 + \sqrt{d}}{2}\right], & \text{if } d \equiv 1 \pmod{4}. \end{cases}$$

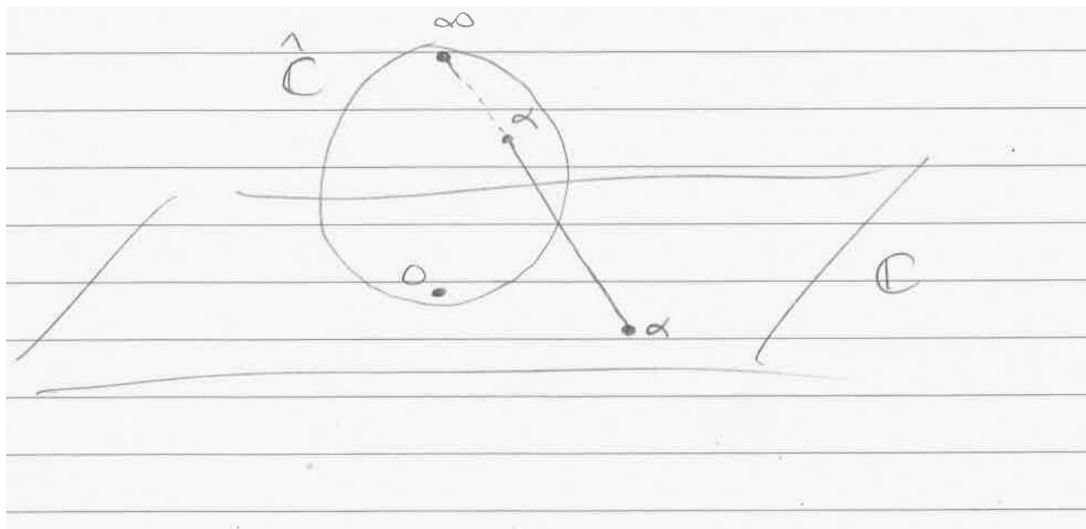
Dedekind's Theorem was the beginning of ring theory. Unfortunately, it didn't lead very quickly to a proof of Fermat's Last Theorem.

The story is not complete, however, without mentioning Dedekind's other amazing work (with Heinrich Weber) extending his technology of rings and fields to the theory of "Riemann surfaces".

Let me temporarily move to the category of "complex manifolds", without defining these. The arrows in this category are "holomorphic maps".

Let $\mathbb{C}$ be the "complex line" and let $\hat{\mathbb{C}}$ be its one point compactification, called the "Riemann sphere". The one new point in $\hat{\mathbb{C}}$ is called the "point at infinity".

Picture:



You may know the following facts from your complex analysis course:

- $\text{Hom}(\mathbb{C}, \mathbb{C}) = \text{holomorphic functions};$
- $\text{Hom}(\mathbb{C}, \hat{\mathbb{C}}) = \text{meromorphic functions};$
- $\text{Hom}(\hat{\mathbb{C}}, \mathbb{C}) = \text{constant functions ("Liouville's Theorem")};$
- $\text{Hom}(\hat{\mathbb{C}}, \hat{\mathbb{C}}) = \text{rational functions}.$

Clearly the last set of functions is the nicest to work with. Let's denote this ring of functions (in fact, a field) by

$$\text{End}(\hat{\mathbb{C}}) = \mathbb{C}(z) = \left\{ \frac{f(z)}{g(z)} : f(z), g(z) \in \mathbb{C}[z], g(z) \neq 0 \right\}.$$

The subring of "polynomials"

$$\mathbb{C}[z] \subseteq \mathbb{C}(z)$$

are the rational functions that have no poles except at ∞ .

Now, consider any polynomial in two complex variables $f(x, y) \in \mathbb{C}[x, y]$. We say that the equation

$$f(x, y) = 0$$

implicitly defines y as an “algebraic function” of x . But we know that this is not really a function.

Example. Let $f(x, y) = y^2 - x$. Then the solution to $f(x, y) = 0$ is

$$y = “\sqrt{x}” ,$$

which is not a real (i.e., “single-valued”) function of x .

Riemann’s great insight in the 1850’s was that the equation $f(x, y) = 0$ does define y as a single-valued function by “analytic continuation” to some new domain, called a “Riemann surface”.

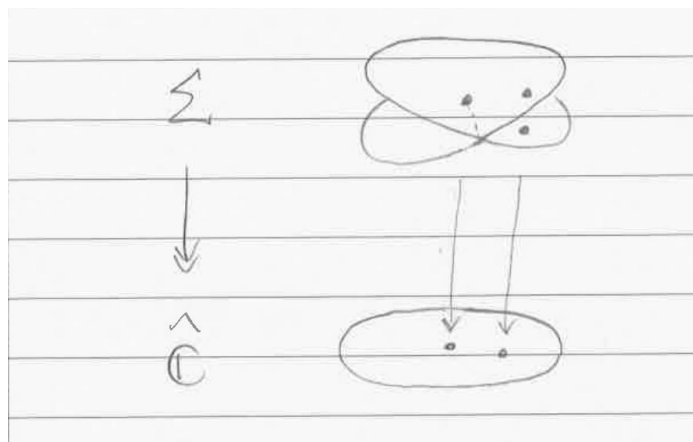
Topologically, a Riemann surface Σ is a finite branched covering of the Riemann sphere $\widehat{\mathbb{C}}$.

Example. The Riemann surface of

$$f(x, y) = y^2 - x$$

is generally a 2:1 covering space, but it is 1:1 (branched) over the points $x = 0$ and $x = \infty$. [The numbers 0 and ∞ each have a unique square root, but every other number has two.]

Near each branch the covering looks like:



Note that the self-intersection is not really there; it’s an artifact of embedding the picture in $\mathbb{R}^3$.

The great insight of Dedekind and Weber (1888) was to realize that the field of meromorphic functions

$$K := \text{Hom}(\Sigma, \widehat{\mathbb{C}})$$

is a finite degree extension of the field

$$\mathbb{C}(z) = \text{Hom}(\widehat{\mathbb{C}}, \widehat{\mathbb{C}}).$$

Picture:

$$\begin{array}{ccc}
 \mathcal{O}_K & \hookrightarrow & K \\
 \uparrow & & \uparrow \\
 \mathbb{C}[z] & \hookrightarrow & \mathbb{C}(z)
 \end{array}
 \qquad
 \begin{array}{c}
 \Sigma \\
 \downarrow \\
 \widehat{\mathbb{C}}
 \end{array}$$

Now Dedekind realized that his technology of rings and ideals allowed him to provide the first rigorous proofs for some of Riemann's theorems (which Riemann had established by appeal to physical intuition from electrostatics).

It's okay if you didn't understand any of that. The point I wanted to make is that ring theory was born from a synthesis of number theory and complex analysis. From this point of view the two most important commutative rings are

$$\mathbb{Z} \quad \text{and} \quad \mathbb{C}[z].$$

The notions of abstract rings and ideals were brought to maturity by Emmy Noether in the 1920s.

However, I always felt that the ring concept is fairly mysterious, as evidenced by the fact that there are still competing definitions of the word "ring" (i.e., with or without "1").

There is also the problem of how noncommutative rings such as $\text{Mat}_n(K)$ should fit into the subject. In my opinion the more recent categorical point of view significantly clarifies the situation:

- The definition of "ring" models the endomorphisms of an abelian group.
- Any ring R (whether commutative or not) should be studied through its category of modules.

On Thursday, I'll return to the careful, systematic (and ahistorical) development of ring theory.

31 Ideals, quotient rings, isomorphism theorems

January 21, 2016

Last time I discussed the origin of ring theory from Dedekind's work on unique factorization and Riemann surfaces; today, back to the dry stuff.

Recall: A ring $(R, +, \circ, 0, 1)$ is

- an abelian group $(R, +, 0)$;
- a monoid $(R, \circ, 1)$;

- distributive laws

$$a \circ (b + c) = (a \circ b) + (a \circ c), \quad (a + b) \circ c = (a \circ c) + (b \circ c).$$

A ring homomorphism $\varphi : R \rightarrow S$ satisfies

- $\varphi(a + b) = \varphi(a) + \varphi(b)$;
- $\varphi(a \circ b) = \varphi(a) \circ \varphi(b)$;
- $\varphi(1_R) = 1_S$.

We define $\ker \varphi$ and $\operatorname{im} \varphi$ as the kernel and image of φ as a homomorphism of abelian groups.

Then we may study how $\ker \varphi$ and $\operatorname{im} \varphi$ interact with the monoid structure. We defined the notion of subring so that

$$\text{"image"} \iff \text{"subring"},$$

but then we found that $\ker \varphi \subseteq R$ is never a subring unless $S = 0$ (the “zero ring”) and $\varphi = 0$ (the “zero morphism”).

To try to capture the “internal properties” of kernels we made the following definition.

Definition. Let R be a ring and let $I \subseteq R$ be an abelian subgroup. We say that I is a (two-sided) *ideal* if for all $a, b \in R$ we have

$$a \in I \text{ or } b \in I \implies a \circ b \in I.$$

Remark. If the converse also holds then we say that I is a *prime ideal*, but this notion is usually only applied in commutative rings.

Claim. This definition works; i.e., we have

$$\text{"kernel"} \iff \text{"ideal"}.$$

Proof. We already proved $\implies$.

So let $I \subseteq R$ be an ideal. We want to construct a ring R' and a ring homomorphism $\varphi : R \rightarrow R'$ such that $\ker \varphi = I$.

We don't need to reinvent the wheel. Since $I \subseteq R$ is a subgroup of an abelian group (hence normal), we obtain a quotient group homomorphism

$$\pi : R \longrightarrow R/I$$

with $\ker \pi = I$. The problem now is to define a monoid structure on R/I so that π becomes a ring homomorphism. And we have no choice: if such a structure exists then it must satisfy

$$\pi(a) \circ \pi(b) = \pi(a \circ b),$$

that is,

$$(a + I) \circ (b + I) = (a \circ b) + I.$$

Such an operation will naturally inherit the monoid and distributive properties from R . The only thing left to check is that the proposed operation is well-defined.

To check this, suppose that $a + I = a' + I$ (i.e., $a - a' \in I$) and $b + I = b' + I$ (i.e., $b - b' \in I$). We want to show that

$$(a \circ b) + I = (a' \circ b') + I.$$

And indeed, we have

$$\begin{aligned} (a \circ b) - (a' \circ b') &= (a \circ b) - (a \circ b') + (a \circ b') - (a' \circ b') \\ &= a \circ (b - b') + (a - a') \circ b'. \end{aligned}$$

Since I is an ideal with $a - a' \in I$ and $b - b' \in I$, this last expression is in I , as desired. $\square$

We can summarize this construction with the following universal property.

Theorem (Universal Property of Ring Quotients). *Let $\varphi : R \rightarrow S$ be a ring homomorphism and let $I \subseteq R$ be an ideal such that $I \subseteq \ker \varphi$. Then there exists a unique ring homomorphism $\bar{\varphi} : R/I \rightarrow S$ such that*

$$\begin{array}{ccc} & R & \\ \pi \swarrow & & \searrow \varphi \\ R/I & \xrightarrow{\quad \bar{\varphi} \quad} & S \end{array}$$

$\exists! \bar{\varphi}$

And this immediately gives us a “First Isomorphism Theorem” for rings. Let’s remind ourselves how it goes.

Theorem (First Isomorphism Theorem for Rings). *Let $\varphi : R \rightarrow S$ be any ring homomorphism. Then the induced map $\bar{\varphi} : R/\ker \varphi \rightarrow S$ restricts to a ring isomorphism*

$$\bar{\varphi} : R/\ker \varphi \xrightarrow{\sim} \text{im } \varphi, \quad a + \ker \varphi \mapsto \varphi(a).$$

Proof. What needs to be checked?

The map is clearly surjective, so we only need to check injectivity. Indeed, we have

$$\begin{aligned} \varphi(a) = \varphi(b) &\implies \varphi(a) - \varphi(b) = 0 \\ &\implies \varphi(a - b) = 0 \\ &\implies a - b \in \ker \varphi \\ &\implies a + \ker \varphi = b + \ker \varphi. \end{aligned}$$

Alternatively, we can show that the map

$$\varphi : R \twoheadrightarrow \text{im } \varphi$$

satisfies the same universal property as the quotient $\pi : R \twoheadrightarrow R/\ker \varphi$. Indeed, given any ring homomorphism $\mu : R \rightarrow S$ such that $\ker \varphi \subseteq \ker \mu$, we will show that there exists a unique $\bar{\mu} : \text{im } \varphi \rightarrow S$ such that

$$\begin{array}{ccc} & R & \\ \varphi \swarrow & & \searrow \mu \\ \text{im } \varphi & \xrightarrow{\exists! \bar{\mu}} & S \end{array}$$

By commutativity, the map $\bar{\mu}$ must satisfy

$$\bar{\mu}(\varphi(a)) = \mu(a),$$

and such a map does exist because

$$\begin{aligned} \varphi(a) = \varphi(b) &\implies \varphi(a) - \varphi(b) = 0 \\ &\implies \varphi(a - b) = 0 \\ &\implies a - b \in \ker \varphi \\ &\implies a - b \in \ker \mu \\ &\implies \mu(a - b) = 0 \\ &\implies \mu(a) - \mu(b) = 0 \\ &\implies \mu(a) = \mu(b). \end{aligned}$$

□

Are there also ring analogues of the Second and Third Isomorphism Theorems?

Well, now we have a problem. When G is a group we let $\mathcal{L}(G)$ be the lattice of subgroups and we think of normal subgroups as (“modular”) elements of this lattice.

Now let R be a ring. If we let $\mathcal{L}(R)$ be the lattice of subrings, then the ideals of R will be invisible, and we don’t want that! So the situation is necessarily more complicated.

Theorem (Third Isomorphism Theorem). *Let R be a ring with ideal I .*

- *If $S \subseteq R$ is a subring containing I , then S/I is a subring of R/I , and this establishes a bijection*

$$\{\text{subrings of } R \text{ containing } I\} \longleftrightarrow \{\text{subrings of } R/I\}.$$

- If $J \subseteq R$ is an ideal containing I , then J/I (defined as an abelian group) is an ideal of R/I , and this establishes a bijection

$$\{\text{ideals of } R \text{ containing } I\} \longleftrightarrow \{\text{ideals of } R/I\}.$$

Furthermore, for each such J we have an isomorphism of rings

$$\frac{R/I}{J/I} \cong \frac{R}{J}.$$

Proof. “Exercise”. ◡ ◡

□

Theorem (Second Isomorphism Theorem). Let R be a ring with ideal $I \subseteq R$.

- If $S \subseteq R$ is a subring, then

$$I + S := \{a + b : a \in I, b \in S\}$$

is a subring of R , $I \cap S$ is an ideal of S , and we have an isomorphism of rings

$$\frac{I + S}{I} \cong \frac{S}{I \cap S}.$$

- If $J \subseteq R$ is an ideal, then

$$I + J := \{a + b : a \in I, b \in J\}$$

is an ideal of R , and $I \cap J$ is an ideal of R .

Proof. “Exercise”. ◡ ◡

□

So we seem to have two Second and two Third “Isomorphism Theorems for Rings”, and in each case only one of them involves an actual isomorphism of rings.

Let’s examine the missing isomorphisms:

- Given an ideal $I \subseteq R$ and a subring $S \subseteq R$ containing I , we have a perfectly good isomorphism of abelian groups

$$\frac{R/I}{S/I} \cong \frac{R}{S},$$

but (as we know) the objects on the left and right are not rings in any reasonable way.

- Given any two ideals $I, J \subseteq R$, we have an isomorphism of abelian groups

$$\frac{I + J}{I} \cong \frac{J}{I \cap J},$$

but, again, neither of these groups has a natural ring structure.

Question. Do the group isomorphisms

$$\frac{R/I}{S/I} \cong \frac{R}{S} \quad \text{and} \quad \frac{I + J}{I} \cong \frac{J}{I \cap J}$$

carry some extra structure related to the ring structure of R ?

The answer is yes, and we will discuss this next time.

32 From ring isomorphism theorems to modules

January 26, 2016

Last time we discussed the First, Second, and Third Isomorphism Theorems for rings.

Theorem (First Isomorphism Theorem). *Let $\varphi : R \rightarrow R'$ be any ring homomorphism. Then we obtain a canonical factorization of φ :*

$$\begin{array}{ccccccc}
 & & \varphi & & & & \\
 & \nearrow & & \searrow & & & \\
 R & \xrightarrow{\pi} & R/\ker \varphi & \xrightarrow[\bar{\varphi}]{\sim} & \operatorname{im} \varphi & \xrightarrow{i} & R' \\
 & & & & & &
 \end{array}$$

In particular, we have an isomorphism of rings

$$R/\ker \varphi \cong \operatorname{im} \varphi.$$

Theorem (Second Isomorphism Theorem). *Let R be a ring with ideal $I \subseteq R$ and subring $S \subseteq R$. Then we have an isomorphism of rings*

$$\frac{I+S}{I} \cong \frac{S}{I \cap S}.$$

Theorem (Third Isomorphism Theorem). *Let R be a ring with ideals $I, J \subseteq R$ such that $I \subseteq J$. Then we have an isomorphism of rings*

$$\frac{R/I}{J/I} \cong \frac{R}{J}.$$

Of course, if $I, J \subseteq R$ are any ideals and $S \subseteq R$ is any subring containing I , then we also have the following isomorphisms of abelian groups:

$$\frac{I+J}{I} \cong \frac{J}{I \cap J} \quad \text{and} \quad \frac{R/I}{S/I} \cong \frac{R}{S}.$$

Unfortunately, these groups do not have any natural ring structure.

Question. Do they have any kind of structure related to the monoid structure of R ?

Answer. Yes.

Recall that for any group G we defined a G -set as a functor

$$F : G \longrightarrow \mathbf{Set}$$

from G (thought of as a category with one object) into the category of sets. We found that G -sets form a category with morphisms given by natural transformations.

Now observe that this construction still works if we replace the group G by a monoid R . [After all, a monoid is nothing but a category with one object.] Thus we define an R -set as a functor

$$F : R \longrightarrow \mathbf{Set}.$$

As before, this is equivalent to choosing a set X and a monoid homomorphism

$$\varphi : R \longrightarrow \text{End}_{\mathbf{Set}}(X).$$

We can also write this concretely by using the notation

$$r(x) := \varphi_r(x) \quad \text{for all } r \in R, x \in X.$$

Then we have the following properties:

- $1_R(x) = x$ for all $x \in X$;
- $r(s(x)) = (rs)(x)$ for all $r, s \in R$ and $x \in X$,

which we could also take as axioms for the concept of a (left) monoid action $R \curvearrowright X$.

The point is this. Consider a ring $(R, +, \circ, 0_R, 1_R)$. Then

- any ideal $I \subseteq R$ carries a natural (left or right) action of the monoid $(R, \circ, 1_R)$;
- any subring $S \subseteq R$ carries a natural (left or right) action of the monoid $(S, \circ, 1_R)$.

Thus we might want to interpret

$$\frac{I+J}{I} \cong \frac{J}{I \cap J} \quad \text{and} \quad \frac{R/I}{S/I} \cong \frac{R}{S}$$

as isomorphisms of R -sets and S -sets, respectively. There is only one problem.

Problem. Is it possible to form a quotient of monoid-sets?

Answer. No, not in general.

However, in our case we don't just have monoid-sets; we have monoid-sets and abelian groups coexisting peacefully.

Hopefully I have motivated the following definition. [If not, then I hope to motivate it by the end of the semester.]

Definition. Let R be a ring. Then a (left) R -module consists of

- an abelian group $(M, +, 0_M)$;
- a (left) monoid action $R \curvearrowright M$;
- a requirement that the action "respects" addition in both R and M .

That is, for all $r, s \in R$ and $m, n \in M$ we require that

$$r(m + n) = r(m) + r(n), \quad (r + s)(m) = r(m) + s(m).$$

[Note that we abuse notation by using “+” for the abelian group operation in both R and M . This is traditional; we will try not to get confused.]

Is it possible to say this definition with fewer words? Certainly. Recall that we already have a natural ring structure on the endomorphisms $\text{End}_{\mathbf{Ab}}(M)$.

Definition (Shorter Definition). Let R be a ring. Then a left R -module consists of an abelian group M and a homomorphism of rings

$$\varphi : R \longrightarrow \text{End}_{\mathbf{Ab}}(M).$$

The homomorphism φ encodes the left monoid action $R \curvearrowright M$ and the fact that it preserves addition. [Sometimes we say that the action is “linear”.]

OK, but is it possible to make the definition even shorter?

Yes, but it will involve me using the term “abelian category” without defining it. [Basically, an abelian category has hom sets that are abelian groups and well-behaved products, coproducts, kernels, cokernels, etc. It is meant to abstract the properties of $\mathbf{Ab}$ (hence the name). Abelian categories are becoming a standard notion in both commutative and noncommutative algebra, so you should be aware of their existence.]

The main point for us is that an abelian category with one object is just a ring.

Definition (Shortest Definition). Let R be a ring, thought of as an abelian category with one object. Then a left R -module is just a functor of abelian categories

$$F : R \longrightarrow \mathbf{Ab}.$$

Oh God, what have we done?

The concept of modules is hard to motivate but we will see in time that it solves many problems.

At the moment, it solves the problem of the isomorphisms

$$\frac{I + J}{I} \cong \frac{J}{I \cap J} \quad \text{and} \quad \frac{R/I}{S/I} \cong \frac{R}{S}.$$

That is: The left is an isomorphism of R -modules and the right is an isomorphism of S -modules, each in a natural way.

To show this we need only observe that a quotient of R -modules is again an R -module.

Given a left R -module M and a submodule $N \subseteq M$ [i.e., N is simultaneously a subgroup and a sub- R -set], consider the abelian group M/N . One can check that the prescription

$$r(m + N) = r(m) + N \quad \text{for all } r \in R, m \in M$$

defines a left R -module structure on M/N and that this structure satisfies the following universal property.

Theorem (Universal Property of R -module Quotients). *Let R be a ring and let $N \subseteq M$ be a left R -submodule. Then for any R -module homomorphism $\varphi : M \rightarrow P$ satisfying $N \subseteq \ker \varphi$, there exists a unique R -module homomorphism $\bar{\varphi} : M/N \rightarrow P$ such that*

$$\begin{array}{ccc} & M & \\ \pi \swarrow & & \searrow \varphi \\ M/N & \xrightarrow{\quad \exists! \bar{\varphi} \quad} & P \end{array}$$

Proof. Everything already works in the category of abelian groups. Just check that the extra R -action doesn't spoil things. $\square$

As is usually the case with abstract algebra, the proof is an afterthought. The real work is in finding the *correct definition*.

33 Motivating modules and the category $R\text{-Mod}$

January 28, 2016

Last time I gave the definition of an “ R -module”. This is a unifying algebraic concept that is above the groups–rings–fields level of generality.

Consequently, it is a difficult concept to motivate. Last time I motivated it through the search for “2nd” and “3rd” Isomorphism Theorems for Rings.

I'll try to motivate it a bit more before proceeding with the basic theory.

Recall that we had two definitions for the “group” concept:

- G1.** A collection of symmetries of an object.
- G2.** A “set with structure” satisfying certain axioms.

The axioms of G2 are intended to model the properties of G1. We then check that our abstract definition is not “too general” as follows.

Theorem (“Cayley’s Theorem”). *Given an abstract group G we have an injective group homomorphism*

$$\varphi : G \hookrightarrow \text{Aut}_{\text{Set}}(G)$$

defined by

$$\varphi_g(h) := g \circ h.$$

Thus we can realize G as a concrete group of symmetries of some “thing”; namely, itself.

The study of all homomorphisms

$$\varphi : G \longrightarrow \text{Aut}_C(X)$$

is called the *representation theory* of G . For any category C we have a category

$$C^G$$

of “representations of G in C ”.

Similarly, we have two definitions for “ring”:

R1. A collection of endomorphisms of an abelian group.

R2. A “set with structure” satisfying certain axioms intended to model R1.

To check that R2 is not “too general” we have another “Cayley-Type Theorem”:

Theorem (“Cayley-Type Theorem”). *Given an abstract ring R , we have an injective ring homomorphism*

$$\lambda : R \hookrightarrow \text{End}_{\text{Ab}}(R)$$

defined by

$$\lambda_a(b) := a \circ b.$$

Thus we can realize R as a concrete ring of endomorphisms of some abelian group; namely, itself.

The study of all homomorphisms

$$\varphi : R \longrightarrow \text{End}_{\text{Ab}}(M)$$

is called the *module theory* of R .

Just as we had a category of G -sets consisting of functors

$$G \longrightarrow \text{Set},$$

we have a category of left R -modules consisting of “additive functors”

$$R \longrightarrow \text{Ab}.$$

[The presence of abelian groups on the ground floor of this theory leads to the perspective of “abelian categories”, which is different from “sets with structure”.]

The two perspectives are often combined by turning the group G into the group ring $R[G]$ and looking at the category of

$$R[G]\text{-modules}.$$

We will do this later.

That was pretty pure. Is there a more applied motivation for modules?

Yes. Again this comes from an analogy between number theory and physics.

Remark. $\mathbb{Z}$ -modules = Abelian groups.

Indeed, if R is any ring then there exists a unique ring homomorphism

$$\varphi : \mathbb{Z} \longrightarrow R$$

defined by $\varphi(1_{\mathbb{Z}}) = 1_R$. [That is, $\mathbb{Z}$ is the initial object in the category **Ring**.]

Thus if M is an abelian group then there is a unique $\mathbb{Z}$ -module structure

$$\varphi : \mathbb{Z} \longrightarrow \text{End}_{\mathbf{Ab}}(M);$$

namely, the obvious one given by “repeated addition” in M . This gives us an isomorphism of categories

$$\mathbb{Z}\text{-Mod} \cong \mathbf{Ab}.$$

[Stronger than just an equivalence.]

Dedekind invented the concept of “module” [his notation] at the same time he invented rings [“ordnungen”]; in his 1871 paper on unique factorization.

To discuss the physics angle, let X be a real manifold. An n -dimensional vector field is a (probably smooth) function

$$\varphi : X \longrightarrow \mathbb{R}^n.$$

Vector fields form an abelian group called, say, $\text{Vec}_n(X)$ under pointwise addition:

$$(\varphi_1 + \varphi_2)(x) := \varphi_1(x) + \varphi_2(x).$$

They do not form a ring, but they do have an important additional structure coming from scalar multiplication in $\mathbb{R}^n$.

Let $\mathbb{R}[X]$ be the ring of (probably smooth) functions $X \rightarrow \mathbb{R}$. We can think of $f \in \mathbb{R}[X]$ as a “scalar” that varies from point to point.

Finally, we are allowed to “scale” vector fields. Given a field $\varphi : X \rightarrow \mathbb{R}^n$ and a “scalar” $f : X \rightarrow \mathbb{R}$ we define the new vector field $f\varphi : X \rightarrow \mathbb{R}^n$ by

$$(f\varphi)(x) := f(x)\varphi(x).$$

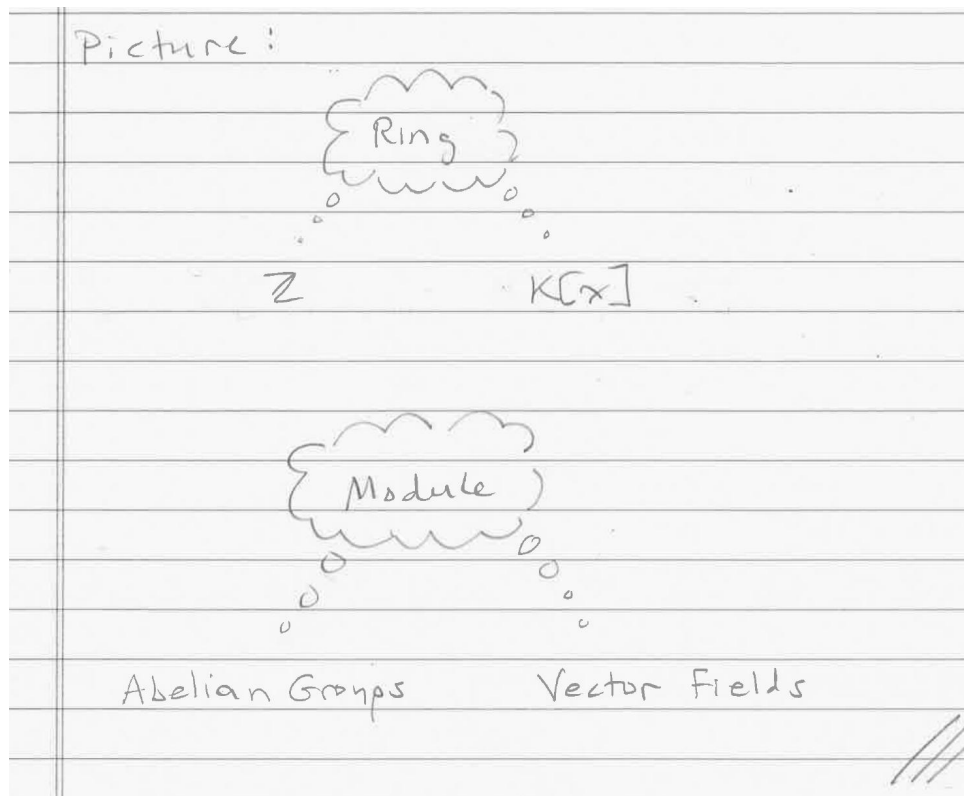
Thus $\text{Vec}_n(X)$ is a module over $\mathbb{R}[X]$.

So if you are physically inclined, you can think of the “module” concept as a generalization of “vector field”.

[Remark: A “vector field” is a “global section of a vector bundle”. But some interesting vector bundles may not have any nowhere-vanishing global sections (e.g., the Möbius band). In this case “modules” must be replaced by “sheaves of modules”.

We will not pursue sheaves in this class but you should be aware of their existence.]

In summary, the “module” concept is a simultaneous generalization of “abelian group” and “vector field”, just as the “ring” concept is a generalization of “integers” and “polynomials over a field”.



Maybe that's enough motivation for today?

Having decided that the definition of module is correct and interesting we will proceed to develop the basic theory.

Definition. Let R be a ring. The collection of left R -modules forms a category which we call $R\text{-Mod}$.

The morphisms in the category are sometimes called “ R -linear maps”. We can guess the definition pretty easily. If left R -modules are “additive functors”

$$R \longrightarrow \mathbf{Ab},$$

then morphisms should be “additive natural transformations”.

To be concrete, let M and N be abelian groups with linear R -actions defined by

$$\alpha : R \longrightarrow \text{End}_{\mathbf{Ab}}(M) \quad \text{and} \quad \beta : R \longrightarrow \text{End}_{\mathbf{Ab}}(N).$$

Then a group homomorphism $\varphi : M \rightarrow N$ is called R -linear if for all $r \in R$ the following diagram commutes:

$$\begin{array}{ccc}
 M & \xrightarrow{\varphi} & N \\
 \alpha_r \downarrow & & \downarrow \beta_r \\
 M & \xrightarrow{\varphi} & N
 \end{array}$$

In symbols, we have

$$\varphi(\alpha_r(m)) = \beta_r(\varphi(m)) \quad \forall m \in M, r \in R.$$

And if you're willing to leave the R -actions implicit we can just write

$$\varphi(r(m)) = r(\varphi(m)) \quad \forall m \in M, r \in R,$$

or even

$$\boxed{\varphi r = r\varphi \quad \forall r \in R.}$$

34 The category $R\text{-Mod}$

February 2, 2016

Let R be a ring. Recall that a left R -module is an additive functor

$$R \longrightarrow \mathbf{Ab}.$$

These naturally form a category with morphisms given by natural transformations.

Concretely: A left R -module consists of an abelian group M and a ring homomorphism

$$\alpha : R \longrightarrow \text{End}_{\mathbf{Ab}}(M).$$

If $\beta : R \rightarrow \text{End}_{\mathbf{Ab}}(N)$ is another module then a function $\varphi : M \rightarrow N$ is called an R -module hom (or an R -linear map) if, for all $r \in R$ and $m, n \in M$,

$$\varphi(m + \alpha_r(n)) = \varphi(m) + \beta_r(\varphi(n)).$$

Examples:

- We saw last time that $\mathbb{Z}$ -modules are just abelian groups:

$$\mathbb{Z}\text{-Mod} = \mathbf{Ab}.$$

- If K is a field then a K -module is just a vector space over K . We will write

$$K\text{-Vec} := K\text{-Mod}$$

for the category of K -vector spaces and K -linear maps.

- More generally, one can use “sheaves of modules” to construct “vector bundles” on a manifold.

If M is an R -module (we will assume “left” unless otherwise stated) then we say that $N \subseteq M$ is an R -submodule if

$$r \in R, m, n \in N \implies m + rn \in N,$$

(i.e., N is closed under addition and scalar multiplication).

Examples:

- subspaces of a vector space;
- Any ring R is a module over itself by left multiplication. The submodules are precisely the left ideals.

If $\varphi : M \rightarrow N$ is a morphism of R -modules then one can check that $\ker \varphi \subseteq M$ and $\operatorname{im} \varphi \subseteq N$ are both R -submodules.

Furthermore, given any submodule $P \subseteq M$ we saw that there is a natural R -module structure on the quotient:

$$r(m + P) := r(m) + P.$$

Thus in the category $R\text{-Mod}$ the following slogan holds:

$$\text{“kernel} \iff \text{submodule} \iff \text{image”}.$$

Note that this is an improvement over the categories **Grp** and **Ring**.

Given a module $M \in R\text{-Mod}$, this inspires us to consider the lattice $\mathcal{L}_R(M)$ of R -submodules.

Note that this is indeed a lattice with top element M and bottom element 0 (the “zero module”). For all $A, B \in \mathcal{L}_R(M)$ the meet and join are given by

$$A \wedge B = A \cap B \quad \text{and} \quad A \vee B = A + B,$$

with the obvious R -module structures.

The lattice $\mathcal{L}_R(M)$ is the correct setting to think about 1st, 2nd and 3rd Isomorphism Theorems and everything works out as nicely as possible.

1st: For all $\varphi : A \rightarrow B$ we have a module isomorphism

$$A/\ker \varphi \cong \operatorname{im} \varphi$$

and lattice isomorphisms

$$\mathcal{L}_R(A, \ker \varphi) \cong \mathcal{L}_R(\operatorname{im} \varphi) \cong \mathcal{L}_R(A/\ker \varphi),$$

where the left isomorphism comes from a Galois connection.

2nd:

For all $A, B \in \mathcal{L}_R(M)$ we have

$$\frac{A + B}{A} \cong \frac{B}{A \cap B}.$$

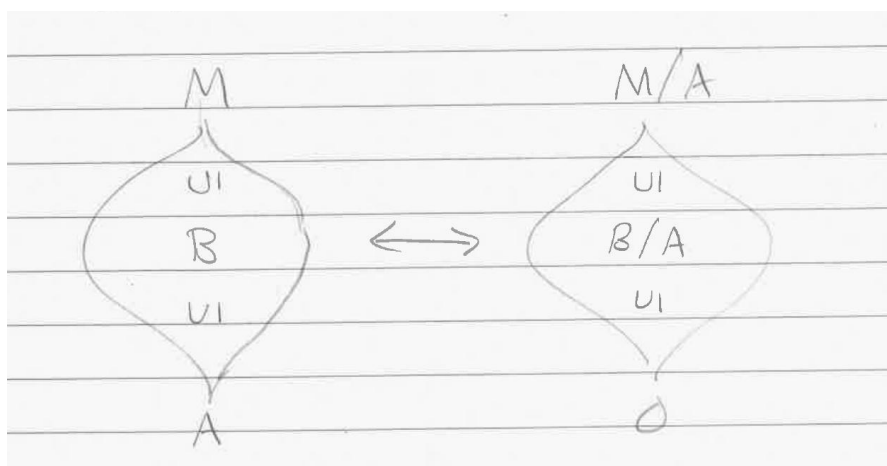
This says in particular that $\mathcal{L}_R(M)$ is a modular lattice [a concept invented by Dedekind in 1900 under the name “dual group of module type”!].

3rd: For all $A, B \in \mathcal{L}_R(M)$ with $A \subseteq B$ we have a module isomorphism

$$\frac{M/A}{B/A} \cong \frac{M}{B}$$

and an isomorphism of lattices

$$\mathcal{L}_R(M, A) \cong \mathcal{L}_R(M/A).$$



Now that lattice theory is working again, maybe we can get the Jordan–Hölder Theorem working?

Definition. We say $M \in R\text{-Mod}$ is a *simple module* if it has no submodules apart from 0 and M .

A *composition series* for $M \in R\text{-Mod}$ is a chain of submodules

$$M = M_0 \supsetneq M_1 \supsetneq M_2 \supsetneq \cdots \supsetneq M_\ell = 0$$

in which each quotient M_i/M_{i+1} is simple (equivalently, each M_{i+1} is a maximal proper submodule of M_i). We call ℓ the length of the series.

Theorem (Jordan–Hölder Theorem). Let $M \in R\text{-Mod}$. If M has a composition series then any two composition series are equivalent. In particular they have the same length ℓ , which we call the length of the module M .

Proof. The proof for groups using the Zassenhaus (Butterfly) Lemma and the Schreier Refinement Lemma goes through with only trivial modification. $\square$

Remarks:

- You already know an example of this theorem. Consider a vector space $V \in K\text{-Vec}$ of finite length. In this case the length is equal to the usual dimension of V (size of a maximal independent set).

Thus we can view “length” as some analogue of “dimension” for more general modules.

- You might wonder if there is some “Master” Jordan–Hölder Theorem containing **Grp** and **R-Mod** as special cases.

Yes there is. If M is a monoid then we define an M -group as a functor

$$M \longrightarrow \mathbf{Grp}.$$

This is also called a “group with operators”. The category $\mathbf{Grp}^M$ has a Jordan–Hölder Theorem simultaneously generalizing the two cases we know.

“Groups with operators” were introduced by Emmy Noether in the 1920s but they have fallen out of style. I don’t know why.

Example (The length of an abelian group). One can check that an abelian group $A \in \mathbb{Z}\text{-Mod}$ is simple if and only if

$$A \cong \mathbb{Z}/p\mathbb{Z}$$

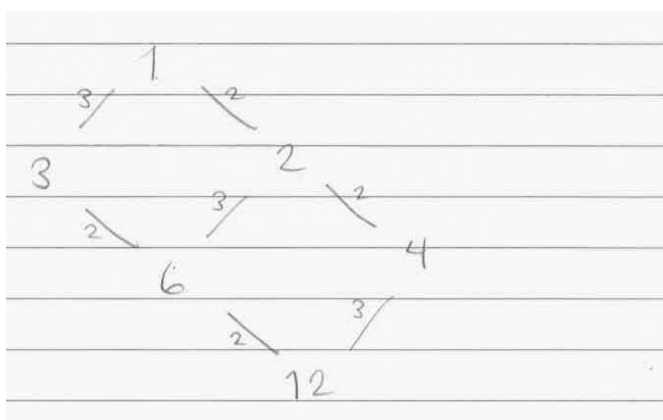
for some prime p . Therefore, an abelian group has finite length if and only if it is finite. What is the length?

Consider $A = \mathbb{Z}/n\mathbb{Z}$. By the 3rd Isomorphism Theorem we have an isomorphism of lattices

$$\mathcal{L}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}) \cong \mathcal{L}_{\mathbb{Z}}(\mathbb{Z}, n\mathbb{Z}) = \mathcal{L}_{\mathbf{Ab}}(\mathbb{Z}, n\mathbb{Z}) \cong \text{Div}(n)^{\text{op}},$$

i.e.,

the lattice of divisors of n under reverse-divisibility. The length of $\mathbb{Z}/n\mathbb{Z}$ is the length of any maximal chain in this lattice.



If

$$n = p_1^{n_1} p_2^{n_2} p_3^{n_3} \cdots$$

for distinct primes $p_1 < p_2 < p_3 < \cdots$, one can see that the length is

$$\ell(\mathbb{Z}/n\mathbb{Z}) = n_1 + n_2 + n_3 + \cdots,$$

i.e., the length of the prime factorization of n .

Note also that $\ell(\mathbb{Z}) = \infty$, so in this case length does not agree with our intuition about “dimension”.

In this case we would do better to look only at chains of prime ideals. The corresponding length is called the “Krull dimension” and we have

$$\text{Kr. dim}(\mathbb{Z}) = 1,$$

which looks nicer.

You see that the word “dimension” is ambiguous. We’ll be careful not to use it without qualification.

35 The category $R\text{-Mod}$ (cont.)

February 4, 2016

Let R be a ring and consider a left module $M \in R\text{-Mod}$. We saw last time that the collection $\mathcal{L}_R(M)$ of submodules of M is a “modular lattice” with

$$A \wedge B = A \cap B \quad \text{and} \quad A \vee B = A + B$$

for all $A, B \in \mathcal{L}_R(M)$. In the special situation that

$$A \cap B = 0 \quad \text{and} \quad A + B = M$$

we will write $M = A \oplus B$ and say that M is the *internal direct sum* of A and B .

We can also characterize the direct sum externally.

Definition. Given $M, N \in R\text{-Mod}$ we define the direct sum as the Cartesian product set

$$M \oplus N := \{(m, n) : m \in M, n \in N\}$$

together with the “direct product” group structure

$$(m_1, n_1) + (m_2, n_2) := (m_1 + m_2, n_1 + n_2)$$

and the left R -action

$$r(m, n) := (rm, rn).$$

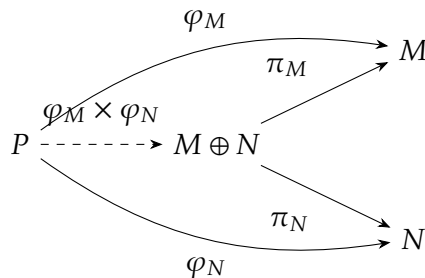
It is easy to check that $\oplus$ is the categorical product in $R\text{-Mod}$: We have projections

$$\begin{aligned} \pi_M : M \oplus N &\longrightarrow M, & (m, n) &\longmapsto m, \\ \pi_N : M \oplus N &\longrightarrow N, & (m, n) &\longmapsto n, \end{aligned}$$

such that for all $P \in R\text{-Mod}$ with maps $\varphi_M : P \rightarrow M$ and $\varphi_N : P \rightarrow N$, the map

$$\begin{aligned} \varphi_M \times \varphi_N : P &\longrightarrow M \oplus N, \\ p &\longmapsto (\varphi_M(p), \varphi_N(p)) \end{aligned}$$

is the unique solution to the diagram



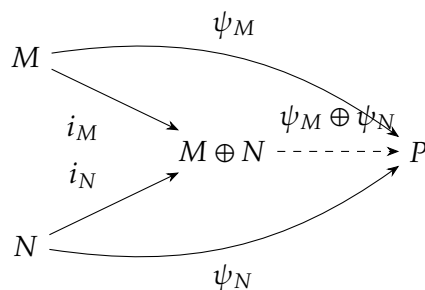
The reason we don't call it " $M \times N$ " is because $\oplus$ is also the coproduct in $R\text{-Mod}$: We have inclusions

$$\begin{aligned} i_M : M &\longrightarrow M \oplus N, & m &\longmapsto (m, 0_N), \\ i_N : N &\longrightarrow M \oplus N, & n &\longmapsto (0_M, n), \end{aligned}$$

such that for all $P \in R\text{-Mod}$ with maps $\psi_M : M \rightarrow P$ and $\psi_N : N \rightarrow P$, the set function

$$\begin{aligned} \psi_M \oplus \psi_N : M \oplus N &\longrightarrow P, \\ (m, n) &\longmapsto \psi_M(m) + \psi_N(n) \end{aligned}$$

is the unique map such that



The fact that the set function $\psi_M \oplus \psi_N$ is actually an R -module homomorphism depends on the commutativity of $(P, +, 0_P)$ and the additivity of the left R -actions.

Thus we will use $\oplus$ for both the product and coproduct in $R\text{-Mod}$.

[However, infinite products and coproducts do not agree; see HW1.]

The category $R\text{-Mod}$ has a zero object given by the "zero module" 0 , hence between any two modules $M, N \in R\text{-Mod}$ we have a unique zero map $0_{MN} : M \rightarrow N$ defined by

$$\begin{array}{ccccc}
 & & 0_{MN} & & \\
 & \curvearrowright & & \curvearrowleft & \\
 M & \xrightarrow{\quad \exists! \quad} & 0 & \xrightarrow{\quad \exists! \quad} & N
 \end{array}$$

This allows us to define kernels and cokernels as follows.

Definition. Let $\varphi : M \rightarrow N$ be any homomorphism of R -modules.

- The *kernel* of φ is a pair $(\ker \varphi, i)$, where $\ker \varphi \in R\text{-Mod}$ and $i : \ker \varphi \rightarrow M$ is a map satisfying

$$\begin{array}{ccccc}
 & & 0 & & \\
 & \curvearrowright & & \curvearrowleft & \\
 P & \xrightarrow{\quad \alpha \quad} & M & \xrightarrow{\quad \varphi \quad} & N \\
 \downarrow \exists! \bar{\alpha} & & \uparrow i & & \\
 \ker \varphi & \xrightarrow{\quad i \quad} & M & \xrightarrow{\quad \varphi \quad} & N \\
 & \curvearrowright & & \curvearrowleft & \\
 & & 0 & &
 \end{array}$$

- The *cokernel* of φ is a pair $(\text{coker } \varphi, \pi)$, where $\text{coker } \varphi \in R\text{-Mod}$ and $\pi : N \rightarrow \text{coker } \varphi$ is a map satisfying

$$\begin{array}{ccccc}
 & & 0 & & \\
 & \curvearrowright & & \curvearrowleft & \\
 M & \xrightarrow{\quad \varphi \quad} & N & \xrightarrow{\quad \beta \quad} & P \\
 & \searrow & \searrow \pi & & \uparrow \exists! \bar{\beta} \\
 & & \text{coker } \varphi & & \\
 & \curvearrowright & & \curvearrowleft & \\
 & & 0 & &
 \end{array}$$

Theorem. *Kernels and cokernels exist in $R\text{-Mod}$.*

Proof. One can check that the set-theoretic kernel satisfies the first universal property and the quotient map

$$\pi : N \longrightarrow N/\text{im } \varphi$$

satisfies the second. □

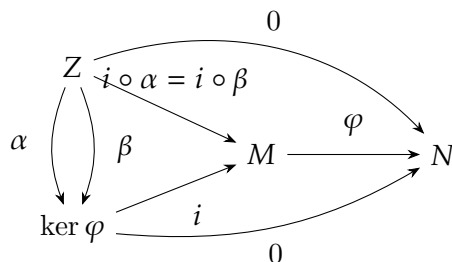
Furthermore, every kernel is a *monomorphism* (“left-cancelable”) in the sense that

$$Z \begin{array}{c} \xrightarrow{\alpha} \\ \xrightarrow{\beta} \end{array} \ker \varphi \xrightarrow{i} M \implies i \circ \alpha = i \circ \beta \implies \alpha = \beta,$$

and every cokernel is an *epimorphism* (“right-cancelable”) in the sense that

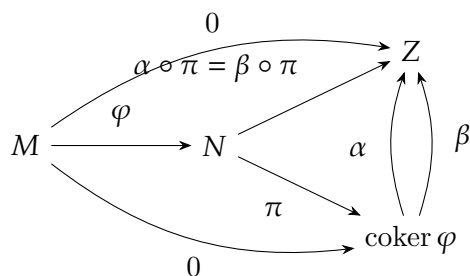
$$N \xrightarrow{\pi} \text{coker } \varphi \begin{array}{c} \xrightarrow{\alpha} \\ \xrightarrow{\beta} \end{array} Z \implies \alpha \circ \pi = \beta \circ \pi \implies \alpha = \beta.$$

Proof. For the first, we have a diagram



By uniqueness, we have $\alpha = \beta$.

For the second, we have a diagram



By uniqueness, we have $\alpha = \beta$. □

Note that this proof works in any category with a zero object. In the category $R\text{-}\mathbf{Mod}$ we will also find that

$$\text{monomorphism} \implies \text{kernel} \quad \text{and} \quad \text{epimorphism} \implies \text{cokernel}.$$

But this is a very special property [in fact, we will find that this property almost characterizes $R\text{-}\mathbf{Mod}$]. To prove it we need to discuss the “enriched” structure of $R\text{-}\mathbf{Mod}$.

Given left modules $M, N \in R\text{-}\mathbf{Mod}$ we will use the notation

$$\operatorname{Hom}_R(M, N) := \operatorname{Hom}_{R\text{-}\mathbf{Mod}}(M, N).$$

We already know that $\operatorname{Hom}_{\mathbf{Ab}}(M, N)$ is an abelian group and in fact

$$\operatorname{Hom}_R(M, N) \subseteq \operatorname{Hom}_{\mathbf{Ab}}(M, N)$$

is a subgroup.

Question: Is $\operatorname{Hom}_R(M, N)$ also an R -module?

Answer: No, not in general. However, if M is also an (R, S) -bimodule [meaning it carries a right S -action commuting with the left R -action] then we can define a left S -module structure on $\operatorname{Hom}_R(M, N)$ by setting

$$(s\varphi)(m) := \varphi(ms) \quad \text{for all } m \in M.$$

If R happens to be commutative then every R -module is an (R, R) -bimodule and so $\operatorname{Hom}_R(M, N)$ is always an R -module.

[If R is a field and M, N are finite-dimensional vector spaces then $\text{Hom}_R(M, N)$ is a vector space of rectangular matrices.]

Furthermore, if $\beta_1, \beta_2 \in \text{Hom}_R(M, N)$ and $\alpha_1, \alpha_2 \in \text{Hom}_R(N, P)$ then we have

$$\begin{aligned}\alpha_1 \circ (\beta_1 + \beta_2) &= \alpha_1 \circ \beta_1 + \alpha_1 \circ \beta_2, \\ (\alpha_1 + \alpha_2) \circ \beta_1 &= \alpha_1 \circ \beta_1 + \alpha_2 \circ \beta_1.\end{aligned}$$

[We say composition is “biadditive”.] And if R is commutative, with $r \in R$ we have

$$\begin{aligned}\alpha_1 \circ (\beta_1 + r\beta_2) &= \alpha_1 \circ \beta_1 + r(\alpha_1 \circ \beta_2), \\ (\alpha_1 + r\alpha_2) \circ \beta_1 &= \alpha_1 \circ \beta_1 + r(\alpha_2 \circ \beta_1).\end{aligned}$$

[We say composition is “ R -bilinear”.]

In summary, we can say that

“ $R\text{-Mod}$ is enriched over $\mathbf{Ab}$ ”,

and when R is commutative,

“ $R\text{-Mod}$ is enriched over $R\text{-Mod}$ ”.

Finally, we can prove the theorem:

Theorem. *In the category $R\text{-Mod}$ we have*

$$\text{monomorphism} \implies \text{kernel} \quad \text{and} \quad \text{epimorphism} \implies \text{cokernel}.$$

Proof sketch. We’ll show the first statement. The proof of the second statement is similar.

So let $\varphi : M \rightarrow N$ be a monomorphism of left R -modules. In fact we will show that φ is the kernel of its cokernel, which is

$$\pi : N \longrightarrow N/\text{im } \varphi.$$

Since φ is a monomorphism we can use the “enriched” structure of $R\text{-Mod}$ to show that $\ker \varphi = 0$ and hence φ is injective [details omitted].

Now let $\alpha : P \rightarrow N$ be any morphism satisfying $\pi \circ \alpha = 0$.

We will show that

$$(*) \quad \begin{array}{ccccc} P & & 0 & & \\ & \searrow \alpha & & \searrow \pi & \\ & N & \xrightarrow{\pi} & N/\text{im } \varphi & \\ & \nearrow \varphi & & \nearrow 0 & \\ M & & 0 & & \end{array}$$

$\exists! \bar{\alpha} : P \rightarrow M$

and it will follow that φ is isomorphic to the kernel of π .

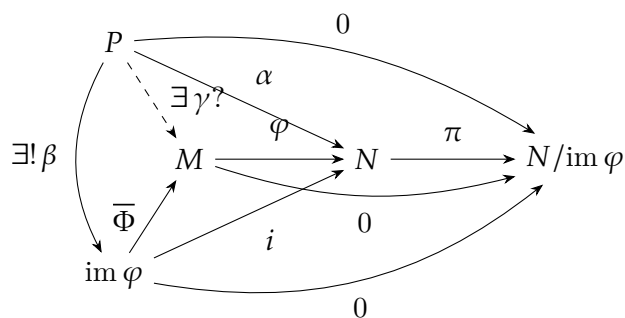
To show this, note that $\ker \pi = \operatorname{im} \varphi$ is the set-theoretic kernel of π , with inclusion

$$i : \operatorname{im} \varphi \longrightarrow N.$$

By the 1st Isomorphism Theorem we obtain an isomorphism

$$\bar{\Phi} : \operatorname{im} \varphi \longrightarrow M = M/\ker \varphi$$

such that $\varphi \circ \bar{\Phi} = i$, and hence a diagram



If γ exists then it must be defined by

$$\gamma := \bar{\Phi} \circ \beta,$$

and we observe that this definition commutes with the rest of the diagram because

$$\varphi \circ \gamma = \varphi \circ (\bar{\Phi} \circ \beta) = (\varphi \circ \bar{\Phi}) \circ \beta = i \circ \beta = \alpha.$$

Finally, if $\bar{\alpha}$ is any map making (*) commute then we must have

$$\begin{aligned} i \circ (\bar{\Phi}^{-1} \circ \bar{\alpha}) &= (i \circ \bar{\Phi}^{-1}) \circ \bar{\alpha} \\ &= \varphi \circ \bar{\alpha} \\ &= \alpha, \end{aligned}$$

and then by uniqueness of β we have

$$\bar{\Phi}^{-1} \circ \bar{\alpha} = \beta, \quad \bar{\alpha} = \bar{\Phi} \circ \beta = \gamma.$$

□

Why am I bothering telling you this?

Because it turns out that the theorem

$$\text{monomorphism} \implies \text{kernel} \quad \text{and} \quad \text{epimorphism} \implies \text{cokernel}$$

is the most important property of $R\text{-Mod}$. This is captured by the following definition and “Cayley-type” theorem.

Definition. A category C is *abelian* if

- it has a zero object;
- it has finite products and coproducts;
- it has kernels and cokernels;
- all monomorphisms are kernels and all epimorphisms are cokernels.

Amazingly, this definition implies that the category C is enriched over $\mathbf{Ab}$. Even more amazingly, we have the following theorem that says that the definition of abelian category faithfully captures the properties of $R\text{-Mod}$.

Theorem (Freyd–Mitchell Embedding Theorem). *Let C be a small abelian category. Then there exists a ring R such that C can be embedded as a full subcategory of $R\text{-Mod}$.*

More generally, if C is any abelian category (small or not) and if $P \in C$ is a “compact, projective generator of C ” (whatever that means) then we have

$$C = R\text{-Mod},$$

where

$$R = \text{End}_C(P).$$

Maybe some day (on some planet) algebra classes will begin with the definition of abelian categories ...

Maybe not!

36 Modules over a PID

February 9, 2016

After discussing the general theory of modules, today we move on to a new topic:

Modules over a PID.

Until further notice, all rings are assumed to be commutative. So let R be a (commutative) ring.

Definition. We say that R is an *integral domain* (“domain” for short) if for all $a, b \in R$ we have

$$ab = 0 \implies a = 0 \text{ or } b = 0.$$

Among domains we have the following chain of implications:

$$\text{Field} \implies \text{Euclidean} \implies \text{PID} \implies \text{UFD}.$$

I assume you've seen this before, but let me quickly sketch the details.

Let $A \subseteq R$ be a subset of a (commutative) ring. We will write

$$(A) := \bigcap_{A \subseteq I} I$$

for the intersection of all ideals $I \subseteq R$ that contain A . We call this the *ideal of R generated by A* .

If $A = \{a_1, a_2, \dots, a_n\}$ we will write

$$(a_1, a_2, \dots, a_n) := (\{a_1, a_2, \dots, a_n\}).$$

Ideals of this form are called *finitely generated*. Ideals of the form $(a) = \{xa : x \in R\}$ are called *principal*.

Definition. An integral domain R is called a *PID* ("principal ideal domain") if for each ideal $I \subseteq R$ there exists an element $a \in R$ such that

$$I = (a).$$

We say that R is a *field* if its group of units satisfies

$$R^\times = R \setminus \{0\}.$$

Equivalently, its only ideals are the trivial ideals

$$0 = (0) \quad \text{and} \quad R = (1).$$

Proof. Let R be a field with ideal $I \subseteq R$. If $I \neq (0)$ then there exists $0 \neq a \in I$. But then $1 = aa^{-1} \in I$ and hence $I = (1)$. Conversely, suppose R has no nontrivial ideals and consider an element $a \neq 0$. Since $(a) \neq (0)$ we must have $(a) = (1)$. Now since $1 \in (a)$, there exists $b \in R$ such that $ab = 1$. $\square$

Thus a field is a PID. More generally, we make the following definition.

Definition. A domain R is called *Euclidean* if there exists a function

$$\delta : R \setminus \{0\} \longrightarrow \mathbb{N}$$

such that for all $a, b \in R$ with $b \neq 0$ there exist $q, r \in R$ with

- $a = qb + r$;
- $r = 0$ or $\delta(r) < \delta(b)$.

Remark: There are really only two interesting examples of Euclidean domains:

- $\mathbb{Z}$ with $\delta(a) = |a|$;
- $K[x]$ with $\delta(f) = \deg(f)$.

The definition is meant to abstract their common properties. But the Euclidean domain concept is actually not used much because of the following theorem.

Theorem. *Euclidean $\implies$ PID.*

Proof. Let (R, δ) be Euclidean and consider an ideal $I \subseteq R$. If $I = (0)$ then we're done, so suppose that $I \neq (0)$. Then the set

$$\delta(I \setminus \{0\}) \subseteq \mathbb{N}$$

is nonempty, so by well-ordering it has a smallest element $n \in \delta(I \setminus \{0\})$. Choose $a \in I \setminus \{0\}$ so that $\delta(a) = n$. Clearly we have $(a) \subseteq I$. I claim that we also have $I \subseteq (a)$. Indeed, consider any $b \in I$. Since $a \neq 0$, there exist $q, r \in R$ such that

$$b = qa + r, \quad r = 0 \text{ or } \delta(r) < \delta(a).$$

The first equation implies that $r = b - qa \in I$. If $r \neq 0$ then by minimality we must have $\delta(a) \leq \delta(r)$, which contradicts the second condition above. Hence $r = 0$ and we conclude that $b = qa \in (a)$. $\square$

Not all PIDs are Euclidean but we prefer to think of $\mathbb{Z}$ and $K[x]$ as PIDs because the language of PIDs is more natural.

Let R be a ring. Given elements $a, b \in R$, we define the notion of "divisibility in R " by

$$a \mid b \iff (b) \subseteq (a).$$

Note that $a \mid 0$ and $1 \mid a$ for all $a \in R$. We will say that a is a *proper divisor* of b if

$$(b) \subsetneq (a) \subsetneq (1).$$

The first inclusion means that a is not associate to b (i.e., we don't have $a = ub$ and $b = u^{-1}a$ for some unit $u \in R^\times$), and the second inclusion says that a itself is not a unit. We say that an element of R is *irreducible* if it has no proper divisors.

Definition. A domain R is called a *UFD* if every nonzero, nonunit element can be written uniquely as a product of irreducible elements.

Theorem. *PIDs $\implies$ UFD.*

Proof. Let R be a PID. There are two steps in the proof.

Step 1: Every element $a \in R$ has a factorization into irreducibles.

If not, then we obtain an infinite increasing chain of ideals

$$(a) \subsetneq (a_1) \subsetneq (a_2) \subsetneq \cdots.$$

Note that the union

$$I := \bigcup_{n \in \mathbb{N}} (a_n)$$

is an ideal. Since R is a PID we must have $I = (b)$ for some $b \in R$.

But then since $b \in \bigcup_n (a_n)$ there exists $m \in \mathbb{N}$ such that $b \in (a_m)$. It follows that

$$(b) \subseteq (a_m) \subsetneq (a_{m+1}) \subseteq I = (b),$$

which is a contradiction.

Step 2: "Euclid's lemma."

We say that an element $p \in R$ is *prime* if for all $a, b \in R$ we have

$$p \mid ab \implies p \mid a \text{ or } p \mid b,$$

or equivalently,

$$ab \in (p) \implies a \in (p) \text{ or } b \in (p).$$

[Note: R is a domain $\iff (0) \subseteq R$ is prime.]

Now let $p \in R$ be irreducible. We will show that p is also prime. Indeed, suppose we have $a, b \in R$ such that

$$ab \in (p) \quad \text{and} \quad a \notin (p).$$

We will show that $b \in (p)$.

Since $a \notin (p)$ we have $(p) \subsetneq (a, p)$, and since R is a PID we have

$$(p) \subsetneq (a, p) = (c)$$

for some $c \in R$. Then since p is irreducible we must have $(a, p) = (c) = (1)$. Since $1 \in (a, p)$ there exist $x, y \in R$ such that

$$ax + py = 1.$$

Multiply both sides by b to get

$$abx + pby = b.$$

Since $ab \in (p)$, write $ab = pd$. Then

$$\begin{aligned} pdx + pby &= b, \\ p(dx + by) &= b. \end{aligned}$$

We conclude that $b \in (p)$.

Finally, consider a nonzero, nonunit $a \in R$ and suppose we have two factorizations into irreducibles

$$a = p_1 p_2 \cdots p_k = q_1 q_2 \cdots q_\ell.$$

Since $p_1 \mid a$ we have $p_1 \mid q_1 q_2 \cdots q_\ell$. Then by Euclid's lemma (i.e., irreducible $\implies$ prime in a PID) and induction there exists i such that $p_1 \mid q_i$. Without loss of generality suppose that $p_1 \mid q_1$. Then since q_1 is irreducible and p_1 is not a unit, there exists a unit $u_1 \in R^\times$ such that $q_1 = u_1 p_1$.

Substituting gives

$$p_1 p_2 \cdots p_k = u_1 p_1 q_2 \cdots q_\ell.$$

Then since R is a domain we can cancel p_1 to get

$$p_2 p_3 \cdots p_k = u_1 q_2 q_3 \cdots q_\ell.$$

At this point we use induction to conclude that $k = \ell$ and that we can reorder the factors such that

$$q_2 = u_2 p_2, \quad q_3 = u_3 p_3, \quad \dots, \quad q_k = u_k p_k$$

for some units $u_2, u_3, \dots, u_k \in R^\times$. □

37 Free modules

February 11, 2016

Among integral domains we have the following chain of implications:

$$\text{Field} \implies \overset{\times}{\text{Euclidean}} \implies \overset{\vee}{\text{PID}} \implies \overset{\times}{\text{UFD}}.$$

This sequence of ideas really goes back to Euclid and the modern form is due to Dedekind. Last time I gave a quick review of the definitions and proofs.

We saw that the definition of “Euclidean domain” is awkward but the language of PIDs is elegant and flexible.

UFDs are also kind of awkward. Last time I taught 762 (then called 662) I pursued the theory of UFDs, which leads to notions of “smoothness” in algebraic geometry. [Notes available on my webpage.]

This time I’m going to look at PIDs, and in particular their module theory, which is very nice. I am aiming for the “fundamental theorem of finitely generated modules over a PID,” which is a simultaneous generalization of

- the classification of finite abelian groups;
- the Jordan canonical form of a matrix.

The simplest kind of PIDs are fields, so we’ll start with vector spaces. And since every vector space is “free” we’ll start with “free modules.” Here’s the idea:

Consider an arbitrary ring R and an arbitrary set A . Our goal is to define the left R -module freely generated by the set A .

If it exists, let’s call it $F_R(A)$. It should satisfy the following properties:

- $A \subseteq F_R(A)$.

- Any R -module homomorphism $\varphi : F_R(A) \rightarrow M$ is determined by its values on the generating set A .

We can formalize this as a universal property.

Universal Property of Free Modules. We have a set function $i : A \rightarrow F_R(A)$ and for any left R -module M and any set function $f : A \rightarrow M$ there exists a unique R -module homomorphism $\varphi : F_R(A) \rightarrow M$ satisfying

$$\begin{array}{ccc} F_R(A) & \xrightarrow{\exists! \varphi} & M \\ & \swarrow i \quad \searrow f & \\ & A & \end{array}$$

We could also see this as a final object in a certain category. If $F_R(A)$ exists then by general nonsense it will be unique and the function $i : A \rightarrow F_R(A)$ will be injective.

Does it exist?

Theorem. *Free modules exist.*

Proof. Consider a ring R and a set A . Following HW1.1(b) we define the R -module

$$R^{\oplus A} := \{\text{functions } A \rightarrow R \text{ with finite support}\}.$$

Now consider the function $i : A \rightarrow R^{\oplus A}$ defined by sending $a \in A$ to the function $i_a : A \rightarrow R$ such that

$$i_a(b) := \begin{cases} 1, & a = b, \\ 0, & a \neq b. \end{cases}$$

For any R -module M and any set function $f : A \rightarrow M$ we will show that there exists a unique R -module map $\varphi : R^{\oplus A} \rightarrow M$ satisfying

$$\begin{array}{ccc} R^{\oplus A} & \xrightarrow{\exists! \varphi} & M \\ & \swarrow i \quad \searrow f & \\ & A & \end{array}$$

and hence $(R^{\oplus A}, i)$ is the free module generated by A .

To do this, note that every element $r \in R^{\oplus A}$ can be written uniquely in the form

$$r = \sum_{a \in A} r_a i_a,$$

and the sum exists because we have $r_a = 0$ for all but finitely many $a \in A$.

Now the assumption $f = \varphi \circ i$ requires that

$$\begin{aligned}
 \varphi(r) &= \varphi\left(\sum_a r_a i_a\right) \\
 &= \sum_a \varphi(r_a i_a) && [\varphi \text{ is additive}] \\
 &= \sum_a r_a \varphi(i_a) && [\varphi \text{ preserves the } R\text{-action}] \\
 &= \sum_a r_a f(a) && [f = \varphi \circ i].
 \end{aligned}$$

This defines a unique function $\varphi : R^{\oplus A} \rightarrow M$ and it is not difficult to check that it is an R -module homomorphism. $\square$

Remarks:

- Given a concrete category $\mathcal{C}$ (i.e., whose objects are sets) and a set A , the free object generated by A consists of
 - an object $F_{\mathcal{C}}(A) \in \mathcal{C}$;
 - a set function $i : A \rightarrow F_{\mathcal{C}}(A)$

such that for all objects $X \in \mathcal{C}$ and set functions $f : A \rightarrow X$ there exists a unique morphism $\varphi : F_{\mathcal{C}}(A) \rightarrow X$ in $\mathcal{C}$ such that

$$\begin{array}{ccc}
 F_{\mathcal{C}}(A) & \xrightarrow{\exists! \varphi} & X \\
 & \nwarrow i \quad \nearrow f & \\
 & A &
 \end{array}$$

There is no reason in general for free objects to exist, but they do exist in most of the categories we care about.

- The proof we gave above is really just a rephrasing of the proof from HW1.1(b) that $R^{\oplus A}$ is a coproduct in $R\text{-Mod}$. [Actually, we only proved it in $\mathbf{Ab}$ but the proof extends to $R\text{-Mod}$.] So we could replace one of the proofs with a verification that the following two pieces of data are equivalent:

$$\begin{array}{ccc}
 R^{\oplus A} & \xrightarrow{\varphi} & M \\
 & \nwarrow i \quad \nearrow f & \\
 & A &
 \end{array}
 \iff
 \begin{array}{ccc}
 R^{\oplus A} & \xrightarrow{\varphi} & M \\
 & \nwarrow i_a \quad \nearrow f_a & \\
 & R_a &
 \end{array}
 \quad \forall a \in A$$

But doing so would probably make the proof less understandable!

- Given an element $r \in R^{\oplus A}$ the notation

$$r = \sum_{a \in A} r_a i_a$$

gets annoying, so we will usually abuse notation and just write

$$r = \sum_{a \in A} r_a a.$$

Here the symbol “ a ” really stands for the function $i_a : A \rightarrow R$. We do this all the time in linear algebra when we confuse “basis vectors” with their corresponding “coordinate functions.”

If $A = \{a_1, a_2, \dots, a_n\}$ is finite then we will further abuse notation by writing

$$r = \sum_{i=1}^n r_i a_i.$$

We might even write this as a vector

$$r = (r_1, r_2, \dots, r_n).$$

If $|A| = n$, we would like to write

$$R^{\oplus A} \cong “R^{\oplus n}.”$$

After all, isn’t $R^{\oplus A}$ just the direct sum of R with itself n times?

Well, we have to be careful.

We are hoping that the following property is true.

Invariant Basis Number Property (IBN). For all sets $A, B \in \mathbf{Set}$ we have

$$R^{\oplus A} \cong R^{\oplus B} \text{ in } R\text{-Mod} \implies A \cong B \text{ in Set}.$$

It turns out that the IBN holds in great generality (e.g., it holds for all commutative rings), but not always.

Luckily the cases where it fails are fairly pathological (e.g., the ring of endomorphisms of an infinite dimensional vector space), so they won’t cause us any trouble.

38 Bases of free modules

February 16, 2016

Let R be a ring and let A be a set. We say that $(F_R(A), i)$ is a *free (left) R -module generated by A* if

- $F_R(A)$ is a (left) R -module;
- $i : A \rightarrow F_R(A)$ is a set function;
- for all (left) R -modules M and set functions $f : A \rightarrow M$ there exists a unique R -module homomorphism $\varphi : F_R(A) \rightarrow M$ such that

$$\begin{array}{ccc}
 F_R(A) & \xrightarrow{\exists! \varphi} & M \\
 & \nwarrow i \quad \nearrow f & \\
 & A &
 \end{array}$$

In other words, $(F_R(A), i)$ is an initial object in a certain category.

If it exists, then by general nonsense it is unique and the function i is injective.

Theorem. *Free modules exist.*

Proof. The coproduct $R^{\oplus A}$ together with the function $i : A \rightarrow R^{\oplus A}$, $a \mapsto i_a$, defined by

$$i_a(b) := \begin{cases} 1_R, & a = b, \\ 0_R, & a \neq b, \end{cases}$$

satisfies the universal property. □

Now we can use free modules to define the concepts of linearly independent and spanning sets in an R -module.

Definition. Consider a module $M \in R\text{-Mod}$ and a set $A \in \text{Set}$. We will call any function $A \rightarrow M$ an *indexed set* of elements of M . Now consider the canonical map $\varphi : R^{\oplus A} \rightarrow M$ such that

$$\begin{array}{ccc}
 R^{\oplus A} & \xrightarrow{\varphi} & M \\
 & \nwarrow i \quad \nearrow & \\
 & A &
 \end{array}$$

- If φ is injective we say that $A \rightarrow M$ is a *linearly independent set* in M .
- If φ is surjective we say that $A \rightarrow M$ is a *spanning set* in M .
- If φ is both injective and surjective [hence an isomorphism of R -modules] we say that $A \rightarrow M$ is a *basis* for M .

Let's unpack this a bit. If we write the function $A \rightarrow M$ as $a \mapsto m_a$ and an element $r \in R^{\oplus A}$ as $\sum_{a \in A} r_a i_a$, recall that the homomorphism $\varphi : R^{\oplus A} \rightarrow M$ is defined by

$$\varphi(r) = \varphi\left(\sum_{a \in A} r_a i_a\right) = \sum_{a \in A} r_a m_a.$$

Now recall that φ is injective if and only if $\ker \varphi = 0$. So we can rephrase the definition of linear independence by saying that $A \rightarrow M$ is linearly independent if and only if

$$\varphi(r) = 0_M \iff r = 0_{R^{\oplus A}},$$

or equivalently,

$$\sum_{a \in A} r_a m_a = 0 \iff r_a = 0 \quad \forall a \in A,$$

which agrees with the familiar definition you learned in linear algebra.

Similarly, the set $A \rightarrow M$ is spanning if for all $m \in M$ there exists $r \in R^{\oplus A}$ such that

$$m = \varphi(r) = \sum_{a \in A} r_a m_a.$$

Theorem. *An R -module has a basis if and only if it is free.*

Proof. First note that $A \rightarrow R^{\oplus A}$ is a basis of $R^{\oplus A}$ because the induced homomorphism $\varphi : R^{\oplus A} \rightarrow R^{\oplus A}$ is the identity map (which is an iso.).

$$\begin{array}{ccc} R^{\oplus A} & \xrightarrow{\varphi} & R^{\oplus A} \\ & \sim & \\ & \swarrow i \quad \searrow i & \\ & A & \end{array}$$

Conversely, suppose that $A \rightarrow M$ is a basis for some R -module M . Then we obtain an isomorphism

$$\begin{array}{ccc} R^{\oplus A} & \xrightarrow{\varphi} & M \\ & \sim & \\ & \swarrow i \quad \searrow i & \\ & A & \end{array}$$

and it follows that $(M, \varphi \circ i)$ is a free module. □

Q: If M has a basis (i.e., if M is free), does every basis have the same cardinality?

A: Not in general. Given a ring R we say that R has the “invariant basis number” property (IBN) if

$$R^{\oplus A} \cong R^{\oplus B} \implies |A| = |B|.$$

In this case, the cardinality $|A|$ is called the *rank* of the module $R^{\oplus A}$.

The ring of endomorphisms over an infinite vector space does *not* have the IBN property, but we will see that most nice rings do have it (including all commutative rings).

To explore this, let’s consider the following three concepts from linear algebra:

1. basis;
2. maximal linearly independent set;
3. minimal spanning set.

Note that (1) $\implies$ (2) & (3).

Proof. Let A be a basis for $M \in R\text{-Mod}$.

To show that A is a maximal linearly independent set, consider any $m \in M \setminus A$. Since A is spanning we can write

$$m = \sum_{a \in A} r_a m_a$$

for some $r_a \in R$. But then the set $A \cup \{m\}$ is not linearly independent.

To show that A is a minimal spanning set consider any $m_a \in A$. Now suppose for contradiction that $A \setminus \{m_a\}$ is a spanning set. It follows that we can write

$$m_a = \sum_{b \neq a} r_b m_b$$

for some $r_b \in R$, which contradicts the fact that A is independent. □

In general, though, we have

$$(2) \not\Rightarrow (1) \text{ or } (3), \quad (3) \not\Rightarrow (1) \text{ or } (2).$$

Proof. Consider $\mathbb{Z}$ as a $\mathbb{Z}$ -module.

Note that $\mathbb{Z}$ is a free $\mathbb{Z}$ -module with basis $\{1\} \subseteq \mathbb{Z}$.

To see that (2) $\not\Rightarrow$ (3), consider the set $\{n\} \subseteq \mathbb{Z}$ for any $n \in \mathbb{Z} \setminus \{0, \pm 1\}$. Note that this is an independent set because for any $a \in \mathbb{Z}$ we have

$$na = 0 \implies a = 0.$$

And it is a maximal independent set since for any $m \in \mathbb{Z} \setminus \{n\}$ we have the nontrivial relation

$$(n)m + (-m)n = 0.$$

However, $\{n\} \subseteq \mathbb{Z}$ is not a spanning set since there exist integers not divisible by n .

To see that (3) $\not\Rightarrow$ (2), consider the set $\{m, n\} \subseteq \mathbb{Z}$ for any coprime $m, n \in \mathbb{Z} \setminus \{0, \pm 1\}$. Since m, n are coprime there exist $x, y \in \mathbb{Z}$ such that

$$1 = xm + yn.$$

Then for any $a \in \mathbb{Z}$ we have

$$a = (ax)m + (ay)n$$

and hence $\{m, n\}$ is a spanning set. It is minimal since neither of $\{m\}$ or $\{n\}$ is a spanning set. And it is not independent because of the relation

$$(n)m + (-m)n = 0.$$

□

Remark. It is not a coincidence that all maximal independent sets of $\mathbb{Z}$ have the same size. We will see that this phenomenon holds in general for modules over an integral domain, even for those that have no basis.

However, when $R = K$ is a field the three concepts (1), (2), (3) are equivalent.

Proof. Let $V \in K\text{-Mod}$ be a vector space.

To prove (2) $\Rightarrow$ (1) let $A \subseteq V$ be a maximal linearly independent set (which exists by Zorn's Lemma). Then for any $v \in V \setminus A$ we have a relation

$$rv + \sum_{a \in A} r_a v_a = 0$$

where not all coefficients are zero. Since A is independent this implies that $r \neq 0$. Then since K is a field we can divide by r to get

$$v = \sum_{a \in A} \left(\frac{-r_a}{r} \right) v_a,$$

which proves that A is a spanning set.

To prove (3) $\Rightarrow$ (1) let $A \subseteq V$ be a minimal spanning set and assume for contradiction that A is not linearly independent. Then there exists a relation

$$\sum_{a \in A} r_a v_a = 0$$

where not all coefficients are zero. Choose $a \in A$ such that $r_a \neq 0$. Then since K is a field we can divide by r_a to get

$$v_a = \sum_{b \neq a} \left(\frac{-r_b}{r_a} \right) v_b.$$

This implies that $A \setminus \{a\}$ is still a spanning set, which contradicts the minimality of A . □

Corollary (assuming Zorn's Lemma). Every vector space is a free module.

To finish, let me give you the original proof that fields have the IBN property. This argument is due to Ernst Steinitz from his paper "Algebraic Theory of Fields" (1910). It was later abstracted by Hassler Whitney and Saunders MacLane into the theory of "matroids."

Lemma (Steinitz Exchange Lemma (1910)). Let K be a field and let $V \in K\text{-Mod}$ be a vector space. Suppose V has a finite spanning set $S \subseteq V$. Then for any independent set $I \subseteq V$ we have

$$|I| \leq |S|.$$

Proof. Let $I = \{i_1, \dots, i_m\}$ and $S = \{s_1, \dots, s_n\}$. Assume for contradiction that $m > n$.

Since S is spanning we can write

$$i_1 = a_1 s_1 + \dots + a_n s_n$$

for some scalars $a_1, \dots, a_n \in K$.

By independence of I we have $i_1 \neq 0$ and hence at least one of $a_1, \dots, a_n$ is nonzero. Without loss, say $a_1 \neq 0$. Then since K is a field we can divide by a_1 to get

$$s_1 = \frac{1}{a_1} i_1 - \frac{a_2}{a_1} s_2 - \dots - \frac{a_n}{a_1} s_n,$$

and it follows that

$$\{i_1, s_2, \dots, s_n\}$$

is a spanning set. Now assume for induction that the set

$$\{i_1, i_2, \dots, i_k, s_{k+1}, \dots, s_n\}$$

is spanning for some $1 \leq k < n$. In this case we have

$$i_{k+1} = b_1 i_1 + \dots + b_k i_k + b_{k+1} s_{k+1} + \dots + b_n s_n$$

for some scalars $b_1, \dots, b_n \in K$. By independence of I we know that at least one of $b_{k+1}, \dots, b_n$ is nonzero. Without loss, say $b_{k+1} \neq 0$.

Then since K is a field we can divide by b_{k+1} to get

$$\begin{aligned} s_{k+1} &= -\frac{b_1}{b_{k+1}} i_1 - \dots - \frac{b_k}{b_{k+1}} i_k + \frac{1}{b_{k+1}} i_{k+1} \\ &\quad - \frac{b_{k+2}}{b_{k+1}} s_{k+2} - \dots - \frac{b_n}{b_{k+1}} s_n, \end{aligned}$$

and it follows that

$$\{i_1, i_2, \dots, i_{k+1}, s_{k+2}, \dots, s_n\}$$

is a spanning set. By induction we conclude that

$$\{i_1, i_2, \dots, i_n\}$$

is a spanning set. Finally, since $m = |I| > |S| = n$ we obtain a relation

$$i_{n+1} = c_1 i_1 + c_2 i_2 + \dots + c_n i_n,$$

contradicting the independence of I . □

Corollary. Let K be a field. Then for all finite sets B_1 and B_2 we have

$$K^{\oplus B_1} \cong K^{\oplus B_2} \implies |B_1| = |B_2|.$$

Proof. Let B_1 and B_2 be two bases of a vector space over K . Applying Steinitz one way gives

$$|B_1| \leq |B_2|$$

and applying it the other way gives

$$|B_2| \leq |B_1|.$$

□

Remark. Maybe this holds for infinite bases too. I don't really care.

39 Independent and spanning sets; fields of fractions

February 18, 2016

If $A \subseteq M$ is a subset of an R -module then there exists a unique R -linear map

$$\varphi_A : R^{\oplus A} \longrightarrow M$$

such that $\varphi_A(i_a) = a$ for all $a \in A$.

- If φ_A is injective we say that $A \subseteq M$ is linearly independent.
- If φ_A is surjective we say that $A \subseteq M$ is a spanning set.
- If φ_A is bijective (hence an isomorphism) we say that $A \subseteq M$ is a basis.

Thus M has a basis if and only if it is isomorphic to a free module.

Now consider the following three properties a subset $A \subseteq M$ could have:

1. $A \subseteq M$ is a basis;
2. $A \subseteq M$ is a maximal independent set;
3. $A \subseteq M$ is a minimal spanning set.

Last time we saw that

$$(1) \implies (2) \text{ and } (3)$$

for general R -modules, and if R is a field then we have

$$(2) \text{ or } (3) \implies (1).$$

Since every R -module has a maximal independent set (by Zorn's Lemma), this implies that every vector space has a basis, hence is free. Then the Steinitz Exchange Lemma shows that every basis has the same cardinality (i.e., fields satisfy the IBN property).

We also saw that $(2) \not\Rightarrow (3)$ and $(3) \not\Rightarrow (2)$ for abelian groups (i.e., $\mathbb{Z}$ -modules).

Example: Consider $\mathbb{Z}$ as a $\mathbb{Z}$ -module. It is free with basis $\{1\}$ or $\{-1\}$.

For all $0 \neq n \in \mathbb{Z}$ the set $\{n\} \subseteq \mathbb{Z}$ is maximal independent but is only spanning if $n = \pm 1$.

For all coprime $m, n \in \mathbb{Z} \setminus \{0, \pm 1\}$ the set $\{m, n\}$ is minimal spanning, but it is never linearly independent.

We can observe from this that the size of a minimal spanning set is not well-defined because $\{1\}$ and $\{2, 3\}$ are both minimal spanning sets of $\mathbb{Z}$. We also observe that every maximal independent set in $\mathbb{Z}$ has size 1.

Furthermore, there exist non-free $\mathbb{Z}$ -modules.

Theorem. *No nontrivial finite abelian group is free.*

Proof. Let M be a nontrivial finite abelian group and consider a subset $A \subseteq M$. If $A \not\subseteq \{0\}$ then choose $0 \neq a \in A$. Then since M is a finite group there exists an integer $n > 0$ such that

$$na = \underbrace{a + a + \cdots + a}_{n \text{ times}} = 0,$$

and it follows that the set A is not independent. Therefore the only independent subset of M is $\emptyset$, which is certainly not a spanning set because M is nontrivial. $\square$

Nevertheless, it will still make sense to discuss the “rank” of an abelian group. Our next goal is to prove the following theorem, which will motivate our interest in integral domains.

Theorem. *Let R be an integral domain and consider a module $M \in R\text{-Mod}$ (possibly not free). Then every maximal independent subset of M has the same cardinality, which we will call the rank of M .*

We already have most of the ingredients to prove this. The only missing piece is the following lemma/construction establishing the close relationship between domains and fields.

Lemma. Let R be a ring. Then

$$R \text{ is an integral domain} \iff R \text{ is isomorphic to a subring of a field.}$$

Proof. First let R be a subring of a field K and consider $a, b \in R$ such that $ab = 0$ and $a \neq 0$. Then a^{-1} exists in K so that

$$b = (a^{-1}a)b = a^{-1}(ab) = a^{-1}0 = 0$$

in K , hence also in R .

Conversely, let R be an integral domain. We need to construct a field containing a subring isomorphic to R . The trick is to mimic the construction of $\mathbb{Q}$ from $\mathbb{Z}$. We define the set of fractions

$$\text{Frac}(R) := \left\{ \left[\frac{a}{b} \right] : a, b \in R, b \neq 0 \right\}$$

together with the equivalence relation

$$\left[\frac{a}{b}\right] = \left[\frac{c}{d}\right] \iff ad = bc$$

and algebraic operations

$$\left[\frac{a}{b}\right] \cdot \left[\frac{c}{d}\right] = \left[\frac{ac}{bd}\right] \quad \text{and} \quad \left[\frac{a}{b}\right] + \left[\frac{c}{d}\right] = \left[\frac{ad + bc}{bd}\right].$$

These exist because $b \neq 0$ and $d \neq 0$ imply $bd \neq 0$.

Now one can check that this defines a field structure on $\text{Frac}(R)$ and that the function

$$R \longrightarrow \text{Frac}(R), \quad r \longmapsto \left[\frac{r}{1}\right]$$

is an injective ring homomorphism. □

The field $\text{Frac}(R)$ constructed in the proof is called the *field of fractions* (or *quotient field*) of the domain R . It satisfies the following universal property.

Theorem (Universal Property of Fractions). *Let R be an integral domain. Then for every field K and every injective ring homomorphism $i : R \hookrightarrow K$ there exists a unique ring homomorphism $\varphi : \text{Frac}(R) \rightarrow K$ satisfying*

$$\begin{array}{ccc} \text{Frac}(R) & \xrightarrow{\exists! \varphi} & K \\ & \nwarrow \quad \nearrow i & \\ & R & \end{array}$$

[Compare to the solution of HW1.6(b).]

I'll sketch the rest of the proof next time and you will fill in the details on HW2.

40 Localization, torsion, free modules

February 23, 2016

Last time we discussed the field of fractions of an integral domain. More generally we have the following construction.

Definition (Localization of a Module). Let R be a commutative ring and let M be an R -module. Let $S \subseteq R$ be a subset satisfying

- $1 \in S$;

- $s, t \in S \implies st \in S$;
- S contains no zero divisors.

Then we will define the set of “fractions”

$$S^{-1}M := \left\{ \left[\frac{m}{s} \right] : m \in M, s \in S \right\}.$$

You will prove on HW2 that the relation

$$\left[\frac{m}{s} \right] = \left[\frac{n}{t} \right] \iff tm = sn$$

is an equivalence and that the following operations are well-defined:

$$\left[\frac{m}{s} \right] + \left[\frac{n}{t} \right] = \left[\frac{tm + sn}{st} \right] \quad \text{and} \quad r \left[\frac{m}{s} \right] = \left[\frac{rm}{s} \right].$$

Furthermore, you will show that this makes $S^{-1}M$ into an R -module with a canonical R -module homomorphism

$$M \longrightarrow S^{-1}M, \quad m \longmapsto \left[\frac{m}{1} \right].$$

We call $S^{-1}M$ a *localization* of M .

We can also localize the ring R by

- defining $S^{-1}R$ as an R -module;
- defining multiplication by

$$\left[\frac{r_1}{s} \right] \left[\frac{r_2}{t} \right] = \left[\frac{r_1 r_2}{st} \right].$$

And then $S^{-1}M$ becomes an $S^{-1}R$ -module.

Picture:

$$\begin{array}{ccc} S^{-1}R & \xrightarrow{\sim} & S^{-1}M \\ \uparrow & & \uparrow \text{ not necessarily} \\ R & \xrightarrow{\sim} & M \\ & & \text{injective} \end{array}$$

Special Case. Let R be an integral domain and take $S = R \setminus \{0\}$ so that $S^{-1}R$ is the field of fractions.

Now let M be an R -module with localization $S^{-1}M$ and consider a subset

$$\begin{array}{ccc} M & \longrightarrow & S^{-1}M \\ \uparrow & & \uparrow \\ A & \longrightarrow & S^{-1}A \end{array}$$

If A is a maximal R -linearly independent set in M then you will show on HW2 that $S^{-1}A$ is a maximal $S^{-1}R$ -linearly independent set in $S^{-1}M$. Then since $S^{-1}R$ is a field (the field of fractions of R),

and since the function $A \rightarrow S^{-1}A$ defined by $a \mapsto [a/1]$ is injective (because A is an independent set), we can apply Steinitz Exchange over $S^{-1}R$ to get:

Theorem. *Let M be a module over an integral domain R . Then any two maximal R -linearly independent subsets of M have the same cardinality, which we call the rank of M .*

Corollary. If R is an integral domain then for any two sets A and B we have

$$R^{\oplus A} \cong R^{\oplus B} \implies |A| = |B|.$$

That is, integral domains have the IBN property.

But recall that not every module over an integral domain is free (has a basis).

Example: Every nontrivial finite abelian group has rank 0 as a $\mathbb{Z}$ -module but is not free because $\emptyset$ is not a spanning set. [By convention we set $\mathbb{Z}^{\oplus \emptyset} := 0$, so the trivial group is free.]

Thus we have seen two kinds of $\mathbb{Z}$ -modules:

- free abelian groups;
- finite abelian groups.

Are there any other kinds?

In a certain sense, NO. If G is a finitely generated abelian group of rank n , then G is isomorphic to a direct sum

$$G \cong \mathbb{Z}^{\oplus n} \oplus T,$$

where T is finitely generated of rank 0, and hence finite.

To make the theorem more useful we'll try to prove it over a general PID.

So let's get started. First we need a definition.

Definition. Let R be a commutative ring and let M be an R -module. We say that $m \in M$ is a *torsion element* if there exists $r \in R \setminus \{0\}$ such that $rm = 0$. We denote the set of torsion elements by

$$\text{Tor}_R(M) := \{m \in M : \exists r \in R \setminus \{0\}, rm = 0\}.$$

If R is a domain then $\text{Tor}_R(M) \subseteq M$ is a submodule.

Proof. Let $m, n \in \text{Tor}_R(M)$, so there exist $r, s \in R \setminus \{0\}$ such that $rm = sn = 0$. Since R is a domain we also have $rs \in R \setminus \{0\}$. Then for all $\alpha \in R$ we have

$$\begin{aligned} rs(m + \alpha n) &= s(rm) + \alpha r(sn) \\ &= s \cdot 0 + \alpha r \cdot 0 \\ &= 0, \end{aligned}$$

and hence $m + \alpha n \in \text{Tor}_R(M)$. □

Furthermore, the quotient module $M/\text{Tor}_R(M)$ is *torsion-free* (i.e., has no nonzero torsion elements).

Proof. Assume for contradiction that we have

$$m + \text{Tor}_R(M) \neq 0 + \text{Tor}_R(M)$$

(i.e., $m \in M \setminus \text{Tor}_R(M)$) and $r \in R \setminus \{0\}$ such that

$$\begin{aligned} 0 + \text{Tor}_R(M) &= r(m + \text{Tor}_R(M)) \\ &= rm + \text{Tor}_R(M). \end{aligned}$$

Hence $rm \in \text{Tor}_R(M)$. It follows that there exists $s \in R \setminus \{0\}$ such that

$$0 = s(rm) = (rs)m.$$

But since R is a domain we have $rs \in R \setminus \{0\}$ and this implies that $m \in \text{Tor}_R(M)$. Contradiction. □

Now I would like to prove the following:

- $M/\text{Tor}_R(M)$ is a free R -module;
- we have a direct sum of R -modules

$$M \cong (M/\text{Tor}_R(M)) \oplus \text{Tor}_R(M).$$

But these results are not true unless we further assume that R is a PID. And to get there we need a slightly better understanding of free modules.

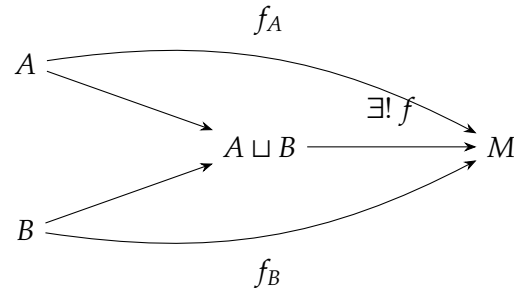
So let's go back to school.

Lemma. Let R be any ring and consider any two sets A, B . Then we have

$$R^{\oplus(A \sqcup B)} \cong R^{\oplus A} \oplus R^{\oplus B}.$$

Proof. Buckle your seatbelts; I'm going to give the correct proof of this.

First I note that the (external) disjoint union is the coproduct in the category of sets. That is, if M is any set with functions $f_A : A \rightarrow M$ and $f_B : B \rightarrow M$ then we have

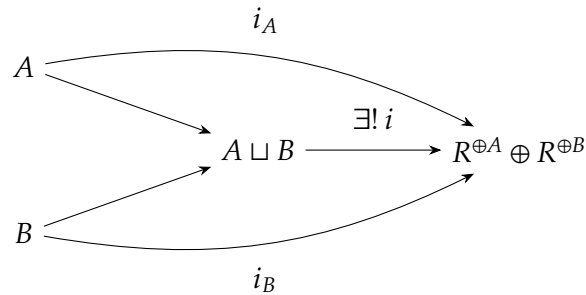


Then in retrospect we can say that $f_A = f|_A$ and $f_B = f|_B$ are the restrictions of the function f to A and B .

Now observe that we have canonical functions

$$i_A : A \longrightarrow R^{\oplus A} \longrightarrow R^{\oplus A} \oplus R^{\oplus B}, \quad i_B : B \longrightarrow R^{\oplus B} \longrightarrow R^{\oplus A} \oplus R^{\oplus B},$$

coming from the free module and direct sum constructions. Thus by the universal property of $A \sqcup B$ we have

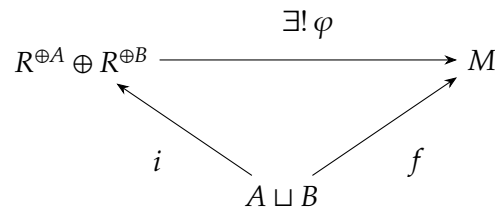


Now I claim that the function

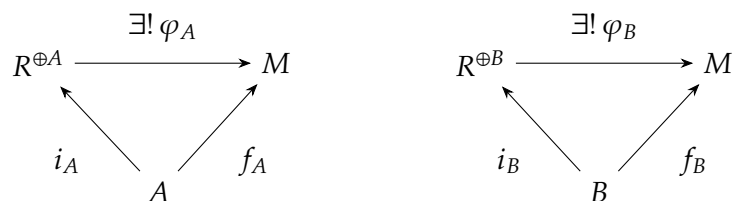
$$i : A \sqcup B \longrightarrow R^{\oplus A} \oplus R^{\oplus B}$$

satisfies the universal property of the free R -module over the set $A \sqcup B$.

In other words, I claim the following: Let M be any R -module. Then for any set function $f : A \sqcup B \rightarrow M$ there exists a unique R -module homomorphism φ such that



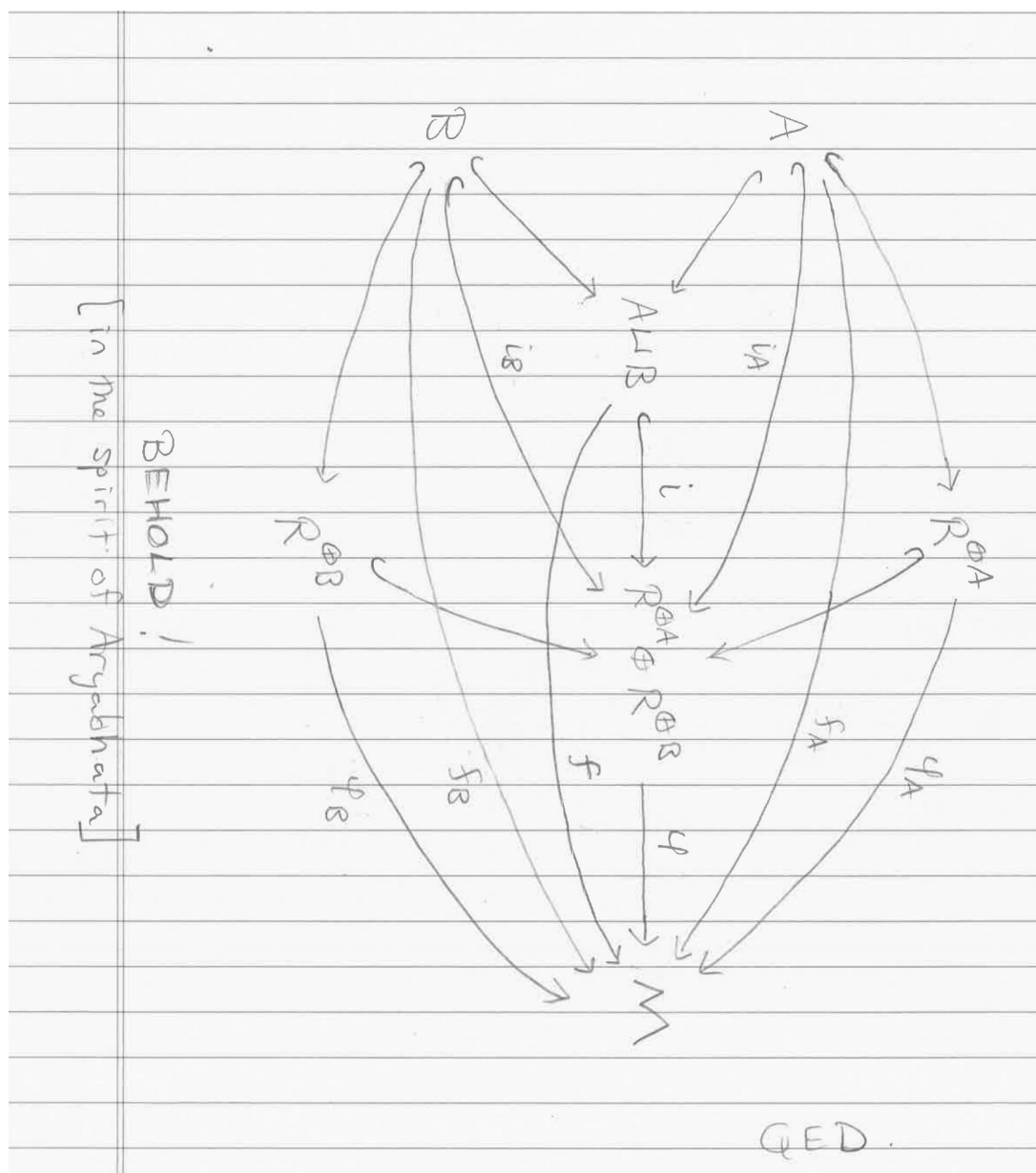
To prove this, consider the restrictions of f to A and B . Then by the universal properties of the free modules $R^{\oplus A}$ and $R^{\oplus B}$ there exist unique R -module homomorphisms φ_A and φ_B such that



Finally, the universal property of coproducts gives a unique R -module homomorphism

$$\varphi : R^{\oplus A} \oplus R^{\oplus B} \longrightarrow M$$

such that the following diagram commutes.



□

Oh my God, what have we done?!

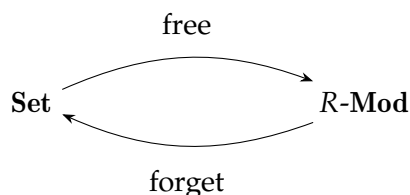
Maybe I could have given a shorter and less correct proof but I wanted to point your attention to the following general situation.

Given an R -module M let $|M|$ denote its underlying set. Then the universal property of free modules can be stated as

$$\mathrm{Hom}_{\mathrm{Set}}(A, |M|) = \mathrm{Hom}_{R\text{-Mod}}(R^{\oplus A}, M).$$

[A set function $A \rightarrow |M|$ is equivalent to an R -module homomorphism $R^{\oplus A} \rightarrow M$.]

In other words the “free functor” $A \mapsto R^{\oplus A}$ and the “forgetful functor” $M \mapsto |M|$ form an adjoint pair



where “free” is the “left adjoint” and “forget” is the “right adjoint.”

Now we can apply a powerful meta-theorem!

Theorem (Meta-Theorem). • *Right adjoints commute with limits.*

- *Left adjoints commute with colimits.*

Since the free functor $A \mapsto R^{\oplus A}$ is left adjoint and since the coproducts $\sqcup$ and $\oplus$ are examples of colimits [see HW2.1] we automatically obtain

$$R^{\oplus(A \sqcup B)} \cong R^{\oplus A} \oplus R^{\oplus B}.$$

[Based on our proof of this fact, you can imagine what the proof of the meta-theorem must look like. Would it make a good homework problem? ◡◡]

Modules over a PID to be continued...

41 Projective modules and the key lemma for PIDs

February 25, 2016

Right now we are proving a structure theorem for modules over a domain.

Recall what we did so far:

If M is a module over an integral domain R then the set of torsion elements

$$\mathrm{Tor}_R(M) := \{m \in M : \exists r \in R \setminus \{0\}, rm = 0\}$$

is actually a submodule of M , and the quotient $M/\mathrm{Tor}_R(M)$ is torsion-free (i.e., has no nonzero torsion elements).

In the general case we can't say much more, but in the special case that R is a PID and M is finitely generated, we will have the following extra properties:

- $M/\mathrm{Tor}_R(M)$ is a free module.
- The canonical surjection $M \rightarrow M/\mathrm{Tor}_R(M)$ splits so that

$$M \cong (M/\mathrm{Tor}_R(M)) \oplus \mathrm{Tor}_R(M).$$

“free part”

“torsion part”

This is an important theorem so I want to give a full proof. The proof is a bit involved but it is edifying! 😊

Lemma. Let R be any ring and let A and B be any sets. Then we have

$$R^{\oplus(A \sqcup B)} \cong R^{\oplus A} \oplus R^{\oplus B}.$$

I gave an explicit proof last time, but this also follows from a general principle. The free functor $A \mapsto R^{\oplus A}$ is left adjoint (to the forgetful functor from $R\text{-Mod}$ to $\mathbf{Set}$), so it commutes with colimits. Since the coproducts $\sqcup$ in $\mathbf{Set}$ and $\oplus$ in $R\text{-Mod}$ are examples of colimits, the result follows.

Remark. Left adjoint functors do not necessarily commute with limits. For example, we will see later that

$$R^{\oplus(A \times B)} \cong R^{\oplus A} \otimes_R R^{\oplus B},$$

where $\times$ is the (Cartesian) product in $\mathbf{Set}$ and $\otimes_R$ is a thing that is not the product in $R\text{-Mod}$.

Lemma. Let A be any set and let $\varphi : M \rightarrow R^{\oplus A}$ be any surjective R -module homomorphism. Then the short exact sequence

$$0 \longrightarrow \ker \varphi \longrightarrow M \xrightarrow{\varphi} R^{\oplus A} \longrightarrow 0$$

splits and we obtain a direct sum

$$M \cong R^{\oplus A} \oplus \ker \varphi.$$

Proof. We will construct a section, i.e., an R -module homomorphism

$$\delta : R^{\oplus A} \longrightarrow M \quad \text{such that} \quad \varphi \circ \delta = \text{id}_{R^{\oplus A}}.$$

Since φ is surjective we can choose for each element $a \in A$ a preimage $m_a \in M$ of the basis element $i_a \in R^{\oplus A}$:

$$\varphi(m_a) = i_a.$$

[This requires the axiom of choice if A is infinite.] Now consider the set function $A \rightarrow M$ defined by $a \mapsto m_a$. Since $R^{\oplus A}$ is free there exists a unique R -module homomorphism $\delta : R^{\oplus A} \rightarrow M$ such that

$$\begin{array}{ccc} R^{\oplus A} & \xrightarrow{\exists! \delta} & M \\ & \swarrow i \quad \searrow a \mapsto m_a & \\ & A & \end{array}$$

Note that for all $a \in A$ we have

$$\varphi \circ \delta(i_a) = \varphi(\delta(i_a)) = \varphi(m_a) = i_a.$$

Then since $\varphi \circ \delta : R^{\oplus A} \rightarrow R^{\oplus A}$ acts like the identity on a basis, it is the identity. We conclude that δ is a section of φ .

Why does this give us a direct sum

$$M \cong R^{\oplus A} \oplus \ker \varphi, ?$$

Well, we already know this holds for abelian groups [MTH 761 HW 3.5], but I'll prove it again to show that it works for R -modules.

The section $\delta : R^{\oplus A} \rightarrow M$ is injective since for all $r_1, r_2 \in R^{\oplus A}$ we have

$$\delta(r_1) = \delta(r_2) \implies \varphi(\delta(r_1)) = \varphi(\delta(r_2)) \implies r_1 = r_2.$$

Now we will show that M is an internal direct sum

$$M = \delta(R^{\oplus A}) \oplus \ker \varphi.$$

First, why is $M = \delta(R^{\oplus A}) + \ker \varphi$?

Pick any $m \in M$. We want to find $n \in \ker \varphi$ and $\delta(r) \in \delta(R^{\oplus A})$ such that

$$m = \delta(r) + n.$$

We will choose

$$r := \varphi(m) \quad \text{and} \quad n := m - \delta(r).$$

Then we check that $n \in \ker \varphi$ since

$$\begin{aligned}\varphi(n) &= \varphi(m - \delta(r)) \\ &= \varphi(m) - \varphi(\delta(r)) \\ &= r - r \\ &= 0.\end{aligned}$$

Next, why is

$$\delta(R^{\oplus A}) \cap \ker \varphi = 0, ?$$

Pick any element $m \in \delta(R^{\oplus A}) \cap \ker \varphi$. Since $m \in \delta(R^{\oplus A})$ there exists $\sum_a r_a i_a \in R^{\oplus A}$ such that

$$m = \delta\left(\sum_a r_a i_a\right) = \sum_a r_a \delta(i_a).$$

Then since $m \in \ker \varphi$ we have

$$\begin{aligned}0 &= \varphi(m) = \varphi\left(\sum_a r_a \delta(i_a)\right) \\ &= \sum_a r_a \varphi(\delta(i_a)) = \sum_a r_a i_a,\end{aligned}$$

and it follows that

$$m = \delta\left(\sum_a r_a i_a\right) = \delta(0) = 0.$$

Finally, since δ is injective we have

$$M = \delta(R^{\oplus A}) \oplus \ker \varphi \cong R^{\oplus A} \oplus \ker \varphi.$$

□

Remark. Just as with Lemma 1, this massive proof can also be shrunk into a general principle.

Definition. Let C be an abelian category. An object $P \in C$ is called *projective* if every short exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow P \longrightarrow 0$$

splits.

Then we can rephrase the lemma.

Lemma. Every free module is projective.

OK fine. Lemmas 1 and 2 are very general facts about free modules that hold over any ring. Here is the very specific fact about PIDs that makes the structure theorem work.

Lemma (Key Lemma About PIDs). Let R be a PID and consider a free module $R^{\oplus A}$ of rank $|A|$. Then any submodule $M \subseteq R^{\oplus A}$ is free of rank at most $|A|$.

Proof (induction on rank). If $|A| = 1$ then $R^{\oplus A} = R$. Then since R is a PID, every submodule (i.e., ideal) $M \subseteq R$ is free (i.e., principal), $M = (r)$. If $r = 0$ then M has rank 0 and if $r \neq 0$ then M has rank 1.

If $|A| = n$ then $R^{\oplus A} = R^{\oplus n}$. Consider the surjective homomorphism

$$\pi : R^{\oplus n} \twoheadrightarrow R^{\oplus(n-1)}$$

defined by

$$\pi(r_1, \dots, r_n) := (r_1, \dots, r_{n-1}).$$

Note that $\ker \pi \cong R$. Now let $M \subseteq R^{\oplus n}$ be any submodule and consider the restricted homomorphism

$$\varphi := \pi|_M : M \longrightarrow R^{\oplus(n-1)}.$$

By induction, the image $\varphi(M) \subseteq R^{\oplus(n-1)}$ is free of rank at most $n - 1$. Then by Lemma 2, the surjective homomorphism

$$\varphi : M \twoheadrightarrow \varphi(M)$$

splits to give

$$M \cong \varphi(M) \oplus \ker \varphi.$$

Then since

$$\ker \varphi = \ker(\pi|_M) \subseteq \ker \pi \cong R$$

we see that $\ker \varphi$ is free of rank at most 1.

Putting these together gives

$$M \cong \varphi(M) \oplus \ker \varphi,$$

where $\varphi(M)$ is free of rank at most $n - 1$ and $\ker \varphi$ is free of rank at most 1. Finally, we use Lemma 1 [a direct sum of free modules is free with basis given by the disjoint union of bases] to conclude that M is free of rank at most

$$(n - 1) + 1 = n.$$

□

Remark. Maybe this holds also for free modules of infinite rank and our induction can be made transfinite; I don't really care.

42 Finitely generated modules over a PID, Part I

March 1, 2016

Modules over a PID continued...

Recall what we have so far.

Lemma. The direct sum of free modules is free:

$$R^{\oplus A} \oplus R^{\oplus B} \cong R^{\oplus (A \sqcup B)}.$$

Lemma. Free modules are projective. That is, every short exact sequence of the form

$$0 \longrightarrow M \longrightarrow N \longrightarrow R^{\oplus A} \longrightarrow 0$$

splits.

Lemmas 1 and 2 are true over any ring. We used them to prove the following key property for modules over a PID.

Lemma (Key Lemma for PIDs). Let R be a PID. Then any submodule of the free module $R^{\oplus A}$ is free of rank at most $|A|$.

Also recall the following. If R is a domain and M is an R -module, then the set of torsion elements

$$\text{Tor}_R(M) := \{m \in M : \exists r \in R \setminus \{0\}, rm = 0\}$$

is a submodule and the quotient module $M/\text{Tor}_R(M)$ is torsion-free.

Now we can prove Part I of the Fundamental Theorem of Finitely Generated Modules over a PID.

Theorem (FTFGMPID, Part I). *Let R be a PID and let M be a finitely generated R -module. Then we can express M as a direct sum of a free module and a torsion module.*

In particular, if $\text{Tor}_R(M) \subseteq M$ is the submodule of torsion elements, then we have

- $M/\text{Tor}_R(M)$ is free, and
-

$$M \cong (M/\text{Tor}_R(M)) \oplus \text{Tor}_R(M).$$

Proof. To show that $M/\text{Tor}_R(M)$ is free, we will actually prove the more general result that any finitely generated torsion-free R -module is free.

So let F be a torsion-free R -module with finite spanning set $S \rightarrow F, s \mapsto f_s$. We will write $F = (S)$ to denote that S spans F . Now let $I \subseteq S$ be a maximal linearly independent subset so that $(I) \cong R^{\oplus I}$ is a free module.

For each $t \in S \setminus I$, I claim that there exists some $0 \neq r_t \in R$ such that

$$r_t f_t \in (I).$$

Indeed, since the set $I \cup \{t\}$ is not linearly independent (by maximality of I), we must have some nontrivial linear relation

$$r_t f_t + \sum_{i \in I} r_i f_i = 0.$$

Since I is independent we must have $r_t \neq 0$. Thus we obtain $0 \neq r_t \in R$ with

$$r_t f_t = - \sum_{i \in I} r_i f_i \in (I),$$

as desired. Now consider the product

$$r := \prod_{t \in S \setminus I} r_t \in R,$$

which exists because S is finite and is nonzero because R is a domain.

Now I claim that

$$rF = r(S) \subseteq (I).$$

Indeed, given an arbitrary element

$$f = \sum_{s \in S} r_s f_s \in (S) = F,$$

we have $r f_s \in (I)$ when $s \in I$ (obviously), and when $s = t \in S \setminus I$ we can use commutativity of R to write

$$r = r' r_t$$

for some $r' \in R$, so that

$$r f_s = r f_t = r' r_t f_t \in (I).$$

By linearity we obtain

$$r f = r \left(\sum_{s \in S} r_s f_s \right) = \sum_{s \in S} r_s r f_s \in (I),$$

as desired.

Now since $rF = r(S) \subseteq (I)$ and since $(I) \cong R^{\oplus I}$ is free, the Key Lemma on PIDs tells us that rF is free.

Finally, since F is torsion-free and since $r \neq 0$, the R -module homomorphism

$$F \xrightarrow{r} F, \quad f \mapsto r f$$

has trivial kernel, so the 1st Isomorphism Theorem implies that

$$F \cong F/\ker(r \cdot) \cong \text{im}(r \cdot) = rF.$$

We conclude that F is free.

In summary, we have shown that if M is a finitely generated module over a PID then

$$M/\text{Tor}_R(M)$$

is a free module. Then it follows from Lemma 2 that the short exact sequence

$$0 \longrightarrow \text{Tor}_R(M) \longrightarrow M \longrightarrow M/\text{Tor}_R(M) \longrightarrow 0$$

splits, so we obtain an isomorphism

$$M \cong (M/\mathrm{Tor}_R(M)) \oplus \mathrm{Tor}_R(M).$$

□

In what sense is the decomposition in the theorem unique?

Suppose that M has rank k and let $S = R \setminus \{0\}$. Using the fact that the “extension functor” $M \mapsto S^{-1}M$ is left adjoint to the “restriction functor” $S^{-1}M \mapsto M$ (details omitted) tells us that the direct sum (a colimit) is preserved under localization:

$$S^{-1}M \cong S^{-1}(M/\mathrm{Tor}_R(M)) \oplus S^{-1}\mathrm{Tor}_R(M).$$

We can also think of this as a direct sum of $S^{-1}R$ -modules. Since $S^{-1}R$ is a field (the field of fractions of R), all three modules are free, so we must have

$$\mathrm{rank}(S^{-1}M) = \mathrm{rank}(S^{-1}(M/\mathrm{Tor}_R(M))) + \mathrm{rank}(S^{-1}\mathrm{Tor}_R(M))$$

by Lemma 1. Finally, we can apply HW2 Problem 4(c) to obtain

$$k = \mathrm{rank}(M) = \mathrm{rank}(M/\mathrm{Tor}_R(M)).$$

Since $M/\mathrm{Tor}_R(M)$ is free, this implies

$$M/\mathrm{Tor}_R(M) \cong R^{\oplus k}.$$

In summary. Let M be a finitely generated module of rank k over a PID R . Then we have

$$M \cong R^{\oplus k} \oplus \mathrm{Tor}_R(M).$$

After the break we’ll prove the FTFGMPID, Part II, which states that the torsion module $\mathrm{Tor}_R(M)$ is a direct sum of cyclic modules.

[Recall the fundamental theorem of finite abelian groups.]

To do this correctly we will first transform the problem into matrix algebra as follows.

Let M be a finitely generated module over a PID R . By definition (of “finitely generated”) we have a surjective homomorphism from a free module:

$$\begin{array}{ccc} R^{\oplus A} & \xrightarrow{\varphi} & M \\ & \nwarrow i \quad \nearrow & \\ & A & \end{array}$$

We can also express the surjectivity of φ by saying that the following sequence is exact:

$$R^{\oplus A} \xrightarrow{\varphi} M \longrightarrow 0.$$

Now consider the kernel

$$\ker \varphi \subseteq R^{\oplus A}.$$

By the Key Lemma for PIDs we know that $\ker \varphi$ is free of rank at most $|A|$; let's say

$$\ker \varphi \cong R^{\oplus B} \quad \text{for some } |B| \leq |A|.$$

Putting everything together gives a short exact sequence

$$0 \longrightarrow R^{\oplus B} \xrightarrow{\psi} R^{\oplus A} \xrightarrow{\varphi} M \longrightarrow 0,$$

where

$$R^{\oplus B} \cong \operatorname{im} \psi = \ker \varphi \subseteq R^{\oplus A}.$$

Note that the map φ was uniquely determined by the inclusion $A \hookrightarrow M$, so the structure of the module is really encoded by the map ψ .

Furthermore, since

$$R^{\oplus B} \xrightarrow{\psi} R^{\oplus A}$$

is a map of free modules, we can encode it as an $|A| \times |B|$ matrix after choosing bases for $R^{\oplus A}$ and $R^{\oplus B}$.

The game is to choose bases so that the matrix $[\psi]$ is as nice as possible and then to read off the structure of the module from the matrix.

After the break I'll give you an algorithm to express any matrix $[\psi]$ over a PID in a standard form called the **Smith Normal Form**. The algorithm will be a generalization of the Euclidean Algorithm.

43 Midterm review

March 3, 2016

Today: Review for Midterm.

The theme for MTH 762 is algebraic structures with more than one binary operation.

To motivate this we looked at the category **Ab** of abelian groups. It differs from the category **Grp** of all groups in two significant ways.

1. Coproduct, cokernel, colimits in general are much nicer in $\mathbf{Ab}$

Given $A, B \in \mathbf{Ab}$, their categorical product is also their coproduct and we call it the direct sum

$$A \oplus B := \{(a, b) : a \in A, b \in B\}.$$

Proof. Consider the injective homomorphisms

$$i_A : A \longrightarrow A \oplus B, \quad a \longmapsto (a, 0_B),$$

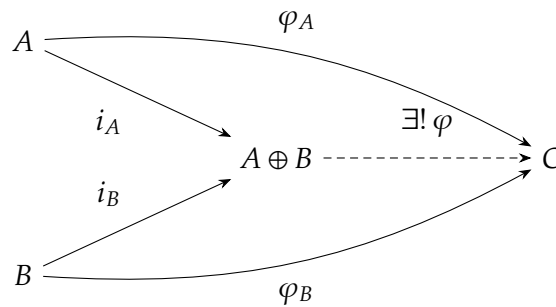
and

$$i_B : B \longrightarrow A \oplus B, \quad b \longmapsto (0_A, b).$$

Now let C be any abelian group and consider two homomorphisms

$$\varphi_A : A \longrightarrow C \quad \& \quad \varphi_B : B \longrightarrow C.$$

We want to show that there exists a unique homomorphism $\varphi : A \oplus B \rightarrow C$ such that



If such a φ exists then it must satisfy

$$\begin{aligned} \varphi(a, b) &= \varphi((a, 0_B) + (0_A, b)) \\ &= \varphi(a, 0_B) + \varphi(0_A, b) \\ &= \varphi(i_A(a)) + \varphi(i_B(b)) \\ &= \varphi_A(a) + \varphi_B(b). \end{aligned}$$

Certainly such a function exists, but is it a *group homomorphism*?

Yes. Since C is abelian we have

$$\begin{aligned} \varphi_A(a_1 + a_2) + \varphi_B(b_1 + b_2) &= (\varphi_A(a_1) + \varphi_A(a_2)) + (\varphi_B(b_1) + \varphi_B(b_2)) \\ &= (\varphi_A(a_1) + \varphi_B(b_1)) + (\varphi_A(a_2) + \varphi_B(b_2)), \end{aligned}$$

as desired. □

2. **Ab** is enriched over **Ab**

Given any group $G \in \mathbf{Grp}$ and any set $S \in \mathbf{Set}$ we can give the hom set

$$\mathrm{Hom}_{\mathbf{Set}}(S, G)$$

the structure of a group by defining

$$(fg)(s) := f(s)g(s).$$

If S is also a group then the product of homomorphisms is not necessarily a homomorphism:

$$\begin{aligned} (fg)(st) &= f(st)g(st) \\ &= f(s)f(t)g(s)g(t) \\ &\neq f(s)g(s)f(t)g(t) \\ &= (fg)(s)(fg)(t). \circ \frown \circ \end{aligned}$$

However, if G is abelian then the above equation becomes true and we conclude that the hom set

$$\mathrm{Hom}_{\mathbf{Grp}}(S, G)$$

is a (necessarily abelian) group. Given any other abelian group H we can also consider the composition function

$$\mathrm{Hom}_{\mathbf{Grp}}(G, H) \times \mathrm{Hom}_{\mathbf{Grp}}(S, G) \longrightarrow \mathrm{Hom}_{\mathbf{Grp}}(S, H), \quad (f, g) \longmapsto f \circ g. \quad (*)$$

Finally, we observe that composition respects the group structures (we say that $(*)$ is “bi-additive”):

- $(f_1 + f_2) \circ g = (f_1 \circ g) + (f_2 \circ g),$
- $f \circ (g_1 + g_2) = (f \circ g_1) + (f \circ g_2).$

In summary, we say that “**Ab** is enriched over **Ab**.” This means that for any abelian groups $A, B \in \mathbf{Ab}$ the hom set

$$\mathrm{Hom}_{\mathbf{Ab}}(A, B)$$

has a natural abelian group structure and composition of morphisms is bi-additive with respect to this structure. $\circ \frown \circ$

What is a ring?

Consider an abelian group $A \in \mathbf{Ab}$ and its set of endomorphisms

$$\mathrm{End}_{\mathbf{Ab}}(A) := \mathrm{Hom}_{\mathbf{Ab}}(A, A).$$

By the above remarks we know that

- $(\text{End}_{\mathbf{Ab}}(A), +, 0_{AA})$ is an abelian group,
- $(\text{End}_{\mathbf{Ab}}(A), \circ, \text{id}_A)$ is a monoid,
- composition distributes over addition.

We will say that an abstract structure

$$(R, +, 0, \cdot, 1)$$

is a ring if it satisfies these three properties.

Theorem (“Cayley Theorem”). *This definition is not too general. That is, given an abstract ring R there exists an abelian group A and an injective ring homomorphism*

$$R \hookrightarrow \text{End}_{\mathbf{Ab}}(A).$$

Proof. Let $R \mapsto |R|$ denote the forgetful functor $\mathbf{Rng} \rightarrow \mathbf{Ab}$. Then the function

$$\lambda : R \longrightarrow \text{End}_{\mathbf{Ab}}(|R|), \quad r \longmapsto \lambda_r,$$

defined by

$$\lambda_r(a) := r \cdot a,$$

is an injective ring homomorphism.

[Why does right multiplication by r not work?] □

What is a module?

There are heuristic and formal ways to motivate this. Today I’ll be formal.

Definition. A left R -module is a pair (A, φ) where A is an abelian group and

$$\varphi : R \longrightarrow \text{End}_{\mathbf{Ab}}(A)$$

is a ring homomorphism.

[The “Cayley Theorem” says that R is a (faithful) left module over itself. Its submodules are called “left ideals.”]

There are two ways to study R -modules.

1. Internal

Let M be an R -module. An R -submodule is a subgroup $N \leq (M, +, 0)$ such that

$$n_1, n_2 \in N \ \& \ r \in R \implies n_1 + rn_2 \in N.$$

Let $\mathcal{L}_R(M)$ denote the collection of R -submodules of M . This is a (modular) lattice with

$$1 = M, \quad 0 = (0_M), \quad \wedge = \cap, \quad \vee = +.$$

In this language, the 1st, 2nd & 3rd Isomorphism Theorems and the Jordan–Hölder Theorem carry over word for word. One needs only to check that the left $(R, \cdot, 1)$ -action doesn't ruin things.

[Remark: The situation for rings is more awkward because there are two distinguished kinds of subobjects:

$$\text{ideals} = \text{kernels}, \quad \text{subrings} = \text{images}.$$

For R -modules we have

$$\text{submodules} = \text{kernels} = \text{images},$$

which is even better than in **Grp**!]

You should know the statements of the Isomorphism Theorems and be able to prove bits and pieces of them. [I won't ask for the hard bits like Zassenhaus or Schreier.]

We define the internal direct sum of R -modules as follows.

Definition. Let $A, B \in \mathcal{L}_R(M)$. If

$$A + B = M \quad \text{and} \quad A \cap B = (0),$$

then we will write $M = A \oplus B$.

2. External

We can rephrase the definition by saying that an R -module is an “additive functor”

$$F : R \longrightarrow \mathbf{Ab}.$$

[We think of R as a category with one object that is “enriched over $\mathbf{Ab}$.”]

In this case it becomes obvious that a morphism of R -modules should be a natural transformation.

In more mundane terms we say that a group homomorphism $\varphi : M \rightarrow N$ is a morphism of R -modules if for all $r \in R$ we have

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & N \\ r \cdot \downarrow & & \downarrow r \cdot \\ M & \xrightarrow{\varphi} & N \end{array} \quad \varphi(r(m)) = r(\varphi(m)) \quad \text{for all } m \in M.$$

The resulting category $R\text{-Mod}$ shares so many properties with $\mathbf{Ab}$ that we go so far as to call it an “abelian category.”

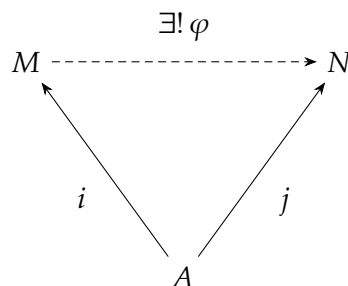
[Remark: $\mathbb{Z}$ is the initial object in $\mathbf{Rng}$ and $\mathbb{Z}\text{-Mod} = \mathbf{Ab}$. This is the sense in which $\mathbf{Ab}$ is the prototype for $R\text{-Mod}$.]

That was the general framework. Then we zoomed in to investigate particularly nice modules and modules over particularly nice rings.

1. Nice Modules

Let R be a ring and consider a set A . The *free R -module generated by A* is a pair (i, M) where

- M is an R -module and $i : A \rightarrow M$ is a function;
- if N is any R -module and $j : A \rightarrow N$ is any function then there exists a unique R -module homomorphism $\varphi : M \rightarrow N$ such that



If such a pair (i, M) exists then the function $i : A \rightarrow M$ is injective and the module M is unique up to isomorphism.

Alternatively, we would like to define a “free functor”

$$F : \mathbf{Set} \longrightarrow R\text{-Mod}$$

that is left adjoint to the “forgetful functor”

$$U : R\text{-Mod} \longrightarrow \mathbf{Set}$$

in the sense that for all $A \in \mathbf{Set}$ and $M \in R\text{-Mod}$ we have a “natural bijection”

$$\mathrm{Hom}_{\mathbf{Set}}(A, U(M)) = \mathrm{Hom}_{R\text{-Mod}}(F(A), M).$$

[A linear function is defined by its values on a basis.]

Theorem. *Free modules exist.*

Proof. The coproduct $R^{\oplus A}$ and the function $i : A \rightarrow R^{\oplus A}$ defined by

$$i_a(b) = \begin{cases} 1_R, & a = b, \\ 0_R, & a \neq b, \end{cases}$$

satisfy the desired universal property. $\square$

Now let M be an R -module and consider an “indexed subset” $A \rightarrow M$ denoted by $a \mapsto m_a$. Let

$$\varphi : R^{\oplus A} \longrightarrow M$$

be the canonical R -module homomorphism from the free module. We say that $A \rightarrow M$ is

- linearly independent,
- spanning,
- a basis,

depending on whether $\varphi : R^{\oplus A} \rightarrow M$ is

- injective,
- surjective,
- bijective,

respectively. Thus a module has a basis if and only if it is free. It is straightforward to check the following general property:

Each basis is a maximal linearly independent set and a minimal spanning set.

But the converse is not true in general.

[Example: Look at the bases, maximal independent sets and minimal spanning sets of $\mathbb{Z}$ as a $\mathbb{Z}$ -module.]

2. Modules over Nice Rings

If $R = K$ is a field then it is straightforward to check that

Each maximal linearly independent set and each minimal spanning set is a basis.

Then assuming Zorn’s Lemma (existence of maximal linearly independent sets) we conclude that every vector space is free.

Furthermore, the Steinitz Exchange Lemma tells us that for all sets A, B we have

$$K^{\oplus A} \cong K^{\oplus B} \implies |A| = |B|,$$

hence the dimension of a vector space is well-defined.

Then, using the trick of “localization,” we can prove that if R is an integral domain then every maximal linearly independent subset of M has the same size, which we call the rank of M .

Remark. The idea is straightforward but there are many details to check [see HW2 solutions].

Remark. The same is not true for minimal spanning sets [for example, $\{1\}$ and $\{2, 3\}$ are minimal spanning in $\mathbb{Z}$ but they don’t have the same size].

But not every maximal linearly independent set is a basis because there may be torsion elements [$m \in M$ such that there exists $r \in R \setminus \{0\}$ with $rm = 0$]. Let $\text{Tor}_R(M)$ be the set of torsion elements. If R is a domain then $\text{Tor}_R(M) \subseteq M$ is a submodule and the quotient $M/\text{Tor}_R(M)$ is torsion-free [has no nonzero torsion elements].

Similarly, if M is a module of rank k over a domain R and if $F \subseteq M$ is a maximal free submodule then we have $F \cong R^{\oplus k}$ and the quotient M/F is torsion [consists of torsion elements].

For a general domain R it is difficult to compare these concepts [the torsion submodule & maximal free submodules], however if R is a PID then we obtain the following:

Theorem (FTFGMPID, Part I). *Let R be a PID and let M be an R -module. If M is finitely generated of rank k then we have*

$$M \cong R^{\oplus k} \oplus \text{Tor}_R(M).$$

You do not need to memorize the proof of this, but you should know that it’s based on the following four lemmas.

Lemma. The direct sum of free modules is free with basis given by the disjoint union of bases:

$$R^{\oplus A} \oplus R^{\oplus B} \cong R^{\oplus (A \sqcup B)}.$$

Lemma. If F is a free R -module, then every short exact sequence

$$0 \longrightarrow M \longrightarrow N \longrightarrow F \longrightarrow 0$$

of R -modules splits.

Lemma. If R is a PID then every submodule of $R^{\oplus A}$ is free of rank at most $|A|$.

Lemma. If R is a PID then every finitely generated torsion-free R -module is free.

[None of these lemmas is trivial to prove from scratch, and I would never ask you to do such a thing on an exam.]

44 Free resolutions and matrix notation

March 17, 2016

Welcome back.

Before the break we proved the

Theorem (F.T.F.G.M.P.I.D., Part I). *Let R be a PID and let M be a finitely generated R -module of rank k . Then we have*

$$M \cong R^{\oplus k} \oplus \operatorname{Tor}_R(M).$$

Part II of the FTFGMPID will explain the structure of the torsion submodule

$$\operatorname{Tor}_R(M) \subseteq M,$$

and to give the correct proof of this we will transform the problem into “matrix algebra,” as follows.

If M is generated by the set $A \rightarrow M$, then by definition there exists a canonical surjective homomorphism of R -modules

$$R^{\oplus A} \xrightarrow{\pi} M.$$

Now consider

$$\ker \pi \subseteq R^{\oplus A}.$$

By the “Key Lemma for PIDs” we know that $\ker \pi$ is a free R -module of rank at most $|A|$. That is, we have a basis $B \rightarrow \ker \pi$ and a canonical isomorphism

$$R^{\oplus B} \xrightarrow{e} \ker \pi,$$

which we can extend to an injective map

$$R^{\oplus B} \xrightarrow{\varphi} R^{\oplus A}$$

with $\text{im } \varphi = \ker \pi$. In summary, we have obtained a (short) exact sequence

$$0 \longrightarrow R^{\oplus B} \xrightarrow{\varphi} R^{\oplus A} \xrightarrow{\pi} M \longrightarrow 0, \quad (*)$$

called a “free resolution of length 1.”

Remark. One can show (but we won’t) that R is a PID if and only if every R -module has a resolution of this form.

This is another good motivation for the PID concept.

Remark. If the short exact sequence $(*)$ splits, then M is isomorphic to a submodule of $R^{\oplus A}$, hence M is free. In particular, this implies that $\text{Tor}_R(M) = 0$. Conversely, if $\text{Tor}_R(M) = 0$ and if $|A| < \infty$, then we proved during the FTFGMPID that M is free, and hence $(*)$ splits.

In summary, the splitting of $(*)$ is controlled by $\text{Tor}_R(M)$ and whether M is finitely generated.

Remark. Using other language [see the Midterm Exam Problem 1] we can say that

$$\pi : R^{\oplus A} \longrightarrow M$$

is the *cokernel* of

$$\varphi : R^{\oplus B} \longrightarrow R^{\oplus A}.$$

Thus the module M is determined up to isomorphism by the map φ .

This leads to a great idea:

We can replace the study of the module M by the study of the homomorphism

$$R^{\oplus B} \xrightarrow{\varphi} R^{\oplus A}.$$

For this reason we will now turn our attention to the study of homomorphisms between free modules.

Q: What is a “matrix”?

A: A homomorphism between finite rank free modules, expressed “in coordinates.”

Let R be a commutative ring and consider two R -modules M, N . We saw previously that the set

$$\text{Hom}_R(N, M)$$

carries a natural R -module structure defined by

$$(\varphi_1 + r\varphi_2)(n) := \varphi_1(n) + r\varphi_2(n)$$

for all $\varphi_1, \varphi_2 \in \text{Hom}_R(N, M)$, $r \in R$, and $n \in N$.

If M and N are both free then we have isomorphisms

$$M \cong R^{\oplus \text{rank}(M)} \quad \text{and} \quad N \cong R^{\oplus \text{rank}(N)},$$

but I want to emphasize that these isomorphisms are *not canonical*; the choice of specific isomorphisms is equivalent to the choice of specific bases for M and N .

Indeed, let

$$A = \{m_a\} \subseteq M \quad \text{and} \quad B = \{n_b\} \subseteq N$$

be bases. Then the corresponding isomorphisms are given by

$$\begin{aligned} R^{\oplus A} &\xrightarrow{\sim} M, & \sum_a r_a i_a &\mapsto \sum_a r_a m_a, \\ R^{\oplus B} &\xrightarrow{\sim} N, & \sum_b r_b i_b &\mapsto \sum_b r_b n_b. \end{aligned}$$

Now consider any R -linear map

$$\varphi : N \longrightarrow M.$$

Composing with the isomorphisms above gives a homomorphism from $R^{\oplus B}$ to $R^{\oplus A}$:

$$R^{\oplus B} \xrightarrow{\sim} N \xrightarrow{\varphi} M \xrightarrow{\sim} R^{\oplus A}.$$

It sends

$$\sum_b r_b i_b \mapsto \sum_b r_b n_b \mapsto \sum_b r_b \varphi(n_b) \mapsto ?$$

To compute the final step we need to express each $\varphi(n_b) \in M$ in terms of the basis $A = \{m_a\} \subseteq M$. Let's just say

$$\varphi(n_b) = \sum_a r_{ab} m_a.$$

Then the final step of the map is given by

$$\sum_b r_b \varphi(n_b) = \sum_b r_b \left(\sum_a r_{ab} m_a \right) = \sum_a \left(\sum_b r_{ab} r_b \right) m_a \mapsto \sum_a \left(\sum_b r_{ab} r_b \right) i_a,$$

and the full map is given by

$$\sum_b r_b i_b \mapsto \sum_a \left(\sum_b r_{ab} r_b \right) i_a.$$

Now we desperately need a good notation to work with this mess. The notation of “matrices” and “matrix multiplication” was invented by Cayley and Sylvester in the 1850s for just this purpose.

Definition. Suppose that $|A| < \infty$ and $|B| < \infty$. Then we will write

$$A = \{m_1, m_2, \dots, m_{|A|}\} \quad \text{and} \quad B = \{n_1, n_2, \dots, n_{|B|}\}.$$

We will express elements of M and N as “column vectors”:

$$\left[\sum_i r_i m_i \right]_A := \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_{|A|} \end{pmatrix}, \quad \left[\sum_j r_j n_j \right]_B := \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_{|B|} \end{pmatrix}.$$

And we will express a homomorphism $\varphi : N \rightarrow M$ as a “matrix”:

$$[\varphi]_{BA} := (r_{ij}) = \begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1|B|} \\ r_{21} & r_{22} & \cdots & r_{2|B|} \\ \vdots & \vdots & \ddots & \vdots \\ r_{|A|1} & r_{|A|2} & \cdots & r_{|A||B|} \end{pmatrix}.$$

Here the entries $r_{ij} \in R$ are defined by

$$\varphi(n_j) = \sum_i r_{ij} m_i.$$

In other words, the j th column of the matrix $[\varphi]_{BA}$ is given by

$$[\varphi(n_j)]_A = \begin{pmatrix} r_{1j} \\ r_{2j} \\ \vdots \\ r_{|A|j} \end{pmatrix}.$$

Now we want to define an operation on matrices so that for all $n \in N$ we have

$$[\varphi(n)]_A = [\varphi]_{BA} ? [n]_B.$$

The correct definition is forced on us by the requirement of “linearity.” If

$$[n]_B = \left[\sum_j r_j n_j \right]_B = \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_{|B|} \end{pmatrix},$$

then we must have

$$\begin{aligned} [\varphi(n)]_A &= \left[\varphi \left(\sum_j r_j n_j \right) \right]_A \\ &= \left[\sum_j r_j \varphi(n_j) \right]_A \\ &= \sum_j r_j [\varphi(n_j)]_A. \end{aligned}$$

And it follows that

$$\begin{aligned} [\varphi]_{BA}[n]_B &:= [\varphi(n)]_A \\ &= \sum_j r_j [\varphi(n_j)]_A \\ &= \sum_j (j\text{th entry of } [n]_B) (j\text{th column of } [\varphi]_{BA}). \end{aligned}$$

[Well, OK. If that's the correct definition then we'll learn to live with it.]

Based on this definition, we will also define the product of two “non-vector” matrices.

Let P be another free R -module with basis

$$C = \{p_1, p_2, \dots, p_{|C|}\} \subseteq P$$

and consider a homomorphism

$$\psi : P \longrightarrow N.$$

Then we will define the matrix $[\varphi]_{BA}[\psi]_{CB}$ such that for all $p \in P$ we have

$$([\varphi]_{BA}[\psi]_{CB})[p]_C = [\varphi]_{BA}([\psi]_{CB}[p]_C).$$

In other words, we will define

$$[\varphi]_{BA}[\psi]_{CB} := [\varphi \circ \psi]_{CA}.$$

Then to compute this matrix, we note that

$$\begin{aligned} j\text{th column of } [\varphi]_{BA}[\psi]_{CB} &= j\text{th column of } [\varphi \circ \psi]_{CA} \\ &= [\varphi(\psi(p_j))]_A \\ &= [\varphi]_{BA}[\psi(p_j)]_B \\ &= [\varphi]_{BA}(j\text{th column of } [\psi]_{CB}). \end{aligned}$$

Remarks:

- Wait a minute! Wasn't this notation supposed to clean up a mess? It looks like we just made the mess bigger.

Well, be patient. Linear algebra was the “category theory” of its day. At first it looks like a mess of abstract nonsense, until you've internalized the technology. Then you'll see that it's an amazingly efficient language.

- I defined matrices and matrix multiplication in terms of *columns*. However, the strength of the notation is that we can also think of it in terms of *rows*.

By convention we think of columns as “vectors” (i.e., lines) and rows as “covectors” (i.e., hyperplanes). The matrix notation balances these concepts in a beautiful way.

45 Matrix multiplication and change of basis

March 22, 2016

Last time I tried to discover the definition of matrix multiplication from scratch. The idea is as follows. Let R be a commutative ring and consider two natural numbers $m, n \in \mathbb{N}$. We know that the hom set

$$\text{Hom}_R(R^{\oplus n}, R^{\oplus m})$$

has a natural R -module structure, and if we choose specific bases for $R^{\oplus m}$ and $R^{\oplus n}$ then each

$$\varphi : R^{\oplus n} \longrightarrow R^{\oplus m}$$

is determined uniquely by $m \times n$ elements of R . How is the structure of $\text{Hom}_R(R^{\oplus n}, R^{\oplus m})$ reflected combinatorially by these $m \times n$ elements?

After playing around we were led to the following definition.

Definition. Let $\text{Mat}_{m \times n}(R)$ denote the set of “ $m \times n$ matrices over R ”

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} =: (a_{ij}).$$

The operations

$$(a_{ij}) + (b_{ij}) := (a_{ij} + b_{ij}) \quad \text{and} \quad r(a_{ij}) := (ra_{ij})$$

make $\text{Mat}_{m \times n}(R)$ into a free R -module of rank $m \times n$, which is canonically isomorphic to

$$(R^{\oplus m})^{\oplus n} \quad \text{or} \quad (R^{\oplus n})^{\oplus m}.$$

[See HW3.]

Given $A = (a_{ij}) \in \text{Mat}_{\ell \times m}(R)$ and $B = (b_{ij}) \in \text{Mat}_{m \times n}(R)$, we define the product matrix

$$AB = (c_{ij}) \in \text{Mat}_{\ell \times n}(R)$$

by

$$c_{ij} := \sum_{k=1}^m a_{ik} b_{kj}.$$

In particular, we have

$$\begin{aligned} (i, j)\text{th entry of } AB &= (i\text{th row of } A)(j\text{th column of } B), \\ i\text{th row of } AB &= (i\text{th row of } A)B, \\ j\text{th column of } AB &= A(j\text{th column of } B). \end{aligned}$$

One can check that the operation $(A, B) \mapsto AB$ is R -bilinear and associative (when defined), and there is a very good reason for this.

Theorem (Justifying the definition). *Choosing bases for $R^{\oplus n}$ and $R^{\oplus m}$ determines an isomorphism of R -modules*

$$\mathrm{Hom}_R(R^{\oplus n}, R^{\oplus m}) \xrightarrow{\sim} \mathrm{Mat}_{m \times n}(R).$$

Furthermore, choosing a basis for $R^{\oplus \ell}$ determines a commutative square

$$\begin{array}{ccc} \mathrm{Hom}_R(R^{\oplus m}, R^{\oplus \ell}) & \xrightarrow{\sim} & \mathrm{Mat}_{\ell \times m}(R) \\ \times \mathrm{Hom}_R(R^{\oplus n}, R^{\oplus m}) & & \times \mathrm{Mat}_{m \times n}(R) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_R(R^{\oplus n}, R^{\oplus \ell}) & \xrightarrow{\sim} & \mathrm{Mat}_{\ell \times n}(R) \end{array}$$

where the left arrow is composition of morphisms and the right arrow is multiplication of matrices.

Special Cases:

- Let M be an R -module. Since R is a cyclic module over itself (generated by $1 \in R$) we have an isomorphism

$$\mathrm{Hom}_R(R, M) \xrightarrow{\sim} M, \quad \varphi \mapsto \varphi(1).$$

If $M \cong R^{\oplus m}$ is free, then any choice of basis determines an isomorphism

$$R^{\oplus m} \cong \mathrm{Hom}_R(R, R^{\oplus m}) \xrightarrow{\sim} \mathrm{Mat}_{m \times 1}(R).$$

Thus we can identify $R^{\oplus m}$ with the module of *vectors* (i.e., column vectors).

- If M is an R -module then it is *not* generally true that

$$\mathrm{Hom}_R(M, R) \cong M.$$

[Example: $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) = 0 \not\cong \mathbb{Z}/2\mathbb{Z}$.]

However, if $M \cong R^{\oplus m}$ is a free module of finite rank, then choosing a basis gives an isomorphism

$$\mathrm{Hom}_R(R^{\oplus m}, R) \xrightarrow{\sim} \mathrm{Mat}_{1 \times m}(R) \cong R^{\oplus m}.$$

Here we are thinking of $R^{\oplus m}$ as a module of *covectors* (i.e., row vectors).

[For a general R -module M we define the dual R -module

$$M^{\vee} := \mathrm{Hom}_R(M, R).$$

More on this later.]

Matrix notation is nice because it is concrete and allows us to work with vectors and covectors simultaneously. However, we must pay for this convenience with the fact that coordinates are *not canonical*.

Change of Basis.

Let M be a free R -module. If $A, B \subseteq M$ are two bases for M then we obtain canonical isomorphisms

$$\begin{array}{ccc} & M & \\ \alpha \nearrow & & \nwarrow \beta \\ R^{\oplus A} & \xrightarrow{\beta^{-1} \circ \alpha} & R^{\oplus B} \end{array}$$

The composition

$$\beta^{-1} \circ \alpha : R^{\oplus A} \longrightarrow R^{\oplus B}$$

is called the *change-of-basis homomorphism*. If

$$A = \{a_1, \dots, a_n\} \quad \text{and} \quad B = \{b_1, \dots, b_m\},$$

then we can express it as a matrix

$$\begin{aligned} [\beta^{-1} \circ \alpha]_{AB} &= \left([\beta^{-1} \circ \alpha(a_1)]_B \cdots [\beta^{-1} \circ \alpha(a_n)]_B \right) \\ &= \begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mn} \end{pmatrix}, \end{aligned}$$

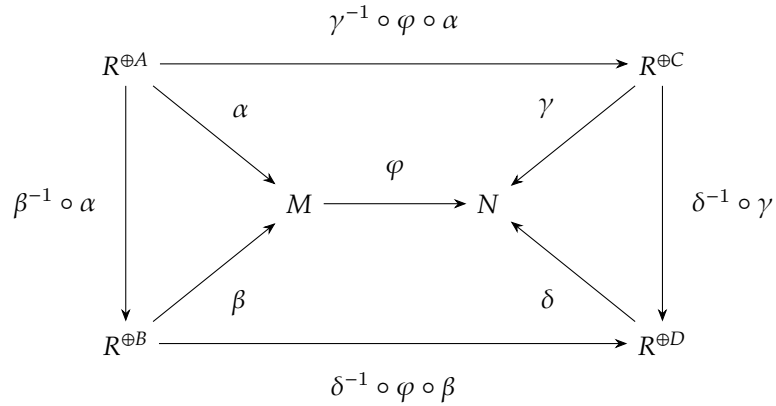
where

$$\beta^{-1} \circ \alpha(a_j) = \sum_i r_{ij} b_i.$$

This matrix is clearly invertible, with

$$[\beta^{-1} \circ \alpha]_{AB}^{-1} = [\alpha^{-1} \circ \beta]_{BA}.$$

Now let N be another free R -module with two bases $C, D \subseteq N$, and let $\varphi : M \rightarrow N$ be any homomorphism. We obtain a diagram:



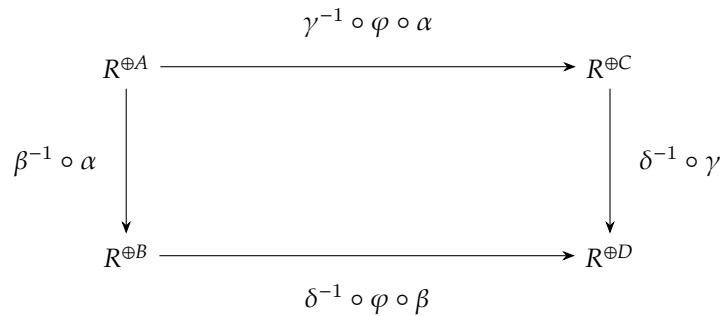
The matrix of φ in the bases A and C is defined by the morphism

$$[\varphi]_{AC} := [\gamma^{-1} \circ \varphi \circ \alpha]_{AC},$$

and the matrix of φ in B and D is defined by

$$[\varphi]_{BD} := [\delta^{-1} \circ \varphi \circ \beta]_{BD}.$$

To see how these matrices are related we just erase the middle part of the diagram:



We obtain

$$\begin{aligned} [\varphi]_{AC} &= [\gamma^{-1} \circ \varphi \circ \alpha]_{AC} \\ &= [(\gamma^{-1} \circ \delta) \circ (\delta^{-1} \circ \varphi \circ \beta) \circ (\beta^{-1} \circ \alpha)]_{AC} \\ &= [\gamma^{-1} \circ \delta]_{DC} [\varphi]_{BD} [\beta^{-1} \circ \alpha]_{AB}. \end{aligned}$$

In summary,

We can think of a matrix $Z \in \text{Mat}_{m \times n}(R)$ as a homomorphism $R^{\oplus n} \rightarrow R^{\oplus m}$ expressed with respect to some choice of bases for $R^{\oplus n}$ and $R^{\oplus m}$.

Furthermore, two such matrices Z_1 and Z_2 represent the same homomorphism with respect to different bases if and only if there exist invertible matrices $X \in GL_m(R)$ and $Y \in GL_n(R)$ such that

$$Z_1 = XZ_2Y.$$

We can also phrase this in the language of last semester (MTH 761).

We can define an action of the group $GL_m(R) \times GL_n(R)$ on the set $\text{Mat}_{m \times n}(R)$ by

$$(A, B) \cdot C := ACB^{-1}.$$

Recall from MTH 761 HW3.4(b) that the corresponding semidirect product

$$\text{Mat}_{m \times n}(R) \rtimes (GL_m(R) \times GL_n(R))$$

is isomorphic to the group of invertible block upper-triangular matrices

$$\left\{ \begin{pmatrix} A & C \\ 0 & B \end{pmatrix} : \begin{array}{l} A \in GL_m(R), \\ B \in GL_n(R), \\ C \in \text{Mat}_{m \times n}(R) \end{array} \right\}.$$

Our next goal is to classify the orbits of the action

$$GL_m(R) \times GL_n(R) \curvearrowright \text{Mat}_{m \times n}(R),$$

i.e., to determine when two $m \times n$ matrices represent the same homomorphism in different coordinates.

One way to do this is to define a “complete normal form”

$$\text{NF} : \text{Mat}_{m \times n}(R) \longrightarrow \text{Mat}_{m \times n}(R)$$

such that

$$\text{NF}(C_1) = \text{NF}(C_2) \iff C_1 \text{ and } C_2 \text{ are in the same orbit.}$$

46 R -algebras and elementary matrices

March 24, 2016

Let R be a commutative ring. Last time we defined the R -module of $m \times n$ matrices $\text{Mat}_{m \times n}(R)$ and discussed how “choosing bases” determines an isomorphism

$$\text{Hom}_R(R^{\oplus n}, R^{\oplus m}) \xrightarrow{\sim} \text{Mat}_{m \times n}(R).$$

In the special case that $m = n$ we will write

$$\text{Mat}_n(R) := \text{Mat}_{n \times n}(R).$$

Then from the R -bilinearity of matrix multiplication we obtain an isomorphism of R -algebras

$$\text{End}_R(R^{\oplus n}) \xrightarrow{\sim} \text{Mat}_n(R).$$

What?

Definition. Let R be a commutative ring. We define an R -algebra as a pair (i, S) , where

- S is an arbitrary ring;
- $i : R \rightarrow S$ is a ring homomorphism such that

$$\text{im } i \subseteq Z(S) := \{a \in S : ab = ba \text{ for all } b \in S\}.$$

We define a morphism of R -algebras

$$\psi : (i_1, S_1) \longrightarrow (i_2, S_2)$$

as a ring homomorphism $\psi : S_1 \rightarrow S_2$ satisfying

$$\begin{array}{ccc} S_1 & \xrightarrow{\psi} & S_2 \\ & \nwarrow i_1 \quad \nearrow i_2 & \\ & R & \end{array}$$

We'll call the resulting category $R\text{-Alg}$.

Remark. Let $i : R \rightarrow S$ be an R -algebra, and for all $r \in R$ define a function

$$\lambda_r : S \longrightarrow S \quad \text{by} \quad \lambda_r(s) := i(r)s.$$

You checked on HW2.5(b) that this defines a ring homomorphism

$$\lambda : R \longrightarrow \text{End}_{\text{Ab}}(S)$$

making S into an R -module.

Then we can say that the monoid structure on S is R -bilinear, in the sense that for all $a, b, c \in S$ and $r \in R$ we have

$$(a + \lambda_r(b))c = ac + \lambda_r(bc) \quad \text{and} \quad a(b + \lambda_r(c)) = ab + \lambda_r(ac).$$

This observation leads to an alternate definition:

Definition. An (associative) R -algebra is an R -module together with an (associative) R -bilinear operation.

Whatever the definition, the prototype of an R -algebra is $\text{Mat}_n(R)$.

Indeed, we know that matrix multiplication is R -bilinear. To see that $\text{Mat}_n(R)$ satisfies the more abstract definition of R -algebra, consider the ring homomorphism

$$R \longrightarrow \text{Mat}_n(R), \quad r \longmapsto \begin{pmatrix} r & & 0 \\ & \ddots & \\ 0 & & r \end{pmatrix} = rI_n,$$

and recall from the end of MTH 761 (specifically, from 12/1/15) that

$$Z(\text{Mat}_n(R)) = \{rI_n : r \in R\} \cong R. \quad (*)$$

Now let me recall the proof of (*).

For all $1 \leq i, j \leq n$ we define the *matrix unit*

$$e_{ij} := (\delta_{ik}\delta_{j\ell})_{k,\ell} \in \text{Mat}_n(R),$$

the matrix with a 1 in position (i, j) and zeros elsewhere. Note that $\{e_{ij}\}$ is a basis for $\text{Mat}_n(R)$ as a free R -module. Then consider any matrix $A \in \text{Mat}_n(R)$. Since matrix multiplication is R -bilinear, we conclude that

$$A \in Z(\text{Mat}_n(R)) \iff Ae_{ij} = e_{ij}A \quad \text{for all } 1 \leq i, j \leq n.$$

The equation $Ae_{ij} = e_{ij}A$ can be written as

$$\underbrace{\begin{pmatrix} 0 & \cdots & a_{1i} & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & a_{ii} & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & a_{ni} & \cdots & 0 \end{pmatrix}}_{\text{only column } j \text{ may be nonzero}} = \underbrace{\begin{pmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ a_{j1} & \cdots & a_{jj} & \cdots & a_{jn} \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{pmatrix}}_{\text{only row } i \text{ may be nonzero}},$$

which implies that

- $a_{ii} = a_{jj}$;
- $a_{ki} = 0$ for $k \neq i$;
- $a_{j\ell} = 0$ for $\ell \neq j$.

Thus if $Ae_{ij} = e_{ij}A$ holds for all $1 \leq i, j \leq n$ we conclude that $A = rI_n$ for some $r \in R$, as desired.

[Actually, I lied; this is not the result that I proved on 12/1/15. To see that result we need two more definitions.]

Definition. Let R be a commutative ring. Since $\text{Mat}_n(R)$ is a ring we can consider its group of units

$$GL_n(R) := \text{Mat}_n(R)^\times \cong \text{Aut}_R(R^{\oplus n}),$$

called the *general linear group*.

Definition. For all $1 \leq i, j \leq n$ we define the *elementary matrices* in $\text{Mat}_n(R)$ as follows:

1. $E_{ij}(r) := I_n + re_{ij}$ for $i \neq j$;
2. $E_{ii}(r) := I_n + (r - 1)e_{ii}$;
3. $P_{ij} := I_n + e_{ij} + e_{ji} - e_{ii} - e_{jj}$ for $i \neq j$.

These matrices act on the left/right of $\text{Mat}_n(R)$ by elementary row/column operations:

1. replace the i th row/column by itself plus r times the j th row/column;
2. multiply the i th row/column by r ;
3. swap the i th and j th rows/columns.

Then here's the result from 12/1/15.

Theorem.

$$Z(GL_n(R)) = \{rI_n : r \in R^\times\} \cong R^\times.$$

Proof. Suppose that $A \in Z(GL_n(R))$. Then for $i \neq j$, since $E_{ij}(1) \in GL_n(R)$ we must have

$$AE_{ij}(1) = E_{ij}(1)A,$$

and hence

$$A(I_n + e_{ij}) = (I_n + e_{ij})A, \quad A + Ae_{ij} = A + e_{ij}A, \quad Ae_{ij} = e_{ij}A.$$

Since this is true for all $i \neq j$, comparing entries as before (and treating $n = 1$ separately) shows that $A = rI_n$ for some $r \in R$. Then since $A \in GL_n(R)$ we must have $r \in R^\times$. The converse is immediate. $\square$

For $i \neq j$ we have $P_{ij}, E_{ij}(r) \in GL_n(R)$ because

$$P_{ij}^{-1} = P_{ij} \quad \text{and} \quad E_{ij}(r)^{-1} = E_{ij}(-r).$$

If $r \in R^\times$ then we also have $E_{ii}(r) \in GL_n(R)$ because

$$E_{ii}(r)^{-1} = E_{ii}(r^{-1}).$$

It is interesting to investigate the subgroups of $GL_n(R)$ generated by certain kinds of elementary matrices.

Examples:

- The subgroup

$$W := \langle P_{ij} \rangle \subseteq GL_n(R),$$

called a *Weyl subgroup*, is isomorphic to the symmetric group S_n .

- If R is a field then we know from Gaussian elimination that every invertible matrix is a product of elementary matrices.
- We will see that this is also true when R is a Euclidean domain, but it fails in general. For example, let

$$R = \mathbb{Z}[\theta], \quad \theta = \frac{1 + \sqrt{-19}}{2}.$$

Then the matrix

$$\begin{pmatrix} 3 - \theta & 2 + \theta \\ -3 - 2\theta & 5 - 2\theta \end{pmatrix}$$

is invertible but it is not a product of elementary matrices.

[Remark: This $\mathbb{Z}[\theta]$ is a PID but it is not Euclidean.]

If R is a Euclidean domain then one can also show that the *special linear group*

$$SL_n(R) := \ker(\det : GL_n(R) \rightarrow R^\times)$$

is generated by the matrices $E_{ij}(r)$ for $i \neq j$, sometimes called “transvections” or “shear mappings.” Again, this is not true in general.

Next time we will use elementary row and column operations (and, more generally, “quasi-elementary operations”) to reduce a matrix over a PID to diagonal form.

This will lead to the FTFGMPID, Part II.

47 Row reduction and normal forms

March 29, 2016

Let R be a commutative ring. Last time we defined the elementary $m \times m$ matrices as follows:

1. $E_{ij}(r) = I_m + r e_{ij}$, where $r \in R$ and $i \neq j$;
2. $E_{ii}(r) = I_m + (r - 1)e_{ii}$, where $r \in R$;
3. $P_{ij} = I_m + e_{ij} + e_{ji} - e_{ii} - e_{jj}$, where $i \neq j$.

These matrices act on the left of $\text{Mat}_{m \times n}(R)$ by row operations and act on the right of $\text{Mat}_{\ell \times m}(R)$ by column operations.

If we restrict ourselves to invertible operations (i.e., we only use $E_{ii}(r)$ when $r \in R^\times$) then we can define group actions. More generally, we define an R -linear action of the group $GL_m(R) \times GL_n(R)$ on the R -module $\text{Mat}_{m \times n}(R)$ by

$$(g, h) \cdot A := gAh^{-1}.$$

[I say this action is “more general” because the group $GL(R)$ is not necessarily generated by elementary matrices. It will be when R is a Euclidean domain.]

We have also seen that the induced action

$$GL_m(R) \times GL_n(R) \curvearrowright \text{Hom}_R(R^{\oplus n}, R^{\oplus m})$$

corresponds to “change of basis” on the modules $R^{\oplus m}$ and $R^{\oplus n}$. We are thus interested in classifying the orbits of this action. I’m sure you’re familiar with the following special case.

Theorem (RREF). *Let K be a field and consider the action*

$$GL_m(K) \curvearrowright \text{Mat}_{m \times n}(K), \quad g \cdot A = gA.$$

By applying invertible row operations we can bring any matrix into reduced row echelon form (RREF), for example

$$\begin{pmatrix} 0 & 0 & 1 & 0 & * & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & * & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

One can show that this defines a function

$$\text{RREF} : \text{Mat}_{m \times n}(K) \longrightarrow \text{Mat}_{m \times n}(K)$$

and that we have

$$\text{RREF}(A) = \text{RREF}(B) \iff \text{there exists } g \in GL_m(K) \text{ with } gA = B.$$

As a corollary of this, if we consider the action

$$GL_m(K) \curvearrowright \text{Mat}_{m \times m}(K),$$

then for all $g \in GL_m(K) \subseteq \text{Mat}_{m \times m}(K)$ we must have

$$\text{RREF}(g) = \text{RREF}(g^{-1}g) = I_m.$$

This means that there exist elementary matrices $E_1, E_2, \dots, E_k$ such that

$$E_k E_{k-1} \cdots E_2 E_1 g = I_m.$$

Then inverting gives

$$g = E_1^{-1} E_2^{-1} \cdots E_{k-1}^{-1} E_k^{-1},$$

and we conclude that every $g \in GL_m(K)$ is a product of elementary matrices.

In summary, the RREF is a “complete invariant” for the action

$$GL_m(K) \curvearrowright \text{Mat}_{m \times n}(K).$$

Remark. This version of RREF was described independently by Wilhelm Jordan and B.-I. Clasen in 1887. The algorithm of “row reduction” is attributed to Gauss, who invented it in order to solve the problem of “least-squares approximation” (and to find the missing dwarf planet Ceres in 1801). However the method already appeared in China (~ 300 BC) in the “Nine Chapters on the Mathematical Art.” Actually the Chinese were interested in systems of linear “Diophantine” equations, so they probably had something like the following result.

Theorem (HNF (Hermite, 1851)). *Consider the action*

$$GL_m(\mathbb{Z}) \curvearrowright \text{Mat}_{m \times n}(\mathbb{Z}), \quad g \cdot A = gA.$$

By applying invertible row operations (i.e., we only consider $E_{ii}(r)$ for $r \in \mathbb{Z}^\times = \{\pm 1\}$) we can bring any matrix into Hermite normal form (HNF). For example,

$$\begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 1 & 0 & 1 \\ 2 & 0 & 4 & 0 \end{pmatrix} \xrightarrow{\text{HNF}} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 2 & -4 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

I won’t define the HNF in detail but I’ll just mention that it is a complete invariant for the action.

More generally, we might try to find complete invariants for the two-sided action

$$GL_m(R) \times GL_n(R) \curvearrowright \text{Mat}_{m \times n}(R).$$

Theorem. *Let K be a field. Then every matrix $A \in \text{Mat}_{m \times n}(K)$ is equivalent to a unique matrix of the form*

$$\begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix},$$

where $k \leq \min(m, n)$ is called the rank of the matrix A .

Proof. Let $A = (a_{ij})$. If $A = 0$ then we’re done. Otherwise, by swapping rows and columns we can assume that $a_{11} \neq 0$. Since K is a field, we can multiply the first row (or column) by a_{11}^{-1} and then we can use row and column operations to obtain a matrix of the form

$$\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A' & \\ 0 & & & \end{pmatrix}.$$

Then by induction we obtain a matrix of the desired form.

To show uniqueness, consider the corresponding linear function

$$\varphi_A : K^{\oplus n} \longrightarrow K^{\oplus m}.$$

We have seen that by choosing bases for $K^{\oplus m}$ and $K^{\oplus n}$ we can write

$$[\varphi_A] = \begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix}.$$

But then observe that k is the dimension of the image $\text{im } \varphi_A \subseteq K^{\oplus m}$ as a K -vector space, which does not depend on the choice of bases. $\square$

If we try to do the same for abelian groups ($\mathbb{Z}$ -modules), we get something called the *Smith normal form* (SNF), which was described by H.J.S. Smith in 1861. To pave the way let's consider an example.

Example. Suppose we want to study the submodule of $\mathbb{Z}$ generated by the two element set

$$\{24, 62\} \subseteq \mathbb{Z}.$$

Then we have a canonical $\mathbb{Z}$ -module homomorphism from the free module $\mathbb{Z}^{\oplus 2}$,

$$\begin{aligned} \varphi : \mathbb{Z}^{\oplus 2} &\longrightarrow \mathbb{Z}, \\ \begin{pmatrix} x \\ y \end{pmatrix} &\longmapsto 24x + 62y, \end{aligned}$$

and we want to study the image $\text{im } \varphi \subseteq \mathbb{Z}$. Note that φ is represented by the matrix

$$[\varphi] = (24 \quad 62),$$

since for all elements $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{Z}^{\oplus 2}$ we have

$$[\varphi] \begin{pmatrix} x \\ y \end{pmatrix} = (24 \quad 62) \begin{pmatrix} x \\ y \end{pmatrix} = 24x + 62y.$$

Now our goal is to find a new basis for $\mathbb{Z}^{\oplus 2}$ in which the matrix $[\varphi]$ is as nice as possible.

And to do this we are allowed to multiply $(24 \quad 62)$ on the right by 2×2 $\mathbb{Z}$ -invertible elementary matrices. We can proceed as follows:

$$(24 \quad 62) \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = (24 \quad 14),$$

$$(24 \quad 14) \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} = (10 \quad 14),$$

$$(10 \quad 14) \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = (10 \quad 4),$$

$$(10 \quad 4) \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = (2 \quad 4),$$

$$(2 \quad 4) \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = (2 \quad 0).$$

Since we're not allowed to divide by 2, this is as simple as it gets. In summary, we have

$$(24 \ 62) \begin{pmatrix} 13 & -31 \\ -5 & 12 \end{pmatrix} = (2 \ 0).$$

[Note that we have

$$\det \begin{pmatrix} 13 & -31 \\ -5 & 12 \end{pmatrix} = 12 \cdot 13 - 31 \cdot 5 = 1 \in \{\pm 1\},$$

as expected because this matrix is invertible over $\mathbb{Z}$.]

In summary, if we make the change of basis

$$\begin{array}{ccc} \mathbb{Z}^{\oplus 2} & \longrightarrow & \mathbb{Z}^{\oplus 2} \\ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} & \longrightarrow & \begin{pmatrix} 13 \\ -5 \end{pmatrix}, \begin{pmatrix} -31 \\ 12 \end{pmatrix} \end{array}$$

then our homomorphism $\varphi : \mathbb{Z}^{\oplus 2} \rightarrow \mathbb{Z}$ is represented by the matrix

$$[\varphi] = (2 \ 0),$$

and we conclude that

$$\begin{aligned} \text{im } \varphi &= \left\{ [\varphi] \begin{pmatrix} x \\ y \end{pmatrix} : \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{Z}^{\oplus 2} \right\} \\ &= \left\{ (2 \ 0) \begin{pmatrix} x \\ y \end{pmatrix} : \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{Z}^{\oplus 2} \right\} \\ &= \{2x : x \in \mathbb{Z}\} \\ &= 2\mathbb{Z} \subseteq \mathbb{Z}. \end{aligned}$$

48 Smith normal form

March 31, 2016

Last time we considered the subgroup (i.e., $\mathbb{Z}$ -submodule) $M \subseteq \mathbb{Z}$ generated by the set

$$\{24, 62\} \subseteq \mathbb{Z}.$$

This generating set defines a canonical morphism

$$\begin{aligned} \varphi : \mathbb{Z}^{\oplus 2} &\twoheadrightarrow M, \\ \begin{pmatrix} x \\ y \end{pmatrix} &\longmapsto 24x + 62y. \end{aligned}$$

We can extend this to a morphism of free modules

$$\varphi : \mathbb{Z}^{\oplus 2} \longrightarrow \mathbb{Z},$$

which is represented by the matrix

$$[\varphi] = \begin{pmatrix} 24 & 62 \end{pmatrix}.$$

Then to compute the isomorphism type of $M = \text{im } \varphi$ we tried to simplify the matrix by changing the basis on $\mathbb{Z}^{\oplus 2}$.

After a sequence of column operations (equivalent to the Euclidean algorithm) we found that

$$\begin{pmatrix} 24 & 62 \end{pmatrix} \begin{pmatrix} 13 & -31 \\ -5 & 12 \end{pmatrix} = \begin{pmatrix} 2 & 0 \end{pmatrix}.$$

[Note that $\det \begin{pmatrix} 13 & -31 \\ -5 & 12 \end{pmatrix} = 1 \in \{\pm 1\} = \mathbb{Z}^\times$, as expected because this matrix is invertible over $\mathbb{Z}$.]

In other words, by changing the basis

$$\begin{array}{ccc} \mathbb{Z}^{\oplus 2} & \longrightarrow & \mathbb{Z}^{\oplus 2} \\ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} & & \begin{pmatrix} 13 \\ -5 \end{pmatrix}, \begin{pmatrix} -31 \\ 12 \end{pmatrix} \end{array}$$

we find that φ is represented by the matrix $[\varphi] = \begin{pmatrix} 2 & 0 \end{pmatrix}$, and we conclude that

$$M = \text{im } \varphi = \left\{ \begin{pmatrix} 2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} : \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{Z}^{\oplus 2} \right\} = \{2x : x \in \mathbb{Z}\} = 2\mathbb{Z}.$$

We also conclude that

$$\mathbb{Z}/M \cong \mathbb{Z}/2\mathbb{Z},$$

which was not obvious from the original definition of M .

For a slightly larger example, consider the submodule $M \subseteq \mathbb{Z}^{\oplus 2}$ generated by the two vectors

$$\left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \end{pmatrix} \right\} \subseteq \mathbb{Z}^{\oplus 2}.$$

We want to compute the isomorphism type of the quotient $\mathbb{Z}^{\oplus 2}/M$. To do this we consider the canonical morphism

$$\begin{aligned} \varphi : \mathbb{Z}^{\oplus 2} &\longrightarrow \mathbb{Z}^{\oplus 2}, \\ \begin{pmatrix} x \\ y \end{pmatrix} &\longmapsto x \begin{pmatrix} 2 \\ 1 \end{pmatrix} + y \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \end{aligned}$$

represented by the matrix

$$[\varphi] = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}.$$

And then we try to simplify the matrix as much as possible by changing the basis on the domain and codomain:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix},$$

$$\begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & -5 \end{pmatrix},$$

$$\begin{pmatrix} 1 & 2 \\ 0 & -5 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -5 \end{pmatrix},$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -5 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}.$$

In summary, we have

$$\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}.$$

This tells us that we have performed the change of basis

$$\begin{pmatrix} a, b \\ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} \longrightarrow \begin{pmatrix} a', b' \\ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix}$$

on the codomain and the change of basis

$$\begin{pmatrix} c, d \\ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} \longrightarrow \begin{pmatrix} c', d' \\ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \end{pmatrix} \end{pmatrix}$$

on the domain. With respect to these new bases we have

$$\begin{aligned} \varphi : \mathbb{Z}^{\oplus 2} &\longrightarrow \mathbb{Z}^{\oplus 2}, \\ \begin{pmatrix} x \\ y \end{pmatrix} &\longmapsto x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 5 \end{pmatrix}, \end{aligned}$$

and now it becomes clear that

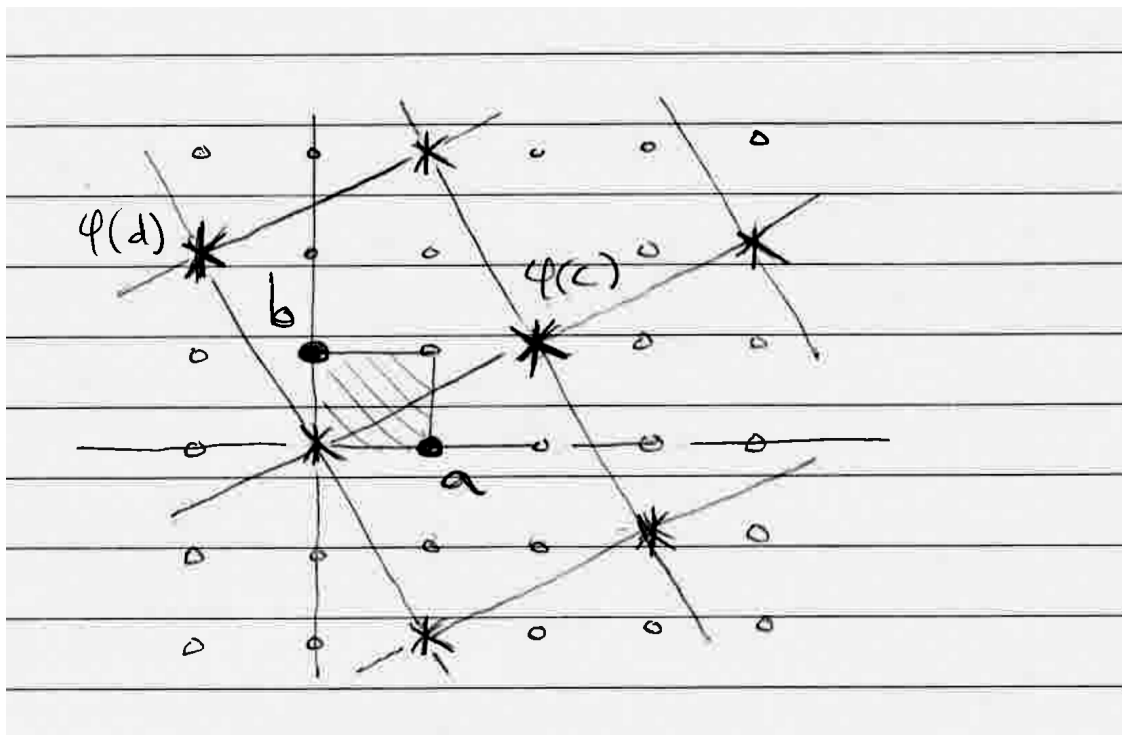
$$M = \text{im } \varphi = \mathbb{Z} \oplus 5\mathbb{Z} \cong \mathbb{Z} \oplus \mathbb{Z} \cong \mathbb{Z}^{\oplus 2}.$$

[We say M is a full rank subgroup of $\mathbb{Z}^{\oplus 2}$.]

Also, since $[\varphi] = \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}$ is diagonal, the quotient $\mathbb{Z}^{\oplus 2}/M$ respects the direct sum in the numerator, so we obtain

$$\mathbb{Z}^{\oplus 2}/M \cong (\mathbb{Z}/\mathbb{Z}) \oplus (\mathbb{Z}/5\mathbb{Z}) \cong 0 \oplus \mathbb{Z}/5\mathbb{Z} \cong \mathbb{Z}/5\mathbb{Z}.$$

It's easier to see what this means in terms of pictures. Note that $\mathbb{Z}^{\oplus 2}$ is a free module with basis a, b and that $M = \text{im } \varphi$ is a free submodule with basis $\varphi(c), \varphi(d)$, as shown below:



We can interpret our calculation as finding new bases for $\mathbb{Z}^{\oplus 2}$ and M that are more obviously related.

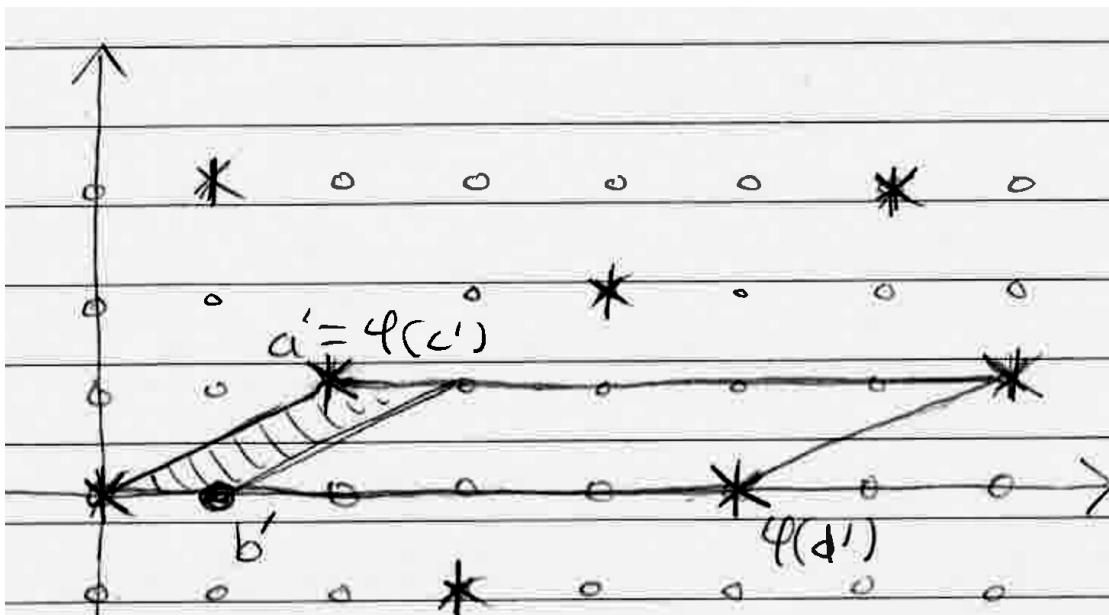
The new basis for $\mathbb{Z}^{\oplus 2}$ is

$$a' = 2a + b, \quad b' = a,$$

and the new basis for M is

$$\begin{aligned} \varphi(c') &= \varphi(c) = 2a + b, \\ \varphi(d') &= \varphi(2c - d) = 2\varphi(c) - \varphi(d) \\ &= 2(2a + b) - (-a + 2b) \\ &= 5a, \end{aligned}$$

as shown below:



Now it's obvious why the quotient is isomorphic to $\mathbb{Z}/5\mathbb{Z}$.

[Because the parallelogram $\phi(c') \times \phi(d')$ is 1×5 times the parallelogram $a' \times b'$.]

Now I will present the general theorem in this subject. It was first described by Henry John Stephen Smith in 1861.

Theorem (Smith Normal Form (SNF)). *Let R be a PID. By applying a sequence of invertible row and column operations we can bring any $A \in \text{Mat}_{m \times n}(R)$ into diagonal form*

$$\begin{pmatrix} d_1 & & & 0 \\ & d_2 & & \vdots \\ & & \ddots & \vdots \\ 0 & \dots & 0 & d_k & 0 \\ & & & 0 & 0 \end{pmatrix},$$

where $k \leq \min(m, n)$ is the rank of A . Furthermore, we can obtain that

$$d_1 \mid d_2 \mid \dots \mid d_k,$$

and in this case the matrix is a complete invariant for the action

$$GL_m(R) \times GL_n(R) \curvearrowright \text{Mat}_{m \times n}(R).$$

Proof. I'll give the proof for Euclidean domains and then mumble some words about how to extend it to PIDs.

So let R be a Euclidean domain with size function

$$\delta : R \setminus \{0\} \longrightarrow \mathbb{N}.$$

Recall the definition: For all $a, b \in R$ with $b \neq 0$ there exist $q, r \in R$ such that

- $a = qb + r$;
- $r = 0$ or $\delta(r) < \delta(b)$.

Now consider a matrix $A = (a_{ij}) \in \text{Mat}_{m \times n}(R)$. I'll describe a (possibly very inefficient) algorithm to put A into SNF.

Step 1. If $A = 0$ then stop. Otherwise we can permute rows and columns so that a_{11} has minimal size.

Step 2. Now we try to clear the first row and first column. If at any point we create an entry smaller than a_{11} , go back to Step 1. (By well-ordering of $\mathbb{N}$ this can only happen finitely many times.)

- If there exists $a_{i1} \neq 0$ with $i > 1$, then we divide by a_{11} to obtain

$$a_{i1} = qa_{11} + r.$$

Subtract $q(\text{row } 1)$ from $(\text{row } i)$ to replace the entry a_{i1} by r . If $r = 0$ then we're happy. Otherwise we have $\delta(r) < \delta(a_{11})$, so we go back to Step 1.

- If there exists $a_{1j} \neq 0$ with $j > 1$, we perform a similar procedure to replace a_{1j} with 0, or with an element smaller than a_{11} , in which case we go back to Step 1.

At the end of Step 2 we obtain a matrix of the form

$$\begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A' & \\ 0 & & & \end{pmatrix},$$

where $a_{11} \neq 0$.

If a_{11} divides every entry of A' , then we can apply recursion to obtain the SNF:

$$\begin{pmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & & & \\ \vdots & \vdots & & A'' & \\ 0 & 0 & & & \end{pmatrix}, \quad \text{etc.}$$

But *does* a_{11} divide every entry of A' ? We need one more step.

Step 3. If there exists an entry b in A' such that a_{11} does not divide b , then we replace (row 1) by itself plus the row containing b :

$$\begin{pmatrix} a_{11} & * & \cdots & * & b & * & \cdots & * \\ 0 & & & & & & & \\ \vdots & & & A' & & & & \\ 0 & & & & & & & \end{pmatrix}.$$

Then we go back to Step 2. Dividing b by a_{11} produces $b = qa_{11} + r$ with $r \neq 0$, hence $\delta(r) < \delta(a_{11})$. And now we're back to Step 1!

I emphasize that this process will stop because if not then we obtain an infinite subset of $\mathbb{N}$ with no smallest element (i.e., the sizes of the various entries a_{11}).

This completes the proof for Euclidean domains. Now what about PIDs? There are two issues to consider.

1. Elementary row/column operations are no longer enough. Instead we will define the pseudo-elementary matrices for $i \neq j$:

$$E_{ij} \begin{pmatrix} a & b \\ c & d \end{pmatrix} := I + (a-1)e_{ii} + be_{ij} + ce_{ji} + (d-1)e_{jj},$$

where $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{Mat}_{2 \times 2}(R)$ is invertible.

If x_i, x_j are the i th and j th rows of A , the i th and j th rows of $E_{ij} \begin{pmatrix} a & b \\ c & d \end{pmatrix} A$ are

$$ax_i + bx_j, \quad cx_i + dx_j,$$

and if y_i, y_j are the i th and j th columns of A , the i th and j th columns of $AE_{ij} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ are

$$ay_i + cy_j, \quad by_i + dy_j.$$

Now consider any two elements $x, y \in R$ (not both zero). Since R is a PID we have $(x, y) = (d)$ for some nonzero $d \in R$, and hence there exist $a, b \in R$ such that $ax + by = d$. It follows that

$$\begin{pmatrix} a & b \\ -y/d & x/d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} d \\ 0 \end{pmatrix}, \quad (x \ y) \begin{pmatrix} a & -y/d \\ b & x/d \end{pmatrix} = (d \ 0),$$

where the two matrices have determinant 1 and hence are invertible.

In this way we can perform all of the necessary reductions in the algorithm by pseudo-elementary operations.

2. To guarantee that the new algorithm will stop, we can no longer rely on the well-ordering of $\mathbb{N}$.

Instead we want to use the fact that a PID is “Noetherian” (i.e., has no infinite increasing chain of ideals). We already used this property when proving that $\text{PID} \Rightarrow \text{UFD}$.

At each step of the algorithm when the “size” of a_{11} decreased we can instead arrange to replace a_{11} by a “proper divisor.” Indeed, if a_{11} divides an element b , then we can get rid of b in the old-fashioned way, and otherwise we replace a_{11} by d , where

$$(a_{11}) \subsetneq (a_{11}, b) = (d).$$

If the algorithm doesn’t terminate, then we will obtain an infinite chain

$$(a_{11}) \subsetneq (a'_{11}) \subsetneq (a''_{11}) \subsetneq \cdots,$$

which is impossible for the usual reason. [See HW2.1(c).] □

Remark. You might ask, is there a more general kind of ring that has Smith Normal Form? The answer is yes: they are called “elementary divisor rings,” and they are basically defined by the property of having SNF.

It has been my experience that in ring theory there is a bijection between theorems and definitions. That is, to the question

“What is the most general kind of ring that satisfies this theorem?”

the answer is always

“We don’t know, so let’s give that kind of ring a special name.” ☺

49 Smith normal form and the structure theorem

April 5, 2016

Last time we proved the existence of Smith normal form over a Euclidean domain and sketched how to extend the proof to PIDs. In summary:

Let R be a PID. For any $A \in \text{Mat}_{m \times n}(R)$ there exist invertible matrices $P \in GL_m(R)$ and $Q \in GL_n(R)$ such that

$$PAQ^{-1} = \begin{pmatrix} d_1 & & & & 0 \\ & d_2 & & & \vdots \\ & & \ddots & & \vdots \\ & & & d_k & 0 \\ 0 & \cdots & 0 & 0 & 0 \end{pmatrix}$$

for some nonzero $d_i \in R$ with

$$d_1 \mid d_2 \mid \cdots \mid d_k.$$

We claimed that this matrix is unique, so it is a complete invariant for the action

$$(GL_m(R) \times GL_n(R)) \curvearrowright \text{Mat}_{m \times n}(R).$$

We call it the *Smith normal form* of A .

Before proving uniqueness, let's examine the implications for the structure theory of modules over a PID.

Let R be a PID and let M be a finitely generated R -module. By definition there is a finite set A and a canonical surjective morphism

$$R^{\oplus A} \xrightarrow{\pi} M.$$

Consider

$$\ker \pi \subseteq R^{\oplus A}.$$

Since $R^{\oplus A}$ is free, our key lemma for PIDs says that $\ker \pi$ is free of rank at most $|A|$.

In particular, there exists a set B with $|B| \leq |A|$ and a canonical bijective homomorphism

$$R^{\oplus B} \xrightarrow[\sim]{\varphi} \ker \pi.$$

Extending the codomain gives a short exact sequence

$$0 \longrightarrow R^{\oplus B} \xrightarrow{\varphi} R^{\oplus A} \xrightarrow{\pi} M \longrightarrow 0.$$

Furthermore,

$$M = \text{im } \pi \cong R^{\oplus A} / \ker \pi = R^{\oplus A} / \text{im } \varphi.$$

Thus the isomorphism type of M is determined by the homomorphism

$$R^{\oplus B} \xrightarrow{\varphi} R^{\oplus A},$$

which is represented in coordinates by an $|A| \times |B|$ matrix. We can change coordinates in both modules to simplify this matrix.

This is exactly what the SNF does for us.

Suppose that $|A| = m$ and $|B| = n \leq m$. Then there exist new bases

$$A' = \{a_1, \dots, a_m\} \subseteq R^{\oplus A}, \quad B' = \{b_1, \dots, b_n\} \subseteq R^{\oplus B},$$

such that

$$[\varphi]_{B'A'} = \begin{pmatrix} d_1 & & & 0 \\ & d_2 & & \vdots \\ & & \ddots & 0 \\ 0 & \cdots & 0 & d_n \\ 0 & \cdots & \cdots & 0 \\ \vdots & & & \vdots \\ 0 & \cdots & \cdots & 0 \end{pmatrix}.$$

In other words,

$$\varphi(b_i) = d_i a_i \quad \text{for all } 1 \leq i \leq n.$$

What does this tell us about M ?

We have

$$\begin{aligned} \text{im } \varphi &= \left\{ \varphi \left(\sum_{j=1}^n r_j b_j \right) : r_j \in R \right\} \\ &= \left\{ \sum_{j=1}^n r_j \varphi(b_j) : r_j \in R \right\} \\ &= \left\{ \sum_{j=1}^n r_j d_j a_j : r_j \in R \right\}. \end{aligned}$$

I claim that

$$M = R^{\oplus A} / \text{im } \varphi \cong R/(d_1) \oplus \cdots \oplus R/(d_n) \oplus R^{\oplus(m-n)}.$$

To see this, consider the obvious homomorphism

$$\Phi : R^{\oplus A} \rightarrow R/(d_1) \oplus \cdots \oplus R/(d_n) \oplus R^{\oplus(m-n)}$$

defined by

$$\Phi \left(\sum_{i=1}^m s_i a_i \right) := (s_1 + (d_1), \dots, s_n + (d_n), s_{n+1}, \dots, s_m).$$

We have $\ker \Phi = \text{im } \varphi$. Indeed,

$$\Phi \left(\sum_{i=1}^m s_i a_i \right) = (0 + (d_1), \dots, 0 + (d_n), 0, \dots, 0)$$

if and only if

$$s_{n+1} = \cdots = s_m = 0 \quad \text{and} \quad s_j \in (d_j) \quad \text{for all } 1 \leq j \leq n,$$

i.e., if and only if

$$\sum_{i=1}^m s_i a_i = \sum_{j=1}^n r_j d_j a_j$$

for some $r_j \in R$. Now the First Isomorphism Theorem gives

$$\begin{aligned} M &\cong R^{\oplus A} / \text{im } \varphi \\ &= R^{\oplus A} / \ker \Phi \\ &\cong \text{im } \Phi \\ &= R/(d_1) \oplus \cdots \oplus R/(d_n) \oplus R^{\oplus(m-n)}. \end{aligned}$$

This (finally) completes the proof of the Fundamental Theorem of Finite Abelian Groups (from MTH 761, 12/3/15), and allows me to state the rest of the F.T.F.G.M.P.I.D.

Theorem (F.T.F.G.M.P.I.D., Part II). *Let R be a PID and let M be a finitely generated R -module of rank k . Then there exist nonzero elements $d_1, \dots, d_n \in R$ with*

$$d_1 \mid d_2 \mid \cdots \mid d_n$$

such that

$$M \cong R^{\oplus k} \oplus R/(d_1) \oplus \cdots \oplus R/(d_n).$$

Furthermore, the elements $d_1, \dots, d_n$ are unique up to associates. They are called the invariant factors of M .

[Compare Part I:

$$\mathrm{Tor}_R(M) \cong R/(d_1) \oplus \cdots \oplus R/(d_n).$$

Thus Part II is really the structure theorem for torsion modules over a PID.]

We have proved existence; it remains to prove uniqueness. This will also complete the proof of uniqueness for the Smith normal form. [The two theorems are basically equivalent.] We'll proceed in four steps.

Step 1. The rank is already well-defined [see HW2], so we can restrict our attention to the torsion submodule

$$T := \mathrm{Tor}_R(M).$$

Step 2. Suppose that

$$T = R/(d_1) \oplus \cdots \oplus R/(d_n),$$

where

$$(d_n) \subseteq (d_{n-1}) \subseteq \cdots \subseteq (d_1).$$

We want to show that this chain of ideals depends only on the isomorphism type of T , not on the generating set.

Define the *annihilator ideal*

$$\mathrm{Ann}_R(T) := \{r \in R : rt = 0 \text{ for all } t \in T\}.$$

I claim that

$$\mathrm{Ann}_R(T) = (d_n).$$

Indeed, suppose that $r \in \mathrm{Ann}_R(T)$. In particular,

$$\begin{aligned} (0 + (d_1), \dots, 0 + (d_n)) &= r(1 + (d_1), \dots, 1 + (d_n)) \\ &= (r + (d_1), \dots, r + (d_n)). \end{aligned}$$

Hence

$$r + (d_n) = 0 + (d_n),$$

which means that $r \in (d_n)$.

Conversely, suppose that $r \in (d_n)$. Since

$$(d_n) \subseteq (d_{n-1}) \subseteq \cdots \subseteq (d_1),$$

we have $r \in (d_i)$ for all i . It follows that

$$r(s_1 + (d_1), \dots, s_n + (d_n)) = (0 + (d_1), \dots, 0 + (d_n))$$

for every element of T . Thus $r \in \text{Ann}_R(T)$.

Therefore (d_n) depends only on the isomorphism type of T . What about the rest of the chain

$$(d_{n-1}) \subseteq \dots \subseteq (d_1)?$$

Step 3. Since R is a PID, it is a UFD, so we have a unique factorization

$$d_n = p_1^{\alpha_{1n}} p_2^{\alpha_{2n}} \dots p_m^{\alpha_{mn}},$$

where $p_1, \dots, p_m$ are distinct primes. Since $d_1 \mid d_2 \mid \dots \mid d_n$, we also have

$$d_j = p_1^{\alpha_{1j}} p_2^{\alpha_{2j}} \dots p_m^{\alpha_{mj}},$$

where, for each i ,

$$\alpha_{i1} \leq \alpha_{i2} \leq \dots \leq \alpha_{in}.$$

On HW4 you will show that

$$R/(d_j) \cong \bigoplus_i R/(p_i^{\alpha_{ij}}),$$

and hence

$$T \cong \bigoplus_{i,j} R/(p_i^{\alpha_{ij}}).$$

The prime powers $p_i^{\alpha_{ij}}$ will be called the *elementary divisors* of T once we prove uniqueness.

Step 4. The primes p_i are determined uniquely up to associates by the module T . Indeed, the ideal (d_n) is unique, and the ideals (p_i) are precisely the prime ideals containing (d_n) . [This is just another way to say unique prime factorization.]

Thus it only remains to show that the natural numbers α_{ij} are uniquely determined by T .

You will finish the proof on HW4. ◡

But first I need to write HW4. ◡

□

50 Applications of the structure theorem

April 7, 2016

We have completed the significant (but unfortunately named) Fundamental Theorem of Finitely Generated Modules Over a Principal Ideal Domain. Here is a summary.

Theorem (F.T.F.G.M.P.I.D.). *Let R be a PID and let M be a finitely generated R -module. Then we have*

$$M \cong R^{\oplus k} \oplus T,$$

where k is the rank of M (the size of a maximal R -linearly independent set) and

$$T = \text{Tor}_R(M)$$

is the collection of R -torsion elements (which is a submodule because R is an integral domain). Furthermore, we can decompose T as a direct sum of cyclic modules

$$T \cong R/(d_1) \oplus R/(d_2) \oplus \cdots \oplus R/(d_n),$$

and if we assume that

$$(d_n) \subseteq \cdots \subseteq (d_1) \neq R,$$

then this collection of ideals is unique. They are called the invariant factors of T (or M).

Alternatively, if

$$d_n = p_1^{\alpha_{1n}} p_2^{\alpha_{2n}} \cdots p_m^{\alpha_{mn}}$$

is the unique prime factorization of d_n [recall that $\text{PID} \Rightarrow \text{UFD}$], then we can use the Chinese Remainder Theorem to decompose T further as

$$T \cong \bigoplus_{i,j} R/(p_i^{\alpha_{ij}}),$$

where

$$d_j = p_1^{\alpha_{1j}} p_2^{\alpha_{2j}} \cdots p_m^{\alpha_{mj}}.$$

The ideals $(p_i^{\alpha_{ij}})$ are also uniquely determined. They are called the elementary divisors of T (or M).

Remarks:

- The Chinese Remainder Theorem and the uniqueness of invariant factors/elementary divisors have not been proved yet.

Hopefully some combination of you & I will prove them on HW4. The proof of uniqueness I have in mind involves “tensoring” with the fields $R/(p_i)$.

- The original version of the FTFGMPID is due to Weierstrass (1868). If Cayley & Sylvester’s “matrix” notation provided the form of Linear Algebra, then Weierstrass’ Theorem provided its content. It is one of the most important theorems of 19th century mathematics.

Now we should discuss applications, I guess. Recall that there are basically two kinds of PIDs: $\mathbb{Z}$ and $K[x]$, for K a field. Applying the F.T. to $\mathbb{Z}$ gives

Theorem (F.T.F.G.A.G.). Recall that abelian groups are just $\mathbb{Z}$ -modules. Thus if G is a finitely generated abelian group then we have

$$G \cong \mathbb{Z}^{\oplus k} \oplus \bigoplus_{i=1}^n \mathbb{Z}/(d_i)$$

for some unique integers $k \geq 0$ and

$$1 < d_1 \mid d_2 \mid \cdots \mid d_n.$$

[I stated this in MTH 761 on 12/3/15 and now it is done.]

Because of this, the structure of finite abelian groups is *much* easier than the structure of all finite groups. A classification such as this often provides shortcuts to theorems, such as the following.

Theorem (Primitive Root Theorem). Let K be a field. Then any finite subgroup of

$$K^\times := K \setminus \{0\}$$

is cyclic. In particular, if K is finite then $K^\times$ itself is cyclic.

Proof. Let $G \subseteq K^\times$ be a finite subgroup. By the F.T. we have

$$G \cong \mathbb{Z}/(d_1) \oplus \cdots \oplus \mathbb{Z}/(d_n),$$

with

$$2 \leq d_1 \mid d_2 \mid \cdots \mid d_n.$$

Observe that for all $g \in G$ we have

$$g^{d_n} = 1$$

(written multiplicatively). If G is not cyclic then we have $n \geq 2$, and hence

$$d_n < d_1 d_2 \cdots d_n = |G|.$$

It follows that

$$\#\{g \in G : g^{d_n} = 1\} = |G| > d_n. \quad (*)$$

But now consider the polynomial $x^{d_n} - 1 \in K[x]$. Since this polynomial has degree d_n , you proved on HW1 [using Descartes' Theorem] that it has at most d_n roots, which contradicts (*).

We conclude that G is cyclic. □

[I really like this proof. All the other proofs I know involve a nontrivial amount of number theory.]

Next we turn to the other PID, namely $K[x]$. This was the case that Weierstrass originally had in mind.

So let K be a field and let M be a $K[x]$ -module defined by a ring homomorphism

$$\lambda : K[x] \longrightarrow \text{End}_{\text{Ab}}(M).$$

Let's look at this carefully. By restricting λ to the subring $K \subseteq K[x]$ we obtain a ring homomorphism

$$\lambda : K \longrightarrow \text{End}_{\text{Ab}}(M),$$

which shows us that M is, in particular, a K -vector space. But note that this is automatically a homomorphism of K -algebras:

$$\begin{array}{ccc} K & \xrightarrow{\lambda} & \text{End}_K(M) \\ & \searrow \text{id} \quad \nearrow \lambda & \\ & K & \end{array}$$

Indeed, for all scalars $a, b \in K$ and all vectors $m, n \in M$ we have

$$\begin{aligned} \lambda_a(m + \lambda_b(n)) &= \lambda_a(m) + \lambda_a(\lambda_b(n)) \\ &= \lambda_a(m) + \lambda_{ab}(n) \\ &= \lambda_a(m) + \lambda_{ba}(n) \\ &= \lambda_a(m) + \lambda_b(\lambda_a(n)), \end{aligned} \tag{*}$$

which shows that $\lambda_a \in \text{End}_K(M)$. For all $\varphi \in \text{End}_K(M)$ we also have (from the definition of K -linearity) that

$$\lambda_a(\varphi(m)) = \varphi(\lambda_a(m)),$$

and hence

$$\lambda_a \in Z(\text{End}_K(M)).$$

Furthermore, note that the image of the full map

$$\lambda : K[x] \longrightarrow \text{End}_{\text{Ab}}(M)$$

is also in the subring $\text{End}_K(M)$. Indeed, since the variable x is (by definition) in the center of $K[x]$, we have

$$\lambda_x(m + \lambda_a(n)) = \lambda_x(m) + \lambda_a(\lambda_x(n)),$$

just as in (*) above.

However, I want to emphasize that the K -linear map

$$\lambda_x : M \longrightarrow M$$

need not be in the center $Z(\text{End}_K(M))$.

We have shown that if M is a $K[x]$ -module then we obtain a homomorphism of K -algebras

$$\begin{array}{ccc}
 K[x] & \xrightarrow{\lambda} & \text{End}_K(M) \\
 & \nwarrow \quad \nearrow & \\
 & K &
 \end{array}$$

where λ_x is some (possibly non-central) K -linear map.

Conversely, if $\varphi \in \text{End}_K(M)$ is any K -linear map, then since $K[x]$ is the “free K -algebra generated by one element,” there exists a unique K -algebra homomorphism

$$\lambda : K[x] \longrightarrow \text{End}_K(M)$$

such that $\lambda_x = \varphi$.

In summary, we have proved the following (tautological) result.

Theorem (Characterization of $K[x]$ -modules). *The data of a $K[x]$ -module is equivalent to a choice of (V, φ) , where V is a K -vector space and*

$$\varphi : V \longrightarrow V$$

is a K -linear endomorphism.

Remarks:

- Actually, the same argument works over any commutative ring R : the data of an $R[x]$ -module is equivalent to a choice of (M, φ) , where M is an R -module and $\varphi : M \rightarrow M$ is an R -linear endomorphism.
- Moreover, the data of an $R[x_1, \dots, x_n]$ -module is equivalent to a choice of R -module M and an n -tuple of commuting R -linear endomorphisms.

However, the case of $K[x]$ -modules is the most interesting to us for two reasons.

1. Consider a $K[x]$ -module corresponding to a pair (V, φ) , where V is a finite dimensional vector space. If $V \cong K^{\oplus n}$, then after choosing a basis we can think of φ as an $n \times n$ matrix $[\varphi]$.
2. If V is finite dimensional then the corresponding $K[x]$ -module will be finitely generated. Then, since $K[x]$ is a PID, we can apply the Fundamental Theorem to obtain some kind of “canonical decomposition” for the matrix $[\varphi]$.

Stay Tuned.

51 Canonical forms for matrices

April 12, 2016

Right now we are applying the FTFGMPID to $K[x]$ -modules when K is a field.

So let V be a $K[x]$ -module with defining ring homomorphism

$$\lambda : K[x] \longrightarrow \text{End}_{\text{Ab}}(V).$$

Restricting λ to the subring $K \subseteq K[x]$ defines a K -vector space structure on the abelian group V (that's why I'm calling it V today). By examining the definitions carefully we see that λ is really a homomorphism of K -algebras

$$\lambda : K[x] \longrightarrow \text{End}_K(V),$$

and then, since $K[x]$ is the “free K -algebra generated by one element,”

we see that λ is uniquely determined by the K -linear endomorphism

$$\lambda_x \in \text{End}_K(V).$$

In summary:

Theorem (Characterization of $K[x]$ -modules). *There is a bijection between $K[x]$ -modules and pairs (V, α) , where V is a K -vector space and*

$$\alpha \in \text{End}_K(V)$$

is a K -linear endomorphism.

Now consider two $K[x]$ -modules

$$\lambda : K[x] \longrightarrow \text{End}_K(U) \quad \& \quad \mu : K[x] \longrightarrow \text{End}_K(V)$$

such that $\lambda_x = \alpha$ and $\mu_x = \beta$. Recall that a $K[x]$ -module homomorphism is defined to be a homomorphism of abelian groups $\varphi : U \rightarrow V$ such that for all $f(x) \in K[x]$ we have a commutative square

$$\begin{array}{ccc} U & \xrightarrow{\varphi} & V \\ \lambda_{f(x)} \downarrow & & \downarrow \mu_{f(x)} \\ U & \xrightarrow{\varphi} & V \end{array}$$

By taking $f(x) \in K \subseteq K[x]$, the diagram above tells us that $\varphi : U \rightarrow V$ is a K -linear map. Then substituting $f(x) = x$ into the diagram gives

$$\begin{array}{ccc} U & \xrightarrow{\varphi} & V \\ \alpha \downarrow & & \downarrow \beta \\ U & \xrightarrow{\varphi} & V \end{array}$$

i.e.,

$$\varphi \circ \alpha = \beta \circ \varphi.$$

If, moreover, φ is an isomorphism of $K[x]$ -modules, we conclude that $\varphi : U \rightarrow V$ is an isomorphism of K -vector spaces such that

$$\varphi \circ \alpha \circ \varphi^{-1} = \beta.$$

In summary:

Theorem (Isomorphism of $K[x]$ -modules). *Two $K[x]$ -modules defined by (U, α) and (V, β) are isomorphic if and only if there exists an isomorphism $\varphi : U \rightarrow V$ of K -vector spaces such that*

$$\varphi \circ \alpha \circ \varphi^{-1} = \beta.$$

This leads to a surprising idea. Let V be a K -vector space and recall that the general linear group

$$\mathrm{GL}_K(V) := \mathrm{Aut}_K(V)$$

acts on endomorphisms by conjugation:

$$\begin{aligned} \mathrm{GL}_K(V) &\curvearrowright \mathrm{End}_K(V), \\ \varphi \cdot \alpha &:= \varphi \circ \alpha \circ \varphi^{-1}. \end{aligned}$$

If V is finitely generated of rank n , then by choosing a basis this gives us an action of the group of invertible $n \times n$ matrices on the algebra of all $n \times n$ matrices:

$$\begin{aligned} \mathrm{GL}_n(K) &\curvearrowright \mathrm{Mat}_n(K), \\ P \cdot A &:= PAP^{-1}. \end{aligned}$$

[Remark: If $K \in \{\mathbb{R}, \mathbb{C}\}$ then we would call this the “adjoint action” of the Lie group $\mathrm{GL}_n(K)$ on its Lie algebra $\mathrm{Mat}_n(K)$.]

In general we say that two matrices in the same orbit of this action are “similar.”

Our goal is now to classify matrices in $\mathrm{Mat}_n(K)$ up to “similarity.” This problem is analogous to the problems solved by the RREF, Hermite & Smith Normal Forms, so we are looking for some new kind of “normal form.”

I consider it surprising that the answer to this problem follows from the Smith Normal Form (via the FTFGMPID).

To be specific, consider any matrix $A \in \mathrm{Mat}_n(K)$. This defines a $K[x]$ -module structure on $V := K^{\oplus n}$ via the map

$$\begin{aligned} \lambda : K[x] &\longrightarrow \mathrm{End}_K(V), \\ x &\longmapsto A. \end{aligned}$$

Since V is finite dimensional this module is finitely generated, and since $K[x]$ is a PID we can apply the FTFGMPID to obtain an isomorphism of $K[x]$ -modules

$$\varphi : V \xrightarrow{\sim} \bigoplus_{i=1}^d K[x]/(f_i(x)), \tag{*}$$

for some polynomials $f_i(x) \in K[x]$. By throwing away zero summands in (*) we can assume that none of these polynomials is a unit. Furthermore, I claim that none of the polynomials is zero. Indeed, the direct sum of $K[x]$ -modules in (*) restricts to a direct sum of K -vector spaces, and we know that dimension adds over direct sums. Since we have assumed that V is finite dimensional, this implies that each direct summand on the right is finite dimensional. Finally, we observe that

$$K[x] = K[x]/(0)$$

is not finite dimensional (e.g., because the set $1, x, x^2, \dots$ is K -linearly independent).

In summary we can assume that the polynomials are nonconstant, monic, and satisfy

$$f_1(x) \mid f_2(x) \mid \cdots \mid f_d(x),$$

and in this case they are unique.

[Jargon: The polynomial $f_d(x)$ is called the minimal polynomial of A and the product

$$f_1(x)f_2(x)\cdots f_d(x)$$

is called the characteristic polynomial.]

Now we can think of the isomorphism φ as a “change of basis” on the vector space V , in which case it is represented by some invertible matrix $P \in \text{GL}_n(K)$. Furthermore, since each of the summands in (*) is a $K[x]$ -submodule, we know in particular that it is preserved by the action of “ x ,” i.e., by the action of the matrix A . Thus, if we choose a K -basis for V by concatenating bases for the summands, we obtain a block-diagonal matrix

$$PAP^{-1} = \begin{pmatrix} B_1 & & & 0 \\ & B_2 & & \\ & & \ddots & \\ & & & \ddots & \\ 0 & & & & B_d \end{pmatrix}.$$

It remains only to choose some “canonical” bases for the summands. I think you’ll agree that the following choice is reasonable.

The Companion Matrix

Let $f(x) \in K[x]$ be a monic polynomial

$$f(x) = r_0 + r_1x + \cdots + r_{m-1}x^{m-1} + x^m,$$

and consider the basis

$$1, x, x^2, \dots, x^{m-1}$$

for the K -vector space $K[x]/(f(x))$. In terms of this basis the action of “ x ” satisfies

$$\begin{aligned} x \cdot 1 &= x, \\ x \cdot x &= x^2, \\ &\vdots \\ x \cdot x^{m-2} &= x^{m-1}, \\ x \cdot x^{m-1} &= x^m = -r_0 - r_1x - \cdots - r_{m-1}x^{m-1}. \end{aligned}$$

Thus, if we think of “multiplying by x ” as a K -linear endomorphism

$$K[x]/(f(x)) \longrightarrow K[x]/(f(x)),$$

its matrix in this basis is the following so-called “companion matrix”:

$$C_{f(x)} := \begin{pmatrix} 0 & 0 & \cdots & 0 & -r_0 \\ 1 & 0 & \cdots & 0 & -r_1 \\ 0 & 1 & \ddots & \vdots & -r_2 \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & 0 & 1 & -r_{m-1} \end{pmatrix}.$$

In summary, we have the following.

Theorem (Rational Canonical Form (RCF)). *Let K be any field and consider any $n \times n$ matrix $A \in \text{Mat}_n(K)$. Then there exist unique monic polynomials*

$$f_1(x) \mid f_2(x) \mid \cdots \mid f_d(x)$$

and an invertible matrix $P \in \text{GL}_n(K)$ such that

$$PAP^{-1} = \begin{pmatrix} C_{f_1(x)} & & & 0 \\ & C_{f_2(x)} & & \\ & & \ddots & \\ 0 & & & C_{f_d(x)} \end{pmatrix}.$$

We call this block-diagonal matrix the rational canonical form of A : $\text{RCF}(A)$. Furthermore, the RCF is a complete invariant for the action of $\text{GL}_n(K)$ on $\text{Mat}_n(K)$ by conjugation.

The RCF essentially corresponds to the “invariant factor” decomposition of the $K[x]$ -module defined by A .

If K is algebraically closed (or if we're willing to pass to an algebraic extension) then we also have a canonical form corresponding to the "elementary divisor" decomposition. Let

$$f_1(x) \mid f_2(x) \mid \cdots \mid f_d(x)$$

be the invariant factors of A and suppose that the minimal polynomial splits as

$$f_d(x) = \prod_i (x - a_i)^{\alpha_{id}}$$

with $a_i \in K$.

Then we have an isomorphism of $K[x]$ -modules

$$V \cong \bigoplus_{i,j} K[x]/((x - a_i)^{\alpha_{ij}}),$$

where the action of A on the left corresponds to "multiplication by x " on the right. Our goal is to choose a canonical basis for each summand on the right. I think you'll agree that the following choice is reasonable.

Jordan Blocks

Consider the basis

$$1, (x - a), \dots, (x - a)^{m-1}$$

for the K -vector space $K[x]/((x - a)^m)$. In terms of this basis, "multiplication by x " is given by

$$\begin{aligned} x \cdot 1 &= x = (x - a) + a \cdot 1, \\ x \cdot (x - a) &= x^2 - ax = (x - a)^2 + a \cdot (x - a), \\ &\vdots \\ x \cdot (x - a)^{m-2} &= (x - a)^{m-1} + a \cdot (x - a)^{m-2}, \end{aligned}$$

and finally

$$\begin{aligned} x \cdot (x - a)^{m-1} &= (x - a)^m + a \cdot (x - a)^{m-1} \\ &= a \cdot (x - a)^{m-1}. \end{aligned}$$

Thus it is represented by the $m \times m$ matrix

$$J_a(m) := \begin{pmatrix} a & 0 & 0 & \cdots & 0 \\ 1 & a & 0 & \ddots & \vdots \\ 0 & 1 & a & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 1 & a \end{pmatrix},$$

which we call a "Jordan block."

In summary, we have the following.

Theorem (Jordan Canonical Form (JCF)). *Let K be an algebraically closed field and consider any $n \times n$ matrix $A \in \text{Mat}_n(K)$. Then there exist elements $a_i \in K$ (the eigenvalues of A), positive integers $\alpha_{ij} \in \mathbb{N}$, and an invertible matrix $P \in \text{GL}_n(K)$ such that*

$$PAP^{-1} = \begin{pmatrix} \ddots & & 0 \\ & \boxed{J_{a_i}(\alpha_{ij})} & \\ 0 & & \ddots \end{pmatrix}.$$

If we order the diagonal Jordan blocks in some reasonable way, then the matrix

$$\text{JCF}(A) := PAP^{-1}$$

is unique. We call it the Jordan Canonical Form of A .

Furthermore, the JCF is a complete invariant for the action of $\text{GL}_n(K)$ on $\text{Mat}_n(K)$ by conjugation.

We'll discuss consequences of the RCF & JCF next time.

52 Minimal and characteristic polynomials

April 14, 2016

Last time we derived the Rational & Jordan Canonical Forms of a matrix from the FTFGM over the PID $K[x]$.

Recall: The ring $\text{Mat}_n(K)$ is actually a K -algebra via the ring homomorphism

$$\lambda : K \longrightarrow \text{Mat}_n(K)$$

defined by

$$\lambda(a) := aI_n \in Z(\text{Mat}_n(K)).$$

Then for any matrix $A \in \text{Mat}_n(K)$ there exists a canonical “evaluation map” from the free algebra

$$\varphi_A : K[x] \longrightarrow \text{Mat}_n(K)$$

defined by $\varphi_A(x) := A$ and $\varphi_A(a) = \lambda(a) = aI_n$ for all $a \in K \subseteq K[x]$. For convenience we will often write

$$\varphi_A(f(x)) = “f(A)”.$$

To be explicit, if

$$f(x) = a_0 + a_1x + \cdots + a_mx^m,$$

then we have

$$f(A) = a_0I_n + a_1A + \cdots + a_mA^m.$$

Now since φ_A is, in particular, a ring homomorphism we know that its kernel

$$\ker \varphi_A = \{f(x) \in K[x] : f(A) = 0\}$$

is an ideal of $K[x]$. Then since $K[x]$ is a PID (recall that $K[x]$ is a Euclidean domain via long division of polynomials and that Euclidean $\Rightarrow$ PID) we conclude that

$$\ker \varphi_A = (m_A(x))$$

for some unique monic polynomial $m_A(x) \in K[x]$ called the *minimal polynomial* of A .

Compare this to the minimal polynomial $m_a(x) \in K[x]$ of an element $a \in L \supseteq K$ in a field extension. If $m_a(x) \neq 0$ (i.e., if a is algebraic over K)

then you proved on HW3 that $m_a(x)$ is irreducible. Proof: Suppose that we have

$$m_a(x) = f(x)g(x)$$

for some $f(x), g(x) \in K[x]$ and evaluate at a to get

$$0 = m_a(a) = f(a)g(a). \quad (*)$$

Since L is a domain we can assume without loss that $f(a) = 0$. This means that $f(x) \in \ker \varphi_a = (m_a(x))$ and hence we have $f(x) = m_a(x)q(x)$ for some $q(x) \in K[x]$. But then since $m_a(x) \neq 0$ and since $K[x]$ is a domain we have

$$\begin{aligned} m_a(x) &= f(x)g(x), \\ m_a(x) &= m_a(x)q(x)g(x), \\ m_a(x)(1 - q(x)g(x)) &= 0, \\ 1 - q(x)g(x) &= 0, \\ 1 &= q(x)g(x), \end{aligned}$$

which implies that $g(x)$ is a unit.

However, the minimal polynomial of a matrix may be reducible. In this case the above proof fails at step (*)

because $\text{Mat}_n(K)$ is not an integral domain. Indeed, even the commutative subring

$$K[A] := \text{im } \varphi_A \subseteq \text{Mat}_n(K)$$

might have zero divisors. In a sense, HW4 is all about exploring this possibility.

Now since $K[x]$ is a unique factorization domain [recall that PID $\Rightarrow$ UFD] we have a unique factorization

$$m_A(x) = p_1(x)^{m_1} \cdots p_d(x)^{m_d},$$

where the polynomials $p_i(x) \in K[x]$ are monic & irreducible.

There is another important polynomial associated to A called the *characteristic polynomial* $\chi_A(x) \in K[x]$, which is defined via the determinant. I was hoping to introduce determinants the correct way (via exterior powers),

but we ran out of time, so here's the quick and dirty way.

Definition (Determinant). Let R be a commutative ring and consider an $n \times n$ matrix $A = (a_{ij}) \in \text{Mat}_n(R)$. We define the *determinant* of A as an alternating sum of products of entries, indexed by permutations $\pi \in S_n$:

$$\det(A) := \sum_{\pi \in S_n} (-1)^\pi \prod_{i=1}^n a_{i, \pi(i)} \in R.$$

Determinants were first mentioned in the year 1683, independently by Seki in Japan and Leibniz in Europe, and they are the reason that matrices were invented. Indeed, Sylvester's word "matrix" means a "womb" that gives birth to determinants!

One can easily check that the determinants of elementary matrices are

$$\det(E_{ij}(r)) = 1 \quad (i \neq j), \quad \det(E_{ii}(r)) = r,$$

and

$$\det(P_{ij}) = -1 \quad (i \neq j).$$

Then for any matrix $A \in \text{Mat}_n(R)$ and for any elementary matrix $E \in \text{Mat}_n(R)$ one can check that

$$\det(EA) = \det(AE) = \det(E) \det(A).$$

When $R = K$ is a field we showed on 3/29/16 that every matrix $A \in \text{Mat}_n(K)$ can be reduced to the form

$$\left(\begin{array}{c|c} I_k & 0 \\ \hline 0 & 0 \end{array} \right)$$

by multiplying on the right and left by elementary matrices. Thus we obtain the following result.

Theorem (Theorem on Determinants). Let K be a field and consider matrices $A, B \in \text{Mat}_n(K)$. Then we have

- $A \in \text{GL}_n(K) \iff \det(A) \neq 0$.
- $\det(AB) = \det(A) \det(B)$.

Proof. There exist $E, F \in \text{GL}_n(K)$ such that E & F are products of (invertible) elementary matrices and such that

$$EAF = \left(\begin{array}{c|c} I_k & 0 \\ \hline 0 & 0 \end{array} \right).$$

Thus

$$A = E^{-1} \left(\begin{array}{c|c} I_k & 0 \\ \hline 0 & 0 \end{array} \right) F^{-1}.$$

It follows that A is invertible if and only if $k = n$. Then from the above observations we have

$$\det(A) = \det(E^{-1}) \det \left(\begin{array}{c|c} I_k & 0 \\ \hline 0 & 0 \end{array} \right) \det(F^{-1}),$$

and it follows that

$$\det(A) \neq 0 \iff k = n.$$

Now if at least one of A & B is not invertible then AB is not invertible and we have

$$\det(AB) = 0 = \det(A) \det(B).$$

So assume that $A, B \in \text{GL}_n(K)$. Then we have $E_1, E_2, F_1, F_2 \in \text{GL}_n(K)$ such that E_1, E_2, F_1, F_2 are products of elementary matrices and such that

$$E_1 A F_1 = I_n \quad \& \quad E_2 B F_2 = I_n.$$

Then we have $A = E_1^{-1} F_1^{-1}$ & $B = E_2^{-1} F_2^{-1}$, and it follows that

$$\begin{aligned} \det(AB) &= \det(E_1^{-1} F_1^{-1} E_2^{-1} F_2^{-1}) \\ &= \det(E_1^{-1} F_1^{-1}) \det(E_2^{-1} F_2^{-1}) \\ &= \det(A) \det(B). \end{aligned}$$

□

Remarks:

- As long as $\text{GL}_n(R)$ is generated by elementary matrices [for example if R is a Euclidean domain] then the same proof works to show that

$$\det(AB) = \det(A) \det(B) \quad \text{and} \quad A \in \text{GL}_n(R) \iff \det(A) \in R^\times.$$

- In fact, the same result holds when R is an arbitrary commutative ring but it requires a new proof. Here's my favorite proof (details omitted):

Let $\bigwedge^k R^{\oplus n}$ be the k -th "exterior power" of the free R -module $R^{\oplus n}$. Then $\bigwedge^k R^{\oplus n}$ is a free R -module of rank

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

In particular we have

$$\bigwedge^n R^{\oplus n} \cong R.$$

Now any R -linear map $\varphi : R^{\oplus n} \rightarrow R^{\oplus n}$ induces a map $\bigwedge^n \varphi : R \rightarrow R$ and, if we choose a basis, then this induced map is "multiplication by the determinant":

$$\begin{array}{ccc}
 R^{\oplus n} & \xrightarrow{A} & R^{\oplus n} \\
 \downarrow & & \downarrow \\
 \bigwedge^n R^{\oplus n} & \xrightarrow{\det(A)} & \bigwedge^n R^{\oplus n}
 \end{array}$$

From this point of view the theorem is obvious, but then one must prove that the combinatorial definition of the determinant is equivalent to (*).

Now we can define the “other polynomial.”

Definition (Characteristic polynomial). Let K be a field and consider a matrix $A \in \text{Mat}_n(K)$. We define the *characteristic polynomial* as a determinant:

$$\chi_A(x) := \det(xI_n - A) \in K[x].$$

Note that this is a monic polynomial of degree n . Its main purpose is for detecting the eigenvalues of A . Recall that $\lambda \in K$ is called an eigenvalue of A [from the German “eigen” = “belonging to”] if there exists some nonzero vector $u \in K^{\oplus n}$ such that

$$Au = \lambda u.$$

In other words, λ is an eigenvalue

$$\begin{aligned}
 &\iff \exists u \neq 0 \text{ with } Au = \lambda I_n u \\
 &\iff \exists u \neq 0 \text{ with } (\lambda I_n - A)u = 0 \\
 &\iff \ker(\lambda I_n - A) \neq 0 \\
 &\iff \lambda I_n - A \text{ is not injective} \\
 &\stackrel{(*)}{\iff} \lambda I_n - A \text{ is not invertible} \\
 &\iff \det(\lambda I_n - A) = 0 \\
 &\iff \chi_A(\lambda) = 0.
 \end{aligned}$$

[Note that the $\Leftarrow$ direction of (*) uses the fact that $\dim(V/W) = \dim V - \dim W$ for vector spaces, which is sometimes called the “rank-nullity theorem.”]

Then as a consequence of the Theorem on Determinants we have for all invertible matrices $P \in \text{GL}_n(K)$ that

$$\begin{aligned}
 \chi_{PAP^{-1}}(x) &= \det(xI_n - PAP^{-1}) \\
 &= \det(xPI_nP^{-1} - PAP^{-1}) \\
 &= \det(P(xI_n - A)P^{-1}) \\
 &= \det(P) \det(xI_n - A) \det(P^{-1}) \\
 &= \det(P) \det(xI_n - A) \det(P)^{-1} \\
 &= \det(xI_n - A) \\
 &= \chi_A(x).
 \end{aligned}$$

From this we conclude that:

- The matrix $A \in \text{Mat}_n(K)$ has at most n distinct eigenvalues.
- The eigenvalues are invariant under change of basis, so they are really a property of K -linear endomorphisms.

Now we have two polynomials associated to the matrix A ,

- $m_A(x)$ related to linear relations among the powers of A ;
- $\chi_A(x)$ related to eigenvalues of A ,

and might wonder how these polynomials are related to each other. Here is the answer.

Theorem (Cayley–Hamilton Theorem).

$$m_A(x) \mid \chi_A(x).$$

There are many ways to prove this; we'll prove it by applying the Rational Canonical Form. But first let's discuss a few consequences.

- Since $\chi_A(x) \neq 0$ the theorem tells us that $m_A(x) \neq 0$. In other words, there is no such thing as a “transcendental matrix”; all matrices are “algebraic.”
- The roots of $m_A(x)$ (if any exist) are eigenvalues of A . Indeed, we have $\chi_A(x) = m_A(x)f(x)$ for some $f(x) \in K[x]$. If α is a root of $m_A(x)$ then

$$\chi_A(\alpha) = m_A(\alpha)f(\alpha) = 0 \cdot f(\alpha) = 0,$$

hence α is an eigenvalue of A .

- Evaluating $\chi_A(x)$ at A gives

$$\chi_A(A) = m_A(A)f(A) = 0 \cdot f(A) = 0.$$

Thus A is annihilated by its own characteristic polynomial. [In fact, this property is equivalent to the statement of the theorem.]

Now the proof.

Proof of Cayley–Hamilton. This will follow directly from the RCF after a few observations.

1. Let V be the $K[x]$ -module defined by

$$\lambda : K[x] \longrightarrow \text{End}_K(V), \quad x \longmapsto A,$$

and let

$$f_1(x) \mid f_2(x) \mid \cdots \mid f_d(x)$$

be the associated monic invariant factors. We showed previously that $(f_d(x))$ equals the annihilator ideal $\text{Ann}_{K[x]}(V) \subseteq K[x]$. But note that we also have

$$\begin{aligned} \text{Ann}_{K[x]}(V) &= \{f(x) \in K[x] : \lambda_{f(x)}(u) = 0 \ \forall u \in V\} \\ &= \{f(x) \in K[x] : f(A)u = 0 \ \forall u \in V\} \\ &= \{f(x) \in K[x] : f(A) = 0\} \\ &= \ker \varphi_A = (m_A(x)). \end{aligned}$$

Since $(f_d(x)) = (m_A(x))$, and since both polynomials are monic, we conclude that

$$m_A(x) = f_d(x).$$

2. Consider any monic polynomial

$$f(x) = r_0 + r_1x + \cdots + r_{m-1}x^{m-1} + x^m \in K[x]$$

with corresponding $m \times m$ companion matrix

$$C_{f(x)} = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 & -r_0 \\ 1 & 0 & \cdots & 0 & 0 & -r_1 \\ 0 & 1 & \ddots & \vdots & \vdots & -r_2 \\ \vdots & \ddots & \ddots & 0 & 0 & \vdots \\ 0 & \cdots & 0 & 1 & 0 & -r_{m-2} \\ 0 & \cdots & \cdots & 0 & 1 & -r_{m-1} \end{pmatrix}.$$

Then one can check by induction that the characteristic polynomial is

$$\chi_{C_{f(x)}}(t) = \det(tI_m - C_{f(x)}) = f(t).$$

3. By the RCF there exists an invertible matrix $P \in \text{GL}_n(K)$ such that

$$PAP^{-1} = \begin{pmatrix} C_{f_1(x)} & & & 0 \\ & C_{f_2(x)} & & \\ & & \ddots & \\ 0 & & & C_{f_d(x)} \end{pmatrix}.$$

Then since the determinant of a block diagonal matrix is equal to the product of the determinants of the blocks we conclude that

$$\begin{aligned} \chi_A(t) &= \chi_{PAP^{-1}}(t) \\ &= \prod_{i=1}^d \chi_{C_{f_i(x)}}(t) \\ &= \prod_{i=1}^d f_i(t) = m_A(t) \prod_{i=1}^{d-1} f_i(t). \end{aligned}$$

□

It is also true that any eigenvalue of A is a root of the minimal polynomial $m_A(x)$.

Proof. Let $\lambda \in K$ be an eigenvalue with eigenvector $u \in K^{\oplus n}$. Then we have $A^k u = \lambda^k u$ for all $k \geq 0$. It follows that for any polynomial $f(x) \in K[x]$ we have $f(A)u = f(\lambda)u$. In particular, taking $f(x) = m_A(x)$ gives

$$0 = 0u = m_A(A)u = m_A(\lambda)u.$$

Since $u \neq 0$ this implies that $m_A(\lambda) = 0$. □

You will explore the relationship between $m_A(x)$ & $\chi_A(x)$ further on HW4.

Here is the most general thing that can be said about the topic. I'll state it without proof.

Theorem (Minimal vs. Characteristic Polynomial). *Let $A \in \text{Mat}_n(K)$ over a general field K and suppose the minimal polynomial has the unique prime factorization*

$$m_A(x) = p_1(x)^{m_1} p_2(x)^{m_2} \cdots p_d(x)^{m_d},$$

where each $p_i(x)$ is irreducible of degree r_i . If we define the dimensions

$$n_i := \dim_K \ker(p_i(A)^{m_i}),$$

then $r_i \mid n_i$ for all i , and we have

$$\chi_A(x) = p_1(x)^{n_1/r_1} p_2(x)^{n_2/r_2} \cdots p_d(x)^{n_d/r_d}.$$

53 Morita equivalence and base change

April 19, 2016

This Week: Epilogue

I was hoping to talk more about noncommutative rings & algebras this semester. In particular, I wanted to discuss the “Artin–Wedderburn Theorem” and then use it to derive the Fundamental Theorem of Galois Theory (via a nice proof by Mitya Boyarchenko). Alas, it was not to be.

Algebra is just too big.

Instead I will spend the final two lectures on the concept of “base change”. This will allow us to give the correct proof of uniqueness in the FTFGMPID.

Let R be a ring (either commutative or noncommutative). Our philosophy in MTH 762 was to replace the study of R by the study of the abelian category $R\text{-Mod}$. There is a deep theorem underlying this philosophy.

Definition. We say that rings R & S are *Morita equivalent* if there exists an equivalence of categories

$$R\text{-Mod} \simeq S\text{-Mod},$$

i.e., a pair of functors

$$\begin{array}{ccc} & F & \\ R\text{-Mod} & \xrightarrow{\quad} & S\text{-Mod} \\ & G & \end{array} \quad (*)$$

with natural isomorphisms

$$G \circ F \cong \text{id}_{R\text{-Mod}} \quad \& \quad F \circ G \cong \text{id}_{S\text{-Mod}}.$$

In this case the functors F, G will automatically be additive, so that $(*)$ is an equivalence of abelian categories. The equivalence $(*)$ also implies the existence of an equivalence of categories of right R & S modules

$$\text{Mod-}R \simeq \text{Mod-}S,$$

so there is no need to distinguish between “left” and “right” Morita equivalence.

Now here’s the theorem.

Theorem (Theorem on Morita Equivalence). *Let R & S be commutative rings. Then*

$$\begin{array}{ccc} R \cong S & \iff & R\text{-Mod} \simeq S\text{-Mod} , \\ \uparrow \text{isomorphism} & & \uparrow \text{equivalence of} \\ \text{of rings} & & \text{categories} \end{array}$$

In other words, the commutative ring R is determined up to isomorphism by its category of modules.

Remarks:

- This is another “Cayley”-type theorem telling us that the definition of the category $R\text{-Mod}$ is not “too general”.
- The theorem is false for noncommutative rings [in fact, for any ring R and for any $n \geq 1$ we have an equivalence

$$R\text{-Mod} \simeq \text{Mat}_n(R)\text{-Mod}].$$

Thus to determine R from the category $R\text{-Mod}$ would require more than just the abelian category structure. The subject of “noncommutative geometry” tries to discover this extra structure.

The theorem on Morita equivalence suggests that homomorphisms of rings should somehow correspond to functors between their categories. To be specific:

Consider any ring homomorphism

$$\varphi : R \longrightarrow S.$$

This determines a fairly obvious functor

$$\varphi_* : S\text{-Mod} \longrightarrow R\text{-Mod}$$

called “restriction of scalars”. To define it consider any S -module M defined by a ring homomorphism

$$\lambda : S \longrightarrow \text{End}_{\text{Ab}}(M).$$

Then pre-composing with φ gives a new ring homomorphism

$$\lambda \circ \varphi : R \longrightarrow \text{End}_{\text{Ab}}(M),$$

which defines an R -module structure on the abelian group M . In the special case that $i : R \hookrightarrow S$ is an inclusion, the functor i_* is literally “restriction of scalars”.

OK, but that was too easy. We really want a way to “extend scalars” from R to S . Since restriction trivially preserves limits and colimits (we say that φ_* is an “exact functor”) we suspect that there might be two functors

$$R\text{-Mod} \longrightarrow S\text{-Mod},$$

one right-adjoint and one left-adjoint to φ_* . One of these is easier to construct, so we’ll do it first.

Let N be an R -module defined by a ring homomorphism

$$\mu : R \longrightarrow \text{End}_{\text{Ab}}(N),$$

and recall that the ring homomorphism $\varphi : R \rightarrow S$ also defines an R -module structure on the abelian group $|S|$ by

$$\begin{aligned} R &\longrightarrow \text{End}_{\text{Ab}}(|S|), \\ r &\longmapsto (s \longmapsto \varphi(r)s). \end{aligned}$$

Now consider the abelian group $\text{Hom}_R(S, N)$ of R -linear maps between these two R -modules. We can define a natural S -module structure on $\text{Hom}_R(S, N)$ as follows:

Given $\alpha : S \rightarrow N$ & $s \in S$ we define the map $s\alpha : S \rightarrow N$ by setting

$$(s\alpha)(s') := \alpha(ss') \quad \forall s' \in S.$$

Aluffi calls the resulting S -module

$$\varphi^!(N) := \text{Hom}_R(S, N)$$

[and I guess this is pronounced as “shriek”]. By chasing definitions(!) one can show that this defines a functor

$$\varphi^! : R\text{-Mod} \longrightarrow S\text{-Mod}$$

and that for all $N \in R\text{-Mod}$ & $M \in S\text{-Mod}$ we have a natural isomorphism

$$\text{Hom}_R(\varphi_*(M), N) \cong \text{Hom}_S(M, \varphi^!(N)).$$

In particular, this tells us that $\varphi^!$ is a right-adjoint functor, so it preserves limits.

That was the easy one. Next, suppose we have some hypothetical functor φ^* from $R\text{-Mod}$ to $S\text{-Mod}$ satisfying the natural isomorphism

$$\text{Hom}_R(N, \varphi_*(M)) \cong \text{Hom}_S(\varphi^*(N), M).$$

What would this mean?

First note that we have an isomorphism of S -modules

$$\begin{aligned} \text{Hom}_S(S, M) &\cong M, \\ \alpha &\longmapsto \alpha(1). \end{aligned}$$

If we think of this as an R -module by restriction then the above isomorphism becomes

$$\text{Hom}_R(N, \text{Hom}_S(S, M)) \cong \text{Hom}_S(\varphi^*(N), M). \quad (*)$$

Where have we seen this before? It was hinted at on HW3 Problem 2. The isomorphism $(*)$ tells us two things.

1. Universal Property of $\varphi^*(N)$:

From HW3.2(a) we have

$$\begin{aligned} \text{Hom}_R(N, \text{Hom}_S(S, M)) &\subseteq \text{Hom}_{\text{Set}}(N, \text{Hom}_{\text{Set}}(S, M)) \\ &\quad \updownarrow \\ &\text{Hom}_{\text{Set}}(N \times S, M). \end{aligned}$$

Thus we can think of each element of the left group as a function $N \times S \rightarrow M$. In fact, the functions on the left have the property of being “ R -bilinear”. Then $(*)$ tells us that for each R -bilinear function $f : N \times S \rightarrow M$ there should be a unique S -unilinear function

$$\bar{f} : \varphi^*(N) \longrightarrow M.$$

2. How to construct it:

From $(*)$ and from HW3.2(b) we have

$$\begin{aligned} \text{Hom}_S(\varphi^*(N), M) &\cong \text{Hom}_R(N, \text{Hom}_S(S, M)) \\ &\subseteq \text{Hom}_{\text{Set}}(N, \text{Hom}_S(S, M)) \\ &\cong \text{Hom}_S(S^{\oplus N}, M), \end{aligned}$$

which suggests that maybe we can construct $\varphi^*(N)$ as a quotient of the (infinitely generated) free S -module $S^{\oplus N}$.

Indeed, this is how the module $\varphi^*(N)$ is usually constructed. By starting with the obvious S -module

$$S^{\oplus N} = \left\{ \sum_i s_i n_i : n_i \in N \right\}$$

and then dividing by all the relations necessary to achieve the desired universal property.

I'll spare you the details and just tell you the name of this thing:

$$\varphi^*(N) = N \otimes_R S.$$

We call this the “tensor product” of N with S over the base ring R .

Basically, a “tensor product” is a machine for turning bilinear functions into unilinear functions. I'm sad that I didn't have more time to motivate this. For now I'll just mention a few properties.

- From the natural isomorphism

$$\mathrm{Hom}_R(N, \mathrm{Hom}_S(S, M)) \cong \mathrm{Hom}_S(N \otimes_R S, M)$$

we see that

$$(-) \otimes_R S : R\text{-Mod} \longrightarrow S\text{-Mod}$$

is a left-adjoint functor, so it commutes with colimits.

- Plugging $N = R$ into the isomorphism gives

$$\mathrm{Hom}_R(R, \mathrm{Hom}_S(S, M)) \cong \mathrm{Hom}_S(R \otimes_R S, M),$$

so that

$$\mathrm{Hom}_S(S, M) \cong \mathrm{Hom}_S(R \otimes_R S, M),$$

and then from Yoneda we conclude that

$$S \cong R \otimes_R S$$

as S -modules.

- Finally, since the construction of the free module $R^{\oplus A}$ is a colimit we conclude that for any set A we have

$$R^{\oplus A} \otimes_R S \cong (R \otimes_R S)^{\oplus A} \cong S^{\oplus A}.$$

This is a very useful property. We will use it next time to complete the proof of uniqueness in the FTFGMPID.

54 Extension of scalars and uniqueness over a PID

April 21, 2016

Last Day of Class

Last time we discussed “restriction and extension of scalars” for commutative rings. The same discussion can be generalized to noncommutative rings but we didn’t have time to go into the details.

Recall: Let R & S be commutative rings and let $\varphi : R \rightarrow S$ be any ring homomorphism. Then we have an adjoint pair of functors between the abelian categories of modules:

$$\begin{array}{ccc} & \varphi^* & \\ R\text{-}\mathbf{Mod} & \xrightleftharpoons{\quad} & S\text{-}\mathbf{Mod} \\ & \varphi_* & \end{array}$$

The functor φ_* is called “restriction of scalars” and it is easy to define:

Given any S -module N defined by a ring homomorphism

$$\mu : S \longrightarrow \text{End}_{\mathbf{Ab}}(N),$$

we let $\varphi_*(N)$ be the R -module structure on the abelian group N defined by the ring homomorphism

$$\mu \circ \varphi : R \longrightarrow \text{End}_{\mathbf{Ab}}(N).$$

The functor

$$\varphi^* : R\text{-}\mathbf{Mod} \longrightarrow S\text{-}\mathbf{Mod}$$

is called “extension of scalars”. It is quite a bit trickier to construct, but if you are willing to just believe in its existence then it is uniquely characterized by being left-adjoint to φ_* . That is, for all modules $M \in R\text{-}\mathbf{Mod}$ & $N \in S\text{-}\mathbf{Mod}$ we have a “natural” isomorphism of abelian groups

$$\text{Hom}_R(M, \varphi_*(N)) \cong \text{Hom}_S(\varphi^*(M), N).$$

[Note that this is yet another example of a “free–forgetful adjunction”, where the forgetful functor is obvious and the free functor is not.]

The word “natural” means that the isomorphism of abelian groups is induced from a “natural isomorphism” of bifunctors

$$\text{Hom}_R(-, \varphi_*(-)) \cong \text{Hom}_S(\varphi^*(-), -).$$

You can imagine this might be hard to check; luckily we don’t have to check it because we’re using it as the *definition* of φ^* .

More often you will see restriction and extension of scalars expressed in the jargon of the “hom–tensor adjunction”: Given a module $N \in S\text{-}\mathbf{Mod}$ recall that we have a natural isomorphism

$$N \cong \text{Hom}_S(S, N).$$

Then we will abuse terminology to write

$$\varphi_*(N) = N = \text{Hom}_S(S, N).$$

For any R -module M we will define the notation

$$M \otimes_R S := \varphi^*(M).$$

[Yes, the symbol $\otimes$ has some extra meaning, but for now we can just take this as a funny notation for extension of scalars.]

Then our adjunction becomes

$$\mathrm{Hom}_R(M, \mathrm{Hom}_S(S, N)) \cong \mathrm{Hom}_S(M \otimes_R S, N). \quad (*)$$

The reason we might want to write it this way is because $(*)$ is a special case of the more general hom–tensor adjunction

$$\mathrm{Hom}_R(M, \mathrm{Hom}_S(P, N)) \cong \mathrm{Hom}_S(M \otimes_R P, N),$$

where P is an arbitrary S -module.

[Mnemonic: Just move P across the comma.]

[Remark: There is a version of the hom–tensor adjunction for noncommutative rings but there we have to be careful about “bimodules” and left vs. right actions. The most important example of the noncommutative version is “restriction and induction” of group representations.]

To show that this abstract stuff is really not so hard, let’s prove that $R \otimes_R S \cong S$. Plugging $M = R$ into $(*)$ gives

$$\begin{aligned} \mathrm{Hom}_S(R \otimes_R S, N) &\cong \mathrm{Hom}_R(R, \mathrm{Hom}_S(S, N)) \\ &\cong \mathrm{Hom}_S(S, N), \end{aligned}$$

and then “by Yoneda” we conclude that

$$R \otimes_R S \cong S.$$

[Note that the homomorphism $\varphi : R \rightarrow S$ was implicit here; that’s one weakness of the notation.] Furthermore, since $(-)\otimes_R S$ is a left-adjoint functor by definition, we know from the “Meta-Theorem” that it commutes with colimits. For example, if A is any set then we obtain an isomorphism of S -modules

$$R^{\oplus A} \otimes_R S \cong (R \otimes_R S)^{\oplus A} \cong S^{\oplus A}.$$

[I think this is where category theory starts to pay out more than we put into it.]

That was enough theory. Let’s finally apply this (relatively speaking, of course) to the theory of modules over an integral domain.

So let R be an integral domain and consider an R -module M . Let $A \subseteq M$ be a maximal R -linearly independent set and consider the corresponding free submodule

$$R^{\oplus A} \subseteq M.$$

By maximality, the quotient $M/R^{\oplus A}$ is a torsion module. [Proof: If not, then there exists $m \notin R^{\oplus A}$ with $rm \notin R^{\oplus A}$ for all $0 \neq r \in R$. But then $A \cup \{m\}$ is R -linearly independent.]

Now consider the short exact sequence

$$0 \longrightarrow R^{\oplus A} \longrightarrow M \longrightarrow M/R^{\oplus A} \longrightarrow 0.$$

Since a cokernel is a colimit it will be preserved under extension of scalars. In particular, consider the inclusion of R in its field of fractions

$$R \longrightarrow K := \text{Frac}(R).$$

Then we obtain an exact sequence of K -vector spaces

$$K^{\oplus A} \longrightarrow M \otimes_R K \longrightarrow (M/R^{\oplus A}) \otimes_R K \longrightarrow 0.$$

In general the left arrow may no longer be injective, but here it will be [because “field of fractions” is an example of localization, which is always exact]. Also, since $M/R^{\oplus A}$ is torsion over R , we have

$$(M/R^{\oplus A}) \otimes_R K = 0.$$

Thus we obtain an exact sequence

$$0 \longrightarrow K^{\oplus A} \longrightarrow M \otimes_R K \longrightarrow 0,$$

which tells us that

$$M \otimes_R K \cong K^{\oplus A}.$$

By Steinitz Exchange over the field K we conclude that the cardinality of A is well-defined. We call it the *rank* of M .

[Yes, we already proved this on HW2, but don't you think this proof is better?]

Next assume that R is a PID and that M is finitely generated, so we have an expansion in terms of elementary divisors,

$$M \cong R^{\oplus k} \oplus \bigoplus_{i,j} R/(p_i^{\alpha_{ij}}). \quad (*)$$

We just saw that the rank k satisfies

$$k = \dim_{\text{Frac}(R)}(M \otimes_R \text{Frac}(R)),$$

and so is well-defined. More generally, if $P \subseteq R$ is any prime ideal then R/P is a domain and we can consider its field of fractions. We define the P -rank of M as the (well-defined) number

$$\dim_{\text{Frac}(R/P)}(M \otimes_R \text{Frac}(R/P)).$$

We already know that the ideals (p_i) from $(*)$ are well-defined (they are the prime ideals containing the annihilator ideal $\text{Ann}_R(M)$). Also, since R is a PID we know that each (p_i) is a maximal ideal, so that

$$\text{Frac}(R/(p_i)) = R/(p_i).$$

By tensoring with the fields $R/(p_i)$ we hope to recover all of the numbers α_{ij} .

First I will state without proof that, for each elementary divisor $a = p_\ell^\alpha$ appearing in (*), we have

$$R/(a) \otimes_R R/(p_i) \cong \begin{cases} R/(p_i), & \ell = i, \\ 0, & \ell \neq i. \end{cases}$$

This implies that the “ (p_i) -rank” of M

$$\dim_{R/(p_i)}(M \otimes_R R/(p_i))$$

equals

$$\#\{j : \alpha_{ij} \neq 0\},$$

so these numbers are well-defined.

Finally, I will state without proof that

$$\frac{p_i^\beta R}{(p_i^\alpha)} \otimes_R R/(p_i) \cong \begin{cases} R/(p_i), & \beta < \alpha, \\ 0, & \beta \geq \alpha. \end{cases}$$

Now consider the (well-defined) “ (p_i) -torsion submodule” of M ,

$$T_{(p_i)} = \bigoplus_j R/(p_i^{\alpha_{ij}}).$$

By applying p_i^β on both sides and then tensoring with the field $R/(p_i)$ we obtain

$$\dim_{R/(p_i)}(p_i^\beta T_{(p_i)} \otimes_R R/(p_i)) = \#\{j : \alpha_{ij} > \beta\}.$$

Then by letting β run over all natural numbers this gives us enough invariants to determine the numbers α_{ij} uniquely.

This completes the proof of the Fundamental Theorem of Finitely Generated Modules over a Principal Ideal Domain.

QED

In MTH 762 I have succeeded in convincing myself that the categorical view of linear algebra is useful, interesting, and intelligible to ordinary human beings. Hopefully I convinced a few of you as well.

55 Final review

April 29, 2016

Today: Review for Final.

Here are the main topics discussed in lecture since the Midterm and in HW3 & HW4.

1. The category $R\text{-Alg}$

Let R be a commutative ring. We define an R -algebra as a pair (S, φ) where

- S is a (possibly noncommutative) ring,
- $\varphi : R \rightarrow S$ is a ring homomorphism,
- $\text{im } \varphi \subseteq Z(S) = \{s \in S : st = ts \text{ for all } t \in S\}$.

We define a morphism of R -algebras

$$\Phi : (S_1, \varphi_1) \longrightarrow (S_2, \varphi_2)$$

as a ring homomorphism $\Phi : S_1 \rightarrow S_2$ such that the following triangle commutes:

$$\begin{array}{ccc} S_1 & \xrightarrow{\Phi} & S_2 \\ & \nwarrow \varphi_1 \quad \nearrow \varphi_2 & \\ & R & \end{array}$$

One can check that this defines a category, which we call $R\text{-Alg}$. Note that we have a forgetful functor

$$U : R\text{-Alg} \longrightarrow R\text{-Mod}$$

defined by “forgetting the monoid structure.” Specifically, if $\varphi : R \rightarrow S$ is an R -algebra then we obtain a ring homomorphism

$$\lambda : R \longrightarrow \text{End}_{\mathbf{Ab}}(|S|)$$

by sending $r \in R$ to the “multiplication map”

$$\lambda_r(s) := \varphi(r)s = s\varphi(r).$$

One can check that this defines an R -module structure on the abelian group $|S|$. [In fact, you did check this on HW2.5(b).] With respect to this structure we can define a subring

$$\text{End}_R(|S|) \subseteq \text{End}_{\mathbf{Ab}}(|S|),$$

and then observe (“trivially” but not necessarily “easily”) that the ring homomorphism λ sends R into the center of this subring:

$$\lambda : R \longrightarrow \text{im } \lambda \subseteq Z(\text{End}_R(|S|)).$$

[We did this in class on 4/7/16.]

Thus we obtain an R -algebra structure on the ring $\text{End}_R(|S|)$ that is compatible with the R -algebra structure on the ring S in the sense that for all $s_1, s_2, s_3 \in S$ and $r \in R$ we have

$$\begin{aligned} s_1(s_2 + \lambda_r(s_3)) &= s_1s_2 + \lambda_r(s_1s_3), \\ (s_1 + \lambda_r(s_2))s_3 &= s_1s_3 + \lambda_r(s_2s_3). \end{aligned}$$

[We say that the monoid structure on S is “ R -bilinear.”]

[Remark: The forgetful functor

$$U : R\text{-}\mathbf{Alg} \longrightarrow R\text{-}\mathbf{Mod}$$

has a left adjoint free functor

$$T : R\text{-}\mathbf{Mod} \longrightarrow R\text{-}\mathbf{Alg}$$

called the “tensor algebra” $M \mapsto T(M)$.

Unfortunately, we didn’t have time to get into this.]

More generally, if M is any R -module defined by a ring homomorphism

$$\lambda : R \longrightarrow \text{End}_{\mathbf{Ab}}(M),$$

then since R is commutative we automatically (“tautologically”) obtain an R -algebra of R -linear endomorphisms

$$\lambda : R \longrightarrow \text{im } \lambda \subseteq Z(\text{End}_R(M)).$$

In this sense, endomorphisms of R -modules are the “concrete” prototype for R -algebras, and thus tell us how we should define “representations” of algebras.

This also motivates the following, more “concrete,” definition of R -algebras.

★ Concrete Definition

Let R be a commutative ring. An R -algebra is a structure $(M, \lambda, \circ, \text{id})$ where

- (M, λ) is an R -module,
- $(M, \circ, \text{id})$ is a monoid structure making M into a ring (i.e. $\circ$ distributes over addition in M),
- the operation $\circ : M \times M \rightarrow M$ is compatible with λ (i.e. it is R -bilinear).

[Any book on representation theory expects you to know this, but no algebra text that I know of actually teaches this.]

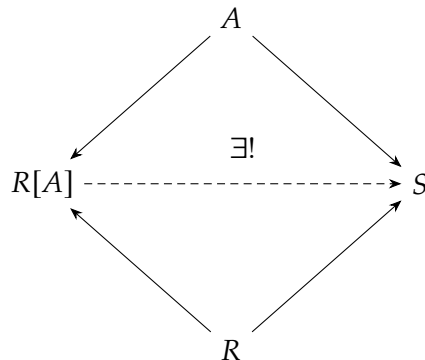
2. The category $R\text{-}\mathbf{CAlg}$

If we restrict our attention to R -algebras $\varphi : R \rightarrow S$ in which S is a commutative ring then we obtain a full subcategory

$$R\text{-}\mathbf{CAlg} \subseteq R\text{-}\mathbf{Alg},$$

called the category of commutative R -algebras. This was the subject of HW3.

Recall that for any set $A \in \mathbf{Set}$ there exists a free commutative algebra $A \rightarrow R[A]$ satisfying the universal property



[There are too many maps here to give them names, so we'll have to be careful!]

If $|A| = n$ then we can identify $R[A]$ with the polynomial algebra in n commuting variables, called

$$R[x_1, x_2, \dots, x_n].$$

[Formally, we can view this as the “symmetric algebra” $S(R^{\oplus A})$ of the free module $R^{\oplus A}$. This

$$S : R\text{-}\mathbf{Mod} \longrightarrow R\text{-}\mathbf{CAlg}$$

is the left adjoint to the forgetful functor $U : R\text{-}\mathbf{CAlg} \rightarrow R\text{-}\mathbf{Mod}$. If $F : \mathbf{Set} \rightarrow R\text{-}\mathbf{Mod}$ is the free module functor then we obtain

$$R[A] = S(F(A)).$$

Unfortunately, we didn't have time to get into this.]

Now suppose that $A = \{s_1, s_2, \dots, s_n\} \hookrightarrow S$. Then we obtain the canonical “evaluation morphism” from the free algebra

$$\begin{aligned} \varphi_A : R[x_1, \dots, x_n] &\longrightarrow S, \\ f(x_1, \dots, x_n) &\longmapsto f(s_1, \dots, s_n). \end{aligned}$$

We say that

- $A \hookrightarrow S$ is R -algebraically independent if φ_A is injective,
- $A \hookrightarrow S$ is R -algebraically generating if φ_A is surjective.

We also sometimes use the notation

$$R[A] = \text{im } \varphi_A \subseteq S,$$

but be warned: this notation conflicts with the previous usage because $\text{im } \varphi_A$ is a free algebra only when $\ker \varphi_A = 0$. We'll try not to get confused.

Furthermore, we say that

- $A \hookrightarrow S$ is R -algebraically almost-generating if S is algebraic over the subring $R[A] = \text{im } \varphi_A$.

[This means that for all $s \in S$ the evaluation morphism

$$\varphi_s : R[A][x] \longrightarrow S$$

is non-injective.]

- $A \hookrightarrow S$ is an R -transcendence basis if it is algebraically independent and almost-generating.

Theorem (★ Steinitz Exchange for Algebras). *Let $S \in R\text{-CAlg}$ be an integral domain, let $A \subseteq S$ be algebraically independent and let $B \subseteq S$ be almost-generating. Then we can replace elements of B by elements of A while maintaining the almost-generating property. It follows that*

$$|A| \leq |B|$$

and hence all R -transcendence bases have the same cardinality, called the R -transcendence degree of S .

No, you do not need to memorize the proof of this theorem, but you should understand the proof [in the HW3 solutions] and be able to reproduce bits and pieces of it.

Here are the key lemmas.

Lemma (1). If $K_1 \subseteq K_2 \subseteq K_3$ are fields then

$$[K_3 : K_1] = [K_3 : K_2][K_2 : K_1].$$

Lemma (2). Let $K \subseteq L$ be fields.

- If $[L : K] < \infty$ then $K \subseteq L$ is algebraic.
- If $K \subseteq L$ is algebraic and finitely generated as a K -algebra then $[L : K] < \infty$.

Lemma (3). Let $K \subseteq L$ be fields. Then the algebraic closure

$$K \subseteq \text{Alg}_K(L) \subseteq L$$

is a subring of L (in fact a subfield).

Lemma (4). Let $K_1 \subseteq K_2 \subseteq K_3$ be fields. If

$$\text{Alg}_{K_1}(K_2) = K_2 \quad \text{and} \quad \text{Alg}_{K_2}(K_3) = K_3,$$

then $\text{Alg}_{K_1}(K_3) = K_3$.

Lemma (5). Lemmas 3 & 4 also hold for integral domains by the trick of “localization.”

[This whole story generalizes to situations where R is an integral domain but S need not be (see “Noether Normalization”). However, the proofs and statements become more difficult.]

3. Matrix Notation

Recall that finite products and coproducts coincide in the category $R\text{-Mod}$ (more generally, in any “additive category”). We denote this “biproduct” by $\oplus$.

Now consider any homomorphism between direct sums of modules

$$\varphi : M_1 \oplus M_2 \longrightarrow N_1 \oplus N_2.$$

By interpreting $N_1 \oplus N_2$ as a product and $M_1 \oplus M_2$ as a coproduct we obtain a diagram:

$$\begin{array}{ccccc}
 M_1 & & & & N_1 \\
 & \searrow i_1 & & \nearrow \pi_1 & \\
 & M_1 \oplus M_2 & \xrightarrow{\varphi} & N_1 \oplus N_2 & \\
 & \nearrow i_2 & & \searrow \pi_2 & \\
 M_2 & & & & N_2
 \end{array}$$

By the universal properties of product & coproduct, the map φ is uniquely determined by its four “components”

$$\varphi_{ij} := \pi_i \circ \varphi \circ i_j : M_j \longrightarrow N_i.$$

Furthermore, we can express φ in terms of the following matrix notation. Given elements

$$m = m_1 + m_2 \in M_1 \oplus M_2 \quad \text{and} \quad n = n_1 + n_2 \in N_1 \oplus N_2$$

we will write

$$m = \begin{pmatrix} m_1 \\ m_2 \end{pmatrix} \quad \& \quad n = \begin{pmatrix} n_1 \\ n_2 \end{pmatrix},$$

and then we have

$$\varphi(m) = \begin{pmatrix} \varphi_{11} & \varphi_{12} \\ \varphi_{21} & \varphi_{22} \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \end{pmatrix} = \begin{pmatrix} \varphi_{11}(m_1) + \varphi_{12}(m_2) \\ \varphi_{21}(m_1) + \varphi_{22}(m_2) \end{pmatrix}.$$

Now suppose we have a homomorphism between free modules

$$\varphi : R^{\oplus A} \longrightarrow R^{\oplus B}.$$

If the bases are $A = \{a_1, \dots, a_m\}$ and $B = \{b_1, \dots, b_n\}$ then we can write this as

$$\varphi : \bigoplus_{j=1}^m (a_j) \longrightarrow \bigoplus_{i=1}^n (b_i),$$

and we can define the component morphisms

$$\varphi_{ij} := \pi_i \circ \varphi \circ i_j : (a_j) \longrightarrow (b_i).$$

If we denote an element $r = \sum_j r_j a_j \in R^{\oplus A}$ as a column vector

$$r = \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{pmatrix},$$

then by iterating the construction above we can express φ in matrix notation as follows:

$$\begin{aligned} \varphi(r) &= \begin{pmatrix} \varphi_{11} & \cdots & \varphi_{1m} \\ \vdots & & \vdots \\ \varphi_{n1} & \cdots & \varphi_{nm} \end{pmatrix} \begin{pmatrix} r_1 \\ \vdots \\ r_m \end{pmatrix} \\ &= \begin{pmatrix} \sum_j \varphi_{1j}(r_j) \\ \vdots \\ \sum_j \varphi_{nj}(r_j) \end{pmatrix}. \end{aligned}$$

Since each component

$$\varphi_{ij} : (a_j) \longrightarrow (b_i)$$

is a homomorphism of rank 1 free modules it is uniquely determined by the ring element $s_{ij} \in R$ such that

$$\varphi_{ij}(a_j) = s_{ij}b_i.$$

Thus it is more meaningful to write

$$\varphi = \begin{pmatrix} s_{11} & \cdots & s_{1m} \\ \vdots & & \vdots \\ s_{n1} & \cdots & s_{nm} \end{pmatrix}.$$

By composing homomorphisms between free modules we recover the familiar definition of matrix multiplication. In summary, if F_1, F_2 are finitely generated free R -modules then any choice of bases

$$F_1 = R^{\oplus A} \quad \& \quad F_2 = R^{\oplus B}$$

determines an isomorphism of abelian groups

$$\Phi_{AB} : \text{Hom}_R(F_1, F_2) \xrightarrow{\sim} \text{Mat}_{|B| \times |A|}(R),$$

an isomorphism of rings

$$\Phi_A : \text{End}_R(F_1) \xrightarrow{\sim} \text{Mat}_{|A|}(R),$$

and an isomorphism of groups

$$\Phi_A : \text{Aut}_R(F_1) \xrightarrow{\sim} \text{GL}_{|A|}(R).$$

[If, moreover, R is commutative then the first is an isomorphism of R -modules & the second is an isomorphism of R -algebras.]

The great benefit of matrix notation is that we can use Gaussian elimination to obtain constructive proofs and a more concrete understanding of modules (at least over nice rings).

Specifically, we define the elementary matrices

$$E_{ij}(r), \quad E_{ii}(r), \quad P_{ij} \in \mathrm{GL}_n(R).$$

If R is a Euclidean domain then these generate the group $\mathrm{GL}_n(R)$.

If R is, more generally, a PID then we also need the pseudo-elementary matrices:

$$E_{ij} \begin{pmatrix} a & b \\ c & d \end{pmatrix} := I + (a-1)e_{ii} + be_{ij} + ce_{ji} + (d-1)e_{jj}.$$

Now we can describe the action of the group $\mathrm{GL}_m(R) \times \mathrm{GL}_n(R)$ on the set $\mathrm{Mat}_{m \times n}(R)$ by left and right multiplication

$$(P, Q) \cdot A := PAQ^{-1}$$

in terms of algorithms involving elementary (and pseudo-elementary) row and column operations. This leads to the following theorem.

Theorem (★ Smith Normal Form). *Let R be a PID and consider a matrix $A \in \mathrm{Mat}_{m \times n}(R)$. Then there exist invertible matrices*

$$P \in \mathrm{GL}_m(R) \quad \& \quad Q \in \mathrm{GL}_n(R)$$

and ideals

$$(r_d) \subseteq \cdots \subseteq (r_2) \subseteq (r_1) \subseteq R$$

such that

$$PAQ^{-1} = \left(\begin{array}{cccc|cc} r_1 & 0 & \cdots & 0 & 0 & \\ 0 & r_2 & \ddots & \vdots & \vdots & \\ \vdots & \ddots & \ddots & 0 & \vdots & \\ 0 & \cdots & 0 & r_d & 0 & \\ \hline 0 & \cdots & \cdots & 0 & 0 & 0 \end{array} \right).$$

Furthermore, the elements $r_1 \mid r_2 \mid \cdots \mid r_d$ are unique up to multiplication by units. We call PAQ^{-1} the Smith Normal Form of A .

4. FTFGMPID, Part II

This allows us to completely determine the structure of finitely generated modules over a PID. Let R be a PID and let $M \in R\text{-Mod}$ be finitely generated. Then by results from before the Spring Break we obtain a short exact sequence

$$0 \longrightarrow R^{\oplus m} \xrightarrow{\varphi} R^{\oplus n} \longrightarrow M \longrightarrow 0,$$

so that

$$M \cong R^{\oplus n} / \text{im } \varphi.$$

By the Smith Normal Form, we can choose bases $A \subseteq R^{\oplus m}$ & $B \subseteq R^{\oplus n}$ so that

$$[\varphi]_{AB} = \underbrace{\begin{pmatrix} r_1 & 0 & \cdots & 0 & 0 \\ 0 & r_2 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & 0 & r_{m-1} & 0 \\ 0 & \cdots & \cdots & 0 & r_m \\ \hline 0 & \cdots & \cdots & 0 & 0 \end{pmatrix}}_m \Bigg\} n.$$

Then it follows from the argument on HW4 Problem 1 that

$$M \cong \frac{R^{\oplus n}}{\text{im } \varphi} \cong R^{\oplus(n-m)} \oplus \bigoplus_{i=1}^m \frac{R}{(r_i)}.$$

In other words, M is isomorphic to a direct sum of cyclic modules. The number $n - m$ (called the rank) and the ideals

$$(r_m) \subseteq \cdots \subseteq (r_2) \subseteq (r_1)$$

(called the invariant factors) are isomorphism invariants of the module M .

5. $K[x]$ -modules

The FTFGMPID has an immediate application to abelian groups ($\mathbb{Z}$ -modules) but I think the more interesting applications relate to the “other” PID (namely, $K[x]$ where K is a field).

Using the fact that $K[x]$ is the free K -algebra generated by one element gives us a bijection between $K[x]$ -modules and pairs (V, φ) where

- V is a K -vector space,
- $\varphi \in \text{End}_K(V)$ is a K -linear endomorphism.

If V is finite dimensional (say $V = K^{\oplus n}$) then by choosing a basis we can think of φ as an $n \times n$ matrix

$$A = [\varphi] \in \text{Mat}_n(K).$$

Furthermore, two such modules (V_1, A_1) & (V_2, A_2) are isomorphic as $K[x]$ -modules if and only if there exists an invertible matrix $P \in \text{GL}_n(K)$ such that

$$PA_1P^{-1} = A_2.$$

Finally, since $K[x]$ is a PID we obtain from the FTFGMPID an isomorphism of $K[x]$ -modules

$$V \cong \bigoplus_{i=1}^d \frac{K[x]}{(f_i(x))}$$

for some unique monic nonconstant polynomials

$$f_1(x) \mid f_2(x) \mid \cdots \mid f_d(x).$$

[Recall: $f_d(x)$ is the minimal polynomial & $\prod_i f_i(x)$ is the characteristic polynomial of the corresponding morphism.]

By choosing the most obvious bases for the direct summands we can then obtain the Rational Canonical Form and the Jordan Canonical Form (when K is algebraically closed) of the corresponding matrix.



There is more to say but it's been a long semester so I think I'll stop here.